MUST-SCORE FACTS

Science and Technology

Consolidated Must-Score Facts and must-score drill cards for every chapter.

UPPCS + UKPCS Study Library · chapter order follows Subject Notes.

Chapters

1. Daily Read — High-Yield Revision Facts
2. Topic 1 — Living World, Cell, Genetics and Biotechnology
3. Topic 2 — Animal Biology and Husbandry
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5. Topic 4 — Digestion, Respiration and Excretion
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27. Chemistry Topic 9 — Discoveries, Discoverers and Miscellaneous Chemistry

Daily Read — High-Yield Revision Facts

1. Complete Vitamin Master Reference & Deficiency Diseases

Vitamin	Chemical / Scientific Name	Fat / Water Soluble	Key Deficiency Disease / Symptoms	Primary Natural Food Sources
Vitamin A	**Retinol**	Fat-soluble	**Night blindness (Nyctalopia)**, **Xerophthalmia** (dry cornea), Keratomalacia	Carrot, Papaya, Mango, Liver, Fish oil, Golden Rice
Vitamin B1	**Thiamine**	Water-soluble	**Beri-beri** (Peripheral neuropathy / cardiac failure), Wernicke-Korsakoff syndrome	Unpolished rice, yeast, whole grains (*lost in polished rice*)
Vitamin B2	**Riboflavin**	Water-soluble	**Cheilosis** (cracking at corners of mouth), Glossitis (red swollen tongue)	Milk, eggs, liver (*gives yellow colour to milk/whey*)
Vitamin B3	**Niacin (Nicotinic acid)**	Water-soluble	**Pellagra** (4Ds: Dermatitis, Diarrhea, Dementia, Death)	Yeast, meat, fish, legumes
Vitamin B5	**Pantothenic acid**	Water-soluble	Burning feet syndrome, fatigue	Mushroom, eggs, broccoli
Vitamin B6	**Pyridoxine**	Water-soluble	Peripheral neuropathy, microcytic anemia, convulsions in infants	Cereals, bananas, meat
Vitamin B7	**Biotin (Vitamin H)**	Water-soluble	Hair loss (alopecia), dermatitis	Egg yolk, nuts, intestinal flora (*avidin in raw egg binds biotin*)
Vitamin B9	**Folic Acid (Folate)**	Water-soluble	**Megaloblastic anemia**, Neural tube defects in fetuses (Spina bifida)	Green leafy vegetables, citrus fruits
Vitamin B12	**Cyanocobalamin**	Water-soluble	**Pernicious anemia**, demyelination of spinal cord (*contains Cobalt metal*)	Meat, fish, eggs, dairy (*Absent in plant foods*)
Vitamin C	**Ascorbic acid**	Water-soluble	**Scurvy** (bleeding gums, petechiae, impaired wound healing); heat-sensitive (*destroyed by cooking*)	Aonla (highest source), Guava, Citrus fruits (Lemon, Orange)
Vitamin D	**Calciferol** (D_2: Ergocalciferol, D_3: Cholecalciferol)	Fat-soluble	**Rickets** in children (bowed legs), **Osteomalacia** in adults (*hormone-like vitamin*)	Sunlight synthesis in skin, Cod liver oil
Vitamin E	**Tocopherol**	Fat-soluble	Infertility / reproductive failure, muscle weakness, hemolytic anemia (*Beauty Vitamin / Anti-oxidant*)	Wheat germ oil (richest), sunflower seeds, almonds
Vitamin K	**Phylloquinone** (K_1), Menaquinone (K_2)	Fat-soluble	**Defective blood clotting**, prolonged bleeding time (needed for synthesis of prothrombin in liver)	Spinach, kale, synthesized by human gut bacteria

> **Fat-Soluble vs Water-Soluble Mnemonic:**
> Fat-soluble: **K - E - D - A** (stored in liver and adipose tissue, hypervitaminosis possible).
> Water-soluble: **B-complex and C** (excreted in urine, regular intake required).

2. Infectious Diseases & Causative Pathogens Master Matrix

A. Viral Diseases

Disease	Causative Virus	Mode of Transmission / Key Diagnostic Feature
COVID-19	SARS-CoV-2 (enveloped positive-sense RNA)	Respiratory droplets; RT-PCR test
Rabies (Hydrophobia)	Lyssavirus / Rhabdovirus (bullet-shaped RNA)	Bite of rabid dogs, bats; affects central nervous system
Poliomyelitis	Poliovirus (Enterovirus, single-stranded RNA)	Fecal-oral route; destroys anterior horn motor neurons of spinal cord; Oral Polio Vaccine (Sabin), Inactivated Polio Vaccine (Salk)
AIDS	Human Immunodeficiency Virus (HIV - Retrovirus with Reverse Transcriptase)	Blood, sexual, maternal; attacks **CD4+ Helper T-cells**; ELISA test, Western Blot confirmatory
Dengue (Break-bone fever)	Flavivirus (DEN-1 to DEN-4, ssRNA)	Transmitted by **Aedes aegypti** mosquito (day-biting); severe thrombocytopenia (platelet drop)
Chikungunya	Chikungunya virus (Alphavirus)	Transmitted by **Aedes aegypti** and *Aedes albopictus*; severe joint pain
Yellow Fever	Flavivirus	*Aedes aegypti* mosquito vector
Zika Virus	Flavivirus	*Aedes* mosquito; causes **Microcephaly** in newborns
Hepatitis B	Hepatitis B Virus (HBV - **DNA virus**, whereas Hep A, C, D, E are RNA)	Blood, body fluids; attacks liver; vaccine available
Smallpox	Variola virus (DNA virus)	Completely **eradicated from world in 1980** (WHO)
Measles, Mumps, Rubella	Morbillivirus, Paramyxovirus, Rubivirus	MMR vaccine

B. Bacterial Diseases

Disease	Causative Bacterium	Key Clinical Feature / Diagnostic Test
Tuberculosis (TB)	*Mycobacterium tuberculosis* (acid-fast bacillus)	Affects lungs; **Mantoux test**; **BCG vaccine** (Bacillus Calmette-Guérin)
Cholera	*Vibrio cholerae* (comma-shaped)	*"Rice water stools"*, massive dehydration; fecal-oral route
Typhoid (Enteric fever)	*Salmonella typhi*	Step-ladder fever pattern, intestinal perforation; **Widal test**
Tetanus (Lockjaw)	*Clostridium tetani* (anaerobic, drumstick-shaped)	Neurotoxin tetanospasmin causes severe muscle spasms; ATS / Tetanus toxoid
Diphtheria	*Corynebacterium diphtheriae*	Grey pseudomembrane over throat / larynx; **Schick test**
Pertussis (Whooping Cough)	*Bordetella pertussis*	Paroxysmal coughing ending in whoop; DPT triple vaccine
Plague (Black Death)	*Yersinia pestis*	Transmitted by **Xenopsylla cheopis** (oriental rat flea); Bubonic, Pneumonic forms
Leprosy (Hansen's Disease)	*Mycobacterium leprae*	Hypopigmented skin patches with loss of sensation; Lepromin test
Anthrax	*Bacillus anthracis* (spore-forming)	Zoonotic disease of cattle; potential bioweapon
Syphilis & Gonorrhea	*Treponema pallidum* & *Neisseria gonorrhoeae*	Sexually transmitted bacterial infections

C. Protozoan & Other Pathogen Diseases

- **Malaria:** Caused by protozoan **Plasmodium** (*P. vivax, P. falciparum* [deadliest malignant tertian], *P. malariae, P. ovale*). Vector: **Female Anopheles mosquito**. Infective stage to humans: **Sporozoite**. Parasite multiplies in liver

cells and destroys red blood cells (**RBCs**), releasing **hemozoin toxin** causing periodic shivering chills and high fever. Discovered by **Sir Ronald Ross** (in Secunderabad, India - Nobel Prize 1902). Treatment: Quinine (extracted from **Cinchona bark**), Artemisinin.

- **Kala-Azar (Visceral Leishmaniasis / Dum-Dum Fever):** Caused by *Leishmania donovani*. Vector: **Phlebotomus sandfly**. Enlarges spleen and liver (hepatosplenomegaly).
- **Sleeping Sickness (African Trypanosomiasis):** Caused by *Trypanosoma brucei*. Vector: **Tsetse fly** (*Glossina*).
- **Amoebic Dysentery:** Caused by *Entamoeba histolytica*.
- **Ringworm (Tinea):** Fungal infection (*Trichophyton, Microsporum, Epidermophyton*).
- **Filariasis / Elephantiasis:** Caused by parasitic nematode worm *Wuchereria bancrofti*. Transmitted by **Culex mosquito**. Blocks lymphatic vessels causing swelling of limbs/scrotum.

3. Human Physiology & Endocrine Hormones

Gland	Hormones Produced	Primary Function & Prelims Exam Triggers
Pituitary (Master Gland)	Growth Hormone (GH), TSH, ACTH, FSH, LH, Prolactin, **Oxytocin**, **Vasopressin (ADH)**	Regulated by Hypothalamus. Hypersecretion of GH causes **Gigantism / Acromegaly**; hyposecretion causes **Dwarfism**. **ADH (Antidiuretic Hormone)** controls water reabsorption in kidneys; deficiency causes **Diabetes Insipidus** (frequent urination without sugar). **Oxytocin** is the milk-ejection & birth hormone.
Thyroid	**Thyroxine (T_4)**, Triiodothyronine (T_3), Calcitonin	Largest endocrine gland; requires **Iodine**. Deficiency causes **Simple Goitre** (enlarged neck), **Cretinism** in infants, **Myxedema** in adults. Hyperthyroidism causes **Grave's Disease** (exophthalmic goitre).
Parathyroid	Parathyroid Hormone (PTH / Collip's hormone)	**Increases blood calcium levels** (hypercalcemic) by dissolving bone; acts antagonistically to thyroid calcitonin.
Pancreas (Islets of Langerhans)	**Insulin** (from \beta-cells), **Glucagon** (from \alpha-cells), Somatostatin (from \delta-cells)	**Insulin lowers blood glucose** by converting glucose to glycogen (glycogenesis). Deficiency causes **Diabetes Mellitus** (hyperglycemia, glycosuria). Discovered by Banting & Best (1921). Glucagon raises blood glucose.
Adrenal Gland	Adrenaline (Epinephrine) & Noradrenaline; Cortisol, Aldosterone	Adrenaline is the **"Emergency Hormone" / "Fight, Flight or Fright" (3F/4S hormone)**; increases heart rate, breathing, blood pressure.
Pineal Gland	**Melatonin**	Regulates circadian rhythm (24-hour sleep-wake cycle).

> **Blood Groups & ABO System:** Discovered by **Karl Landsteiner** (1900).
> - **Universal Donor: O^- (O Negative)** — has neither A nor B antigens and no Rh factor on RBCs.
> - **Universal Recipient: AB^+ (AB Positive)** — has both A and B antigens and Rh factor; has no anti-A or anti-B antibodies in plasma.
> - **Normal Blood Pressure: 120/80 mm Hg** (Systolic/Diastolic), measured by **Sphygmomanometer**. Blood pH: **7.35–7.45 (slightly basic)**.

4. Optics & Physics Principles (High-Frequency Exam Traps)

Phenomenon	Underlying Optical Principle	Exam Application / Examples
Mirage in deserts / Looming in cold	**Total Internal Reflection (TIR)**	Optical illusion; light travels from denser to rarer medium at angle > critical angle.
Sparkling of Diamond	**Total Internal Reflection (TIR)**	Very small critical angle of diamond (□pprox 24.4^{\circ}) due to high refractive index (□pprox 2.42).

Phenomenon	Underlying Optical Principle	Exam Application / Examples
Endoscope / Optical Fibre Cables	**Total Internal Reflection (TIR)**	Zero signal loss communication; invented by Narinder Singh Kapany.
Blue colour of sky / Red colour of sunset	**Rayleigh Scattering of Light**	Scattering $\propto \frac{1}{\lambda^4}$. Blue light (shorter wavelength) scatters most; red light (longest wavelength) scatters least and reaches eyes at horizon.
Twinkling of Stars	**Atmospheric Refraction**	Continuously fluctuating refractive index of atmospheric air layers.
Coin appears raised / Stick bent in water	**Refraction of Light**	Light bends away from normal when entering air from water.
Rainbow Formation	**Dispersion + Refraction + Total Internal Reflection**	Water droplets act as tiny prisms. Primary rainbow: 2 refractions + 1 internal reflection; Secondary rainbow: 2 refractions + 2 internal reflections.
Soap bubble shining with brilliant colours	**Interference of Light**	Destructive and constructive interference of light reflected from upper and lower thin film surfaces.

Common Eye Defects & Corrective Lenses

1. **Myopia (Short-sightedness):** Nearby objects seen clearly, distant objects blurred. Image formed **in front of retina**. Corrected by **Concave Lens** (diverging lens).
2. **Hypermetropia (Long-sightedness):** Distant objects seen clearly, nearby objects blurred. Image formed **behind retina**. Corrected by **Convex Lens** (converging lens).
3. **Presbyopia (Old age vision):** Loss of accommodation power of ciliary muscles. Corrected by **Bifocal Lens** (upper part concave for distance, lower part convex for reading).
4. **Astigmatism:** Inability to focus simultaneously on horizontal and vertical lines due to irregular cornea curvature. Corrected by **Cylindrical Lens**.

5. Commercial Names of Common Chemical Compounds

Chemical / Common Commercial Name	Chemical Formula	Scientific / IUPAC Name	High-Yield Application
Baking Soda	NaHCO_3	Sodium hydrogen carbonate	Baking, antacid, fire extinguisher
Washing Soda	$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$	Sodium carbonate decahydrate	Glass, soap, water softening
Caustic Soda	NaOH	Sodium hydroxide	Soap manufacturing, petroleum refining
Caustic Potash	KOH	Potassium hydroxide	Soft soaps, alkaline batteries
Bleaching Powder	CaOCl_2	Calcium hypochlorite	Disinfectant for water, bleaching cotton
Quicklime	CaO	Calcium oxide	Cement manufacture, white washing
Slaked Lime	Ca(OH)_2	Calcium hydroxide	Mortar, neutralizing soil acidity
Plaster of Paris (POP)	$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$	Calcium sulphate hemihydrate	Bone fractures, statues, false ceilings
Gypsum	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	Calcium sulphate dihydrate	Retarder in cement setting, soil reclamation
Laughing Gas	N_2O	Nitrous oxide	Dental anaesthetic, propellant

Chemical / Common Commercial Name	Chemical Formula	Scientific / IUPAC Name	High-Yield Application
Tear Gas	\text{CCl}_3\text{NO}_2 (Chloropicrin) / CS gas	2-chlorobenzalmalononitrile	Riot control
Heavy Water	\text{D}_2\text{O}	Deuterium oxide	Moderator in nuclear reactors
Dry Ice	\text{Solid CO}_2	Solid carbon dioxide	Refrigerant, sublimes without melting
Epsom Salt	\text{MgSO}_4 \cdot 7\text{H}_2\text{O}	Magnesium sulphate heptahydrate	Laxative, bath salt
Blue Vitriol	\text{CuSO}_4 \cdot 5\text{H}_2\text{O}	Copper sulphate pentahydrate	Fungicide (Bordeaux mixture), electroplating
Green Vitriol	\text{FeSO}_4 \cdot 7\text{H}_2\text{O}	Ferrous sulphate heptahydrate	Ink manufacturing
White Vitriol	\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}	Zinc sulphate heptahydrate	Eye drops, pigments

6. Major Alloys & Their Compositions

Alloy	Major Constituents (%)	Primary Properties & Applications
Brass (Pital)	**Copper (60–70%) + Zinc (30–40%)**	Utensils, musical instruments, cartridge cases (*No Tin*)
Bronze (Kansa)	**Copper (88%) + Tin (12%)**	Statues, medals, coins, bells (*No Zinc*)
German Silver	**Copper (50%) + Zinc (30%) + Nickel (20%)**	**Contains ZERO Silver!** Utensils, artificial jewellery
Stainless Steel	**Iron + Chromium (18%) + Nickel (8%) + Carbon (0.1–1%)**	Rust-resistant; Chromium gives corrosion resistance
Solder (Tanka)	**Lead (50%) + Tin (50%)**	Low melting point; joining electrical wires
Duralumin	**Aluminium (95%) + Copper (4%) + Magnesium (0.5%) + Manganese (0.5%)**	Light and strong; aircraft bodies, pressure cookers
Gun Metal	**Copper (88%) + Tin (10%) + Zinc (2%)**	Heavy machinery gears, gun barrels
Bell Metal	**Copper (78%) + Tin (22%)**	High resonance for casting bells
Amalgam	**Mercury + Any other metal** (except Iron, Platinum, Tungsten)	Dental fillings (Silver-Mercury amalgam); **Iron does not form amalgam**
Nichrome	**Nickel (60%) + Chromium (40%)**	High resistance & high melting point; electric heater coils

7. Indian Space Programme (ISRO) & Defence Tech

- **ISRO Founded: 15 August 1969**; Headquarters: **Bengaluru**; Space Commission & Department of Space (DoS) formed in 1972.
- **Launch Port:** Satish Dhawan Space Centre (SDSC - SHAR), **Sriharikota (Andhra Pradesh)**, on Pulicat Lake barrier island. Second spaceport under construction at **Kulasekarapattinam (Tamil Nadu)** for SSLV launches.
- **Key Historic Missions:**
- **Aryabhata:** India's first satellite, launched on **19 April 1975** by Soviet Kosmos-3M rocket.
- **Chandrayaan-1:** Launched on **22 October 2008** via PSLV-C11; Moon Impact Probe detected water molecules (H_2O/OH) on lunar surface.

- **Mangalyaan (Mars Orbiter Mission - MOM):** Launched on **5 November 2013** via PSLV-C25; inserted into Martian orbit on **24 September 2014** on maiden attempt.
- **Chandrayaan-3:** Launched on **14 July 2023** via LVM3-M4; successfully achieved historic **soft landing near lunar South Pole** on **23 August 2023** (celebrated as **National Space Day**). Lander: *Vikram*, Rover: *Pragyan*. Landing site named **Shiv Shakti Point**.
- **Aditya-L1:** India's first solar observatory launched on **2 September 2023** via PSLV-C57; inserted into halo orbit around **Lagrange Point 1 (L1)** on 6 January 2024.
- **DRDO Missiles (Integrated Guided Missile Development Programme - IGMDP, 1983 - Dr. APJ Abdul Kalam):**
- **P - Prithvi:** Surface-to-Surface short-range ballistic missile (Liquid fuel).
- **A - Agni:** Surface-to-Surface medium-to-intercontinental ballistic missile (Agni I to V, Solid fuel, MIRV tech).
- **T - Trishul:** Surface-to-Air short-range missile.
- **N - Nag:** Anti-Tank guided missile (*"Fire and Forget"*); helicopter-launched version is **Helina / Dhruvastra**.
- **A - Akash:** Surface-to-Air medium-range missile with Rajendra multi-target phased array radar.
- **Nuclear Power Plants in India:** 1. **Narora Atomic Power Station (NAPS): Bulandshahr, Uttar Pradesh** (PHWR, only nuclear plant in UP) 2. **Tarapur:** Maharashtra (Oldest nuclear plant in India, 1969) 3. **Rawatbhata:** Rajasthan (near Kota, Chambal river) 4. **Kakrapar:** Gujarat (Tapi river) 5. **Kaiga:** Karnataka (Kali river) 6. **Kalpakkam (MAPS):** Tamil Nadu (Indira Gandhi Centre for Atomic Research) 7. **Kudankulam:** Tamil Nadu (Largest nuclear power capacity in India, VVER Russian reactors) 8. **Gorakhpur Nuclear Plant:** Under construction in **Fatehabad, Haryana** (Not in Gorakhpur, UP!)

Topic 1 — Living World, Cell, Genetics and Biotechnology

Consolidated — 50 Must-Score Facts

1. The word **'Science'** originates from the Latin word *scientia*, meaning **'knowledge'**; science is the systematic empirical knowledge of the physical world gained through observation and experimentation.
2. The term **'Biology'** was coined independently by French biologist **Jean-Baptiste Lamarck** and German naturalist **Gottfried Reinhold Treviranus** in **1802**; **Aristotle** is the Father of Biology and Zoology; **Theophrastus** is the Father of Botany (*Historia Plantarum*); **Hippocrates** is the Father of Medicine.
3. Defining properties of life are **Cellular organisation**, **Metabolism** (sum of all chemical reactions), and **Consciousness** (response to stimuli); growth and reproduction are **not** absolute defining properties (mules, worker bees, infertile human couples do not reproduce).
4. **Key Specialized Biological Branches**:
5. **Key Agricultural & Applied Culture Terminologies**:
6. **Origin of Life (Oparin-Haldane Hypothesis)**:
7. **The Primitive Earth Atmosphere**:
8. **The Miller-Urey Experiment (1953)**:
9. **Geological Chronology of Cenozoic Epochs (Oldest to Most Recent)**:
10. **Lamarckism (Theory of Inheritance of Acquired Characteristics)**:
11. **Darwinism (Theory of Natural Selection)**:
12. **Neo-Darwinism & Modern Synthetic Theory**:
13. **Hugo de Vries' Mutation Theory (1901)**:
14. **Landmark Evolutionary Fossils & Missing Links**:
15. **Augustin Pyramus de Candolle (A. P. de Candolle)**: Swiss botanist who **first coined the term 'Taxonomy'** and was the **first to use vascular tissue characteristics** to classify plants into *Vasculares* (with vascular bundles) and *Cellulares* (without vascular bundles).
16. **Carolus Linnaeus** (*Systema Naturae*, 1758) established the **Binomial Nomenclature** and is the Father of Modern Taxonomy; scientific names consist of **Genus** (capitalised) and **species epithet** (lowercase), printed in *italics*.
17. The taxonomic hierarchy in ascending order is: **Species → Genus → Family → Order → Class → Phylum / Division → Kingdom**; the basic and real unit of classification is **Species** (group of individuals with fundamental similarities capable of interbreeding).
18. **R. H. Whittaker (1969)** proposed the **Five-Kingdom System**: **Monera** (prokaryotes), **Protista** (unicellular eukaryotes), **Fungi** (chitinous heterotrophs), **Plantae** (autotrophs with cellulose wall), and **Animalia** (heterotrophs without cell wall).
19. **Carl Woese (1990)** introduced the **Three-Domain System** based on 16S rRNA sequences: **Bacteria**, **Archaea**, and **Eukarya**; Archaebacteria have branched ether lipids allowing them to survive in extreme environments (methanogens in ruminant guts, halophiles in salt lakes, thermoacidophiles in hot springs).
20. **Antony van Leeuwenhoek (1632–1723)** is the Father of Microbiology; discovered bacteria ("animalcules") using single high-power lenses; bacteria are 0.5–5.0 \mu m in size and visible under a compound microscope.
21. The smallest known living cell capable of autonomous growth and reproduction is ***Mycoplasma gallisepticum*** (0.1–0.3 \mu m, lacks a cell wall, naturally insensitive to penicillin and beta-lactam antibiotics); the largest single cell is an **Ostrich egg**; the longest human cell is the **Neuron**.
22. **Prokaryotes** (bacteria, cyanobacteria) have naked circular DNA in a **nucleoid**, lack membrane-bound organelles, and possess **70S ribosomes** (50S + 30S); **Eukaryotes** have membrane-bound nuclei, linear DNA with histones, and cytoplasmic **80S ribosomes** (60S + 40S).
23. **Cell Wall** chemical facts: Plants = **Cellulose + Hemicellulose + Pectin**; Fungi = **Chitin** (polymer of N-acetylglucosamine); Bacteria = **Peptidoglycan / Murein**; Algae = **Cellulose + Galactans + Mannans + $CaCO_3$**.

24. **Singer and Nicolson (1972)** proposed the **Fluid Mosaic Model** of the plasma membrane: a quasi-fluid bilayer of phospholipids with protein "icebergs floating in a lipid sea"; selectively permeable.
25. **Mitochondria** (Kolliker 1857, Benda 1898) are the powerhouses of the cell (ATP generation via oxidative phosphorylation on inner membrane cristae); contain circular DNA and 70S ribosomes (semi-autonomous); inherited maternally.
26. **Plastids** (Schimper): **Chloroplasts** (photosynthesis, contain chlorophyll and circular DNA); **Chromoplasts** (pigment storage: Lycopene in tomato, Capsanthin in red chilli, Carotene in carrot, Betanin in beetroot); **Leucoplasts** (storage: Amyloplasts = starch, Elaioplasts = oils/fats, Aleuroplasts = proteins).
27. **Endoplasmic Reticulum (ER)**: **Rough ER (RER)** has 80S ribosomes on its surface and synthesises proteins; **Smooth ER (SER)** lacks ribosomes and synthesises lipids, steroid hormones, and detoxifies drugs in the liver.
28. **Golgi Apparatus** (Camillo Golgi, 1898) packages and secretes macromolecules, modifies proteins (glycosylation to form glycoproteins and glycolipids), and forms the **acrosome** of sperm and primary lysosomes.
29. **Lysosomes** (Christian de Duve, 1955) are "suicidal bags" containing hydrolytic enzymes (acid hydrolases active at pH 4.5–5.0) that digest cellular waste and foreign pathogens.
30. **Ribosomes** (George Palade, 1953) are non-membrane-bound ribonucleoprotein particles (rRNA + protein) responsible for protein synthesis; known as the "protein factories" of the cell.
31. **Centrosome / Centrioles** (Boveri, 1888) have a 9+0 cartwheel microtubule arrangement; present in animal cells to form the mitotic spindle and basal bodies; **absent in higher plant cells**.
32. **Peroxisomes** contain catalase and urate oxidase to break down toxic H_2O_2 and perform photorespiration in plants; **Glyoxysomes** convert fats into carbohydrates via the glyoxylate cycle in germinating fatty seeds (castor, groundnut).
33. **Nucleus** was discovered by **Robert Brown (1831)**; the **Nucleolus** (Fontana 1781) is the site of active ribosomal RNA (**rRNA**) synthesis; **Chromatin** was named by Walther Flemming.
34. Human somatic cells have **46 chromosomes (23 pairs)**; gametes have **23 chromosomes (haploid, n)**; 22 pairs are autosomes and 1 pair is sex chromosomes (allosomes: XX in female, XY in male).
35. **Watson and Crick (1953)** proposed the **double-helix model of DNA** based on Rosalind Franklin and Maurice Wilkins' X-ray crystallography; won the Nobel Prize in 1962.
36. In DNA, **Adenine pairs with Thymine** (A = T, two hydrogen bonds) and **Guanine pairs with Cytosine** (G \equiv C, three hydrogen bonds); purines (A, G) always pair with pyrimidines (C, T, U).
37. **Chargaff's Rule**: In double-stranded DNA, the ratio of Adenine to Thymine and Guanine to Cytosine is equal, i.e., A + G = T + C or (A+G)/(T+C) = 1.
38. **RNA** is typically single-stranded, contains **Ribose sugar**, and replaces Thymine with **Uracil**; three functional types: mRNA (messenger), tRNA (transfer / soluble adaptor with cloverleaf secondary structure), and rRNA (ribosomal, structural and catalytic).
39. Nitrogen is an essential constituent of **DNA, RNA, proteins, and chlorophyll**, but is **not** an essential constituent of ordinary **carbohydrates** (composed solely of C, H, and O).
40. The **Cell Cycle** consists of Interphase (G_1 \rightarrow S \rightarrow G_2) and M-Phase; **DNA replication and centriole duplication occur strictly during the S (Synthesis) phase**; chromosome number remains 2n while DNA content doubles from 2C to 4C.
41. **Mitosis** (equational division) occurs in somatic cells, yielding 2 genetically identical diploid (2n) daughter cells; metaphase is the best stage to study chromosome morphology and count; anaphase is the stage where sister chromatids separate.
42. **Colchicine** (alkaloid from *Colchicum autumnale*) inhibits mitotic spindle microtubule formation, arresting cell division at **metaphase**; widely used to induce artificial **polyploidy** in plant breeding.
43. **Meiosis** (reductional division) occurs in germ cells, producing 4 genetically diverse haploid (n) daughter cells; chromosome number is halved during **Anaphase-I**.
44. **Prophase-I of Meiosis has 5 stages**: Leptotene (bouquet stage) → **Zygotene** (synapsis, synaptonemal complex formation) → **Pachytene** (**crossing over** between non-sister chromatids via enzyme **recombinase**) → **Diplotene**

(**chiasmata** formation) → Diakinesis (terminalisation of chiasmata).

45. **Gregor Johann Mendel (1822–1884)**, Father of Genetics, performed hybridisation on garden pea (*Pisum sativum*) for 7 years (1856–1863), tracking 7 contrasting characters; work rediscovered in **1900** by **Hugo de Vries, Carl Correns, and Erich von Tschermak**.
46. Mendel's phenotypic ratios: Monohybrid = **3:1** (genotypic **1:2:1**); Dihybrid = **9:3:3:1**; Test cross ratios = **1:1** (monohybrid) and **1:1:1:1** (dihybrid).
47. **Non-Mendelian inheritance**: **Incomplete dominance** gives an intermediate phenotype (e.g., pink 4 o'clock plant *Mirabilis jalapa* and snapdragon *Antirrhinum* with 1:2:1 phenotypic ratio); **Codominance** expresses both alleles fully (e.g., ABO blood group I^A I^B producing AB blood); **Multiple alleles** (ABO system has 3 alleles I^A, I^B, i, yielding 6 genotypes and 4 phenotypes).
48. **Pleiotropy** is when a single gene influences multiple phenotypic traits (e.g., Phenylketonuria, Sickle-cell anaemia); **Polygenic inheritance** is when a single trait is controlled by three or more genes (e.g., human skin colour, human height).
49. **Sickle-cell anaemia** is an autosomal recessive point mutation on chromosome 11: substitution of **Glutamic acid by Valine** at position 6 of the \beta-globin chain (GAG \rightarrow GUG).
50. Sex-linked recessive disorders carried on the X chromosome include **Haemophilia** (bleeder's / royal disease, factor VIII or IX deficiency) and **Red-Green Colour Blindness** (Daltonism); males are affected far more frequently (XY) while females are usually carriers (X^h X).

📋 Must-Score Drill — Scientists, Organelles & Biotech Tools

Key Biological Scientists & Landmark Milestones

Scientist	Milestone / Contribution	Direct Must-Score
Aristotle	Father of Biology & Zoology	First classified animals into *Anaima* (no red blood) and *Enaima* (red blood)
Theophrastus	Father of Botany	Wrote *Historia Plantarum*, classified 480 plants
Carolus Linnaeus	Father of Taxonomy; Binomial Nomenclature	Authored *Systema Naturae* (1758) and *Species Plantarum* (1753)
Robert Hooke	Discovered the cell (1665)	Observed empty cork compartments in honey-comb pattern; coined term "Cell" in *Micrographia*
Anton van Leeuwenhoek	First living free cell (1674)	Discovered bacteria, protozoa, spermatozoa, and RBCs using handcrafted single lenses
Robert Brown	Discovered cell nucleus (1831)	Identified dark circular body in orchid root cells
Schleiden & Schwann	Cell Theory (1838–1839)	Schleiden (botanist) + Schwann (zoologist): "All living organisms are made of cells"
Rudolf Virchow	Expanded Cell Theory (1855)	Stated ***Omnis cellula-e cellula***: New cells arise only from division of pre-existing cells
Singer & Nicolson	Fluid Mosaic Model (1972)	Structure of plasma membrane (phospholipid bilayer + mosaic of intrinsic/extrinsic proteins)
Camillo Golgi	Discovered Golgi Apparatus (1898)	Used silver nitrate staining method (Black Reaction) on owl nerve cells
Christian de Duve	Discovered Lysosomes (1955)	Identified hydrolytic digestive bags; won Nobel Prize in 1974
George Palade	Discovered Ribosomes (1953)	Discovered non-membranous ribonucleoprotein particles under electron microscope
Watson & Crick	DNA Double Helix Model (1953)	Derived B-DNA structure using Rosalind Franklin & Maurice Wilkins' X-ray data
Gregor Johann Mendel	Father of Genetics	Formulated Laws of Inheritance using garden pea (*Pisum sativum*, 1856–1863)

Scientist	Milestone / Contribution	Direct Must-Score
Hugo de Vries	Mutation Theory (1901)	Coined term "Mutation" from experiments on evening primrose (*Oenothera lamarckiana*)
T. H. Morgan	Father of Experimental Genetics	Discovered sex linkage and gene mapping using fruit fly (*Drosophila melanogaster*)
Kary Mullis	Invented PCR (1983)	Amplified DNA using thermophilic *Taq* polymerase; Nobel Prize in 1993
Alec Jeffreys	Invented DNA Fingerprinting (1984)	Used Variable Number Tandem Repeats (VNTRs) for individual forensic identification
Köhler & Milstein	Hybridoma Technology (1975)	Fused B-lymphocytes with myeloma cells to produce **Monoclonal Antibodies (mAbs)**
Charpentier & Doudna	CRISPR-Cas9 Gene Editing (2012)	Molecular tool for precise genome editing; Nobel Prize in Chemistry 2020

One-Liner Revision – Cell, Genetics and Biotechnology

#	Fast Fact
1	Cell theory = Schleiden (1838) + Schwann (1839) + Virchow (1855)
2	Smallest cell = Mycoplasma (0.1-0.3 micron); Largest single cell = Ostrich egg
3	Prokaryote = **NO nucleus**, NO membrane-bound organelles; e.g., bacteria
4	Eukaryote = **TRUE nucleus** + membrane-bound organelles; e.g., plants, animals
5	Mitochondria = "Powerhouse of cell" (ATP synthesis)
6	Chloroplast = "Kitchen of cell" (photosynthesis); has own DNA + 70S ribosomes
7	Ribosome: Prokaryote = **70S** (50S+30S); Eukaryote = **80S** (60S+40S)
8	Golgi body = "Post Office / Traffic police" of cell (protein packaging and dispatch)
9	Lysosome = "Suicide bag" (autolysis); contains hydrolytic enzymes
10	DNA bases: **A-T** (adenine-thymine) and **G-C** (guanine-cytosine)
11	RNA has **Uracil** instead of Thymine; ribose sugar instead of deoxyribose
12	DNA double helix discovered by **Watson and Crick** (1953); X-ray by Rosalind Franklin
13	Human chromosomes = **46** (23 pairs); sex chromosomes = XY (male), XX (female)
14	Mitosis = 2 identical daughter cells (diploid); for growth and repair
15	Meiosis = 4 haploid daughter cells; for gamete formation; crossing over in Prophase I
16	Mutation = permanent change in DNA; caused by radiation, chemicals, viruses
17	Restriction enzymes = "Molecular scissors"; EcoRI is most famous (cuts at GAATTC)
18	PCR invented by **Kary Mullis (1983)**; Nobel Chemistry 1993
19	First recombinant DNA = Cohen and Boyer (1973); using EcoRI
20	First transgenic plant = tobacco (1983); first transgenic animal = mouse
21	Dolly the sheep = first cloned mammal from adult somatic cell (Wilmut, 1996, UK)
22	Bt gene from **Bacillus thuringiensis**; Bt crops resist Lepidoptera/Coleoptera pests
23	Human Genome Project completed = **April 2003** (~3.2 billion base pairs, ~20,000-25,000 genes)
24	CRISPR-Cas9 = Nobel Chemistry **2020** (Emmanuelle Charpentier + Jennifer Doudna)
25	Golden Rice = transgenic rice with beta-carotene gene; developed to address Vitamin A deficiency
26	Insulin produced by GM bacteria = **Humulin** (Eli Lilly, 1982); first biotech drug

#	Fast Fact
27	Chakrabarty's superbug (1980) = 1st GMO to get US patent; *Pseudomonas putida*
28	RNA interference (RNAi) = Nobel 2006 (Fire and Mello); gene silencing mechanism
29	Stem cells: Totipotent (zygote) > Pluripotent (embryo) > Multipotent (adult)
30	Bioremediation = microorganisms clean pollutants; Chakrabarty's *Pseudomonas* = oil spill

Topic 2 — Animal Biology and Husbandry

Consolidated — 50 Must-Score Facts

1. Animals are eukaryotic, multicellular, heterotrophic organisms lacking cell walls and photosynthetic pigments; store reserve food as **glycogen and fat**.
2. **Porifera** (sponges) exhibit cellular level of organisation, have thousands of minute incurrent pores (**Ostia**) and a single large excurrent opening (**Osculum**), and maintain a unique water canal system lined by flagellated **Choanocytes / Collar cells**.
3. **Cnidaria (Coelenterata)** possess stinging capsules called **Cnidoblasts / Nematocysts** for defence and prey capture; show radial symmetry and tissue-level organisation; exhibit alternation of generation (**Metagenesis**) between sessile cylindrical **Polyp** and free-swimming umbrella-like **Medusa** (e.g. *Hydra*, *Aurelia* jellyfish, *Physalia* Portuguese man-of-war). Corals (*Anthozoa*) secrete massive CaCO_3 skeletons of organic origin. *Hydra* has **no blood but respires** via simple diffusion.
4. **Platyhelminthes** (flatworms) are triploblastic, acoelomate, dorso-ventrally flattened worms; excrete and osmoregulate via specialized **Flame cells (Protonephridia)** (e.g. *Planaria*, *Taenia solium* pork tapeworm, *Fasciola hepatica* liver fluke). Tapeworms are **hermaphrodites** (monoecious).
5. **Aschelminthes / Nematoda** (roundworms) possess a **Pseudocoelom** (false body cavity); sexually dimorphic (separate sexes); examples: *Ascaris lumbricoides* (intestinal roundworm), *Wuchereria bancrofti* (filarial worm causing elephantiasis transmitted by *Culex* mosquito), *Dracunculus medinensis* (guinea worm, a nematode, NOT an annelid!).
6. **Annelida** (segmented worms) exhibit true **metameric segmentation**, a closed circulatory system, and excrete via coiled tubular **Nephridia**; earthworms breathe cutaneously across moist glandular skin; possess **4 to 5 pairs of lateral hearts** (aortic arches in segments 7–11 or 7, 9, 12, 13).
7. Earthworm (*Pheretima posthuma*) blood is **red** because haemoglobin (**erythrocruorin**) is dissolved in the blood plasma; **lacks red blood cells (RBCs)**; known as the "friend of the farmer" for aerating soil and increasing organic matter.
8. **Arthropoda** is the **largest phylum** of the animal kingdom (over two-thirds of all named species, >1 million insects); characterized by **chitinous exoskeleton**, jointed appendages, open circulatory system with a haemocoel, and excretion via **Malpighian tubules** (insects), **Green glands / Antennal glands** (crustaceans/prawns), or **Coxal glands** (spiders/scorpions).
9. **Insects (Class Insecta / Hexapoda)**: 3 main body segments (**Head, Thorax, Abdomen**), **3 pairs of legs (6 legs)** attached to the thorax, usually 2 pairs of wings, 1 pair of antennae, tracheal respiration via spiracles. Examples: Cockroach, Honeybee, Mosquito, Butterfly, Cicadas, Froghoppers (spittlebugs), Pond skaters (*Gerridae*), Termites (white ants, Isoptera).
10. **Arachnids (Class Arachnida)**: 2 body regions (**Cephalothorax and Abdomen**), **4 pairs of walking legs (8 legs)**, **zero wings and zero antennae**; mouth appendages are **chelicerae** (fangs) and **pedipalps**. Examples: Spiders, Scorpions, Ticks, Mites. Ticks and mites have unsegmented fused bodies.
11. **Spider Silk (Gossamer Silk)**: Protein fibre spun by spiders, renowned for being both strong and elastic (often described as stronger than steel). Spiders practice **sexual cannibalism** (female kills and eats the male before, during, or after copulation).
12. **Peacock Tarantula (*Poecilotheria metallica*)**: Critically Endangered, arboreal spider endemic to deciduous forests of Andhra Pradesh / Pakkam Malai; it is an **insectivorous arachnid, NOT a crustacean**.
13. Insect blood is termed **Haemolymph**; it is **colourless / white** in cockroaches because it completely lacks respiratory pigments like haemoglobin; does not transport oxygen (oxygen is delivered directly to cells by tracheae).
14. **Mosquito Mouthparts & Posture**: Only **female mosquitoes suck blood** (mouthparts modified to pierce and suck: labium sheath, labrum feeding tube, paired mandibles and maxillae forming piercing stylet); **males have**

mouthparts adapted to suck flower/plant juices. Females have larger wings. **Female *Anopheles rests at an angle (~45°)* to the surface with abdomen pointing up;*** Culex *rests parallel** to the surface.

15. **Bioluminescence in Glowworms**: Larvae and adults of *Arachnocampa luminosa* / fireflies emit blue-green light from abdominal light organs via the enzyme **Luciferase** acting on luciferin substrate.
16. **Beneficial vs Harmful Insects**: *Bombyx mori* (silk), *Laccifer / Kerria lacca* (lac resin), *Apis mellifera* (honey/wax) are beneficial. ***Helicoverpa armigera*** (cotton bollworm/corn earworm) is a polyphagous destructive agricultural pest.
17. **Parasitoid species**: Larvae develop inside/on a host and ultimately kill it (e.g. parasitic wasps, parasitic flies, certain carabid beetles). Centipedes are generalist predators; termites are detritivores.
18. **Detritivores & Filter Feeders**:
19. **Ant Biology**: Ants inject **Formic acid** (HCOOH, simple carboxylic acid) when they bite. Ants have the **largest brain relative to body size** in the animal kingdom. Leafcutter ants (*Attini*) actively **cultivate fungi** for food.
20. **Mollusca** is the second largest animal phylum; soft-bodied, unsegmented animals usually enclosed in a calcareous shell (CaCO_3); possess a sensory rasping tongue-like feeding organ called **Radula** and a fleshy muscular foot; true pearls are harvested from the marine pearl oyster (***Pinctada vulgaris***, class Bivalvia/Pelecypoda).
21. **Cephalopods**: **Octopus (Devilfish)** has 8 arms with suckers, camera eyes, and no external shell. **Colossal Squid (*Mesonychoteuthis hamiltoni*)** and **Giant Squid (*Architeuthis dux*)** are the largest living invertebrates.
22. **False Fish vs True Fish (Must-Score Fact)**:
23. **Fishes in Frozen Lakes**: Lakes freeze only at the surface; because water reaches maximum density at 4^\circ\text{C}, the bottom water remains liquid (>0^\circ\text{C}), allowing fish and aquatic life to survive under the ice shield. Fish die outside water because gills collapse and cannot absorb gaseous oxygen.
24. **Echinodermata**: Exclusively marine spiny-skinned animals; exhibit secondary radial symmetry (bilateral in free-swimming larvae, pentamerous radial in adults); possess an ambulacral **Water Vascular System** operating suction-cup **Tube feet (Podia)**; **all echinoderms are NOT viviparous — they are typically oviparous with external fertilisation**.
25. **Animal heart chambers**: **Fish** = 2 chambers (1 atrium + 1 ventricle; single venous circulation); **Amphibians & most Reptiles** = 3 chambers (2 atria + 1 ventricle; mixed blood tolerated); **Crocodiles, Birds & Mammals** = **4 chambers** (2 atria + 2 ventricles; complete double circulation).
26. **Amphibian Metamorphosis**: Tadpoles are aquatic and herbivorous; metamorphosis into carnivorous frogs involves **dramatic shortening of the intestine** (controlled by thyroid hormone) to process meat diets. Frogs are cold-blooded and breathe via skin and lungs.
27. **Reptiles & Snakes**:
28. **Poisonous Species Across Taxa**: Poisonous species occur among **butterflies** (African Giant Swallowtail, Monarch — sequester plant alkaloids), **fishes** (Puffer fish Tetraodontidae with tetrodotoxin; Reef stonefish most venomous fish), and **frogs** (poison dart frogs with skin alkaloid secretions).
29. **Herbivorous Exceptions**: Adult **Green sea turtles** (*Chelonia mydas*) are strictly herbivorous; **Parrot fish** graze on algae; **Dugong / Sea cow** is a large herbivorous marine mammal. (Dogs are carnivores/omnivores, NOT herbivores).
30. **Flightless Birds & Aves**: **Kiwi** (*Apteryx*) is a small flightless bird endemic to New Zealand. **White Leghorn** is an exotic Mediterranean breed of fowl (chicken).
31. **Nocturnal Animals**: Highly developed hearing, smell, and adapted eyesight: Mosquitoes, Bats, Owls, Kiwis, Aye-aye (largest nocturnal primate using echolocation), **Malabar civet** (*Viverra civettina*). (Lion-tailed macaque is strictly diurnal; Sambar deer is crepuscular/nocturnal).
32. **Poikilothermic (Cold-blooded)**: Fish, Amphibians, Reptiles (body temperature fluctuates with environment).
33. **Homeothermic (Warm-blooded)**: Birds and Mammals (maintain stable internal body temperature via homeostasis).

34. **Respiratory organs across animal taxa**: Cutaneous (Earthworms, Frogs), Branchial (Fish, Tadpoles), Tracheal (Insects), Book lungs (Spiders, Scorpions), Book gills (Horseshoe crab *Limulus*), Pulmonary (Adult frogs, Reptiles, Birds, Mammals via blowholes in Cetaceans).
35. **Kangaroo rat** (*Dipodomys*) can survive its entire life without drinking water; produces the **most concentrated, almost solid urine** among all mammals and relies on metabolic water.
36. **Camels**: Store water in their **intestine** (can drink >100\text{ litres} at once) and store energy as **fat in the hump**; have broad padded 2-toed feet; lack an Omasum (pseudoruminant).
37. **Gallbladder Absence**: Animals that do **not possess a gallbladder** include: all members of the deer family (except musk deer), equines (horses, zebras), camels, giraffes, elephants, rhinoceroses, whales, rats, and doves.
38. **Hooves vs Claws**: Horses, zebras, deer, cattle have hooves; **Jackal (*Canis aureus*)** belongs to Canidae and has **padded feet with nails/claws, NO hooves**.
39. **Nilgai (*Boselaphus tragocamelus*)**: Largest Asian **antelope** (family Bovidae, deer-like appearance, permanent hollow horns); NOT a cow.
40. **Wild Animal Scientific Names**: Leopard = ***Panthera pardus***; Lion = *Panthera leo*; Tiger = *Panthera tigris*; Snow Leopard = *Panthera uncia*; Jaguar = *Panthera onca*; Asiatic wild ass = *Equus hemionus*; Barasingha = *Cervus duvaucelii*; Chinkara = *Gazella bennettii*.
41. **Big Cat Behaviour**: Lions have **no specific breeding season**; **Cheetahs do NOT roar** (chirp, purr, growl); both male and female leopards mark territory by scent/urine.
42. **Predator Defense Mechanisms**: **Hedgehog** rolls into a tight ball with quills pointing outwards; **Pangolin** rolls into an armoured ball with sharp overlapping scales; **Marmots do NOT roll up** (they live in burrows and hide).
43. **Indian Squirrels**: Primarily arboreal, **build nests in trees** (not burrows in ground), hoard/bury nuts and seeds in ground, and are **omnivorous**.
44. **Primate Intelligence & Tool Use**: **Orangutan** uses sticks as tools to scrape insects from tree holes, poke logs for honey, and open *Neesia* fruits. Chimpanzee, Parrot, and Dolphin have been taught language with success; dogs cannot learn language. Humans evince a unique desire to learn and self-express.
45. **Apes vs Monkeys**: Apes (Superfamily Hominoidea) are tailless: Gibbons (lesser apes), Chimpanzees, Gorillas, Orangutans, Humans (great apes). **Gibbon** is the smallest, cleanest, gentlest ape. **Langurs** have long tails and are Old World monkeys (Colobinae), NOT apes.
46. **Brain Capacity**: Modern human average cranial capacity is **~1,350 cc** (range 1,000–2,000 cc); fossil **Cro-Magnon man had ~1,600 cc** (larger than modern average). **Ant has the largest brain in proportion to its body size**. Sperm whale has largest absolute brain (~7 kg).
47. **Panda**: Giant Panda (*Ailuropoda melanoleuca*) belongs to the **Bear family (Ursidae)**.
48. **Dingo (*Canis lupus dingo*)**: Australian wild dog that **cannot bark**; instead, it howls like a wolf.
49. **Sloth bear**: The shaggy bear commonly seen performing on Indian streets.
50. **Otter**: Semiaquatic carnivorous mammal (subfamily Lutrinae, family Mustelidae); fierce, aggressive predatory behaviour.

📋 Must-Score Drill — Phyla, Breeds & Products

Major Invertebrate & Vertebrate Phyla

Phylum / Class	Diagnostic Anatomical Feature	Classic Classic Representative
Porifera	Cellular level, ostia, osculum, collar cells (choanocytes)	*Sycon*, *Spongilla* (freshwater), *Euspongia* (bath sponge)
Cnidaria	Cnidoblasts, radial symmetry, metagenesis (polyp & medusa)	*Hydra*, *Aurelia* (jellyfish), *Metridium* (sea anemone)
Platyhelminthes	Flat body, acoelomate, flame cells (protonephridia)	*Planaria*, *Taenia* (tapeworm), *Fasciola* (liver fluke)
Nematoda	Cylindrical, pseudocoelomate, sexual dimorphism	*Ascaris* (roundworm), *Wuchereria* (filarial worm)

Phylum / Class	Diagnostic Anatomical Feature	Classic Classic Representative
Annelida	True metameric segments, nephridia, closed blood system	*Pheretima* (earthworm), *Hirudinaria* (cattle leech)
Arthropoda	Jointed appendages, chitinous cuticle, haemolymph, Malpighian tubules	Cockroach, Butterfly, Prawn, Scorpion, Spider, Centipede
Mollusca	Soft body, calcareous shell, mantle, radula	*Pila* (apple snail), *Pinctada* (pearl oyster), *Octopus*
Echinodermata	Spiny skin, water vascular system, tube feet, radial adult	*Asterias* (starfish), *Echinus* (sea urchin), *Cucumaria*
Chondrichthyes	Cartilaginous skeleton, ventral mouth, placoid scales, no air bladder	*Scoliodon* (dogfish shark), *Torpedo* (electric ray), *Pristis* (sawfish)
Osteichthyes	Bony skeleton, terminal mouth, cycloid scales, operculum, air bladder	*Labeo rohita* (rohu), *Catla*, *Hippocampus* (sea horse)
Amphibia	Dual life, scale-less moist skin, 3-chambered heart, cold-blooded	*Rana tigrina* (frog), *Bufo* (toad), *Salamandra*, *Ichthyophis*
Reptilia	Epidermal scales/scutes, cold-blooded, 3 chambers (crocodile = 4)	Cobra, Krait, Chameleon, Wall lizard, Crocodile, Turtle
Aves	Feathers, hollow pneumatic bones, homeothermic, 4-chambered heart	*Pavo cristatus* (peacock), *Struthio* (ostrich), Penguin
Mammalia	Mammary glands, hair, diaphragm, 7 cervical vertebrae, warm-blooded	Platypus, Kangaroo, Bat, Blue whale, Elephant, Human

Complete Silkworm & Silk Classification

Silk Variety	Silkworm Species	Primary Host Plant Leaves	Primary Producing States	Key Distinction
Mulberry	*Bombyx mori*	Mulberry (*Morus alba*)	Karnataka, Andhra Pradesh, West Bengal	Accounts for >70% of total Indian silk
Tasar	*Antheraea mylitta*	Arjun (*Terminalia arjuna*), Asan	Jharkhand, Chhattisgarh, Odisha	Wild forest vanya silk; copperish colour
Muga	*Antheraea assamensis*	Som (*Machilus*), Soalu (*Litsaea*)	Assam (holds GI Tag)	Lustrous golden-yellow; highest durability
Eri	*Samia ricini*	Castor (*Ricinus communis*)	Assam, Meghalaya, Nagaland	Spun silk; Ahimsa silk (moths spared)

Elite Livestock Breeds of India

Animal Group	Breed Name	Native Home Tract	Economic Role / Distinctive Trait
Cattle (Milch)	**Sahiwal**	Montgomery (Punjab tract)	Highest milk yielder among indigenous zebu cows; reddish dun
Cattle (Milch)	**Gir**	Gir hills, Kathiawar (Gujarat)	Convex forehead, long pendulous drooping ears; hardy
Cattle (Milch)	**Red Sindhi**	Karachi and Hyderabad (Sindh)	Deep red colour; compact frame, high heat tolerance
Cattle (Dual)	**Hariana**	Rohtak, Hisar, Karnal (Haryana)	Outstanding work bullocks and good milk yield
Cattle (Dual)	**Tharparkar**	Tharparkar desert (Rajasthan)	Deep white/grey coat; thrives on sparse desert scrub
Cattle (Draught)	**Amritmahal**	Hassan, Mysore (Karnataka)	Fiercest, fastest, most spirited draught bullock in India
Cattle (Draught)	**Kankrej**	Rann of Kutch (Gujarat)	Heaviest Indian breed; famous graceful *Sawai Chal* gait
Cattle (Exotic)	**Holstein-Friesian**	Netherlands	World's highest milk production volume; black-and-white

Animal Group	Breed Name	Native Home Tract	Economic Role / Distinctive Trait
Cattle (Exotic)	**Jersey**	Jersey Island (UK)	Milk has exceptionally rich golden fat (~5.0–5.5%)
Buffalo	**Murrah**	Rohtak, Hisar (Haryana)	"Black Gold"; tightly curled horns; leading dairy buffalo
Buffalo	**Bhadawari**	Etawah, Agra (UP) / Gwalior	Copper-coloured coat; highest milk fat content (up to 12–13%)
Buffalo	**Jaffarabadi**	Gir, Kathiawar (Gujarat)	Massive drooping flat horns; heaviest buffalo breed
Goat	**Jamunapari**	Etawah (Chambal ravines, UP)	Largest Indian goat; Roman nose ("parrot mouth"), long ears
Goat	**Barbari**	UP plains / Delhi	Compact, early maturing, prolific breeder; "City Goat"
Goat	**Black Bengal**	West Bengal	Prolific (multiple kids); premium tender meat and fine skin
Goat	**Changthangi**	Changthang plateau (Ladakh)	Yields world-famous luxurious undercoat wool (**Pashmina**)
Poultry	**Kadaknath**	Jhabua, Dhar (Madhya Pradesh)	Melanin pigment turns meat, bones, and organs jet black (GI tag)
Poultry	**Aseel**	Andhra Pradesh / Telangana	Muscular game bird renowned for courage and stamina in fights
Poultry (Exotic)	**White Leghorn**	Italy	Most popular commercial egg-laying bird (layer) globally

One-Liner Revision – Animal Biology and Husbandry

#	Fast Fact
1	Largest animal = Blue whale (Balaenoptera musculus)
2	Fastest land animal = Cheetah (~120 km/h)
3	Largest land animal = African elephant
4	Tallest animal = Giraffe
5	Only mammal to fly = Bat (Order Chiroptera)
6	Mammals that lay eggs (Monotremes) = Duck-billed platypus + Echidna
7	4-chambered heart = Mammals, Birds, Crocodile
8	3-chambered heart = Most amphibians and reptiles
9	2-chambered heart = Fish (Pisces)
10	Ruminant stomach chambers: Rumen, Reticulum, Omasum, Abomasum
11	Largest egg (non-extinct) = Ostrich egg
12	Silk from = **Bombyx mori** (mulberry silkworm)
13	Lac produced by = **Kerria lacca** (lac insect)
14	Honey bee: Queen = fertilised egg; Drone = **unfertilised egg** (parthenogenesis)
15	Largest bee colony queen = **Queen bee** (lives 3-5 years); Workers live 6 weeks
16	Pearl produced by = Oyster (Mollusca) – NOT an insect product
17	Coral = formed by **polyps** (phylum Cnidaria); calcium carbonate exoskeleton
18	Sponges = phylum Porifera; simplest multicellular animals; no tissues
19	Earthworm = hermaphrodite; no legs; locomotion by setae (bristles)
20	Starfish (Sea star) = NOT a true fish; phylum Echinodermata
21	Whale = mammal (not fish); breathes air; warm-blooded
22	Bat = mammal; only true flying mammal; uses echolocation (ultrasound)

#	Fast Fact
23	Snake has no legs; moves using ribs + ventral scales; no eyelids
24	Chameleon = changes colour for temperature regulation + communication
25	India's only ape = **Hoolock Gibbon** (NE India; Assam)
26	National Animal of India = **Bengal Tiger** (Panthera tigris tigris)
27	Asiatic Lion = found ONLY in **Gir Forest, Gujarat**
28	One-horned Rhinoceros = **Kaziranga NP, Assam**
29	Project Tiger launched = **1973** (first reserve = Jim Corbett NP)
30	Amphibians = first vertebrates to live on land; "biological thermometers" of ecosystem

Topic 3 — Nutrition, Vitamins, Diseases and Medicine

Consolidated — 50 Must-Score Facts

1. By mass, **Oxygen** is the most abundant element in the human body (~65%), followed by Carbon (~18.5%), Hydrogen (~9.5%), and Nitrogen (~3.2%); by total number of atoms, **Hydrogen** is the most abundant.
2. **Calcium** is the most abundant mineral element in the human body (~1.5% of body weight; 99% in bones and teeth as hydroxyapatite crystals); **Iron** is the most abundant essential micromineral (3–4 grams, primarily inside haemoglobin).
3. The adult human brain accounts for only **~2% of total body weight**, yet consumes **~20% of total resting oxygen and glucose**.
4. Energy values of macronutrients: **Carbohydrates yield \mathbf{\sim 4.0\text{ kcal/g}}**; **Proteins yield \mathbf{\sim 4.0\text{ kcal/g}}**; **Fats / Lipids yield \mathbf{\sim 9.0\text{ kcal/g}}** (highest physiological fuel value).
5. **Glucose** (dextrose / grape sugar) and **Fructose** (fruit sugar / levulose) are functional isomers with identical empirical formula (C_6H_{12}O_6); **Fructose is the sweetest naturally occurring sugar**.
6. **Sucrose** (table / cane sugar) is a non-reducing disaccharide composed of **Glucose + Fructose**; **Lactose** (milk sugar) is **Glucose + Galactose**; **Maltose** (malt sugar) is **Glucose + Glucose**.
7. **Starch** (plant storage carbohydrate) and **Glycogen** ("animal starch", stored in liver and skeletal muscles) are polymers of \alpha-D-glucose digested by human amylases; **Cellulose** is a structural polymer of \beta-D-glucose that humans **cannot digest** due to the total absence of the enzyme cellulase (acts as roughage / dietary fibre).
8. Proteins are polymers of amino acids linked by peptide bonds; of the 20 standard amino acids, **9 are essential amino acids** in adults (must be supplied in diet: Histidine, Isoleucine, Leucine, Lysine, Methionine, Phenylalanine, Threonine, Tryptophan, Valine).
9. **Kwashiorkor** is caused by severe **protein deficiency with adequate calorie intake**; characterized by pitting **oedema**, swollen "pot belly", and "flaky-paint" dermatosis; **Marasmus** is caused by combined **protein-energy malnutrition (PEM)** in infants <1\text{ year}; characterized by extreme emaciation, dry wrinkled skin ("little old man" face), and **absence of oedema**.
10. **Fat-Soluble Vitamins**: **A, D, E, K**; dissolve in organic solvents and fats, stored in the liver and adipose tissues; excessive intake causes toxic **Hypervitaminosis**.
11. **Water-Soluble Vitamins**: **B-Complex group and Vitamin C**; not stored in large amounts in the human body (readily excreted in urine, require daily dietary intake); **Vitamin B12 is the unique exception** (stored in the human liver for 3–5 years).
12. **Vitamin A (Retinol)**: Essential for rhodopsin synthesis in retinal rods; deficiency causes **Nyctalopia (Night Blindness)**, **Xerophthalmia** (dry, keratinized cornea), and **Bitot's spots**; rich sources: Cod-liver oil, liver, egg yolk, and plant precursors like **\beta-Carotene** in carrots, mangoes, and papayas.
13. **Vitamin D (Calciferol / Sunshine Vitamin)**: Synthesised in human skin upon exposure to solar UV-B radiation from 7-dehydrocholesterol; acts as a pro-hormone promoting intestinal absorption of calcium and phosphorus; deficiency causes **Rickets** in children (bowed legs, pigeon chest) and **Osteomalacia** in adults (soft, fragile bones).
14. **Vitamin E (Tocopherol / Anti-Sterility Vitamin)**: Powerful lipid-soluble biological **antioxidant** protecting cell membranes from free radical oxidation; richest natural dietary source is **Wheat-germ oil**, almonds, and sunflower oil.
15. **Vitamin K (Phylloquinone / Naphthoquinone)**: Required in the liver as a cofactor for the post-translational carboxylation of clotting factors **II (Prothrombin), VII, IX, and X**; deficiency causes prolonged bleeding time and haemorrhage; synthesized by symbiotic colon bacteria (*E. coli*); rich plant source: **Alfalfa** and dark green leafy vegetables.
16. **Vitamin C (Ascorbic Acid / Anti-Scurvy Vitamin)**: Required for the enzymatic hydroxylation of proline and lysine during **Collagen synthesis**; deficiency causes **Scurvy** (bleeding and spongy gums, loose teeth, poor

wound healing, petechial haemorrhages); destroyed by prolonged boiling and cooking heat; richest sources: **Amla (Indian gooseberry)**, guava, citrus fruits (lemon, orange); **absent in milk, meat, and eggs**.

17. **Vitamin B1 (Thiamine)**: Active coenzyme TPP (Thiamine Pyrophosphate) required in carbohydrate metabolism; deficiency causes **Beriberi** (Dry Beriberi = peripheral neuropathy; Wet Beriberi = congestive heart failure and oedema); **eating polished / milled parboiled rice strips the thiamine-rich outer aleurone layer**, precipitating beriberi.
18. **Vitamin B2 (Riboflavin)**: Precursor of coenzymes FMN and FAD; imparts characteristic yellow colour to cow milk and whey; deficiency causes **Ariboflavinosis**: **Cheilosis** (painful cracks at the corners of the mouth), **Glossitis** (magenta-red inflamed tongue), and keratitis.
19. **Vitamin B3 (Niacin / Nicotinic Acid)**: Precursor of NAD and NADP; synthesised in small quantities from the essential amino acid **Tryptophan**; deficiency causes **Pellagra**, classically remembered by the **"4 Ds"**: **Dermatitis** (Casal's necklace pattern on sun-exposed skin), **Diarrhoea**, **Dementia**, and **Death**; prevalent in populations relying exclusively on maize / corn diets (maize is deficient in tryptophan and contains bound niacytin).
20. **Vitamin B5 (Pantothenic Acid)**: Essential constituent of **Coenzyme-A (CoA)**; deficiency causes "Burning Feet Syndrome" (*Gopalan's syndrome*).
21. **Vitamin B6 (Pyridoxine)**: Coenzyme for amino acid transamination and decarboxylation; required for synthesis of haemoglobin, serotonin, and GABA; deficiency causes peripheral neuropathy, microcytic anaemia, and convulsions in infants; *Lucent note*: **Vitamin B6 helps in remembering dreams**.
22. **Vitamin B7 (Biotin / Vitamin H)**: Sulfur-containing coenzyme for carboxylation reactions; raw egg white contains the protein **Avidin**, which binds biotin tightly and blocks absorption, causing induced biotin deficiency.
23. **Vitamin B9 (Folic Acid / Folate)**: Required for DNA synthesis and purine/thymidylate production; deficiency causes **Megaloblastic Anaemia**; maternal folate deficiency in periconceptional period causes fatal **Neural Tube Defects (Spina Bifida and Anencephaly)** in newborns.
24. **Vitamin B12 (Cyanocobalamin)**: The only vitamin containing a metallic trace element (**Cobalt**); absent in ordinary plant foods; synthesized exclusively by microorganisms; absorbs in the terminal ileum mediated by **Castle's Intrinsic Factor** secreted by gastric parietal cells; deficiency causes **Pernicious Anaemia** (megaloblastic anaemia + irreversible subacute combined degeneration of the spinal cord).
25. **Minamata Disease**: Severe neurological disease caused by methylmercury poisoning from consuming contaminated fish and shellfish in Minamata Bay, Japan; damages sensory cortex and cerebellum, producing ataxia, sensory numbness, and tremors.
26. **Itai-Itai Disease ("Ouch-Ouch" Disease)**: Chronic **Cadmium (Cd)** poisoning occurring in Toyama Prefecture, Japan; causes severe painful osteomalacia, multiple bone fractures, and renal tubular dysfunction.
27. **Blue Baby Syndrome (Infantile Methemoglobinemia)**: Caused by high **Nitrate (NO_3^-)** concentrations in rural drinking water (>45\text{ mg/L}); intestinal microbes convert nitrate to nitrite, which oxidizes hemoglobin Fe^{2+} into Fe^{3+} **Methemoglobin**, incapable of releasing oxygen to tissues, producing cyanotic slate-blue skin discoloration.
28. **Blackfoot Disease & Arsenicosis**: Chronic **Arsenic (As)** toxicity from contaminated tube-well groundwater (prevalent in the Ganga-Brahmaputra alluvial plains of West Bengal and eastern UP); causes severe peripheral vascular gangrene (Blackfoot disease) and hyperkeratosis.
29. **Fluorosis (Skeletal and Dental)**: Chronic ingestion of excess **Fluoride (F^-)** in groundwater (>1.5\text{ mg/L}); causes mottling of dental enamel (Dental Fluorosis), calcification of spinal ligaments, and crippling skeletal deformities ("Knock-knee" syndrome / Genu valgum).
30. **Wilson's Disease**: Autosomal recessive inborn error of copper metabolism leading to toxic accumulation of **Copper (Cu)** in the liver, brain basal ganglia, and cornea (pathognomonic golden-brown **Kayser-Fleischer rings** in the cornea).
31. **Cholera**: Caused by the comma-shaped bacterium ***Vibrio cholerae***; transmitted via fecal-oral contamination of water/food; secretes cholera enterotoxin stimulating massive intestinal chloride secretion, resulting in profuse painless **"rice-water" diarrhea** and severe lethal dehydration; treated with Oral Rehydration Solution (ORS).

32. **Tuberculosis (TB)**: Chronic infectious airborne disease caused by acid-fast bacillus ***Mycobacterium tuberculosis*** (discovered by **Robert Koch, 1882**); primarily attacks lungs (pulmonary TB); confirmed by sputum smear, chest X-ray, and **Mantoux Tuberculin Skin Test**; protected by the **BCG (Bacille Calmette-Guérin)** live attenuated vaccine; treated via WHO **DOTS (Directly Observed Treatment, Short-course)**.
33. **Typhoid Fever (Enteric Fever)**: Caused by the rod-shaped Gram-negative bacterium ***Salmonella typhi***; transmitted via contaminated food and water; classic signs include step-ladder remittent fever, coated tongue, relative bradycardia, and intestinal ulceration; confirmed by the **Widal Test** (detects agglutinating antibodies against O and H antigens).
34. **Tetanus (Lockjaw / धनुस्तंभ)**: Caused by anaerobic, spore-forming, drumstick-shaped soil bacterium ***Clostridium tetani***; enters via contaminated lacerations and rustic puncture wounds; releases the neurotoxin **Tetanospasmin**, which blocks inhibitory glycine/GABA release, causing painful involuntary spasms of masseter muscles (trismus / lockjaw) and arched body posture (opisthotonos); prevented by tetanus toxoid (TT) vaccine.
35. **Diphtheria**: Acute bacterial infection caused by ***Corynebacterium diphtheriae***; forms a thick, grey, leathery pseudomembrane over the tonsils and pharynx, causing lethal airway asphyxiation; confirmed by the **Schick Test**; prevented as part of the **DPT (Diphtheria, Pertussis, Tetanus)** trivalent pediatric vaccine.
36. **Leprosy (Hansen's Disease / कुष्ठ रोग)**: Chronic mycobacterial infection caused by ***Mycobacterium leprae*** (discovered by **Gerhard Hansen, 1873**); targets cooler peripheral nerves and skin; produces hypopigmented anaesthetic skin macules and claw hands; cured by standard **Multi-Drug Therapy (MDT)** (Rifampicin, Dapsone, and Clofazimine).
37. **Poliomyelitis (Polio)**: Enteroviral infection transmitted via fecal-oral route; the **Poliovirus** destroys anterior horn motor neurons of the spinal cord, causing asymmetric flaccid paralysis; prevented by **Salk Vaccine** (inactivated injectable polio vaccine, Jonas Salk, 1952) and **Sabin Vaccine** (live attenuated oral polio vaccine drops, Albert Sabin, 1961).
38. **Rabies (Hydrophobia / अलर्क रोग)**: Fatal neurotropic bullet-shaped rhabdovirus infection transmitted through saliva via bites of rabid mammals (dogs, wolves, bats); virus ascends peripheral nerves into the CNS, causing intense pharyngeal spasm upon attempting to swallow water (**Hydrophobia**); pathognomonic diagnostic **Negri bodies** form inside cerebral neurons; post-exposure rabies vaccine was developed by **Louis Pasteur (1885)**.
39. **Dengue Fever ("Breakbone Fever")**: Caused by the **Dengue flavivirus** (4 serotypes: DENV 1–4); transmitted by the diurnal day-biting mosquito ***Aedes aegypti*** (the tiger mosquito); hallmarks are acute retro-orbital headache, high fever, severe arthralgia/myalgia ("breakbone"), and precipitous drop in **blood platelet count (thrombocytopenia)**; also transmits Chikungunya and Zika viruses.
40. **Malaria**: Mosquito-borne parasitic disease caused by protozoan parasites of the genus ***Plasmodium*** (*P. vivax, P. falciparum, P. malariae, P. ovale*); transmitted exclusively by the bite of infected female ***Anopheles*** mosquitoes; **Sir Ronald Ross (1897, Secunderabad, India)** discovered the mosquito transmission vector; *P. falciparum* causes lethal **Malignant Tertian / Cerebral Malaria**; natural treatment is **Quinine** (extracted from the bark of the **Cinchona tree**, *Cinchona calisaya*).
41. **Kala-Azar (Visceral Leishmaniasis / Dumdum Fever)**: Caused by flagellated protozoan ***Leishmania donovani***; transmitted by the vector **Sandfly (*Phlebotomus argentipes*)**; endemic in Bihar, Jharkhand, and eastern UP; infects the reticuloendothelial system, causing massive hepatosplenomegaly, severe anaemia, and dark skin hyperpigmentation.
42. **Sleeping Sickness (African Trypanosomiasis)**: Caused by protozoan flagellate ***Trypanosoma brucei***; transmitted by the bite of the **Tsetse fly (*Glossina palpalis*)**; attacks the central nervous system, producing daytime sleepiness and fatal coma.
43. **Amoebiasis (Amoebic Dysentery)**: Protozoan intestinal infection caused by ***Entamoeba histolytica***; invades large intestinal mucosa producing characteristic "flask-shaped" ulcers; causes frequent bloody stools with mucus, abdominal cramps, and potential metastatic amoebic liver abscesses.
44. **Cancer Biology & Classification**: Malignant neoplasia characterized by uncontrolled cellular proliferation and loss of contact inhibition: **Carcinoma** (cancers of epithelial origin; accounts for ~85% of all human cancers, e.g., breast, lung, cervical, colon cancer); **Sarcoma** (cancers of mesenchymal/connective tissue origin, e.g.,

bone/osteosarcoma, muscle, cartilage); **Leukemia** (cancer of hematopoietic white blood cells in bone marrow); **Lymphoma** (cancer of lymphatic nodal tissue, e.g., Hodgkin's disease).

45. **Oncogenes & Metastasis**: Normal cells contain regulatory genes called **Proto-oncogenes**, which mutate into active cancer-causing **Oncogenes** under exposure to physical, chemical, or biological carcinogens; **Metastasis** (detachment, bloodstream dissemination, and colonisation of distant organ sites by malignant tumor cells) is the single most definitive clinical hallmark of malignancy.
46. **Diabetes Mellitus**: Metabolic disorder caused by absolute deficiency of insulin (**Type 1 Diabetes**, autoimmune destruction of pancreatic beta cells) or peripheral tissue insulin resistance (**Type 2 Diabetes**); characterized by fasting hyperglycemia, **Glycosuria** (glucose in urine), polyuria (excessive urination), polydipsia (excessive thirst), and polyphagia; diagnosed by Fasting Blood Sugar and **HbA_{1c} test** (\ge 6.5\% indicates diabetes).
47. **Diabetes Insipidus**: Rare endocrinological disorder completely unrelated to insulin or blood sugar levels; caused by deficiency of **Antidiuretic Hormone (ADH / Vasopressin)** secreted by the posterior pituitary or renal insensitivity to ADH; characterized by massive excretion of extremely dilute, hypotonic urine (up to 15–20 litres/day) and extreme thirst; urine is non-sweet (lacks glucose entirely).
48. **Vaccine Platforms**: **Inactivated / Killed Vaccines**: Pathogen destroyed by heat or chemical formalin (e.g., Covaxin, Salk Polio vaccine, Rabies vaccine); **Live Attenuated Vaccines**: Weakened living pathogen inducing strong lifelong immunity (e.g., BCG, MMR, Sabin Oral Polio vaccine, Yellow Fever vaccine); **Subunit / Recombinant Vaccines**: Specific antigenic fragments (e.g., Hepatitis B recombinant surface antigen, Corbevax); **mRNA Vaccines**: Synthetic mRNA in lipid nanoparticles (e.g., Pfizer-BioNTech, Moderna); **DNA Vaccines**: Plasmid DNA encoding viral antigen (e.g., ZyCoV-D, world's first needle-free plasmid DNA vaccine).
49. **Landmark Clinical Diagnostic Tests**: **ELISA (Enzyme-Linked Immunosorbent Assay)** and **Western Blot**: Standard screening and confirmatory tests for HIV / AIDS antibodies; **Widal Test**: Diagnostic agglutination test for Typhoid fever; **Mantoux Test / Tuberculin Skin Test**: Diagnostic delayed hypersensitivity test for Tuberculosis exposure; **Schick Test**: Susceptibility test for Diphtheria; **Biopsy / Histopathology**: Gold-standard definitive diagnostic test for cancer malignancy; **Amniocentesis**: Prenatal testing of amniotic fluid for fetal chromosomal abnormalities (e.g., Down syndrome), strictly regulated in India under the PCPNDT Act to prevent female foeticide.
50. **Clinical Diagnostic Instruments**: **Electrocardiogram (ECG)**: Graphical recording of cardiac electrical conduction; **Electroencephalogram (EEG)**: Recording of cerebral cortex neuroelectrical activity (diagnoses epilepsy, coma, brain death); **Electromyogram (EMG)**: Recording of skeletal muscle and peripheral nerve electrical potential; **Endoscopy**: Uses flexible fiber-optic bundles applying **Total Internal Reflection** to inspect hollow internal viscera (stomach, duodenum).

📋 Must-Score Drill — Vitamins, Diseases & Diagnostic Tests

Comprehensive Vitamin Master Reference Table

Vitamin	Chemical Name	Solubility	Classical Deficiency Syndrome	Key Dietary Sources & Notes
Vitamin A	Retinol / Axerophthol	Fat	**Night Blindness (Nyctalopia)**, Xerophthalmia, Bitot's spots	Cod-liver oil, liver, carrots (\beta-carotene), mango, papaya
Vitamin D	Calciferol (D_2: Ergocalciferol, D_3: Cholecalciferol)	Fat	**Rickets** (children), **Osteomalacia** (adults)	Synthesised in sunlit skin; cod-liver oil, egg yolk, fortified milk
Vitamin E	Tocopherol (\alpha, \beta, \gamma)	Fat	Mild haemolytic anaemia, sterility in animals	**Wheat-germ oil**, almonds, vegetable seed oils; antioxidant
Vitamin K	Phylloquinone (K_1), Menaquinone (K_2)	Fat	**Haemorrhage / delayed blood clotting**	Alfalfa, spinach, cabbage, synthesised by gut bacteria
Vitamin C	Ascorbic Acid	Water	**Scurvy** (bleeding spongy gums, poor wound healing)	**Amla**, citrus fruits, guava, tomatoes; destroyed by heat/cooking

Vitamin	Chemical Name	Solubility	Classical Deficiency Syndrome	Key Dietary Sources & Notes
Vitamin B1	Thiamine	Water	**Beriberi** (dry = neuropathy, wet = heart failure)	Whole unpolished cereals, yeast; stripped by polishing rice
Vitamin B2	Riboflavin	Water	**Cheilosis** (cracked mouth corners), Glossitis	Cow milk (gives yellow tinge), eggs, liver; not in cod-liver oil
Vitamin B3	Niacin / Nicotinic acid	Water	**Pellagra** (Dermatitis, Diarrhoea, Dementia)	Meat, fish, peanuts; synthesised from amino acid tryptophan
Vitamin B5	Pantothenic acid	Water	Burning Feet Syndrome (*Gopalan's syndrome*)	Eggs, meat, mushrooms; integral part of Coenzyme A
Vitamin B6	Pyridoxine	Water	Microcytic anaemia, convulsions, helps remember dreams	Rice bran, whole grains, bananas, nuts
Vitamin B7	Biotin (Vitamin H)	Water	Hair loss, dermatitis, muscle pain	Liver, egg yolk; blocked by raw egg white avidin protein
Vitamin B9	Folic acid / Folate	Water	**Megaloblastic Anaemia**, Neural Tube Defects	Green leafy vegetables (foliage), legumes, liver
Vitamin B12	Cyanocobalamin	Water	**Pernicious Anaemia**, neurological degeneration	Meat, fish, eggs, dairy; **contains Cobalt**; stores in liver

Infectious Diseases & Causative Pathogens Master Table

Disease	Pathogen Class	Specific Pathogen Name	Primary Transmission Vector / Vehicle	Key Clinical Feature / Diagnostic Fact
Tuberculosis	Bacterium	*Mycobacterium tuberculosis*	Airborne droplets	Mantoux test; BCG vaccine; DOTS therapy
Typhoid	Bacterium	*Salmonella typhi*	Contaminated food and water	Widal test; step-ladder fever; rose spots
Cholera	Bacterium	*Vibrio cholerae*	Contaminated water / flies	Profuse painless "rice-water stools"; ORS
Diphtheria	Bacterium	*Corynebacterium diphtheriae*	Droplet infection	Schick test; tough grey throat pseudomembrane
Tetanus	Bacterium	*Clostridium tetani*	Spores in soil / dirty wounds	Lockjaw (trismus); opisthotonos; TT toxoid
Plague	Bacterium	*Yersinia pestis*	Rat flea (*Xenopsylla cheopis*)	Painful swollen lymph node buboes
Leprosy	Bacterium	*Mycobacterium leprae*	Prolonged close contact	Hansen's disease; anaesthetic skin patches; MDT
Pertussis	Bacterium	*Bordetella pertussis*	Droplets	Whooping cough; DPT vaccine
Malaria	Protozoan	*Plasmodium falciparum / vivax*	Female *Anopheles* mosquito	Ronald Ross; haemozoin chills; Quinine
Kala-azar	Protozoan	*Leishmania donovani*	Sandfly (*Phlebotomus*)	Hepatosplenomegaly, skin darkening
Amoebiasis	Protozoan	*Entamoeba histolytica*	Water/food with cyst	Stools with blood and mucus; abdominal cramps
Sleeping Sickness	Protozoan	*Trypanosoma brucei*	Tsetse fly (*Glossina*)	Disrupted circadian cycle, coma
Filariasis	Helminth	*Wuchereria bancrofti*	Female *Culex* mosquito	Elephantiasis; blocked lymphatic vessels
Dengue	Virus	Dengue virus (Flavivirus)	Female *Aedes aegypti* mosquito	Breakbone fever, thrombocytopenia
Rabies	Virus	Rhabdovirus	Bite of rabid dog/animal	Hydrophobia, Negri bodies; Pasteur vaccine

Disease	Pathogen Class	Specific Pathogen Name	Primary Transmission Vector / Vehicle	Key Clinical Feature / Diagnostic Fact
Polio	Virus	Poliovirus (Enterovirus)	Faecal-oral route	Salk (injected killed) vs Sabin (oral live)
AIDS	Virus	Human Immunodeficiency Virus	Blood, sexual, perinatal	Destroys CD4+ T_H cells; ELISA + Western blot

One-Liner Revision – Nutrition, Diseases and Medicine

#	Fast Fact
1	Carbohydrate = primary energy source; **4 kcal/g**
2	Fat = most concentrated energy source; **9 kcal/g**
3	Protein = **4 kcal/g**; essential amino acids = 9 (must come from diet)
4	Vitamin A (Retinol) = Night blindness; sources: carrot, liver, egg
5	Vitamin B1 (Thiamine) = Beriberi; found in rice bran, cereals
6	Vitamin B2 (Riboflavin) = Ariboflavinosis (cracks at mouth corners)
7	Vitamin B3 (Niacin) = Pellagra (4Ds: Dermatitis, Diarrhoea, Dementia, Death)
8	Vitamin B9 (Folic acid) = Neural tube defects in foetus if deficient
9	Vitamin B12 (Cobalamin) = Pernicious anaemia; only in animal foods
10	Vitamin C (Ascorbic acid) = Scurvy; richest source = Amla (Indian gooseberry)
11	Vitamin D (Calciferol) = Rickets (children), Osteomalacia (adults); sunlight synthesis
12	Vitamin E (Tocopherol) = antioxidant; deficiency = sterility/muscle weakness
13	Vitamin K (Phylloquinone) = blood clotting; deficiency = excessive bleeding
14	Fat-soluble vitamins = **A, D, E, K** (stored in liver/fat)
15	Water-soluble vitamins = **B-complex + C** (excreted in urine; daily needed)
16	Iron deficiency = **Anaemia** (haemoglobin low; pale, fatigued)
17	Iodine deficiency = **Goitre** (enlarged thyroid)
18	Calcium deficiency = **Tetany / Osteoporosis**
19	Fluoride excess = **Fluorosis** (mottled teeth, skeletal damage)
20	Penicillin discovered = **Alexander Fleming (1928)**; Nobel 1945 (Fleming+Chain+Florey)
21	First antibiotic = **Penicillin** (from *Penicillium notatum* mould)
22	Malaria vector = **Female Anopheles mosquito**; caused by Plasmodium
23	Dengue vector = **Aedes aegypti** mosquito
24	Filaria (Lymphatic filariasis) vector = **Culex mosquito**
25	Plague (Black Death) vector = rat flea (*Xenopsylla cheopis*); bacterium *Yersinia pestis*
26	Cholera = *Vibrio cholerae*; rice-water stools; foul-smelling; water-borne
27	Tuberculosis = *Mycobacterium tuberculosis*; Mantoux test; BCG vaccine
28	Vaccine for smallpox = **Edward Jenner (1796)**; smallpox eradicated **1980** (WHO)
29	ORS (Oral Rehydration Solution) = key treatment for diarrhoea; WHO/UNICEF
30	DOTS = Directly Observed Treatment Short-course; for **tuberculosis** control

Topic 4 — Digestion, Respiration and Excretion

Consolidated — 50 Must-Score Facts

1. Starch digestion begins in the **mouth (buccal cavity)**; salivary glands secrete **Ptyalin (Salivary Amylase)** at optimum pH 6.8, hydrolysing ~30% of dietary starch into the disaccharide **Maltose**; protein and fat digestion do **not** occur in the mouth.
2. The adult human dental formula is **\frac{2123}{2123} \times 2 = 32** (2 Incisors, 1 Canine, 2 Premolars, 3 Molars per quadrant); milk / deciduous dental formula of children is **\frac{2102}{2102} \times 2 = 20** (**Premolars are completely absent in milk dentition**).
3. **Tooth Enamel** is the **hardest substance in the human body**; secreted by specialized epithelial cells called **Ameloblasts**; composed of 96% inorganic mineral crystals of **Calcium Hydroxyapatite**; dentine is secreted by **Odontoblasts**.
4. Protein digestion begins in the **stomach**; gastric chief (peptic) cells secrete inactive **Pepsinogen**, which is activated into **Pepsin** by **Hydrochloric Acid (HCl)** at a strongly acidic **pH of 1.5–2.0**.
5. Gastric juice also contains **Rennin (Chymosin)** in infants, which coagulates soluble milk protein **Casein** into insoluble **Calcium Paracaseinate** in the presence of calcium ions; adults digest milk protein via pepsin.
6. **Trypsin** and **Chymotrypsin** are proteolytic enzymes produced by the **pancreas** and active in the alkaline medium of the **small intestine**; they are **never secreted by the stomach**.
7. Inactive pancreatic **Trypsinogen** is activated into active **Trypsin** by the brush-border enzyme **Enterokinase (Enteropeptidase)** secreted by the duodenal intestinal mucosa.
8. The **Liver** is the **largest gland of the human body** (weighing 1.2–1.5 kg); synthesises and secretes **Bile**; stores glycogen, synthesises urea via the ornithine cycle, detoxifies drugs, and produces prothrombin and fibrinogen.
9. **Bile contains NO digestive enzymes whatsoever**; contains bile salts (**Sodium Glycocholate and Sodium Taurocholate**) which **emulsify fats** (breaking large lipid globules into micro-droplets) and bile pigments (**Bilirubin and Biliverdin**).
10. Bile is stored and concentrated up to 10-fold inside the **Gall Bladder**; removal of the gall bladder (cholecystectomy) does **not** stop bile production (bile continues to trickle directly from liver hepatocytes into the duodenum).
11. Complete digestion of carbohydrates, proteins, and lipids terminates in the **Small Intestine (Duodenum, Jejunum, and Ileum)**; the small intestine is the longest segment of the alimentary canal (~6 metres).
12. Intestinal mucosal projections called **Villi and Microvilli** expand the internal absorptive surface area of the small intestine by **over 600-fold**; each villus houses blood capillaries and a central lymphatic vessel called a **Lacteal**.
13. Dietary fats are absorbed uniquely: fatty acids and monoglycerides form water-soluble **Micelles**, enter mucosal enterocytes, are re-esterified into triglycerides, coated with a protein jacket to form microscopic globules called **Chylomicrons**, and released into **Lacteals (Lymph)**, not directly into blood.
14. Humans **cannot digest Cellulose** because the human digestive tract does not secrete the enzyme **Cellulase**; cellulose passes through the gut undigested as dietary fibre / roughage, providing mechanical bulk to stimulate peristalsis.
15. The **Large Intestine** (Caecum, Colon, Rectum) does **not** secrete digestive enzymes; absorbs water, mineral salts, and certain drugs, and consolidates undigested residue into faeces.
16. The **Epiglottis** is a cartilaginous elastic flap that covers the glottis (opening of the larynx/windpipe) during swallowing to prevent food boluses from entering the respiratory tract.
17. The human vocal cords are housed inside the **Larynx (Voice Box)**; the prominent anterior projection of the thyroid cartilage in adult human males is called the **Adam's Apple**.
18. The **Trachea (Windpipe)** is supported by 16–20 incomplete **C-shaped rings of Hyaline Cartilage** that prevent airway collapse during negative-pressure inspiration.

19. Microscopic **Alveoli** (~300 million in human lungs) are the ultimate structural and functional units of gas exchange; alveolar walls are lined by simple squamous epithelium and wrapped in capillary networks.
20. Quiet normal **Inspiration** is an active muscular process: contraction and downward flattening of the dome-shaped **Diaphragm** (increases vertical thoracic volume) coupled with contraction of **External Intercostal Muscles** (lifts ribs outward and upward); intra-alveolar pressure drops below atmospheric pressure, drawing air into lungs.
21. Normal quiet **Expiration** is a passive process resulting from the elastic recoil of lungs and relaxation of the diaphragm and intercostal muscles.
22. **Tidal Volume (TV)**: Volume of air inspired or expired during a normal quiet breath = **\mathbf{\sim 500\text{ mL}}** in healthy adults.
23. **Inspiratory Reserve Volume (IRV)**: Extra volume inspired forcefully after normal tidal inspiration = 2500–3000\text{ mL}.
24. **Expiratory Reserve Volume (ERV)**: Extra volume expired forcefully after normal tidal expiration = 1000–1100\text{ mL}.
25. **Residual Volume (RV)**: Volume of air remaining permanently inside lungs even after maximal forced expiration = **\mathbf{1100–1200\text{ mL}}**; prevents alveolar collapse; **cannot be measured by a spirometer**.
26. **Vital Capacity (VC)**: Maximum volume of air exhaled after maximal forced inspiration: VC = TV + IRV + ERV \approx \mathbf{3500–4500\text{ mL}}.
27. **Oxygen Transport in Blood**: **\mathbf{97\%} is transported bound to Haemoglobin as Oxyhaemoglobin** inside red blood cells; only **\mathbf{3\%}** is dissolved in blood plasma.
28. Each molecule of haemoglobin (Hb) contains 4 iron atoms (Fe^{2+}) and can reversibly bind a maximum of **4 molecules of Oxygen (O_2)**.
29. **Carbon Dioxide Transport in Blood**:
    - **\mathbf{\sim 70\%} is transported as Bicarbonate ions (\mathbf{HCO_3^-})** dissolved in blood plasma.
    - **\mathbf{\sim 20–25\%} is transported bound to haemoglobin as Carbamino-haemoglobin**.
    - **\mathbf{\sim 7\%} is transported dissolved in blood plasma**.
30. The conversion of CO_2 and H_2O into carbonic acid (H_2CO_3) and bicarbonate inside RBCs is catalysed by the zinc-containing enzyme **Carbonic Anhydrase**; balanced by the inward migration of chloride ions from plasma into RBCs (**Chloride Shift / Hamburger Phenomenon**).
31. **Bohr Effect**: Lowering of blood pH (acidosis) or elevated carbon dioxide tension (pCO_2) shifts the oxyhaemoglobin dissociation curve to the **right**, decreasing haemoglobin's oxygen affinity and promoting rapid unloading of oxygen to respiring tissues.
32. **Carbon Monoxide (CO) Poisoning**: Carbon monoxide has an affinity for haemoglobin **\mathbf{200–250\text{ times greater}} than oxygen**; forms extremely stable **Carboxyhaemoglobin (COHb)**, permanently blocking oxygen binding sites, causing severe cellular asphyxiation, tissue hypoxia, and cherry-red skin discoloration.
33. **Regulation of Respiration**: Primary respiratory rhythm centre is located in the **Medulla Oblongata** of the brain stem; **Pneumotaxic centre** is in the **Pons Varolii**; central chemoreceptors are exquisitely sensitive to arterial **excess of \mathbf{CO_2} and Hydrogen ions (\mathbf{H^+})**, **not to low oxygen**.
34. Aerobic cellular respiration yields **36 or 38 molecules of ATP** per oxidized glucose molecule: Glycolysis in cytoplasm (net 2 ATP + 2 NADH) \rightarrow Link reaction in mitochondrial matrix (2 NADH) \rightarrow Krebs Cycle in mitochondrial matrix (2 ATP + 6 NADH + 2 FADH2) \rightarrow Electron Transport System (ETS) on inner mitochondrial cristae.
35. Anaerobic glycolysis in over-exerted skeletal muscles produces **Lactic Acid**; accumulation causes acute muscle fatigue and cramp; lactic acid is transported via bloodstream to the liver and resynthesised into glucose via the **Cori Cycle**.
36. Nitrogenous excretion types:
    - **Ammonotelic**: Excrete highly toxic **Ammonia** requiring copious water (300–500\text{ mL/g}); e.g. bony fish, aquatic amphibians (tadpoles), aquatic invertebrates.

- **Ureotelic**: Excrete less toxic **Urea**; synthesised in the liver via the **Ornithine Cycle / Krebs-Henseleit Cycle** from ammonia and CO_2; e.g. mammals, adult amphibians, marine sharks.
- **Uricotelic**: Excrete non-toxic, insoluble **Uric Acid** as a semi-solid paste/pellet with minimal water loss; e.g. birds, terrestrial reptiles, insects, land snails.

37. The structural and functional filtration unit of the human kidney is the **Nephron** (~1 to 1.2 million nephrons per kidney).
38. **Malpighian Body (Renal Corpuscle)**: Composed of a tuft of capillaries called the **Glomerulus** encased inside a double-walled cup called **Bowman's Capsule**.
39. **Ultrafiltration**: Glomerular capillary blood pressure forces water, glucose, amino acids, ions, and urea across the trilaminar filtration membrane into Bowman's capsule, forming **Glomerular Filtrate (180 litres/day; GFR = 125 mL/min)**; blood cells and plasma proteins (albumin) are retained.
40. **Selective Reabsorption in the Proximal Convoluted Tubule (PCT)**: PCT is lined by simple cuboidal brush-border epithelium with microvilli; **reabsorbs \mathbf{70–80\%} of all filtered water and electrolytes, and \mathbf{100\%} of filtered glucose and amino acids** via active transport.
41. **Loop of Henle & Counter-Current Multiplier**: Hairpin loop maintaining high osmotic medullary gradient; descending limb is permeable to water and impermeable to salts; ascending limb is impermeable to water and actively pumps out NaCl; operates alongside **Vasa Recta** to produce concentrated hypertonic urine.
42. **Hormonal Regulation of Excretion**:
    - **Antidiuretic Hormone (ADH / Vasopressin)**: Secreted by posterior pituitary; acts on DCT and collecting ducts to insert aquaporins, promoting water reabsorption; deficiency causes **Diabetes Insipidus** (profuse dilute urine).
    - **Renin-Angiotensin-Aldosterone System (RAAS)**: Juxtaglomerular apparatus (JGA) releases **Renin** in response to low blood pressure; renin converts angiotensinogen \rightarrow angiotensin-I \rightarrow angiotensin-II (powerful vasoconstrictor); stimulates adrenal cortex to release **Aldosterone**, increasing Na^+ and water reabsorption.
    - **Atrial Natriuretic Factor (ANF)**: Secreted by heart atria in response to high blood pressure; promotes sodium excretion (natriuresis) and vasodilation, antagonising RAAS.
43. Normal human urine is slightly acidic (**pH \mathbf{\sim 6.0}**); pale yellow colour is due to the pigment **Urochrome** (derived from the metabolic breakdown of haemoglobin).
44. **Renal Calculi (Kidney Stones)**: Solid crystalline deposits formed inside renal pelvis and calyces, primarily composed of insoluble **Calcium Oxalate (CaC_2O_4)** crystals.
45. **Gout**: Painful metabolic arthritis caused by the deposition of sharp **Monosodium Urate / Uric Acid crystals** in synovial joints (most commonly the metatarsophalangeal joint of the big toe) due to hyperuricaemia.
46. **Juxtaglomerular Apparatus (JGA) & Macula Densa**: Specialized cellular complex located where the distal convoluted tubule contacts the afferent arteriole; macula densa senses low sodium/chloride delivery, triggering JG granular cells to release the enzyme **Renin**, activating the systemic vasoconstrictor and salt-retaining cascade.
47. **Counter-Current Multiplier & Exchanger**: Descending limb of the Loop of Henle is permeable to water but impermeable to electrolytes; the ascending limb is impermeable to water but actively transports NaCl into the medullary interstitium; paired with hairpin **Vasa Recta** capillaries, this maintains a steep osmotic gradient (300\text{ mOsm/L} at the cortex to 1200\text{ mOsm/L} in deep renal medulla), enabling humans to excrete hypertonic urine up to 4 times more concentrated than blood plasma.
48. **Bile Salts & Micelle Formation**: Bile salts (sodium glycocholate and taurocholate) possess amphipathic structure (hydrophilic head, hydrophobic tail); they emulsify large dietary lipid droplets into microscopic droplets and form spherical aggregates called **Micelles** (4–7\text{ nm}), which ferry monoglycerides and free fatty acids to the apical brush-border membrane of intestinal enterocytes for passive diffusion.
49. **Gastrointestinal Sphincters & Valves**: Flow of ingested matter is regulated by specialized muscular sphincters: **Lower Esophageal / Gastroesophageal (Cardiac) Sphincter** prevents gastric acid reflux into the esophagus; **Pyloric Sphincter** meters chyme release into the duodenum; **Ileocaecal Valve** prevents backflow of fecal

contents from the caecum into the ileum; **Internal (involuntary) and External (voluntary) Anal Sphincters** control defecation.

50. **Physiological vs Anatomical Dead Space**: **Anatomical dead space** is the volume of the conducting respiratory airways (nose, pharynx, larynx, trachea, bronchi, bronchioles \approx 150\text{ mL} in adults) where gas exchange cannot occur; in healthy individuals, anatomical dead space equals **physiological dead space**; elevated physiological dead space occurs when alveoli are ventilated but poorly perfused (e.g., in emphysema or pulmonary embolism).

📋 Must-Score Drill — Digestive Juices, Lung Volumes & Nephron Functions

Complete Alimentary Canal Enzymes Master Reference

Digestive Gland / Source	Secreted Juice / Enzyme	Substrate Acted Upon	End Products Formed	Optimum pH
Salivary Glands	**Ptyalin (Salivary Amylase)**	Cooked Starch (~30%)	Maltose, Isomaltose, \alpha-dextrins	6.8 (Slightly acidic)
Gastric Glands (Chief cells)	**Pepsin** (from Pepsinogen)	Native Proteins	Proteoses and Peptones (Peptides)	1.5–2.0 (Strongly acidic)
Gastric Glands (Infants)	**Rennin** (from Prorennin)	Milk protein Casein	Insoluble Calcium Paracaseinate	1.5–2.0
Liver (Hepatocytes)	**Bile** (NO enzymes)	Large fat globules	Emulsified fat micro-droplets	7.6–8.6 (Alkaline)
Pancreas (Exocrine Acini)	**Pancreatic Amylase (Amylopsin)**	Remaining Starches	Maltose and Isomaltose	7.5–8.5
Pancreas (Exocrine Acini)	**Trypsin & Chymotrypsin**	Proteins, Peptones	Short Dipeptides & Tripeptides	7.5–8.5
Pancreas (Exocrine Acini)	**Steapsin (Pancreatic Lipase)**	Emulsified Triglycerides	Diglycerides, Monoglycerides, Fatty acids	7.5–8.5
Pancreas (Exocrine Acini)	**Carboxypeptidases**	Peptides at C-terminus	Free Amino Acids	7.5–8.5
Small Intestine (Succus Entericus)	**Enterokinase**	Trypsinogen	Active Trypsin (Master activator)	7.8
Small Intestine (Succus Entericus)	**Maltase**	Maltose	Glucose + Glucose	7.8
Small Intestine (Succus Entericus)	**Sucrase (Invertase)**	Sucrose	Glucose + Fructose	7.8
Small Intestine (Succus Entericus)	**Lactase**	Lactose	Glucose + Galactose	7.8
Small Intestine (Succus Entericus)	**Erepsin (Peptidases)**	Dipeptides	Individual Free Amino Acids	7.8

Respiratory Volumes & Capacities Breakdown

Parameter	Clinical Definition	Standard Adult Volume
Tidal Volume (TV)	Volume of air inspired or expired in a normal quiet respiration	**\sim 500\text{ mL}**
Inspiratory Reserve Volume (IRV)	Maximum extra volume inspired forcefully after quiet inspiration	**2500–3000\text{ mL}**
Expiratory Reserve Volume (ERV)	Maximum extra volume expired forcefully after quiet expiration	**1000–1100\text{ mL}**

Parameter	Clinical Definition	Standard Adult Volume
Residual Volume (RV)	Volume of air remaining permanently inside lungs after maximal expiration	**1100–1200\text{ mL}**
Vital Capacity (VC)	Maximum air exhaled after maximal forced inspiration (TV + IRV + ERV)	**3500–4500\text{ mL}**
Total Lung Capacity (TLC)	Total volume of air in lungs after maximal inspiration (VC + RV)	**5000–6000\text{ mL}**

Nephron Segments & Functional Architecture

Nephron Segment	Histological Lining	Primary Transport Process	Key Substances Moved
Glomerulus & Bowman's Capsule	Podocytes with filtration slits	**Ultrafiltration** (180\text{ L/day})	Filters water, glucose, salts, urea; retains blood cells and large proteins
Proximal Convoluted Tubule (PCT)	Simple cuboidal brush border	**Selective Reabsorption** (70–80\%)	100\% Glucose, 100\% Amino acids, 70–80\% Na^+, K^+, Cl^-, HCO_3^-, Water
Descending Limb of Henle	Thin squamous epithelium	**Water reabsorption**	Highly permeable to water; completely impermeable to electrolytes
Ascending Limb of Henle	Thick cuboidal epithelium	**Active electrolyte pumping**	Impermeable to water; actively pumps Na^+, K^+, Cl^- into medullary interstitium
Distal Convoluted Tubule (DCT)	Simple cuboidal epithelium	**Conditional reabsorption**	Facultative water reabsorption under **ADH**; Na^+ reabsorption under **Aldosterone**
Collecting Duct	Simple columnar epithelium	**Final urine concentration**	Water reabsorption under ADH; secretes H^+, K^+ to regulate blood pH

One-Liner Revision – Digestion, Respiration and Excretion

#	Fast Fact
1	Digestion starts in the **mouth** (salivary amylase/ptyalin digests starch)
2	Stomach secretes = HCl + Pepsin (protein digestion) + Mucin (protects lining)
3	Liver = largest **gland** of the body; also largest internal organ
4	Bile secreted by = **liver**; stored in = **gallbladder**
5	Main site of digestion and absorption = **Small intestine** (duodenum/jejunum/ileum)
6	Large intestine main function = **water absorption**; no digestion
7	Longest part of alimentary canal = **Small intestine** (~6-7 metres)
8	Pancreas = both **exocrine** (digestive enzymes) and **endocrine** (insulin/glucagon)
9	Aerobic respiration: Glucose + O2 = CO2 + H2O + **38 ATP**
10	Anaerobic in muscles = **lactic acid** (causes muscle cramp/fatigue)
11	Anaerobic in yeast = **ethanol + CO2** (fermentation; used in brewing)
12	Breathing rate (adults) = 15-18 breaths/min at rest
13	Tidal volume = air breathed in/out in one normal breath (~500 mL)
14	Vital capacity = maximum air exhaled after maximum inhalation (~4.5 L)
15	Gas exchange in lungs = in **alveoli** (O2 diffuses in; CO2 diffuses out)
16	Haemoglobin carries **O2** as oxyhaemoglobin; also carries some CO2
17	CO poisoning = CO binds Hb **240x more tightly** than O2 (cherry-red colour)

#	Fast Fact
18	Mammals excrete **Urea** (ureotelic); Birds/reptiles excrete **Uric acid**
19	Fish excrete **Ammonia** (ammonotelic; highly toxic; needs lots of water)
20	Nephron = functional unit of kidney; Bowman's capsule + tubules + loop of Henle
21	Urine formation: Filtration + Selective reabsorption + Tubular secretion
22	Normal urine output = ~1-1.5 L/day; urine pH ~6 (slightly acidic)
23	Kidney stone = **calcium oxalate** (most common); treated by lithotripsy
24	Dialysis = used when kidneys fail; hemodialysis filters blood through machine
25	Skin is also an excretory organ = excretes sweat (water, NaCl, urea)
26	Lungs excrete = CO2 and water vapour
27	Liver excretes = bile pigments (bilirubin from dead RBCs)
28	ADH (Anti-diuretic hormone) = controls water reabsorption in kidney tubules
29	Jaundice = yellow skin/eyes due to accumulation of **bilirubin**
30	Appendix = vestigial organ in large intestine; removed in appendicitis

Topic 5 — Blood, Heart, Circulation and Lymph

Consolidated — 50 Must-Score Facts

1. Blood is a specialized fluid connective tissue consisting of **~55% liquid Plasma** and **~45% cellular Formed Elements**; slightly alkaline with a normal physiological **pH of 7.35–7.45**; total blood volume in a healthy adult is **5 to 6 litres** (~7–8% of body weight).
2. Blood plasma is 90–92% water; plasma proteins include **Albumin** (maintains colloidal osmotic / oncotic pressure, ~4 g/dL), **Globulins** (\alpha, \beta, \gamma — \gamma-globulins act as circulating **antibodies**), and **Fibrinogen** (essential for blood clotting); **\text{Serum} = \text{Plasma} - \text{Clotting Factors (Fibrinogen)}**.
3. **Erythrocytes (Red Blood Cells / RBCs)**: Normal count is **5.0–5.5 million / \mathbf{mm^3}** in males and 4.5–5.0 million / \text{mm}^3 in females; mature mammalian RBCs are **biconcave, circular, and enucleated** (lack nucleus, mitochondria, and Golgi bodies; **Camel and Llama are exceptions: their RBCs are oval and nucleated**).
4. Absence of mitochondria forces RBCs to generate ATP exclusively through **anaerobic glycolysis**, preventing them from consuming any of the oxygen they transport.
5. RBCs are produced in the red bone marrow (**Erythropoiesis**); stimulated by the glycoprotein hormone **Erythropoietin (EPO)** secreted by renal juxtaglomerular cells in response to hypoxia.
6. The average lifespan of a human red blood cell is **\mathbf{\sim 120\text{ days}}**; worn-out aged RBCs are destroyed by macrophages in the **Spleen**, which is universally designated as the **"Graveyard of RBCs"** (and acts as a reserve "Blood Bank").
7. **Haemoglobin (Hb)**: Conjugated chromoprotein; normal concentration is **14–17\text{ g/dL}** in adult men and **12–15\text{ g/dL}** in women; consists of the protein globin and 4 iron-porphyrin **Haem** groups containing **ferrous iron (\mathbf{Fe^{2+}})**; imparts the red colour to blood.
8. Each molecule of haemoglobin can bind a maximum of **4 molecules of Oxygen (O_2)** to form bright red **Oxyhaemoglobin**; deoxygenated blood is darker bluish-red.
9. **Methemoglobin & Blue Baby Syndrome**: When ferrous iron (Fe^{2+}) of haemoglobin is oxidized into ferric iron (Fe^{3+}), it forms **Methemoglobin**, which cannot bind or carry oxygen; infants ingesting groundwater contaminated with **Nitrates (NO_3^-)** suffer from **Methemoglobinemia (Blue Baby Syndrome)**, presenting with acute cyanosis and chocolate-brown blood.
10. **Leukocytes (White Blood Cells / WBCs)**: Total count is **6,000–8,000/\mathbf{mm^3}**; nucleated, lack haemoglobin; divided into **Granulocytes** and **Agranulocytes**; the normal ratio of RBC to WBC is **600:1**.
11. **Granulocytes**:
    - **Neutrophils (60–65%)**: Most numerous; multi-lobed nucleus; primary phagocytic first-responders that engulf bacteria and cell debris.
    - **Eosinophils / Acidophils (2–3%)**: Bilobed nucleus; release histaminase; counts surge in **allergic reactions and intestinal helminth parasitic infections**.
    - **Basophils (0.5–1%)**: Least numerous; secrete **Histamine** (vasodilator in inflammation), **Serotonin**, and **Heparin** (natural anticoagulant).
12. **Agranulocytes**:
    - **Monocytes (6–8%)**: Largest WBCs; bean/kidney-shaped nucleus; enter tissues to transform into massive phagocytic **Macrophages**.
    - **Lymphocytes (20–25%)**: Large rounded nucleus; responsible for specific adaptive immunity: **B-lymphocytes** (produce humoral antibodies) and **T-lymphocytes** (mature in the thymus, execute cell-mediated immunity).
13. **Thrombocytes (Blood Platelets)**: Normal count is **\mathbf{1.5–4.5\text{ lakh}}/\mathbf{mm^3}**; non-nucleated protoplasmic cell fragments budded from giant marrow **Megakaryocytes**; lifespan 7–10 days;

aggregate at wound sites and release **Thromboplastin** to trigger blood clotting; depleted drastically in dengue fever.

14. **Blood Clotting (Coagulation Cascade)**:
    - Step 1: Ruptured tissues and platelets release **Thromboplastin / Thrombokinase**.
    - Step 2: Prothrombin is converted to active **Thrombin** in the presence of **Calcium ions (\mathbf{Ca^{2+}} / Factor IV)** and thrombokinase.
    - Step 3: Thrombin converts soluble **Fibrinogen** into insoluble **Fibrin polymer threads**, forming a network that traps blood cells to produce a clot (**Thrombus**).
15. **Vitamin K** is indispensable for hepatic synthesis of four essential clotting factors: **Factor II (Prothrombin), Factor VII, Factor IX, and Factor X**.
16. **Heparin** is a natural conjugated polysaccharide anticoagulant secreted by basophils and mast cells; prevents intravascular clotting inside healthy blood vessels.
17. In clinical blood banks, blood clotting in collection bags is prevented by adding calcium-chelating chemical anticoagulants: **Sodium Citrate, EDTA, or CPD (Citrate Phosphate Dextrose)**.
18. **ABO Blood Group System**: Discovered by Austrian biologist **Karl Landsteiner (1900–1901)**; blood groups (A, B, AB, O) are determined by specific oligosaccharide **antigens on the surface of RBC membranes** and complementary **antibodies in blood plasma**:
    - **Group A**: Has A antigen on RBCs; anti-B antibody in plasma.
    - **Group B**: Has B antigen on RBCs; anti-A antibody in plasma.
    - **Group AB**: Has **both A and B antigens** on RBCs; **NEITHER anti-A nor anti-B antibody** in plasma.
    - **Group O**: Has **NEITHER A nor B antigen** on RBCs; **BOTH anti-A and anti-B antibodies** in plasma.
19. Transfusion rules: **O Negative (\mathbf{O^-})** is the **Universal Donor** for red-cell transfusions (lacks A, B, and Rh antigens, so donor RBCs cannot be attacked by recipient antibodies); **AB Positive (\mathbf{AB^+})** is the **Universal Recipient** (lacks all anti-A, anti-B, and anti-Rh antibodies in plasma, so cannot attack transfused red cells).
20. **Rh Factor (Rhesus Antigen / Antigen D)**: Discovered by **Karl Landsteiner and Alexander S. Wiener (1940)** in the Rhesus macaque monkey; present on RBCs of ~85% of humans (Rh-positive); ~15% are Rh-negative.
21. **Erythroblastosis Fetalis (Haemolytic Disease of the Newborn)**: Occurs when an **Rh-negative mother** conceives an **Rh-positive fetus**; fetal Rh antigens leak into maternal blood during first delivery, sensitizing the mother to produce maternal anti-Rh antibodies (**IgG class**); during a subsequent Rh-positive pregnancy, maternal anti-Rh IgG crosses the placenta and hemolyses fetal erythrocytes, causing severe fatal anaemia, kernicterus, and hydrops fetalis; prevented by administering **Anti-D Rh-immunoglobulin (RhoGAM)** to the mother within 72 hours of delivering an Rh-positive baby.
22. **Bombay Blood Group (hh Phenotype)**: Rare blood phenotype discovered in Mumbai by **Dr. Y. M. Bhende (1952)**; individuals lack the universal precursor **H-gene**, so cannot biosynthesise A or B antigens; tests as Group O in routine ABO grouping, but serum contains anti-A, anti-B, and **anti-H antibodies**; can donate blood to anyone in the ABO system, but can **receive blood ONLY from another Bombay blood group donor**.
23. The human heart is a hollow, muscular, myogenic organ situated in the mediastinum of the thoracic cavity between the lungs, tilted slightly to the left; enclosed within a double-walled fibro-serous sac called the **Pericardium** filled with lubricating pericardial fluid.
24. The human heart contains **\mathbf{4\text{ chambers}}**: two thin-walled superior receiving chambers called **Atria (Auricles)** and two thick-walled inferior pumping chambers called **Ventricles**.
25. The **Left Ventricle** possesses the thickest, most muscular myocardial wall (about 3 times thicker than the right ventricle) because it must pump blood at high pressure into the **Aorta** across the entire systemic body circuit.
26. Internal Heart Valves:
    - **Tricuspid Valve**: Possesses 3 muscular cusps; guards the right atrio-ventricular orifice between the right atrium and right ventricle.

- **Bicuspid Valve (Mitral Valve)**: Possesses 2 muscular cusps; guards the left atrio-ventricular orifice between the left atrium and left ventricle.
- **Semilunar Valves**: Guard the exit of the right ventricle into the **Pulmonary Artery** (Pulmonary semilunar valve) and the exit of the left ventricle into the **Aorta** (Aortic semilunar valve); prevent backflow into ventricles during diastole.

27. Valve cusps are tethered to papillary muscles in the ventricular walls by tough, inextensible fibrous cords called **Chordae Tendineae ("Heart Strings")**, preventing cusps from inverting into atria during high-pressure ventricular systole.
28. **Nodal Conduction System (Myogenic Autonomic Rhythmicity)**:
    - **SA Node (Sino-Atrial Node / "Heart of Hearts")**: Located in the upper-right corner of the right atrium near the entry of the superior vena cava; generates intrinsic rhythmic action potentials at **70–75 beats/minute**; functions as the natural **Pacemaker of the Heart**.
    - **AV Node (Atrio-Ventricular Node)**: Located in the lower-left corner of the right atrium near the inter-atrial septum; acts as a relay station, introducing a crucial **0.1-second delay** to allow atria to empty completely before ventricular contraction begins; called the **Pacesetter**.
    - **Bundle of His (AV Bundle)**: Specialized muscle fascicle descending down the interventricular septum, bifurcating into Right and Left Bundle Branches.
    - **Purkinje Fibres**: Microscopic conduction fibres spreading rapidly throughout the ventricular myocardium to trigger coordinated contraction from the cardiac apex upward.
29. If the natural SA node pacemaker fails, an artificial battery-powered **Electronic Pacemaker** (invented by Wilson Greatbatch, containing a lithium-iodine battery) is surgically implanted under the skin to stimulate the heart electrically.
30. The **Cardiac Cycle** lasts **\mathbf{0.8\text{ seconds}}** at a normal resting rate of 72 beats/minute:
    - **Atrial Systole**: 0.1 second (atria contract, topping up ventricles).
    - **Ventricular Systole**: 0.3 second (ventricles contract at high pressure, pumping blood into aorta and pulmonary artery).
    - **Joint Diastole (Complete Cardiac Diastole)**: 0.4 second (all four chambers relax simultaneously and refill passively).
31. **Stroke Volume**: Volume of blood pumped out by each ventricle during each individual cardiac cycle = **\mathbf{\sim 70\text{ mL}}**.
32. **Cardiac Output**: Volume of blood pumped by each ventricle per minute: \text{Cardiac Output} = \text{Stroke Volume (70 mL)} \times \text{Heart Rate (72 bpm)} \approx \mathbf{5040\text{ mL/min}} \approx \mathbf{5\text{ Litres/min}}
33. **Heart Sounds Auscultated via Stethoscope**:
    - **First Heart Sound ("LUBB")**: Low-pitched, booming, longer duration (~0.15s); produced by the sudden, simultaneous **closure of the Atrioventricular Valves (Tricuspid and Bicuspid)** at the onset of ventricular systole.
    - **Second Heart Sound ("DUPP")**: Higher-pitched, sharp, shorter duration (~0.10s); produced by the rapid **closure of the Semilunar Valves** at the onset of ventricular diastole.
    - **Logic:** Heart sounds are produced by the snapping shut of heart valves, **not** by the contraction of muscle tissue itself.
34. **Electrocardiogram (ECG)**: Graphical recording of the electrical vector voltages generated by the heart during cardiac cycles; invented by Dutch physician **Willem Einthoven (Father of ECG, Nobel Prize 1924)**:
    - **P Wave**: Represents **Atrial Depolarization** (electrical excitation spreading from SA node across atria, preceding atrial contraction).
    - **QRS Complex**: Represents **Ventricular Depolarization** (rapid electrical spread triggering ventricular systole).

- **T Wave**: Represents **Ventricular Repolarization** (return of ventricles from excitation to resting electrical state).

35. **Blood Pressure (BP)**: Lateral hydrostatic pressure exerted by circulating blood against the elastic walls of major systemic arteries; measured in **mm Hg** using a **Sphygmomanometer** placed over the brachial artery at the level of the heart:
    - **Systolic Blood Pressure (SBP)**: Peak arterial pressure during left ventricular contraction = **$\mathbf{120\text{ mm Hg}}$**.
    - **Diastolic Blood Pressure (DBP)**: Minimum baseline arterial pressure during ventricular relaxation = **$\mathbf{80\text{ mm Hg}}$**.
    - **Standard Normal Resting BP**: **$\mathbf{120/80\text{ mm Hg}}$**.
    - **Pulse Pressure**: $\text{Systolic} - \text{Diastolic} = 120 - 80 = \mathbf{40\text{ mm Hg}}$.
    - **Hypertension**: Sustained resting arterial pressure $\ge 140/90\text{ mm Hg}$; damages kidneys, retina, and coronary vessels.
36. **Arteries vs Veins Anatomical Distinctions**:
    - **Arteries**: Carry blood **away** from the heart under high pulsatile pressure; walls are thick, muscular, and rich in elastic fibres (thick tunica media); deep-seated; **lack internal valves**; carry bright red oxygenated blood (**Exception: The Pulmonary Artery carries deoxygenated blood from the right ventricle to the lungs**).
    - **Veins**: Carry blood **toward** the heart under low continuous pressure; walls are thin with a wide lumen; superficial; **contain crescent-shaped Semilunar Valves** to prevent backflow of blood under gravity; carry dark deoxygenated blood (**Exception: The Pulmonary Veins carry fully oxygenated blood from the lungs into the left atrium**).
    - **Capillaries**: Microscopic microscopic vessels whose walls consist of a single layer of endothelial cells; site of all nutrient, gas, and metabolic exchange between blood and tissue interstitial fluid.
37. **Double Circulation in Humans**: Blood passes through the heart **twice** to complete one full systemic circuit:
    - **Pulmonary Circuit**: $\text{Right Ventricle} \xrightarrow[\text{Deoxygenated}]{\text{Pulmonary Artery}} \text{Lungs} \xrightarrow[\text{Oxygenated}]{\text{Pulmonary Veins}} \text{Left Atrium}$.
    - **Systemic Circuit**: $\text{Left Ventricle} \xrightarrow[\text{Oxygenated}]{\text{Aorta}} \text{Body Tissues} \xrightarrow[\text{Deoxygenated}]{\text{Venae Cavae}} \text{Right Atrium}$.
38. **The Hepatic Portal System**: Specialized venous pathway in which venous blood draining the stomach, pancreas, spleen, and intestines does **not** return directly to the inferior vena cava, but is collected by the **Hepatic Portal Vein** and delivered to capillary sinusoids of the **Liver**; ensures absorbed toxins, glucose, and nutrients are processed and detoxified before reaching the systemic heart.
39. **Coronary Circulation**: The heart myocardium does not absorb nutrients or oxygen directly from blood passing through its chambers; it is supplied exclusively by the **Right and Left Coronary Arteries** (the very first branches arising from the root of the ascending aorta); coronary venous blood returns directly to the right atrium via the **Coronary Sinus**.
40. **Cardiovascular Pathology**:
    - **Atherosclerosis**: Deposition of cholesterol, calcium, and fibrous tissue forming hard atheromatous plaques within arterial walls, narrowing the lumen.
    - **Arteriosclerosis**: Hardening and loss of elasticity of arterial walls due to calcification.
    - **Angina Pectoris**: Severe acute substernal chest pain radiating to the left shoulder and arm caused by transient cardiac ischemia (inadequate oxygen supply to heart muscle).
    - **Myocardial Infarction (Heart Attack)**: Sudden acute thrombotic occlusion of a coronary artery, causing irreversible ischemic necrosis (death) of a patch of myocardium.
41. **Lymphatic System & Lymph (Tissue Fluid / Interstitial Fluid)**:

- As blood flows through arterial capillary beds, hydrostatic pressure forces fluid, water-soluble nutrients, and ions out through capillary pores into intercellular spaces, forming **Interstitial / Tissue Fluid**.
- Interstitial fluid drained by specialized blind-ended **Lymphatic Capillaries** is termed **Lymph**.
- **Chemical Composition of Lymph**: Straw-coloured, transparent fluid identical to blood plasma **minus red blood cells, platelets, and large plasma proteins (albumin/fibrinogen)**; contains abundant **Lymphocytes**.

42. **Circulation of Lymph**: Lymphatic capillaries coalesce into lymphatic vessels resembling veins with numerous valves; drain into two large terminal trunks: the **Right Lymphatic Duct** (drains right upper quadrant) and the **Thoracic Duct (Left Lymphatic Duct)** (drains remaining 75% of body); both empty directly into the **Subclavian Veins** at the base of the neck.
43. **Lymph Nodes**: Small oval glandular encapsulated filter organs distributed along lymphatic channels (in neck, armpits, groin, mesentery); filter lymph, trap circulating bacteria, viruses, and metastasizing cancer cells; packed with lymphocytes and plasma cells that mount adaptive immune responses.
44. **The Spleen (Pliha / तिल्ली)**:
    - Largest single mass of lymphatic tissue in the human body; located in the left hypochondriac region of the abdomen beneath the diaphragm.
    - Contains **Red Pulp** (vascular sinusoids that filter blood, remove old worn-out erythrocytes and blood-borne pathogens) and **White Pulp** (lymphoid sheaths housing T and B lymphocytes).
    - Functions: **Graveyard of RBCs**, **Blood Bank** (can contract to release stored red blood cells into systemic circulation during acute haemorrhage), and major antibody-producing lymphoid organ.
45. **Cardiopulmonary Resuscitation (CPR)**: Emergency life-saving procedure comprising chest compressions (depth 5–6 cm at a rate of 100–120 compressions/minute) combined with artificial rescue breaths to manually preserve blood circulation to the brain and vital organs during cardiac arrest.
46. **Frank-Starling Law of the Heart**: States that within physiological limits, the stroke volume of the heart increases in response to an increase in the end-diastolic volume (venous return); greater myocardial stretch before contraction lengthens sarcomeres closer to optimal overlap, generating greater contractile force during subsequent ventricular systole.
47. **Coronary Sinus & Thebesian Veins**: Venous blood draining the left ventricular myocardium collects in the great cardiac vein, empties into the wide **Coronary Sinus**, and discharges directly into the right atrium; minute **Thebesian Veins (Venae Cordis Minimae)** drain inner myocardial layers directly into all four cardiac chambers, bypassing major coronary veins.
48. **Hematopoietic Stem Cell Lineages**: Pluripotent hematopoietic stem cells (HSCs) in the red bone marrow differentiate into two primary progenitor lineages: the **Myeloid Lineage** (gives rise to erythrocytes, megakaryocytes/platelets, monocytes, and granulocytes) and the **Lymphoid Lineage** (gives rise to T-lymphocytes, B-lymphocytes, and Natural Killer cells).
49. **Inherited Coagulation Factor Deficiencies (Hemophilia)**: **Hemophilia A (Classic Hemophilia)** is caused by a congenital deficiency of clotting **Factor VIII** (anti-hemophilic globulin); **Hemophilia B (Christmas Disease)** is caused by a deficiency of clotting **Factor IX**; both are X-linked recessive genetic traits transmitted by asymptomatic carrier mothers predominantly to male offspring.
50. **Sphygmomanometer & Auscultatory Korotkoff Sounds**: Arterial blood pressure is measured clinically using an inflatable arm cuff and mercury or aneroid **Sphygmomanometer** (invented by Scipione Riva-Rocci); the physician inflates the cuff above systolic pressure and auscultates over the brachial artery with a stethoscope; the onset of tapping **Korotkoff sounds** marks Systolic BP (120 mm Hg), and their total disappearance marks Diastolic BP (80 mm Hg).

📋 Must-Score Drill — Chambers, Blood Groups & Vessels

Vertebrate Heart Chambers Master Summary

Animal Taxon	Heart Chambers	Structural Separation	Direction of Blood Flow
Fish (Pisces)	**2 Chambers** (1 Atrium + 1 Ventricle)	Single circuit; no partition	Venous blood only \rightarrow Gills \rightarrow Body tissues
Amphibians (Frogs)	**3 Chambers** (2 Atria + 1 Ventricle)	Incomplete double circulation	Mixed blood in single ventricle
Reptiles (Snakes, Lizards)	**3 Chambers** (Incompletely 4)	Incomplete interventricular septum	Partial mixing of oxygenated/deoxygenated blood
Crocodiles (Reptile Exception)	**4 Chambers** (2 Atria + 2 Ventricles)	Complete septum	Complete separation of blood streams
Birds (Aves)	**4 Chambers** (2 Atria + 2 Ventricles)	Complete double circulation	Right systemic aortic arch
Mammals (Human, Whale, Bat)	**4 Chambers** (2 Atria + 2 Ventricles)	Complete double circulation	Left systemic aortic arch

Comprehensive ABO & Rh Compatibility Table

Blood Group	Antigens on Erythrocytes	Antibodies in Plasma	Can Donate Red Cells To	Can Receive Red Cells From
A^+	A, Rh	Anti-B	A^+, AB^+	A^+, A^-, O^+, O^-
A^-	A only	Anti-B, (Anti-Rh if exposed)	A^+, A^-, AB^+, AB^-	A^-, O^-
B^+	B, Rh	Anti-A	B^+, AB^+	B^+, B^-, O^+, O^-
B^-	B only	Anti-A, (Anti-Rh if exposed)	B^+, B^-, AB^+, AB^-	B^-, O^-
AB^+	A, B, Rh	**NONE**	**AB^+ only**	**ALL GROUPS (Universal Recipient)**
AB^-	A, B	**NONE**	AB^+, AB^-	AB^-, A^-, B^-, O^-
O^+	Rh only	Anti-A, Anti-B	O^+, A^+, B^+, AB^+	O^+, O^-
O^-	**NONE**	Anti-A, Anti-B	**ALL GROUPS (Universal Donor)**	**O^- only**
Bombay (hh)	None (lacks H-antigen)	Anti-A, Anti-B, **Anti-H**	All ABO groups	**Bombay (hh) only**

Complete Vessel Anatomy & Circulation Map

Blood Vessel Name	Oxygenation Status	Origin Chamber / Structure	Destination Organ / Structure	Unique Physiological Feature
Ascending Aorta	Fully Oxygenated	Left Ventricle	Entire Systemic Body	Largest artery; highest blood pressure
Superior Vena Cava	Deoxygenated	Head, Neck, Upper Limbs	Right Atrium	Drains venous blood from upper body
Inferior Vena Cava	Deoxygenated	Abdomen, Lower Limbs	Right Atrium	Largest vein in the human body
Pulmonary Artery	**Deoxygenated**	Right Ventricle	Lungs (Alveoli)	**Only artery carrying deoxygenated blood**

Blood Vessel Name	Oxygenation Status	Origin Chamber / Structure	Destination Organ / Structure	Unique Physiological Feature
Pulmonary Veins (4)	**Oxygenated**	Lungs (Alveoli)	Left Atrium	**Only veins carrying oxygenated blood**
Hepatic Portal Vein	Deoxygenated (Nutrient-rich)	Gastrointestinal Tract	Liver Sinusoids	Connects two capillary networks
Coronary Arteries	Oxygenated	Root of Aorta	Myocardium of Heart	Supplies heart muscle directly

One-Liner Revision – Blood, Heart and Circulation

#	Fast Fact
1	Blood = **55% plasma** + 45% formed elements (RBC, WBC, Platelets)
2	RBC lifespan = **120 days**; WBC lifespan = a few hours to several years
3	RBC count (normal) = **45-55 lakh/mm3** in men; 40-50 lakh/mm3 in women
4	WBC count (normal) = **4,000-11,000/mm3**
5	Platelet count (normal) = **1.5-3.5 lakh/mm3**
6	Most abundant WBC = **Neutrophil** (60-70%); first line of defence
7	Antibody-producing WBC = **Lymphocyte** (B-lymphocytes)
8	Haemoglobin = iron (Fe2+) + globin protein; made in **bone marrow**
9	ABO blood groups discovered = **Karl Landsteiner (1901)**; Nobel 1930
10	Universal Donor = **O-** (O negative); no A, B antigens; no Rh antigen
11	Universal Recipient = **AB+** (AB positive); has all antigens
12	Erythroblastosis foetalis = Rh incompatibility between Rh- mother and Rh+ baby
13	Heart chambers: 2 Atria (upper) + 2 Ventricles (lower) = **4 chambers**
14	Heart rate (normal resting adults) = **60-100 beats/min**
15	Heartbeat sound “Lub-Dub” = closing of **AV valves** (Lub) + semilunar valves (Dub)
16	ECG = Electrocardiogram; records electrical activity of heart
17	Blood pressure normal = **120 mmHg** (systolic) / **80 mmHg** (diastolic)
18	Hypertension = BP > 140/90 mmHg (Silent killer)
19	Arteries carry blood FROM heart; thicker walls; no valves (except at heart base)
20	Veins carry blood TO heart; thinner walls; have valves to prevent backflow
21	Pulmonary artery = ONLY artery carrying deoxygenated blood (from heart to lungs)
22	Pulmonary vein = ONLY vein carrying oxygenated blood (from lungs to heart)
23	Coronary arteries = supply blood to heart muscle itself
24	Heart attack (Myocardial infarction) = blockage of coronary artery
25	Clotting factor = **Fibrinogen** (in plasma) converts to fibrin; **Vitamin K** needed
26	Haemophilia = genetic disorder; blood doesn’t clot; X-linked recessive
27	Spleen = “Graveyard of RBCs”; also filters blood and stores platelets
28	Bone marrow = site of blood cell production (haematopoiesis)
29	Lymph = plasma that leaks from capillaries into tissue; carried in lymphatic system

#	Fast Fact
30	Carotid arteries = main arteries supplying blood to the brain

Topic 6 — Nervous System, Hormones, Reproduction and Support

Consolidated — 50 Must-Score Facts

1. The structural and functional unit of the nervous system is the **Neuron**; adult neurons reside permanently in the quiescent G_0 phase of the cell cycle and **do not divide** (lack active centrioles).
2. A typical neuron consists of a **Cell Body (Soma/Cyton)** containing Nissl's granules, short branched receptive **Dendrites**, and a single long conducting **Axon** terminating in synaptic knobs; myelin sheath in the peripheral nervous system is secreted by **Schwann cells**.
3. Two neurons communicate across a microscopic gap called a **Synapse** via chemical messengers called **Neurotransmitters** (e.g. **Acetylcholine**, Noradrenaline, Dopamine, GABA); electrical sparks do not jump across ordinary synapses.
4. The central nervous system (CNS) is wrapped in three protective fibro-vascular membranes called **Meninges**: outer tough **Duramater**, middle web-like **Arachnoid mater**, and inner delicate vascular **Piamater** (Mnemonic: **DAP**); space between arachnoid and piamater is filled with shock-absorbing **Cerebrospinal Fluid (CSF)**.
5. **Cerebrum**: Largest part of the human brain (~80% of brain mass); seat of intelligence, conscious thought, memory, reasoning, voluntary motor control, speech (Broca's area), and sensation; right and left cerebral hemispheres are joined internally by a broad transverse curved band of white nerve fibres called the **Corpus Callosum**.
6. **Hypothalamus**: Situated at the base of the diencephalon; acts as the body's **Thermostat** (regulates core body temperature); controls hunger, thirst, satiety, sleep-wake cycles, emotional responses (anger, fear, pleasure via the limbic system), and regulates the pituitary gland via releasing and inhibiting neurohormones.
7. **Cerebellum**: Second largest part of the brain; displays an internal tree-like branching pattern of white matter called **Arbor Vitae** ("Tree of Life"); coordinates voluntary muscular contractions, maintains posture, and controls **physical equilibrium and body balance**; alcohol intoxication temporarily depresses the cerebellum, causing muscular uncoordination and a staggering gait.
8. **Medulla Oblongata**: Continues inferiorly into the spinal cord; houses vital autonomous reflex centres controlling **respiratory rhythm, cardiac rate, arterial blood pressure (vasomotor centre)**, as well as protective reflexes: swallowing, coughing, sneezing, vomiting, and salivation.
9. Peripheral nerves: Humans possess **$\mathbf{12\text{ pairs}}$ of Cranial Nerves** (arising from brain; Vagus nerve / CN X is the longest and most widely distributed) and **$\mathbf{31\text{ pairs}}$ of Spinal Nerves** (8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal; all spinal nerves are mixed nerves).
10. Autonomic Nervous System (ANS): **Sympathetic System** prepares the body for intense physical exertion ("Fight or Flight": dilates pupils, accelerates heart, dilates bronchi, inhibits gut peristalsis); **Parasympathetic System** restores calm ("Rest and Digest": constricts pupils, slows heart, stimulates digestive peristalsis and salivation).
11. A **Reflex Arc** is an involuntary, rapid, automated, stereotyped response to a sensory stimulus that does not require conscious intervention by the cerebral cortex; sequential pathway: **Receptor → Sensory (Afferent) Neuron → Spinal Cord Interneuron → Motor (Efferent) Neuron → Effector Muscle / Gland**.
12. **The Human Eye Anatomy**: The eyeball wall possesses three layers: outer fibrous tunic (**Sclera** anteriorly continuous with the transparent **Cornea**; cornea lacks blood vessels, absorbs oxygen directly from atmospheric air, making corneal grafts highly successful without immune rejection); middle vascular tunic (**Choroid**, **Ciliary Body**, and pigmented **Iris** with central pupil); inner sensory tunic (**Retina** containing photoreceptor cells).
13. **Photoreceptors: Rods vs Cones**: **Rods** (~120 million) contain the purplish-red pigment **Rhodopsin** (visual purple, requiring Vitamin A/retinol), operate in twilight/dim light (**Scotopic vision**), and cannot perceive colors; **Cones** (~6 million) contain **Iodopsin**, operate in bright light (**Photopic vision**), and mediate color vision (Trichromatic: red, green, blue); the **Fovea Centralis (Yellow Spot / Macula Lutea)** contains exclusively densely packed cones and provides the highest visual acuity.

14. **Common Ophthalmic Defects**: **Myopia (Short-sightedness)**: Eyeball too long or corneal curvature excessive, image forms in front of retina, corrected by **Concave Lens**; **Hypermetropia (Far-sightedness)**: Eyeball too short, image forms behind retina, corrected by **Convex Lens**; **Presbyopia**: Age-related loss of ciliary muscle flexibility, corrected by **Bifocal Lens**; **Astigmatism**: Asymmetrical corneal curvature, corrected by **Cylindrical Lens**; **Cataract**: Progressive opacity/clouding of the crystalline lens, treated by surgical phacoemulsification and intraocular lens (IOL) implant.
15. **The Human Ear Anatomy & Hearing**: Outer ear (Pinna and external auditory meatus); middle ear containing three auditory ossicles that amplify sound vibrations ~20-fold: **Malleus (Hammer)**, **Incus (Anvil)**, and **Stapes (Stirrup)** — the **Stapes is the smallest bone in the human body**; Eustachian tube connects the middle ear cavity to the nasopharynx, equalizing atmospheric air pressure on either side of the tympanic membrane; inner ear contains the fluid-filled coiled **Cochlea** (organ of hearing, Organ of Corti) and **Semicircular Canals** (dynamic and static body balance).
16. **Endocrine Glands (Ductless Glands)**: Secretions are poured directly into circulating blood; target distant organs; e.g. Pituitary, Thyroid, Parathyroid, Adrenal, Pineal, Thymus.
17. **Exocrine Glands**: Possess tubular ducts that deliver secretions locally onto an epithelial surface; e.g. Salivary, Sweat, Sebaceous, Lacrimal (tear), Mammary glands.
18. **Heterocrine / Mixed Glands**: The **Pancreas** acts as both an exocrine gland (acinar cells secrete digestive enzymes via the pancreatic duct) and an endocrine gland (Islets of Langerhans secrete insulin and glucagon into blood).
19. **Pituitary Gland (Hypophysis / Master Gland)**: Located in a depression of the sphenoid bone called the **Sella Turcica**; connected to the hypothalamus via the hypophyseal infundibulum stalk; anterior pituitary (Adenohypophysis) secretes **Growth Hormone (GH)**, **Prolactin (PRL)**, **TSH**, **ACTH**, **FSH**, and **LH**; hypersecretion of GH causes **Gigantism** in children and **Acromegaly** in adults (gorilla-like facial enlargement); hyposecretion causes **Dwarfism** (Pituitary dwarf / Midgets).
20. **Posterior Pituitary Hormones (Neurohypophysis)**: Stores and releases two peptide neurohormones synthesized by hypothalamic neurons: **Oxytocin** (stimulates powerful uterine contractions during parturition/labor and milk ejection / "let-down" from mammary glands upon infant suckling) and **Antidiuretic Hormone (ADH / Vasopressin)** (promotes water reabsorption in renal tubules; deficiency causes Diabetes Insipidus).
21. **The Thyroid Gland**: Largest endocrine gland in the human body; butterfly-shaped with two lateral lobes situated on either side of the trachea, joined by the **Isthmus**; follicular cells require **Iodine** to synthesize **Thyroxine (T_4)** and **Triiodothyronine (T_3)**, which regulate basal metabolic rate (BMR); also secretes **Calcitonin** from parafollicular C-cells, which lowers blood calcium levels.
22. **Thyroid Pathologies**: **Simple / Endemic Goitre**: Enlargement of the thyroid gland due to dietary iodine deficiency (prevalent in sub-Himalayan regions); **Cretinism**: Congenital infant hypothyroidism causing severe mental retardation, dwarfism, and pot belly; **Myxoedema**: Severe adult hypothyroidism causing sluggish metabolism, puffiness, and cold intolerance; **Grave's Disease (Exophthalmic Goitre)**: Hyperthyroid autoimmune disorder characterized by high BMR, tachycardia, and bulging protruding eyeballs (**Exophthalmos**).
23. **The Parathyroid Glands**: Four pea-sized glands embedded on the posterior surface of the thyroid gland lobes; secrete **Parathyroid Hormone (PTH / Collip's Hormone)**, a hypercalcemic peptide hormone that raises blood calcium levels by stimulating bone resorption (osteoclast activity) and renal/intestinal calcium reabsorption; acts as an antagonist to calcitonin; deficiency causes painful tetanic muscle spasms (**Hypocalcemic Tetany**).
24. **Adrenal Cortex (Outer 80%)**: Derived from mesoderm; organized into three zones (Mnemonic: **GFR**): Zona Glomerulosa secretes **Mineralocorticoids** (**Aldosterone**, regulates sodium/potassium balance and blood pressure); Zona Fasciculata secretes **Glucocorticoids** (**Cortisol**, "stress hormone", stimulates gluconeogenesis, anti-inflammatory and immunosuppressive); Zona Reticularis secretes small amounts of sex corticoids/androgens.
25. **Adrenal Medulla (Inner 20%)**: Derived from ectoderm (sympathetic neural crest cells); secretes catecholamines **Adrenaline (Epinephrine)** and **Noradrenaline (Norepinephrine)**; known as the **"Fight-or-Flight" / Emergency**

/ 3F (Fright, Fight, Flight) Hormones; rapidly increases heart rate, cardiac output, respiratory rate, pupillary dilation, and glycogen breakdown into glucose during acute emotional or physical stress.

26. **The Pineal Gland (Epiphysis Cerebri)**: Located on the dorsal roof of the diencephalon; secretes **Melatonin**, which regulates the 24-hour diurnal **Circadian Rhythm (Sleep-Wake Cycle)**, core body temperature, and delays premature puberty.
27. **The Thymus Gland**: Located behind the sternum in the anterior mediastinum; secretes **Thymosins**, which stimulate the maturation and differentiation of **T-lymphocytes**; acts as the primary "training school" of cellular immunity; prominent in infants, undergoes progressive age-related atrophy after puberty, leading to weakened immunity in the elderly.
28. **Endocrine Pancreas (Islets of Langerhans)**: Microscopic clusters of endocrine cells scattered throughout the pancreas: **Alpha (□lpha) cells** (~20%) secrete **Glucagon** (hyperglycemic hormone, stimulates hepatic glycogenolysis); **Beta (□eta) cells** (~70%) secrete **Insulin** (hypoglycemic 51-amino-acid peptide, facilitates cellular glucose uptake and glycogenesis; discovered by **Banting and Best, 1921**); **Delta (\delta) cells** (~5%) secrete **Somatostatin** (inhibits both insulin and glucagon secretion).
29. **Male Reproductive System Anatomy**: Primary sex organs are the **Testes**; suspended outside the abdominal cavity in the **Scrotum**, which maintains an optimal testicular temperature **2–2.5^\circ\text{C} lower** than normal core body temperature, essential for viable **Spermatogenesis**; failure of testes to descend into the scrotum is called **Cryptorchidism**; microscopic **Seminiferous Tubules** produce spermatozoa, while **Interstitial Cells of Leydig** secrete **Testosterone** (male androgen).
30. **Male Accessory Ducts & Glands**: Sperm travel from seminiferous tubules through Rete Testis \rightarrow Vasa Efferentia \rightarrow **Epididymis** (site of physiological sperm maturation and motility acquisition) \rightarrow **Vas Deferens** \rightarrow Ejaculatory Duct \rightarrow Urethra; accessory glands include paired **Seminal Vesicles** (secrete alkaline fluid rich in **Fructose**, providing energy for sperm), single **Prostate Gland** (secretes milky slightly alkaline fluid), and paired **Bulbourethral (Cowper's) Glands** (secrete lubricating pre-ejaculatory fluid).
31. **Female Reproductive System Anatomy**: Primary sex organs are the paired **Ovaries**; produce female gametes (ova) and secrete steroid hormones (**Estrogen** from ovarian Graafian follicles and **Progesterone** from the Corpus Luteum); paired **Fallopian Tubes (Oviducts)** consist of Fimbriae, Infundibulum, **Ampulla** (the normal site of **Fertilisation** where sperm meets ovum), and Isthmus; lead into the pear-shaped **Uterus (Womb)** composed of perimetrium, thick smooth muscle myometrium, and glandular inner endometrium.
32. **The Menstrual Cycle Phases**: Normal human cycle lasts ~28 days; **Menstrual Phase** (days 1–5, shedding of functional endometrial lining due to sharp decline in progesterone and estrogen); **Follicular / Proliferative Phase** (days 6–13, FSH stimulates Graafian follicle development and estrogen production, rebuilding the endometrium); **Ovulatory Phase** (day 14, rapid surge in **Luteinizing Hormone / LH Surge** triggers rupture of the mature Graafian follicle and release of the secondary oocyte); **Luteal / Secretory Phase** (days 15–28, ruptured follicle transforms into the **Corpus Luteum**, which secretes large amounts of **Progesterone** to maintain endometrial thickness for pregnancy).
33. **Fertilisation, Blastocyst Implantation & Placenta**: Fertilisation occurs in the **Ampulla of the fallopian tube**; the single-cell zygote undergoes rapid mitotic cleavage divisions into a morula and then a hollow sphere of cells called the **Blastocyst**; blastocyst embeds into the uterine endometrium (**Implantation**, occurs 6–8 days post-fertilisation); outer trophoblast cells develop chorionic villi, interdigitating with uterine tissue to form the **Placenta** (acts as a structural and physiological exchange organ and temporary endocrine gland).
34. **Human Chorionic Gonadotropin (hCG)**: Secreted by placental syncytiotrophoblasts immediately after blastocyst implantation; rescues and maintains the corpus luteum, preventing menstrual breakdown and sustaining progesterone secretion; excreted in maternal urine and detected by home monoclonal antibody pregnancy test kits (**Prega News / Gravindex test**).
35. **Contraception & Reproductive Health**: **Barrier Methods**: Condoms (prevent STI/HIV transmission and pregnancy), diaphragms, cervical caps; **Intrauterine Devices (IUDs)**: Non-medicated (Lippes Loop), Copper-releasing (**Copper-T / CuT**, Cu ions suppress sperm motility and fertilising capacity), and Hormone-releasing (LNG-20, Progestasert); **Oral Contraceptive Pills**: Combined estrogen-progestin pills inhibit ovulation; **Saheli**

(Centchroman) is a once-a-week non-steroidal oral contraceptive pill developed by the **Central Drug Research Institute (CDRI), Lucknow**; **Surgical Sterilization**: Permanent contraception via **Vasectomy** in males (ligation of vas deferens) and **Tubectomy** in females (ligation of fallopian tubes).

36. **Human Adult Skeleton Breakdown (206 Bones)**: Composed of **Axial Skeleton (80 bones)**: Skull (22 bones: 8 cranial + 14 facial), Hyoid bone (1), Ear ossicles (6: 3 in each ear), Vertebral Column (26 bones: 7 cervical, 12 thoracic, 5 lumbar, 1 fused sacrum, 1 fused coccyx), Sternum (1), and Ribs (24 bones / 12 pairs); and **Appendicular Skeleton (126 bones)**: Pectoral girdle (4: 2 clavicles + 2 scapulae), Upper limbs (60: 30 in each arm), Pelvic girdle (2 hip / coxal bones), and Lower limbs (60: 30 in each leg).
37. **Special Cranial and Thoracic Bone Facts**: The **Hyoid bone** is a U-shaped solitary bone located at the base of the buccal cavity above the larynx; it does not articulate with any other bone in the human skeleton; the human neck has **7 Cervical Vertebrae** (true for almost all mammals, including the giraffe); first cervical vertebra (C_1) is the **Atlas** (articulates with occipital condyles for "Yes" nodding movement); second vertebra (C_2) is the **Axis** (has odontoid peg for "No" rotational head movement).
38. **Human Rib Cage Anatomy**: 12 pairs of ribs categorized into three classes: **True Ribs (Pairs 1–7)** articulate posteriorly with thoracic vertebrae and anteriorly attach directly to the sternum via costal hyaline cartilages; **False Ribs (Pairs 8–10)** do not attach directly to the sternum, but join the costal cartilage of the 7th rib; **Floating Ribs (Pairs 11 and 12)** do not attach anteriorly at all, having free anterior cartilaginous tips that shield kidneys.
39. **Upper and Lower Limb Skeletal Anatomy**: Upper limb (30 bones): **Humerus** (arm) \rightarrow **Radius & Ulna** (forearm) \rightarrow **8 Carpals** (wrist) \rightarrow **5 Metacarpals** (palm) \rightarrow **14 Phalanges** (fingers: 2 in thumb, 3 in each finger); Lower limb (30 bones): **Femur (Thigh bone, longest, heaviest, and strongest bone in the human body)** \rightarrow **Patella (Kneecap, a sesamoid bone formed within the quadriceps tendon)** \rightarrow **Tibia & Fibula** (shin and calf; tibia is the main weight-bearing shin bone) \rightarrow **7 Tarsals** (ankle; heel bone is calcaneus) \rightarrow **5 Metatarsals** (sole) \rightarrow **14 Phalanges** (toes).
40. **Bone Histology & Composition**: Bone is a specialized rigid vascular connective tissue; matrix consists of organic collagen fibers (~33%) and inorganic mineral salts (~67%), predominantly crystalline **Calcium Phosphate / Hydroxyapatite**; concentric cylindrical functional units of compact bone are **Haversian Systems (Osteons)** traversed by central Haversian canals transmitting blood vessels and nerves; bone cells include **Osteocytes** (mature bone cells inside lacunae), **Osteoblasts** (bone-forming cells), and **Osteoclasts** (multinucleated bone-resorbing cells).
41. **Joints of the Human Skeleton**: Three functional and structural classes: **Fibrous / Synarthrodial Joints (Immovable)**: Bones joined by dense collagen fibers with no articular cavity (e.g., **Sutures of the skull**, teeth in sockets/gomphosis); **Cartilaginous / Amphiarthrodial Joints (Slightly Movable)**: Bones connected by hyaline or fibrocartilage allowing limited movement (e.g., **Intervertebral discs** between adjacent vertebrae, **Pubic symphysis**); **Synovial / Diarthrodial Joints (Freely Movable)**: Separated by a fluid-filled synovial cavity, lined by synovial membrane secreting lubricating **Synovial Fluid**, and cushioned by articular hyaline cartilage.
42. **Synovial Joint Subtypes**: **Ball and Socket Joint**: Multiaxial, maximum range of movement in all planes (e.g., Shoulder joint between head of humerus and glenoid cavity of scapula; Hip joint between head of femur and acetabulum); **Hinge Joint**: Uniaxial movement like a door hinge (e.g., **Elbow joint, Knee joint, Interphalangeal finger joints**); **Pivot Joint**: Rotational movement around a central axis (e.g., **Atlanto-axial joint** between atlas and axis, proximal radio-ulnar joint); **Gliding Joint**: Flat surfaces sliding over one another (e.g., **Intercarpal joints** between wrist bones, intertarsal joints); **Saddle Joint**: Biaxial joint like rider on saddle (e.g., **Carpometacarpal joint of the human thumb**, enabling opposable thumb grip).
43. **Connective Support Tissues: Tendon vs Ligament**: **Tendon**: Dense regular white fibrous connective tissue composed of tough, inelastic, parallel collagen bundles; **attaches muscle to bone**; strain or rupture causes muscle tendonitis; **Ligament**: Dense regular yellow elastic connective tissue composed of elastic fibers and collagen; **attaches bone to bone** across a joint; provides joint stability; excessive stretching or tearing of a ligament is clinically defined as a **Sprain** (sprain = ligament tear; strain = tendon/muscle tear).
44. **Cartilage Characteristics**: Flexible, non-vascular supporting connective tissue; extracellular matrix consists of chondroitin sulfate proteoglycans synthesized by **Chondrocytes** situated inside lacunae; lacks direct blood

vessels (nutrients diffuse slowly through the fibrous perichondrium, explaining extremely slow healing of cartilaginous injuries); **Hyaline Cartilage** is the most abundant type (covers articular ends of long bones, costal cartilages, rings of trachea, embryonic skeleton).

45. **Three Types of Human Muscle Tissues**: **Skeletal Muscle (Striated / Voluntary)**: Cylindrical, unbranched, multinucleated fibers with peripheral nuclei; displays alternating light and dark transverse striations; attached to bones, under conscious somatic motor control; prone to fatigue; **Smooth Muscle (Non-striated / Involuntary / Visceral)**: Spindle-shaped (fusiform), unbranched, single central nucleus; lacks striations; lines walls of internal hollow organs (stomach, intestines, blood vessels, bronchi); autonomous, slow sustained contractions, fatigue-resistant; **Cardiac Muscle (Striated / Involuntary)**: Found exclusively in the myocardial wall of the heart; cylindrical, branched, uninucleated fibers; displays cross-striations and unique **Intercalated Discs** (gap junctions allowing synchronous wave-like contraction of the syncytium); myogenic, highly resistant to fatigue.
46. **Microscopic Structure of Skeletal Muscle Fibers**: Functional contractile unit of a muscle myofibril is the **Sarcomere** (segment between two successive transverse **Z-lines**); composed of overlapping thin and thick myofilaments: **Thin Filaments** are composed of **Actin**, surrounded by regulatory protein filaments **Tropomyosin** and calcium-binding **Troponin** complexes; **Thick Filaments** are composed of the motor protein **Myosin** with protruding cross-bridge heads containing ATPase enzymes.
47. **Sliding Filament Theory of Muscle Contraction**: Proposed by **H.E. Huxley and A.F. Huxley (1954)**; states that muscle contraction occurs not by shortening of myofilaments, but by the **sliding of thin actin filaments past thick myosin filaments toward the center of the sarcomere (H-zone)**; when a motor nerve action potential depolarizes the muscle membrane (sarcolemma), the sarcoplasmic reticulum floods the cytoplasm with **Calcium ions (\mathbf{Ca^{2+}})**; Ca^{2+} binds to troponin, displacing tropomyosin and exposing active binding sites on actin; myosin heads bind actin, hydrolyze ATP, and execute a power stroke; during contraction, the **Sarcomere shortens, the I-band narrows, the H-zone disappears, while the A-band length remains completely unchanged**.
48. **Rigor Mortis**: State of muscular rigidity and stiffening of the body that sets in 3 to 4 hours after biological death; caused by cessation of cellular respiration, leading to complete depletion of **ATP**; without ATP, myosin cross-bridge heads cannot detach from actin filaments, locking muscles in a permanent contracted state until autolytic lysosomal enzymes decompose muscle proteins (~15–24 hours later).
49. **Muscle Fatigue & Lactic Acidosis**: During prolonged intense anaerobic muscular exertion, oxygen supply to skeletal muscle tissues becomes insufficient for aerobic oxidative phosphorylation; muscle cells switch to anaerobic glycolysis, converting pyruvate into **Lactic Acid** via lactate dehydrogenase; accumulation of lactic acid decreases intracellular pH, impairs calcium binding to troponin, and precipitates **Muscle Fatigue / Cramps**; lactic acid is transported via blood to the liver and converted back into glycogen via the **Cori Cycle**.
50. **Key Skeletal and Muscular Pathologies Drill**: **Osteoporosis**: Age-related metabolic bone disorder characterized by decreased bone mineral density and deterioration of bone microarchitecture, leading to fragility fractures (common in post-menopausal women due to deficiency of **Estrogen**); **Rickets & Osteomalacia**: Defective bone mineralization due to deficiency of **Vitamin D or Calcium**; **Arthritis**: Inflammatory inflammation of joints; **Rheumatoid Arthritis**: Autoimmune disorder where immune antibodies attack the synovial membrane, forming invasive inflammatory pannus tissue; **Gout**: Deposition of needle-shaped monosodium urate crystals in joints due to elevated uric acid; **Myasthenia Gravis**: Autoimmune neuromuscular disorder where autoantibodies block acetylcholine receptors at motor endplates, causing progressive skeletal muscle weakness and fatigue.

📋 Must-Score Drill — Glands, Lenses & Bone Counts

Master Gland Secretions & Clinical Pathology

Endocrine Gland	Primary Hormones	Physiological Target & Action	Clinical Hypo / Hyper Disorder
Anterior Pituitary	**Growth Hormone (GH)**	Bone and muscle protein synthesis	Hypo: Pituitary Dwarfism; Hyper: Gigantism (child), Acromegaly (adult)
Posterior Pituitary	**Oxytocin**	Uterine myometrium & mammary alveoli	Triggers labour contractions & milk ejection reflex
Posterior Pituitary	**ADH (Vasopressin)**	Collecting ducts of nephron (water reabsorption)	Hypo: **Diabetes Insipidus** (profuse dilute urine)
Thyroid Gland	**Thyroxine (T_4) & T_3**	Systemic cells (controls Basal Metabolic Rate)	Hypo: Goitre, Cretinism (child), Myxedema; Hyper: Graves' Disease
Thyroid Gland	**Calcitonin**	Bone osteoblasts (lowers blood calcium)	Hypocalcemic regulation
Parathyroid Gland	**Parathormone (PTH)**	Bone osteoclasts, kidneys (raises blood calcium)	Hypo: **Tetany** (painful muscle spasms); Hyper: Osteoporosis
Adrenal Medulla	**Adrenaline (Epinephrine)**	Heart, blood vessels, bronchi (3F emergency)	"Fight, Fright or Flight" response
Adrenal Cortex	**Cortisol (Glucocorticoid)**	Gluconeogenesis, suppresses inflammation	Hypo: **Addison's Disease**; Hyper: **Cushing's Syndrome**
Adrenal Cortex	**Aldosterone**	Distal nephron (Na^+ and water retention)	Maintains blood pressure; regulates electrolyte balance
Pancreatic Islets	**Insulin (\beta-cells)**	Skeletal muscle, liver (lowers blood glucose)	Hypo: **Diabetes Mellitus** (Hyperglycaemia)
Pancreatic Islets	**Glucagon (\alpha-cells)**	Liver (stimulates glycogenolysis, raises sugar)	Antagonistic to insulin
Pineal Gland	**Melatonin**	Brain circadian clock (sleep-wake diurnal rhythm)	Regulates sleep cycle and body temperature
Thymus Gland	**Thymosins**	T-lymphocytes (promotes immune maturation)	Atrophies after puberty

Complete Human Skeleton 206 Bones Breakdown

Skeletal Division	Sub-region / Group	Exact Bone Count	Named Bones to Memorize
Axial (80)	Cranium (Braincase)	**8**	Frontal (1), Parietal (2), Temporal (2), Occipital (1), Sphenoid (1), Ethmoid (1)
Axial (80)	Facial Bones	**14**	Maxilla (2), Zygomatic (2), Mandible (1 — only movable skull bone), Palatine (2), etc.
Axial (80)	Hyoid Bone	**1**	Horseshoe bone in throat base; tongue attachment
Axial (80)	Auditory Ossicles	**6**	Malleus (2), Incus (2), **Stapes (2 — smallest bone)**
Axial (80)	Vertebral Column	**26**	**Cervical (7)**, **Thoracic (12)**, **Lumbar (5)**, **Sacrum (1)**, **Coccyx (1)**
Axial (80)	Thoracic Cage	**25**	**Sternum (1)** + **Ribs (24: 7 true pairs, 3 false pairs, 2 floating pairs)**
Appendicular (126)	Pectoral Girdle	**4**	Scapula (2), Clavicle (2 collar bones)
Appendicular (126)	Upper Limbs	**60**	Humerus (2), Radius (2), Ulna (2), Carpals (16), Metacarpals (10), Phalanges (28)

Skeletal Division	Sub-region / Group	Exact Bone Count	Named Bones to Memorize
Appendicular (126)	Pelvic Girdle	**2**	Coxal / Hip bones (2: Ilium + Ischium + Pubis fused)
Appendicular (126)	Lower Limbs	**60**	**Femur (2 — longest bone)**, Patella (2), Tibia (2), Fibula (2), Tarsals (14), Metatarsals (10), Phalanges (28)

One-Liner Revision – Control, Reproduction and Support

#	Fast Fact
1	Human body has **206 bones** (adult); newborn has ~270-300 (fuse over time)
2	Smallest bone = **Stapes** (stirrup bone in ear)
3	Largest bone = **Femur** (thigh bone)
4	Hardest substance in body = **Enamel** (of teeth)
5	Brain weight (adult) = ~**1.4 kg** (approximately 2% of body weight; uses 20% energy)
6	Cerebrum = thinking, memory, voluntary movement, speech
7	Cerebellum = balance, coordination, posture
8	Medulla oblongata = controls heartbeat, breathing, BP (vital centres)
9	Reflex arc: Stimulus -> Receptor -> Sensory nerve -> Spinal cord -> Motor nerve -> Effector
10	Pituitary = "**Master gland**"; controls thyroid, adrenal, gonads via trophic hormones
11	Thyroid secretes **Thyroxine** (T4); requires iodine; deficiency = goitre/cretinism
12	Parathyroid = secretes **PTH** (parathormone); raises blood calcium
13	Adrenal medulla = **Adrenaline** (epinephrine); emergency "fight-or-flight" hormone
14	Adrenal cortex = **Cortisol** (stress); Aldosterone (Na+ retention)
15	Pancreas islets: Alpha cells = Glucagon; Beta cells = **Insulin**
16	Diabetes mellitus = insufficient insulin; high blood glucose (hyperglycaemia)
17	Diabetes insipidus = ADH deficiency (NOT insulin); excessive urination
18	Pineal gland = secretes **Melatonin**; regulates sleep-wake cycle (circadian rhythm)
19	Thymus = secretes **Thymosin**; maturation of T-lymphocytes; largest in childhood
20	Human ovum = largest cell in human body; sperm = smallest
21	Gestation period (human) = **266-280 days** (~9 months)
22	In vitro fertilisation (IVF) = test-tube baby; first = **Louise Brown (1978, UK)**
23	Amniocentesis = prenatal diagnosis of chromosomal abnormalities (amniotic fluid)
24	Menstrual cycle = **28 days** (average); ovulation on ~Day 14
25	Testosterone = male sex hormone; secreted by testes (Leydig cells)
26	Oestrogen + Progesterone = female sex hormones; secreted by ovaries
27	Joint types: Hinge (elbow/knee), Ball-and-socket (hip/shoulder), Pivot (neck/atlas-axis)
28	Cartilage = flexible connective tissue; no blood vessels; found in ears, nose, joints
29	Osteoporosis = low bone density; post-menopausal women; calcium + Vit D needed
30	EEG = Electroencephalogram (brain waves); ECG = Electrocardiogram (heart waves)

Topic 7 — Plants, Agriculture and Plant Diseases

Consolidated — 50 Must-Score Facts

1. **Plant Taxonomy Hierarchy**: Categorized from higher to lower: **Kingdom \rightarrow Phylum (Division) \rightarrow Class \rightarrow Order \rightarrow Family \rightarrow Genus \rightarrow Species**. **Species** is the basic taxonomic unit consisting of individuals with fundamental morphological similarities capable of interbreeding.
2. **Augustin Pyramus de Candolle (A.P. de Candolle)**: Swiss botanist who **first coined the term "Taxonomy" (1813)** and first used internal **vascular tissue characteristics** to divide plants into *Vasculares* (with vascular bundles) and *Cellulares* (without vascular bundles). (Linnaeus is the "Father of Modern Taxonomy").
3. **Plant Tissues**: **Apical Meristem** (vertical height / primary growth); **Intercalary Meristem** (internode bases, re-growth of grazed grasses); **Lateral Meristem / Cambium** (secondary growth, increases **girth / diameter** of stems).
4. **Simple Permanent Tissues**: **Parenchyma** (living packing tissue, modified into **Aerenchyma** with large air chambers in aquatic hydrophytes for buoyancy); **Collenchyma** (living with pectin corner thickenings, provides mechanical flexibility); **Sclerenchyma** (dead, heavily lignified; commercial jute/hemp fibres and gritty sclereids/stone cells of guava and pear).
5. **Complex Vascular Tissues (Xylem and Phloem)**: **Xylem** conducts water and dissolved mineral salts unidirectionally from roots to leaves; composed of Tracheids, Vessels (unique to angiosperms), Xylem Fibres, and Xylem Parenchyma (only living xylem cell); **Phloem** translocates photosynthetic organic solutes bidirectionally from source to sink; composed of Sieve Tubes (enucleated at maturity), Companion Cells, Phloem Fibres (bast fibres), and Phloem Parenchyma.
6. **Ascent of Sap**: Driven by **Transpiration Pull** based on the **Cohesion-Tension Theory** of **Dixon and Joly (1894)**. Positive **Root Pressure** causes **Guttation** through marginal leaf **Hydathodes** in moist early mornings.
7. **Modified Underground Stems (Food Storage)**: **Rhizome**: Prostrate horizontal underground stem with distinct nodes, internodes, and scale leaves (e.g., **Ginger, Turmeric, Banana**); **Tuber**: Swollen distal end of underground branch bearing eyes/axillary buds (e.g., **Potato**); **Bulb**: Reduced disc-like stem surrounded by concentric fleshy scale leaves (e.g., **Onion, Garlic**); **Corm**: Solid, unbranched, vertically growing swollen underground stem (e.g., **Colocasia / Arbi, Amorphophallus / Zamikand, Saffron / Crocus**).
8. **Modified Storage Roots (NOT Stems)**: Derived from the root system, lack nodes, internodes, and scale leaves: **Tuberous Adventitious Root** (e.g., **Sweet Potato / Shakarkand**); **Conical Taproot** (e.g., **Carrot**, rich in \beta-carotene); **Fusiform Taproot** (e.g., **Radish**); **Napiform Taproot** (e.g., **Beetroot, Turnip / Shalgam**).
9. **Aerial Stem Storage**: In **Sugarcane (*Saccharum officinarum*, Family Gramineae/Poaceae)**, sucrose is stored predominantly in the **aerial stalk / stem** (propagated by stem cuttings with nodes).
10. **The Pea Family (Fabaceae / Leguminosae)**: Dicotyledonous plants with root nodules harbouring nitrogen-fixing bacteria: **Pea (*Pisum sativum*, herb with leaf tendrils)**, **Groundnut / Peanut (*Arachis hypogaea*)**, **Horse-gram / Kulthi (*Macrotyloma uniflorum*)**, **Soybean (*Glycine max*)**.
11. **Spices & Floral Anatomy**: **Clove (*Laung*)** is the dried unopened **flower bud** of *Syzygium aromaticum* (contains eugenol); **Saffron (*Kesar*)** consists of the dried **stigma and style** of *Crocus sativus*; **Cinnamon (*Dalchini*)** is the dried **inner bark** of *Cinnamomum verum*; **Black Pepper** is the dried unripe **berry/drupe** fruit of *Piper nigrum*; **Cardamom (*Elaichi*)** is the dried **capsule fruit and seeds** of *Elettaria cardamomum*.
12. **True Fruits vs False Fruits**: **True Fruits** develop exclusively from the fertilized ovary of a flower (e.g., **Mango, Grape, Tomato, Banana**); **False Fruits (Pseudocarps)** develop from other floral parts like the fleshy receptacle / thalamus in addition to the ovary (e.g., **Apple and Pear** are pomes where the edible part is the fleshy thalamus; **Cashew Nut** where the fleshy stalk is the swollen peduncle; **Strawberry** where the edible part is the fleshy thalamus).

13. **Fruit Classifications & Edible Morphology**: **Drupe (Stone Fruit)**: Single seed enclosed in a hard stony endocarp (e.g., **Mango** where edible part is fleshy **mesocarp**; **Coconut** where edible part is the cellular and liquid **endosperm**); **Berry**: Fleshy throughout with multiple seeds (e.g., Tomato, Guava); **Hesperidium**: Berry with leathery glandular peel and endocarpic juicy hair segments (e.g., Orange, Lemon); **Syconus / Sorosis (Composite Fruits)**: Derived from whole inflorescence (e.g., **Fig, Pineapple, Mulberry, Jackfruit**); **Litchi**: Edible part is the fleshy succulent **Aril** (outgrowth of the funicle).
14. **Plant Pigments Master Fact**: **Chlorophyll**: Green photosynthetic pigment containing central **Magnesium (Mg^{2+})**; **Lycopene**: Bright red carotenoid in ripe **tomatoes**; **Anthocyanin**: Water-soluble vacuolar pigment responsible for red/purple/blue colours in **apple skin, beetroot, purple cabbage, and rose petals**; **Carotene**: Orange pigment in **carrots**; **Betacyanin**: Red-violet pigment in **beetroot**; **Curcumin**: Yellow phenolic compound in **turmeric**; **Capsanthin**: Red carotenoid pigment in **ripe red chillies**; **Xanthophyll / Lutein**: Yellow pigment in egg yolk and autumn leaves.
15. **Plant Odour & Pungency Chemistry**: **Pungency of Chillies**: Caused by the alkaloid **Capsaicin**; **Pungency of Mustard Oil**: Caused by the sulfur glucoside **Allyl Isothiocyanate** (hydrolyzed from sinigrin by myrosinase); **Pungency of Garlic**: Caused by the sulfur-containing amino acid derivative **Allicin**; **Pungency & Tearing in Onions**: Caused by volatile **Syn-propanethial-S-oxide** (formed by the enzyme lachrymatory factor synthase acting on S-alkenyl-L-cysteine sulfoxides); **Bitterness in Bitter Gourd (Karela)**: Caused by tetracyclic triterpene glucosides called **Momordicines**.
16. **Commercial Fibres & Economic Products**: **Cotton**: Surface epidermal hairs of seeds (*Gossypium*), composed of nearly pure **Cellulose (~90%)**; **Jute and Sunn Hemp**: Phloem / bast sclerenchyma fibres extracted by biological water retting; **Cork**: Harvested from the outer protective bark (*phellem*) of the Mediterranean cork oak tree ***Quercus suber***; **Natural Rubber**: Polyisoprene polymer harvested as milky latex from the bark of ***Hevea brasiliensis***; **Quinine**: Powerful antimalarial alkaloid extracted from the dried stem bark of the ***Cinchona tree*** (*Cinchona calisaya / ledgeriana*); **Triticale**: First man-made artificial cereal hybrid developed by cross-breeding **Wheat (*Triticum*) \times Rye (*Secale*)**.
17. **Ecological Plant Adaptations**: **Hydrophytes**: Aquatic plants with poorly developed roots, reduced vascular tissue, thin cuticle, and extensive **Aerenchyma** tissue for buoyancy (e.g., *Hydrilla, Eichhornia, Lotus*); **Xerophytes**: Desert drought-adapted plants with sunken stomata, thick waxy cuticle, reduced scale-like leaves or spines to minimize transpiration, and fleshy photosynthetic stems (**Phylloclade**, e.g., *Opuntia, Cactus*); **Halophytes**: Salt-tolerant mangrove vegetation growing in coastal saline marshes; develop specialized negatively geotropic breathing roots called **Pneumatophores** that emerge vertically out of saline mud to absorb oxygen (e.g., *Rhizophora, Avicennia*).
18. **Epiphytes & Velamen Tissue**: **Epiphytes** are space-parasites that grow physically supported upon branches of other trees for sunlight without drawing nutrition or water from the host (e.g., **Vanda, Orchids**); they absorb moisture directly from humid atmospheric air using specialized spongy, hygroscopic aerial root tissue called **Velamen**.
19. **Forensic Utility of Diatoms**: **Diatoms** are unicellular photosynthetic algae with indestructible silica shells (*frustules*); inhalation of water during drowning draws diatoms into the bloodstream and internal organs (bone marrow, liver); **presence of diatoms in closed organs is a definitive forensic test confirming death by drowning**.
20. **Insectivorous & Carnivorous Plants**: Grow in acidic, waterlogged bogs **severely deficient in nitrates/nitrogen**; photosynthesise carbohydrates normally, but trap and digest small insects using proteases to fulfill nitrogen demands: **Pitcher Plant (*Nepenthes*)** (leaf lamina modified into hollow pitcher, leaf apex into lid), **Venus Flytrap (*Dionaea*)**, **Sundew (*Drosera*)**, and **Bladderwort (*Utricularia*)**.
21. **Photosynthesis Light Reactions (Photochemical Phase)**: Occurs in the **Thylakoid membranes / Grana** of chloroplasts; light energy absorbed by chlorophyll pigments excites electrons in Photosystems II and I; driven by the **Hill Reaction / Photolysis of Water**: water molecules are split into protons, electrons, and **Oxygen gas (O_2)**: 2H_2O \rightarrow 4H^+ + 4e^- + O_2\uparrow; **all oxygen liberated during photosynthesis**

originates from water (H_2O), NOT from carbon dioxide (CO_2) (proven by Ruben and Kamen using heavy oxygen isotope ^{18}O); generates assimilatory power in the form of **ATP and NADPH**.

22. **Photosynthesis Dark Reactions (Calvin Cycle / Biosynthetic Phase)**: Occurs in the **Stroma** of chloroplasts; does not directly require light, but uses ATP and NADPH produced during light reactions; atmospheric CO_2 is fixed by the primary carboxylating enzyme **RuBisCO (Ribulose-1,5-bisphosphate carboxylase-oxygenase)**, the most abundant protein on Earth; primary stable product is 3-phosphoglyceric acid (3-PGA, a 3-carbon compound, hence **C_3 pathway**).
23. **C_3 vs C_4 vs CAM Pathways**: **C_3 Plants**: Moderate climates (e.g., Wheat, Rice, Barley, Soybean); suffer from wasteful **Photorespiration** under high temperature; **C_4 Plants**: Tropical grasses (e.g., **Maize, Sugarcane, Sorghum / Jowar, Bajra**); exhibit specialized **Kranz Anatomy** (bundle sheath cells surrounded by mesophyll); fix CO_2 initially using **PEP Carboxylase** into 4-carbon Oxaloacetic Acid (OAA); show **zero photorespiration** and high water-use efficiency; **CAM Plants**: Succulents and epiphytes (e.g., **Pineapple, Agave, Cactus, Opuntia**); open stomata at night (**Scotoactive stomata**) to fix CO_2 into malic acid, keeping stomata closed during scorching daylight to conserve water.
24. **Action Spectrum of Photosynthesis**: Photosynthesis absorbs light in the Photosynthetically Active Radiation (PAR) spectrum (400–700\text{ nm}); maximal photosynthetic rate occurs under **Red Light** (\sim 660\text{ nm}) followed by **Blue Light**; **Green Light** is reflected by chlorophyll and is least effective for photosynthesis.
25. **Plant Respiration & Cold Storage**: Plant respiration is an oxidative catabolic process converting glucose and oxygen into CO_2, H_2O, and ATP; in agricultural post-harvest storage, cold storage extends fruit and vegetable shelf life because **low temperatures drastically slow down the rate of cellular respiration, transpiration, and ethylene synthesis** (UPPCS 2021).
26. **Auxins (Indole-3-Acetic Acid / IAA)**: Synthesised in stem apical meristems; promotes **Apical Dominance** (inhibits growth of lateral axillary buds); stimulates root initiation in stem cuttings; synthetic auxin **2,4-D (2,4-Dichlorophenoxyacetic acid)** is widely used as a selective weedicide / herbicide to eradicate broad-leaved dicot weeds in cereal crop fields without harming monocot crops.
27. **Gibberellins (Gibberellic Acid / GA_3)**: Discovered by **Eiichi Kurosawa (1926)** from the fungus *Gibberella fujikuroi* causing "foolish seedling" (*Bakanae*) disease in rice; promotes stem internode elongation (bolting in rosette plants like cabbage), breaks seed and bud dormancy, and induces **parthenocarpy** (seedless fruit production) in grapes and apples.
28. **Cytokinins (Zeatin / Kinetin)**: Discovered by Skoog and Miller; naturally abundant in regions of rapid cell division (coconut milk, corn kernels); promotes **cell division (cytokinesis)**, delays leaf senescence (**Richmond-Lang Effect**), and breaks apical dominance by promoting lateral bud outgrowth.
29. **Ethylene (C_2H_4)**: The **only gaseous plant hormone**; synthesised in ripening fruits and aging tissues; acts as the primary natural **Fruit Ripening Hormone**; commercial fruit ripening uses **Ethephon (Ethrel)**, which releases ethylene; *Lucent trap*: artificial ripening with calcium carbide (CaC_2) releases toxic **Acetylene gas (C_2H_2)**, which mimics ethylene but leaves dangerous arsenic and phosphorus residues.
30. **Abscisic Acid (ABA / "Stress Hormone")**: Naturally occurring sesquiterpene plant growth inhibitor; induces **closure of stomata** during acute water deficit / drought to prevent transpirational water loss; promotes seed dormancy and accelerates abscission of leaves and flowers; acts as an antagonist to gibberellins.
31. **Photoperiodism & Vernalization**: **Photoperiodism**: Physiological response of plants to the relative length of daylight and darkness; mediated by the photoreceptor pigment **Phytochrome** (exists in red-absorbing P_r and far-red-absorbing P_{fr} forms); **Vernalization**: Promotion of flowering by exposure to a prolonged period of low winter temperatures (1–7^\circ\text{C}); prevents precocious reproductive development in autumn, common in winter varieties of wheat and barley.
32. **Biological Nitrogen Fixation (BNF)**: Conversion of inert atmospheric dinitrogen (N_2) into ammonia (NH_3) catalyzed by the oxygen-sensitive microbial enzyme complex **Nitrogenase**; nitrogenase contains **Molybdenum and Iron (Mo-Fe protein)**; protected from oxygen poisoning inside legume root nodules by the oxygen-scavenging pink-red pigment **Leghaemoglobin**.

33. **Symbiotic Nitrogen Fixers**: ***Rhizobium leguminosarum*** forms symbiotic root nodules with leguminous crops (Pea, Gram, Lentil, Soybean); ***Frankia*** is an actinomycete that fixes nitrogen in non-leguminous trees (e.g., *Casuarina, Alnus*); ***Anabaena azollae*** is a cyanobacterium living in the leaf cavities of the tiny free-floating freshwater fern ***Azolla pinnata*** (widely applied as an exceptionally productive biofertiliser in flooded wetland **paddy / rice fields**).
34. **Free-Living & Associative Biofertilisers**: **Free-living Aerobic Nitrogen Fixers**: ***Azotobacter*** (applied in wheat, maize, cotton crops); **Free-living Anaerobic Fixers**: ***Clostridium***; **Associative Symbionts**: ***Azospirillum*** (colonizes rhizosphere of sorghum, maize, sugarcane); **Cyanobacteria / Blue-Green Algae (BGA)**: *Nostoc, Anabaena, Tolypothrix, Aulosira* fix nitrogen via specialized thick-walled cells called **Heterocysts** in alkaline flooded soils.
35. **Mycorrhiza & Phosphorus Solubilization**: **Mycorrhiza** is a mutualistic symbiotic association between fungal hyphae and roots of higher plants (e.g., *Glomus* genus); fungal hyphae vastly expand the root absorption surface, selectively absorbing immobile soil **Phosphorus (P)** and water for the plant, while the host plant provides carbohydrates to the fungus; **VAM (Vesicular-Arbuscular Mycorrhiza)** is an apex biofertiliser for phosphate nutrition.
36. **Macronutrients in Plant Nutrition**: Plants require 9 essential macronutrients absorbed in large quantities (>10\text{ mmol/kg} dry matter): **Carbon (C), Hydrogen (H), Oxygen (O)** (framework elements from air and water); primary fertilizer elements **Nitrogen (N), Phosphorus (P), Potassium (K)**; and secondary elements **Calcium (Ca), Magnesium (Mg), Sulfur (S)**.
37. **Micronutrients in Plant Nutrition**: Essential trace elements required in minute quantities (<10\text{ mmol/kg} dry matter): **Iron (Fe), Manganese (Mn), Copper (Cu), Zinc (Zn), Boron (B), Molybdenum (Mo), Chlorine (Cl), Nickel (Ni)**; **Zinc** is required for auxin (IAA) synthesis; **Boron** is essential for pollen germination and sugar translocation; **Molybdenum** is a component of nitrate reductase and nitrogenase; **Magnesium** is the central metal ion of chlorophyll.
38. **Classic Plant Nutritional Deficiency Disorders**: **Khaira Disease of Rice**: Caused by severe **Zinc (Zn) deficiency**; symptoms include reddish-brown chlorotic spots on young leaves; treated by spraying zinc sulfate (ZnSO_4); discovered by Y.L. Nene at Pantnagar; **Marsh Spot of Pea**: Caused by **Manganese (Mn) deficiency**; **Whiptail of Cauliflower**: Caused by **Molybdenum (Mo) deficiency**; **Black Heart of Potato**: Caused by **Oxygen deficiency / hypoxia** in poorly ventilated cold storage warehouses; **Hollow Heart of Potato**: Caused by excessive nitrogen fertilizer and rapid growth; **Browning / Hollow Stem of Cauliflower**: Caused by **Boron (B) deficiency**.
39. **Fungal Diseases of Cereal Crops**: **Late Blight of Potato**: Caused by the oomycete fungus ***Phytophthora infestans***; devastated Ireland's potato harvest, causing the catastrophic **Irish Famine (1845–1849)**; **Ergot of Rye and Bajra**: Caused by ***Claviceps purpurea***; replaces grain with dark poisonous sclerotia containing ergotamine and lysergic acid alkaloids; ingestion causes painful gangrenous and convulsive ergotism ("St. Anthony's Fire"); **Loose Smut of Wheat**: Internally seed-borne fungal disease caused by ***Ustilago tritici***; transforms ears into a mass of black powdery spores; **Black Stem Rust of Wheat**: Caused by heteroecious fungus ***Puccinia graminis tritici*** (requires wheat and barberry host plants to complete lifecycle; pioneered in India by Dr. K.C. Mehta).
40. **Bacterial & Viral Crop Diseases Drill**: **Bacterial Blight of Rice**: Caused by ***Xanthomonas oryzae***; produces translucent water-soaked lesions and kresek seedling wilt; **Citrus Canker**: Caused by ***Xanthomonas axonopodis pv. citri***; produces raised corky necrotic lesions on leaves and fruit of lemons and limes; **Crown Gall Disease**: Plant tumors caused by the soil bacterium ***Agrobacterium tumefaciens*** (harbours the tumor-inducing Ti plasmid used as a natural genetic engineer in biotechnology); **Tobacco Mosaic Disease (TMD)**: Caused by Tobacco Mosaic Virus (TMV, first isolated in crystalline form by Wendell Stanley, 1935); **Bunchy Top of Banana**: Caused by Banana bunchy top virus transmitted by the banana aphid *Pentalonia nigronervosa*; **Yellow Vein Mosaic of Bhindi (Okra)**: Devastating viral disease transmitted by the insect vector **Whitefly (*Bemisia tabaci*)**.
41. **Nematode Crop Infestations**: **Root-knot disease** in tobacco, tomato, and eggplant crops is caused by the microscopic endoparasitic soil nematode ***Meloidogyne incognita***; infects roots inducing characteristic giant multinucleated gall cells, stunting plant growth; controlled modernly using **RNA interference (RNAi) / gene silencing** technology.

42. **Organic Farming & Vermicomposting**: Organic farming avoids synthetic fertilizers, chemical pesticides, and GMOs; relies on crop rotations, green manures, and **Vermicompost**; vermicompost is produced by earthworm species like ***Eisenia foetida*** (red wigglers) digesting animal dung and farm biomass; earthworm castings are enriched with available nitrogen, phosphorus, exchangeable potassium, and beneficial soil microbes.
43. **Green Manuring Crops**: Sowing and plowing green succulent leguminous crops into the soil at flowering stage to enrich organic matter and nitrogen: apex green manure crops are **Dhaincha (*Sesbania aculeata*)**, **Sunn hemp (*Crotalaria juncea*)**, **Cowpea / Lobia (*Vigna unguiculata*)**, and **Guar / Cluster bean (*Cyamopsis tetragonoloba*)**; Dhaincha is exceptionally tolerant of waterlogging and alkaline/saline soils.
44. **Major Cash Crops & Agricultural Classification**: **Kharif Crops**: Sown in June–July at onset of monsoon, harvested in September–October (e.g., **Rice/Paddy, Maize, Jowar, Bajra, Cotton, Groundnut, Soybean, Jute, Arhar/Tur**); **Rabi Crops**: Sown in October–November (winter), harvested in March–April (e.g., **Wheat, Barley, Mustard / Rapeseed, Gram / Chickpea, Pea, Potato, Linseed**); **Zaid Crops**: Grown between March and June summer months (e.g., **Watermelon, Muskmelon, Cucumber, Gourd, Fodder crops**).
45. **Genetically Modified (GM) Crops in India**: **Bt Cotton** is the **only GM commercial crop permitted for commercial cultivation in India** (approved in 2002 by the Genetic Engineering Appraisal Committee / GEAC); incorporates the *Cry1Ac* and *Cry2Ab* endotoxin genes from soil bacterium *Bacillus thuringiensis* targeting American bollworm; **Bt Brinjal** was placed under commercial moratorium in 2010; **GM Mustard (DMH-11)** is a hybrid mustard developed by Delhi University using the barnase-barstar system.
46. **Pesticides & Bio-Pesticides Drill**: **DDT (Dichlorodiphenyltrichloroethane)**: Persistent organochlorine insecticide; bioaccumulates and biomagnifies up trophic levels, causing thinning of eggshells in raptor birds; largely banned globally under the Stockholm Convention; **Bio-pesticides**: Environmentally safe formulations derived from natural organisms, including **Neem-based formulations** containing the triterpenoid limonoid **Azadirachtin** (antifeedant and insect growth regulator from *Azadirachta indica*), and fungal bio-agents like ***Trichoderma viride*** (controls fungal root pathogens).
47. **Hydroponics & Aeroponics**: **Hydroponics**: Technique of cultivating terrestrial plants in a sterile **soil-less water culture** containing precisely calibrated aqueous mineral nutrient solutions (developed by Julius von Sachs, 1860); **Aeroponics**: Growing plants suspended in an air or mist environment without soil or aggregate medium, roots misted periodically with nutrient-rich sprays; maximizes root oxygenation and water efficiency.
48. **Plant Cell Totipotency & Tissue Culture**: **Totipotency** is the intrinsic cellular capacity of any nucleated plant cell / explant to regenerate into a complete, fertile multicellular plant organism (concept formulated by Gottlieb Haberlandt, 1902); utilized in commercial **Micropropagation** to mass-produce pathogen-free clonal plantlets (e.g., Banana, Orchids, Potato) in sterile synthetic media containing balanced auxin and cytokinin ratios.
49. **Polyembryony & Parthenocarpy**: **Polyembryony**: Phenomenon of occurrence of more than one embryo in a single seed (discovered by Antony van Leeuwenhoek, 1719 in citrus seeds); prominent in **Citrus (Orange, Lemon)** and **Mango**; **Parthenocarpy**: Formation and development of fruits without fertilization of ovules, yielding naturally seedless fruits (e.g., **Banana, Pineapple, Seedless Grapes**); can be induced artificially by applying auxins or gibberellins.
50. **Key Botanical Taxonomy & Economic Families Drill**: **Poaceae / Gramineae (Grass Family)**: Rice, Wheat, Maize, Sugarcane, Bamboo (fastest-growing woody grass); **Fabaceae / Leguminosae (Pulse Family)**: Gram, Pea, Arhar, Soybean, Groundnut; **Solanaceae (Nightshade Family)**: Potato, Tomato, Brinjal, Chilli, Tobacco (*Nicotiana tabacum*, leaves rich in nicotine), Belladonna (*Atropa belladonna*, source of atropine); **Brassicaceae / Cruciferae (Mustard Family)**: Mustard, Radish, Cabbage, Cauliflower, Turnip; **Cucurbitaceae (Gourd Family)**: Bottle gourd, Bitter gourd, Cucumber, Pumpkin, Watermelon; **Malvaceae (Mallow Family)**: Cotton, Okra / Bhindi, Hibiscus.

📋 Must-Score Drill — Edible Parts, Hormones & Diseases

Master Edible Parts & Commercial Products Table

Commercial Product / Plant	Botanical Name	Plant Family	Morphological Nature / Edible Part	High-Yield Chemistry / Note
Clove (*Laung*)	*Syzygium aromaticum*	Myrtaceae	**Dried unopened Flower Bud**	Contains **Eugenol** volatile essential oil
Saffron (*Kesar*)	*Crocus sativus*	Iridaceae	**Dried Stigma and Style**	Crocin pigment; Karewas of Kashmir
Potato	*Solanum tuberosum*	Solanaceae	**Modified Underground Stem (Tuber)**	Starch; bears axillary buds (“eyes”)
Sweet Potato	*Ipomoea batatas*	Convolvulaceae	**Modified Adventitious Root**	Starch and carotene storage
Ginger	*Zingiber officinale*	Zingiberaceae	**Modified Underground Stem (Rhizome)**	Gingerol; nodes and internodes present
Turmeric (*Haldi*)	*Curcuma longa*	Zingiberaceae	**Modified Underground Stem (Rhizome)**	**Curcumin** yellow antioxidant dye
Onion / Garlic	*Allium cepa / sativum*	Amaryllidaceae	**Fleshy Scale Leaves**	Pungent allyl sulfide / allicin sulfur
Apple / Pear	*Malus domestica*	Rosaceae	**Fleshy Thalamus (False Fruit / Pome)**	Edible receptacle surrounds core
Coconut	*Cocos nucifera*	Arecaceae	**Endosperm** (liquid water + solid copra)	Drupe with fibrous mesocarp (coir)
Litchi	*Litchi chinensis*	Sapindaceae	**Fleshy Aril**	Outgrowth from funicle/integument
Mango	*Mangifera indica*	Anacardiaceae	**Fleshy Mesocarp**	Stone fruit (Drupe)
Quinine	*Cinchona calisaya*	Rubiaceae	**Dried Stem Bark**	Antimalarial quinoline alkaloid
Cinnamon (*Dalchini*)	*Cinnamomum verum*	Lauraceae	**Dried Inner Bark**	Cinnamaldehyde aromatic oil
Tulsi	*Ocimum sanctum*	Lamiaceae	**Leaves**	**Phenols & Flavonoids (Eugenol)**
Cotton	*Gossypium hirsutum*	Malvaceae	**Epidermal Seed Hairs**	Pure **Cellulose** (Zero protein/nitrogen)
Jute / Sunn hemp	*Corchorus / Crotalaria*	Malvaceae / Fabaceae	**Phloem Sclerenchyma (Bast Fibres)**	Lignified structural stem fibres
Okra (Bhindi)	*Abelmoschus esculentus*	Malvaceae	**Whole Green Seed Pod (Capsule)**	Pentagonal cross-section fruit
Singhara / Water Chestnut	*Trapa bispinosa*	Trapaceae	**Edible Seed of Nut**	Nut fruit; rich in starch
Cashew Nut	*Anacardium occidentale*	Anacardiaceae	**Seed of Nut (fleshy stalk = false fruit)**	True nut attached to fleshy peduncle
Cork	*Quercus suber*	Fagaceae	**Outer Bark of Stem**	Native to Mediterranean region
Triticale	*Triticum \times Secale*	Poaceae	**Whole Cereal Grain**	First man-made hybrid cereal crop

Master Plant Underground Modifications: Stems vs Roots

Botanical Classification	Specialized Structure	Distinctive Anatomical Signature	Commercial Plant Examples	Must-Score Fact
Modified Stem	**Rhizome**	Prostrate, horizontal fleshy axis; bears distinct nodes, internodes, terminal buds, and scaly leaves	**Ginger** (*Zingiber officinale*), **Turmeric** (*Curcuma longa*), Lotus	Possesses nodes & internodes (unlike roots)
Modified Stem	**Tuber**	Swollen distal tip of underground branch; bears spirally arranged axillary buds ("eyes") at nodes	**Potato** (*Solanum tuberosum*)	Member of Solanaceae; eyes are vegetative buds
Modified Stem	**Bulb**	Reduced, flattened disc-like stem surrounded by concentric fleshy storage scale leaves	**Garlic** (*Allium sativum*), **Onion** (*Allium cepa*), Tulip, Lily	Fleshy scale leaves store food; disc stem at base
Modified Stem	**Corm**	Stout, solid, unbranched vertically swollen fleshy underground stem with circular nodes	**Jimikand / Suran** (*Amorphophallus*), Colocasia (*Arbi*), Crocus	Vertically oriented solid stem
Modified Root	**Tuberous Adventitious Root**	Swollen storage root arising from stem nodes or radicle; lacks nodes, internodes, and eyes	**Sweet Potato** (*Ipomoea batatas*, Convolvulaceae)	Edible part is root, **not** stem
Modified Root	**Conical Taproot**	Broad at base and tapering uniformly toward apex	**Carrot** (*Daucus carota*)	Rich in provitamin A (\beta-Carotene)
Modified Root	**Napiform Taproot**	Globular, spherical base abruptly tapering into a thread-like tail	**Turnip** (*Shaljam*), **Beetroot** (*Beta vulgaris*)	Storage organ is root
Modified Root	**Fusiform Taproot**	Swollen in middle, tapering toward both base and apex	**Radish** (*Raphanus sativus*)	Pungent mustard oil isothiocyanates

Botanical Classification of Fruits: True vs False & Structural Types

Fruit Category	Botanical Definition	Morphological Fruit Type	Key Plant Examples	Edible Part / Must-Score
True Fruit	Derived exclusively from fertilized ovary	**Drupe (Stone Fruit)**	**Mango** (*Mangifera indica*), Peach, Plum	Edible **Mesocarp**; stony endocarp protects seed
True Fruit	Derived exclusively from fertilized ovary	**Berry**	**Grape, Tomato, Date, Banana**	Entire pericarp and placenta edible
True Fruit	Derived from single flower with hard rind	**Pepo**	**Melon, Watermelon, Squash, Cucumber**	Fleshy placenta and mesocarp
True Fruit	Special drupe with arillode	**Drupe with Aril**	**Lychee** (*Litchi chinensis*)	Sweet, juicy, translucent **Fleshy Aril**
False Fruit	Fleshy receptacle/thalamus encloses core	**Pome**	**Apple** (*Malus*), **Pear** (*Pyrus*)	Edible **Thalamus**; ovary core discarded
False Fruit	Swollen peduncle (fruit stalk) + nut	**Pseudocarp + Nut**	**Cashew Nut** (*Anacardium*)	Fleshy peduncle (cashew apple) + nut seed
False Fruit	Derived from entire compact inflorescence	**Sorosis (Multiple Fruit)**	**Mulberry, Pineapple, Jackfruit, Ficus**	Derived from catkin, spike, or spadix

Master Plant Pigments, Flavours & Volatile Chemistry

Commercial Crop / Organ	Visual Colour / Sensory Trait	Active Chemical / Pigment Responsible	Chemical Class / Mechanism	Must-Score Fact
Tomato	Deep vibrant red	**Lycopene** (C_{40}H_{56})	Carotenoid tetraterpene	Antioxidant; does not convert to Vitamin A

Commercial Crop / Organ	Visual Colour / Sensory Trait	Active Chemical / Pigment Responsible	Chemical Class / Mechanism	Must-Score Fact
Apple Skin	Crimson red blush	**Anthocyanin**	Water-soluble flavonoid glycoside	Formed in peel cells exposed to autumn sun
Papaya Pulp	Rich golden-yellow	**Caricaxanthin**	Xanthophyll carotenoid	Contains proteolytic enzyme **Papain**
Carrot	Red-orange colour	**\beta-Carotene**	Carotenoid provitamin A	Cleaved in intestine into retinal / Vitamin A
Beetroot (Chukandar)	Purplish-red colour	**Betanin**	Betalain nitrogenous pigment	Excreted in urine (beeturia)
Turmeric (*Haldi*)	Bright yellow colour	**Curcumin** (diferuloylmethane)	Polyphenolic curcuminoid	Powerful anti-inflammatory & antioxidant
Chilli Peppers	Burning heat / Pungency	**Capsaicin** (C_{18}H_{27}NO_3)	Lipophilic alkaloid / capsaicinoid	Binds TRPV1 receptors; Scoville scale
Crushed Garlic	Sharp pungent odour	**Allicin** (diallyl thiosulfinate)	Organosulfur compound	Formed from alliin by alliinase; antibacterial
Garlic / Onion Breath	Persistent body odour	**Allyl methyl sulphide (AMS)**	Volatile organosulfur compound	Absorbed into blood; exhaled via lungs & sweat
Cut Onion	Tear induction (crying)	**(Z)-propanethial S-oxide**	Lachrymatory factor (LF) gas	1-propenyl sulfenic acid \rightarrow tears
Ripe Fruits	Natural sweet taste	**Fructose (Fruit Sugar)**	Monosaccharide ketohexose	Sweetest natural sugar; low glycemic index

Ecological Plant Groups & Specialized Morphological Adaptations

Ecological Class	Primary Habitat / Environmental Stress	Diagnostic Morphological / Anatomical Adaptation	Classical Plant Representatives	Distinctive Must-Score
Xerophytes	Arid deserts, extreme water deficit	Sunken stomata, thick waxy cuticle, reduced leaves (spines), succulent water storage	Cactus, *Opuntia*, *Aloe*, *Calotropis*, Pineapple	Stems perform photosynthesis (**Cladode**)
Phreatophytes	Hyper-arid desert plains	**Extremely long taproots (20–30\text{ metres} / 65–100\text{ ft})** reaching deep groundwater	Mesquite (*Prosopis*), Desert Tamarisk	Independent of seasonal topsoil rain
Hydrophytes	Aquatic lakes, ponds, swamps	Large internal air cavities (**Aerenchyma**), poorly developed xylem, reduced roots	Lotus, *Hydrilla*, *Eichhornia*, *Vallisneria*	Aerenchyma provides buoyancy & gas exchange
Halophytes	Saline coastal marshes, mangrove lagoons	Salt exclusion glands, succulent leaves, **pneumatophores** (breathing roots) with lenticels	*Rhizophora*, *Avicennia*, *Sonneratia*	Viviparous germination (seed sprouts on tree)
Lithophytes	Bare rocky surfaces, boulders	Cling directly to rock without soil via rhizoids or sticky pads; feed on rain and detritus	**Lichens**, **Mosses**, rock ferns	Pioneer species initiating primary succession
Epiphytes	Humid tropical forest tree canopies	Cling to host bark purely for **mechanical support**; aerial roots with **Velamen tissue** absorb air moisture	**Orchids**, *Vanda*, Spanish moss	Derive **zero nutrition** from host tree
Insectivorous	Acidic bogs, peaty swamps, **nitrogen-deficient**	Leaves modified into traps (pitchers, tentacles, bladders) to capture and digest insects	***Nepenthes khasiana*** (Meghalaya), *Drosera*, *Utricularia*	Trap insects for **Nitrogen**, not carbohydrates

Phytohormone Functions & Commercial Applications

Hormone Class	Primary Discovery Background	Key Physiological Roles	Commercial Agricultural Use
Auxin (IAA)	Went (1928), *Avena* curvature	Phototropism; Apical dominance; cell elongation	Rooting hormone (IBA); Selective weedicide (**2,4-D**)
Gibberellin (\mathbf{GA_3})	Kurosawa (1926), Bakanae disease	Internodal elongation; **Bolting**; breaks dormancy	Lengthens grape bunches; brewing malting
Cytokinin	Skoog & Miller (1955), Kinetin	Promotes cell division; **delays senescence**	Tissue culture shoot induction with auxin
Ethylene (\mathbf{C_2H_4})	Gaseous hormone; Burg (1962)	**Fruit ripening**; promotes abscission/senescence	Artificial ripening of bananas/mangoes (**Ethephon**)
Abscisic Acid (ABA)	Addicott (1963), Dormin	**Stress hormone**; closes stomata; seed dormancy	Drought-hardening antitranspirant

Master Reference: Biofertilizers, Nitrogen Fixation & Microbial Agricultural Inputs

Biofertilizer Category	Specific Microorganism / Vector	Target Crops & Target Soil Ecosystem	Biological Mechanism & Must-Score
Symbiotic N_2 Bacteria	***Rhizobium leguminosarum***, ***Rhizobium japonicum***	**Legumes (Fabaceae)**: Chickpea, Gram, Pea, Soybean, Alfalfa, Clover, Lentil, Pigeon pea	Colonizes root hairs, induces root nodule formation; synthesizes **Leg-haemoglobin** (pink/red heme pigment) to buffer O_2 and protect oxygen-sensitive **Nitrogenase** enzyme. *R. japonicum* is specialized for **Soybean**.
Free-Living Aerobic N_2 Bacteria	***Azotobacter chroococcum***	Non-legumes: **Wheat, Cotton, Mustard, Maize**	Fixes free atmospheric nitrogen non-symbiotically in the rhizosphere. **Common Trap:** Not a bioinsecticide!
Associative N_2 Bacteria	***Azospirillum brasilense***	**Sugarcane, Sorghum, Maize, Millets, Wheat**	Colonizes root cortical intercellular spaces; fixes N_2 and secretes growth-promoting gibberellins/auxins.
Cyanobacteria (Blue-Green Algae)	***Nostoc*, *Anabaena*, *Oscillatoria***	**Paddy (Water-logged Rice fields)**	Photosynthetic prokaryotes; fix nitrogen under oxic light conditions in specialized thick-walled cells called **Heterocysts** (which lack Photosystem II to maintain anoxia).
Aquatic Fern Symbiosis	***Azolla pinnata*** + ***Anabaena azollae***	**Flooded Rice Fields & Bovine Cattle Feed**	*Azolla* is a floating **aquatic pteridophyte (fern)** harboring endophytic cyanobacterium *Anabaena* in dorsal leaf cavities. Supplies 30–40 kg N/ha; also utilized as **supplementary high-protein feed for milking cows/buffaloes** to boost milk yield.
Phosphate-Solubilizing Bacteria (PSB)	*Pantoea agglomerans* (P5), *Pseudomonas putida* (P13), *Bacillus megaterium var. phosphaticum*	Universal across field crops	Secrete organic acids (gluconic, citric) that chelate calcium and iron, solubilizing insoluble mineral phosphates into bioavailable H_2PO_4^-.
Phosphatic Mycorrhizal Fungi	**Vascular Arbuscular Mycorrhiza (VAM)** (*Glomus spp.*)	Forestry, Horticultural & Field Crops	Symbiotic fungal hyphae penetrating root cortical cells; dramatically enhances root absorption of poorly mobile **Phosphorus (P)**, water, and zinc; boosts drought and salinity tolerance.
Soil Conditioner & Microbial Habitat	**Biochar**	Vertical farming, dryland agriculture	Stable, porous carbonaceous charcoal synthesized via **Pyrolysis** (high-temperature biomass heating under oxygen deprivation). Retains moisture/nutrients, provides microhabitat for BNF microorganisms.

Biofertilizer Category	Specific Microorganism / Vector	Target Crops & Target Soil Ecosystem	Biological Mechanism & Must-Score
Biological Soil Aerator	**Earthworms (*Pheretima posthuma*)**	Vermicomposting & Organic farming	Termed **"Ploughers of the soil"**; burrows aerate soil and accelerate pit decomposition of agricultural waste (livestock excreta, crop straw) into nutrient-dense vermicompost.

- **Agronomic Management Facts for Nitrogen Use Efficiency (NUE)**:
- NUE can be significantly maximized through:
  1. **Split application** of nitrogenous fertilizers matching crop physiological demand stages.
  2. Incorporating **Nitrification inhibitors** (e.g. nitrapyrin, dicyandiamide) to prevent rapid conversion of ammonium to leachable nitrates.
  3. Applying **Slow-release nitrogen fertilizers** (e.g., Neem-coated urea, sulphur-coated urea).
- **Non-Nitrogen-Fixing Crops Trap**: Paddy, wheat, sugarcane, amaranth, spinach, and purslane (kulfa) do **not** fix atmospheric nitrogen on their own and require microbial inoculants or chemical nitrogen supplementation!

One-Liner Revision – Plants, Agriculture and Diseases

#	Fast Fact
1	Photosynthesis equation: 6CO2 + 6H2O + light -> C6H12O6 + 6O2
2	Chlorophyll absorbs **red** and **blue-violet** light; reflects **green** (hence green colour)
3	Light reactions occur in **thylakoid** membranes (grana of chloroplast)
4	Dark reactions (Calvin cycle) occur in **stroma** of chloroplast
5	Transpiration = evaporation of water through stomata; creates suction for water uptake
6	Xylem = water + minerals transport (root to leaves); dead cells; one-way
7	Phloem = food (sucrose) transport; living cells; bidirectional
8	Rhizobium = nitrogen-fixing bacteria in legume root nodules; mutualistic
9	Azotobacter = free-living nitrogen-fixing bacteria in soil
10	Cyanobacteria (blue-green algae) = nitrogen fixers; e.g., Anabaena in rice fields
11	Auxin = phototropism (bending towards light); produced at shoot tip
12	Gibberellin = promotes stem elongation; used to break seed dormancy
13	Ethylene = fruit ripening gas; only gaseous plant hormone
14	Abscisic acid = stress hormone ("stress hormone of plant"); causes wilting + dormancy
15	Green Revolution: M.S. Swaminathan (India) + Norman Borlaug (Mexico; Nobel 1970)
16	HYV wheat = Sonalika, Kalyan Sona; HYV rice = IR8 ("Miracle Rice")
17	Vermiculture = use of earthworms to decompose organic waste (vermicompost)
18	Algal bloom = excessive algae growth due to eutrophication (excess nitrate/phosphate)
19	Eutrophication = excess nutrients in water -> algae bloom -> O2 depletion -> fish death
20	Most common plant disease agent = **Fungi** (then bacteria, then viruses)
21	Rust of wheat = *Puccinia graminis* (fungus)
22	Blast of rice = *Pyricularia oryzae* (fungus) – most destructive rice disease
23	Smut of wheat = *Ustilago tritici* (fungus)
24	Citrus canker = *Xanthomonas citri* (bacteria)

#	Fast Fact
25	Tobacco Mosaic Virus (TMV) = first virus discovered (Dmitri Ivanovsky, 1892)
26	Potato virus = causes “late blight of potato” (*Phytophthora infestans*) = Irish Famine
27	Hardwood (dicot wood) = Teak, Sal, Rosewood; Softwood (conifer) = Pine, Spruce, Deodar
28	Bryophytes = “amphibians of plant kingdom” (e.g., moss, liverwort); no vascular tissue
29	Gymnosperms = naked seeds (no fruit); e.g., Pine, Cycas, Ginkgo
30	Angiosperms = enclosed seeds in fruit; most advanced; ~2.5 lakh species

Topic 8 — Microbiology, Environment and Applied Science

Consolidated — 50 Must-Score Facts

1. **The Four Microbial Groups**: Microorganisms span four major distinct groups: **Bacteria** (prokaryotic unicellular organisms with peptidoglycan walls and 70S ribosomes), **Fungi** (eukaryotic heterotrophs with chitinous walls and 80S ribosomes), **Protozoa** (unicellular animal-like protists), and **Viruses** (acellular obligate intracellular nucleoprotein complexes consisting of DNA or RNA inside a capsid).
2. **Bacterial Morphology & Shapes**: Classified by morphological shape: **Coccus** (spherical, e.g., *Streptococcus pneumoniae*); **Bacillus** (rod-shaped, e.g., *Lactobacillus, Bacillus anthracis*); **Spirillum** (spiral/helical, e.g., *Treponema pallidum*); and **Vibrio** (comma-shaped, e.g., *Vibrio cholerae*).
3. **Gram Staining Differential (Christian Gram, 1884)**: Distinguishes bacteria based on cell wall architecture: **Gram-Positive** bacteria have a thick peptidoglycan cell wall containing teichoic acids, retaining the primary crystal violet dye (**Purple/Violet**); **Gram-Negative** bacteria have a thin peptidoglycan layer enclosed by an outer lipopolysaccharide membrane, losing crystal violet upon ethanol decolourisation and counterstaining with safranin (**Pink/Red**).
4. **Bacterial Endospores**: Highly dehydrated, metabolically dormant survival structures produced by genera *Bacillus* and *Clostridium* (e.g., *Clostridium tetani, Bacillus anthracis*); contain high levels of **Calcium Dipicolinate** in the spore core, rendering them resistant to boiling, desiccation, chemical disinfectants, and ionizing radiation (destroyed only by pressurized autoclaving at 121^\circ\text{C} for 15 minutes).
5. **Mycoplasma (PPLO — Pleuropneumonia-Like Organisms)**: The **smallest known living cells capable of autonomous growth and reproduction** (0.1–0.3\ \mu\text{m}); completely **lack a cell wall**, enclosed only by a sterol-rich plasma membrane; naturally **resistant to penicillin and \beta-lactam antibiotics** because penicillin acts specifically by inhibiting peptidoglycan synthesis.
6. **Viruses, Bacteriophages & Sub-Viral Agents**: **Viruses** are non-cellular, obligate intracellular parasites possessing genetic material (either DNA or RNA, **never both**) enclosed within a protein coat (**Capsid** composed of capsomeres); **Bacteriophages** are viruses that infect and destroy bacteria, possessing double-stranded DNA; **Viroids** (discovered by **T.O. Diener, 1971**) are infectious naked low-molecular-weight circular RNA molecules lacking any protein capsid (cause potato spindle tuber disease); **Prions** are infectious, misfolded neurotoxic protein molecules lacking nucleic acids entirely (cause bovine spongiform encephalopathy / "Mad Cow Disease" and Creutzfeldt-Jakob disease).
7. **Viral vs Bacterial Culturing**: Bacteria and fungi possess autonomous metabolic machinery and can be readily cultured in synthetic cell-free artificial nutrient broths/agar; **Viruses lack metabolic machinery and cannot grow on artificial cell-free media**, strictly requiring living host tissue cells (*in vivo* or live cell cultures *in vitro*) to replicate.
8. **Industrial Microbes: Curd & Dairy**: ***Lactobacillus acidophilus*** **(Lactic Acid Bacteria / LAB)** ferments lactose into lactic acid, coagulating milk protein casein; LAB also improves nutritional quality by dramatically synthesizing **Vitamin B12** and checking disease-causing microbes in the human gut.
9. **Brewer's & Baker's Yeast**: ***Saccharomyces cerevisiae*** is a single-celled eukaryotic fungus; used in baking to leaven dough (ferments carbohydrates releasing **Carbon Dioxide (CO_2)**, causing bread to puff up with spongy pores) and in brewing to ferment malted cereals and fruit juices into **Ethanol**.
10. **Cheese Microbiology**: **Swiss Cheese** displays large, characteristic holes ("eyes") produced by the vigorous release of **CO_2 gas** by the propionic acid bacterium ***Propionibacterium shermanii***; **Roquefort Cheese** is ripened by inoculating the mould fungus ***Penicillium roqueforti***, giving it a distinctive sharp blue-veined flavour.
11. **Industrial Microbial Organic Acids**: **Citric Acid** is produced by the aerobic fungal mould ***Aspergillus niger***; **Acetic Acid (Vinegar)** is produced by the aerobic bacterium ***Acetobacter aceti*** (oxidizes ethanol into acetic acid); **Butyric Acid** is produced by the obligate anaerobic bacterium ***Clostridium butyricum***; **Lactic Acid** is produced by ***Lactobacillus bulgaricus***.

12. **Microbial Industrial Enzymes**: **Lipases** (produced by *Candida lipolytica*) are added to laundry detergent formulations to digest and remove oily/greasy fabric stains; **Pectinases and Proteases** (from *Aspergillus* species) are used to clarify bottled fruit juices, making commercial juices clear rather than cloudy.
13. **Wonder Drug: Streptokinase ("Clot Buster")**: Enzyme obtained from culture filtrates of the bacterium ***Streptococcus*** (modified by genetic engineering); acts as a potent fibrinolytic agent / **"clot buster"** injected clinically to dissolve obstructing coronary thrombi in patients experiencing acute myocardial infarction (heart attack).
14. **Immunosuppressant: Cyclosporin A**: Bioactive 11-amino-acid cyclic peptide isolated from the soil fungus ***Trichoderma polysporum***; acts as a selective **immunosuppressive agent** that blocks T-cell receptor activation, preventing organ graft rejection in human kidney, liver, and heart transplant recipients.
15. **Cholesterol Lowering: Statins**: Produced by the red yeast ***Monascus purpureus***; acts as a competitive inhibitor of the rate-limiting enzyme **HMG-CoA reductase** in human hepatocytes, effectively lowering circulating blood cholesterol levels.
16. **Discovery of Antibiotics**: **Alexander Fleming (1928)** accidentally discovered the first true antibiotic, **Penicillin**, observing that the fungal mould ***Penicillium notatum*** inhibited the growth of *Staphylococcus* bacteria; purified and established as an effective wartime therapy by **Howard Florey and Ernst Chain** (Nobel Prize in Medicine, 1945); **Selman Waksman (1944)** isolated **Streptomycin** from the soil actinomycete *Streptomyces griseus* (first effective antibiotic against Tuberculosis) and coined the term "Antibiotic".
17. **Milestone Medical Pioneers Drill**: **Edward Jenner (1796)**: Father of Immunology; developed the world's first successful vaccine using cowpox pus to confer cross-immunity against lethal **Smallpox**; **Louis Pasteur**: Disproved spontaneous generation, developed **Pasteurization** (heating liquid to 63^\circ\text{C} for 30 minutes or 72^\circ\text{C} for 15 seconds), established the Germ Theory of Disease, and invented rabies and anthrax vaccines; **Robert Koch (1843–1910)**: Formulated Koch's Postulates, discovered the causative bacterial agents of **Anthrax (*Bacillus anthracis*)**, **Tuberculosis (*Mycobacterium tuberculosis*, 1882)**, and **Cholera (*Vibrio cholerae*)** (Nobel Prize 1905).
18. **Sewage Treatment Plants (STP) Primary Treatment**: Physical mechanical removal of large and small floating debris and grit through sequential **filtration and sedimentation**; floating solids trapped by bar screens, while grit (soil, pebbles) settles in primary settling tanks to form primary sludge; the supernatant liquid forms the primary effluent.
19. **STP Secondary Treatment (Biological Treatment)**: Primary effluent is transferred to large **aeration tanks** constantly agitated mechanically and pumped with air; allows vigorous growth of beneficial aerobic heterotrophic microbes into **Flocs** (masses of bacteria associated with fungal filaments); flocs consume dissolved organic matter, drastically reducing the **Biochemical Oxygen Demand (BOD)** of sewage water.
20. **Biochemical Oxygen Demand (BOD)**: Represents the amount of dissolved oxygen consumed by aerobic microorganisms to biologically oxidize all biodegradable organic matter present in one litre of water; **high BOD indicates high organic pollution and severely depleted dissolved oxygen**; clean drinking water has a BOD <1\text{ mg/L}, while raw domestic sewage has a BOD of 200–400\text{ mg/L}.
21. **Chemical Oxygen Demand (COD)**: Measure of the oxygen equivalent required to chemically oxidize both biodegradable and non-biodegradable organic pollutants in a water sample using a strong chemical oxidant (potassium dichromate, K_2Cr_2O_7 in acid); **COD is always greater than BOD (COD > BOD)** because it measures total oxidizable matter.
22. **Biogas (Gobar Gas) Technology**: Anaerobic digestion of wet cattle dung (*gobar*) and agricultural biomass inside an airtight digester tank; carried out by anaerobic methanogenic bacteria (principally ***Methanobacterium***); produces Biogas composed primarily of **Methane (\mathbf{CH_4}, 50–70%)**, **Carbon Dioxide (\mathbf{CO_2}, 30–40%)**, and trace gases (H_2S, H_2); high calorific value, burns with a clean smokeless flame leaving no ash; spent slurry serves as an enriched organic biofertiliser rich in N and P.
23. **Drinking Water Quality & Biological Contamination Markers**: Presence of the enteric bacterium ***Escherichia coli* (Coliform bacteria)** in well or tap water is the universally recognized biological index confirming recent

fecal contamination of the water source; presence of *E. coli* indicates dangerous contamination with human or animal sewage and the probable presence of waterborne pathogens (Salmonella, Vibrio, Hepatitis A).

24. **Drinking Water Disinfection & Purification**: Conventional urban purification involves sedimentation, sand-filtration, and chemical disinfection; dissolved pathogenic microbes are inactivated by **Chlorination** (dosage leaves 0.2–0.5\text{ mg/L} free residual chlorine forming germicidal hypochlorous acid, HOCl), **Ozonation (O_3)** (powerful oxidant leaving no chemical residue), and **Ultraviolet (UV) Irradiation** (UV-C at 254\text{ nm} scrambles microbial DNA/RNA, rendering pathogens unable to replicate without altering water taste).
25. **Reverse Osmosis (RO) Desalination**: When external hydrostatic pressure exceeding the natural osmotic pressure (\Pi) is applied to concentrated saline water, pure water molecules are forced backward through a semi-permeable cellulose acetate or polyamide membrane against the natural concentration gradient; widely utilized for domestic water purification and seawater desalination.
26. **Biodegradable vs Non-Biodegradable Pollutants**: **Biodegradable Pollutants**: Organic substances decomposed into simple harmless inorganic elements by natural saprophytic microorganisms within a short timeframe (e.g., **Cotton, Jute, Wool, Cellulose paper, Animal dung, Agricultural biomass**); **Non-Biodegradable Pollutants**: Synthetic or inorganic substances that microorganisms lack enzymes to break down, persisting indefinitely in the biosphere (e.g., **Polythene plastics, PVC, Glass bottles, Aluminium cans, DDT, BHC / Lindane, Heavy metals like Mercury, Lead, Cadmium**).
27. **Biomagnification vs Bioaccumulation**: **Bioaccumulation**: Gradual increase in the concentration of a persistent toxic chemical within the fatty tissues of a single biological organism over its lifespan; **Biomagnification**: Progressive, sequential increase in the concentration of a non-biodegradable, fat-soluble toxin (such as **DDT or Methylmercury**) at each successively higher trophic level along a food chain; top carnivores (e.g., fish-eating raptor birds, humans) accumulate the highest lethal concentrations.
28. **Bioremediation & Superbug**: Utilization of microorganisms to metabolize and detoxify environmental pollutants in soil and water: **In-situ bioremediation** (treating contaminated soil on site via bioventing and bioaugmentation); **Ex-situ bioremediation** (excavating soil to treat in bioreactors or biopiles); **Pseudomonas putida ("Superbug")**: First patented genetically engineered microorganism, developed by Indian-American scientist **Dr. Ananda Mohan Chakrabarty (1971)**, containing multiple plasmids engineered to break down toxic petroleum hydrocarbon slicks (octane, hexane, xylene, camphor) during maritime oil spills.
29. **Oilzapper Technology**: Consortium of five naturally occurring bacterial strains immobilized on a carrier material, developed by **The Energy and Resources Institute (TERI), New Delhi**; applied directly to terrestrial crude oil spills and oil refinery sludge, biologically degrading complex hydrocarbons into harmless CO_2 and water within a few months without chemical toxins.
30. **Biomedical Waste Color Coding Rules**: Mandated under the Biomedical Waste Management Rules by the Ministry of Environment, Forest & Climate Change: **Yellow Bag/Bin**: Highly infectious human and animal anatomical waste, soiled cotton bandages, expired cytotoxic drugs (disposed by high-temperature **Incineration** or plasma pyrolysis); **Red Bag/Bin**: Contaminated recyclable plastic waste, IV tubes, catheters, syringes without needles (disposed by **Autoclaving/Microwaving followed by shredding and recycling**); **White Translucent Container (Puncture-proof)**: Waste sharps, needles, scalpels, blades (disposed by autoclaving and metal shredding); **Blue Cardboard Box/Bin**: Glassware, medicine vials, ampoules, and metallic implants (disposed by disinfection with sodium hypochlorite and recycling).
31. **Major Research Institutes of Uttar Pradesh (UP-PCS Match Pairs)**: **Central Drug Research Institute (CDRI)**: Lucknow (developed *Saheli* / Centchroman); **National Botanical Research Institute (NBRI)**: Lucknow; **Indian Institute of Sugarcane Research (IISR)**: Lucknow; **Birbal Sahni Institute of Palaeosciences (BSIP)**: Lucknow; **Indian Institute of Pulses Research (IIPR)**: Kanpur; **National Sugar Institute (NSI)**: Kanpur; **Indian Institute of Vegetable Research (IIVR)**: Varanasi; **Central Institute for Subtropical Horticulture (CISH)**: Rahman Khera, Lucknow; **Central Institute of Medicinal and Aromatic Plants (CIMAP)**: Lucknow; **Central Avian Research Institute (CARI)**: Izatnagar, Bareilly; **Indian Veterinary Research Institute (IVRI)**: Izatnagar, Bareilly; **Central Institute for Research on Goats (CIRG)**: Makhdoom, Mathura; **Indian Grassland and Fodder Research Institute (IGFRI)**: Jhansi; **Indian Institute of Carpet Technology (IICT)**: Bhadohi.

32. **Apex National & International Scientific Institutes**: **Indian Agricultural Research Institute (IARI / Pusa Institute)**: New Delhi; **Central Rice Research Institute (CRRI)**: Cuttack, Odisha; **Central Potato Research Institute (CPRI)**: Kufri, Shimla (Himachal Pradesh); **Central Arid Zone Research Institute (CAZRI)**: Jodhpur, Rajasthan; **Central Plantation Crops Research Institute (CPCRI)**: Kasaragod, Kerala; **National Dairy Research Institute (NDRI)**: Karnal, Haryana; **National Institute of Nutrition (NIN)**: Hyderabad; **National Institute of Virology (NIV)**: Pune, Maharashtra; **International Crops Research Institute for the Semi-Arid Tropics (ICRISAT)**: Patancheru, Hyderabad (international apex center); **International Rice Research Institute (IRRI)**: Los Baños, Philippines.
33. **Indian Scientific Legends & Inventions**: **Sir Jagadish Chandra Bose (J.C. Bose)**: Invented the **Crescograph** (instrument capable of magnifying and recording minute movements of plant growth up to 10,000 times); proved physiological excitability and nervous response in living plants; pioneer of millimeter microwave radio optics; **Acharya Prafulla Chandra Ray (P.C. Ray)**: Father of Indian Chemical Science; synthesised Mercurous Nitrite (Hg_2(NO_2)_2, 1896); established Bengal Chemical and Pharmaceutical Works; **Dr. Homi Jehangir Bhabha**: Father of the Indian Nuclear Program; conceived India's three-stage nuclear power program; established TIFR and AEC; **Dr. Vikram Sarabhai**: Father of the Indian Space Program; established ISRO and PRL Ahmedabad; **Birbal Sahni**: Renowned Indian palaeobotanist; discovered the fossil plant group *Pentoxylales* in the Rajmahal hills of Jharkhand.
34. **The Six Traditional Indian Seasons (*Shad Ritus*)**: Categorized based on solar movement and the Hindu lunar calendar: **Vasanta (Spring)**: Chaitra–Vaishakha (mid-Feb to mid-Apr, moderate warmth); **Grishma (Summer)**: Jyeshtha–Ashadha (mid-Apr to mid-Jun, intense heat, *Loo* winds); **Varsha (Monsoon Rains)**: Shravana–Bhadrapada (mid-Jun to mid-Aug, South-West Monsoon); **Sharad (Autumn)**: Ashwina–Kartika (mid-Aug to mid-Oct, monsoon retreat, clear skies); **Hemanta (Pre-Winter)**: Margashirsha–Pausha (mid-Oct to mid-Dec, pleasant cold); **Shishira (Late Winter)**: Magha–Phalguna (mid-Dec to mid-Feb, extreme cold, frost).
35. **Important Commemorative Science and Health Days**: **National Science Day**: **28th February** (commemorates Sir C.V. Raman's discovery of the **Raman Effect** in 1928, Nobel Prize 1930); **World Health Day**: **7th April** (marks founding of WHO in 1948); **World Malaria Day**: **25th April**; **National Technology Day**: **11th May** (commemorates Pokhran-II nuclear tests in 1998); **World Environment Day**: **5th June**; **World Hepatitis Day**: **28th July** (birthday of Dr. Baruch Blumberg, discoverer of Hepatitis B virus); **Ozone Day**: **16th September** (commemorates Montreal Protocol, 1987); **World Rabies Day**: **28th September** (death anniversary of Louis Pasteur); **World AIDS Day**: **1st December**; **National Energy Conservation Day**: **14th December**.
36. **Master Spectrum of Biological Interactions**: **Mutualism (+/+)**: Both species derive obligate or facultative benefit (e.g., Lichens = fungus + alga; Mycorrhiza = fungus + plant root; Figs and pollinating fig wasps); **Commensalism (+/0)**: One species benefits while the host is neither helped nor harmed (e.g., Epiphytic orchids on mango branch; Barnacles on whale skin; Cattle egrets foraging beside grazing cattle); **Parasitism (+/-)**: Parasite derives nutrition at direct expense and detriment of the host (e.g., *Cuscuta / Amarbel* total stem parasite; Liver flukes; Ticks); **Amensalism (-/0)**: One species is inhibited or killed while the other is unaffected (e.g., *Penicillium* mould secreting penicillin inhibiting nearby *Staphylococcus* bacteria); **Predation (+/-)**: Predator captures, kills, and consumes prey.
37. **Hyperparasitism**: Biological phenomenon where a secondary parasite lives upon or inside another primary parasite: classic example is a **Bacteriophage** (viral parasite) infecting and replicating inside a pathogenic **Bacterium** (e.g., *Vibrio cholerae*), which is itself parasitizing a human host organism.
38. **Lichens: Pioneer Organisms & Air Quality Indicators**: Composite mutualistic organisms consisting of a photosynthetic **Phycobiont** (cyanobacterium or green alga, manufactures food) and a heterotrophic **Mycobiont** (fungus, absorbs water and minerals, provides shelter); act as ecological **pioneers** on bare rock in primary ecological succession by secreting oxalic and lichenic acids that etch rock into soil; **extremely sensitive to atmospheric Sulfur Dioxide (\mathbf{SO_2})** pollution, functioning as natural **bio-indicators of clean air** (absent in polluted industrial cities); litmus paper dye is chemically extracted from the lichen ***Roccella tinctoria***.
39. **Fungi Kingdom Characteristics**: Non-vascular, achlorophyllous eukaryotic heterotrophs; reproduce by spores; cell wall is constructed of **Chitin** (a polymer of N-acetylglucosamine) and glucans; store excess carbohydrates as

Glycogen and Oil droplets (identical to animals, **never as starch**); obtain nutrients as saprobes (absorbing dissolved nutrients via extracellular hyphal enzymes), parasites, or mutualists.

40. **Edible vs Deadly Toxic Mushrooms**: Fleshy macroscopic fruiting bodies of higher fungi (mostly Basidiomycota); **Edible Mushrooms**: White button mushroom (*Agaricus bisporus*), Oyster mushroom (*Pleurotus*), Paddy straw mushroom (*Volvariella*), and prized wild **Gucchi / Morel (*Morchella esculenta*, an Ascomycete with honeycomb cap)**; **Deadly Toxic Mushrooms**: **Death Cap (*Amanita phalloides*)** contains deadly heat-stable oligopeptide **Amatoxins and Phalloidins** that irreversibly inhibit human RNA Polymerase II, causing acute liver and renal failure within days.
41. **Thermal Dimorphism in Pathogenic Fungi**: Unique biological capability of certain human fungal pathogens to alter their morphological growth form depending on environmental temperature: they grow as **filamentous multicellular mycelial moulds at cooler room temperature (25^\circ\text{C})** in nature, but convert into **unicellular pathogenic budding yeasts at human body temperature (37^\circ\text{C})** inside tissues (e.g., *Histoplasma capsulatum, Blastomyces dermatitidis*).
42. **Ecosystem Trophic Architecture & 10% Law**: Trophic levels represent sequential nutritional feeding positions in an ecosystem: First Trophic Level (T_1, Primary Producers / Green plants) \rightarrow Second Trophic Level (T_2, Primary Consumers / Herbivores) \rightarrow Third Trophic Level (T_3, Secondary Consumers / Primary Carnivores) \rightarrow Fourth Trophic Level (T_4, Tertiary Consumers / Top Carnivores); **Lindeman's 10% Energy Transfer Law (1942)**: Only approximately **10% of energy** entering a trophic level is incorporated into biomass and transferred to the next higher level; 90% is lost as metabolic respiratory heat and waste.
43. **Ecological Pyramids**: Graphical representation of trophic parameters: **Pyramid of Energy is ALWAYS Upright** in every natural ecosystem without exception (governed by the Second Law of Thermodynamics); **Pyramid of Biomass in Aquatic / Marine Ecosystems is INVERTED** (standing crop biomass of microscopic phytoplankton primary producers is small at any given moment, supported by a much larger biomass of long-lived zooplankton and fishes); **Pyramid of Numbers in a Tree Ecosystem is Inverted** (a single large oak/banyan tree supports hundreds of herbivorous birds and thousands of hyperparasitic lice).
44. **Oceanic Primary Producers & Global Oxygen Budget**: Oceans cover ~71% of Earth's surface and generate over 50% of the planetary atmospheric oxygen; dominant primary photosynthetic producers in marine pelagic zones are microscopic unicellular **Phytoplankton**, principally **Diatoms (Bacillariophyceae)** with silica frustules, **Dinoflagellates**, and marine cyanobacteria (e.g., ***Prochlorococcus***, the most abundant photosynthetic organism on Earth).
45. **Algal Blooms & Eutrophication**: Excessive enrichment of freshwater bodies with agricultural runoff containing dissolved **Nitrates and Phosphates** triggers explosive population growth of surface phytoplankton, termed an **Algal Bloom**; decaying algae consume virtually all dissolved oxygen, creating hypoxic **"dead zones"** and causing catastrophic fish kills; some blooms produce neurotoxins (e.g., marine dinoflagellates *Gymnodinium* produce saxitoxin, creating toxic **Red Tides**).
46. **Microbial Decomposers & Humification**: Soil saprotrophic bacteria and fungi (actinomycetes, moulds) execute final nutrient mineralization; they decompose recalcitrant plant lignocellulose and animal chitin through **Humification**, forming dark amorphous colloidal **Humus**; humus acts as an essential soil conditioner with immense cation exchange capacity and reservoir of plant nutrients.
47. **Bioleaching / Microbial Mining**: Industrial extraction of valuable metals from low-grade ores using living acidophilic bacteria (principally ***Acidithiobacillus ferrooxidans***); bacteria oxidize insoluble metal sulfides into water-soluble metal sulfates (e.g., leaching **Copper, Uranium, and Gold** from mining tailings with low energy cost and minimal atmospheric pollution).
48. **Biosensors & Bioluminescence**: **Biosensors**: Analytical devices integrating a biological recognition element (enzymes, whole microbial cells, or antibodies) with a physicochemical transducer to detect minute concentrations of pesticides, pollutants, or blood glucose; **Bioluminescence**: Production of cold light by living organisms through the oxidation of the substrate **Luciferin** catalyzed by the enzyme **Luciferase** in the presence of ATP and oxygen (seen in fireflies, deep-sea fishes, and luminous bacteria like *Vibrio fischeri*).

49. **Cryopreservation & Ex-Situ Germplasm Conservation**: Ultra-low temperature preservation of living biological materials (viable microbial cultures, plant shoot tips, animal embryos, human sperm/ova) by storage in **Liquid Nitrogen at -196^\circ\text{C}**; completely arrests all enzymatic and biochemical cellular metabolic processes, maintaining viability for decades without genetic mutation; apex technique for modern gene banks and biodiversity germplasm conservation.
50. **Biological Unity of the Human Species**: From an anthropological, anatomical, biochemical, and genetic standpoint, all living human beings belong strictly to a single biological species: ***Homo sapiens***; all fertile human races share identical chromosomal karyotype (2n = 46), identical ABO/Rh blood grouping systems, identical protein architectures, and can interbreed freely with 100% biological fertility; superficial ethnic phenotypic variations (skin pigmentation, eye fold, hair texture) reflect adaptive micro-evolutionary climatic variations and account for less than 0.1\% of total genomic sequence differences.

📋 Must-Score Drill — Microbes, Waste, Institutes & Seasons

1. Master Table of Microbes & Industrial Products

Commercial Product	Producing Microorganism	Microbial Class	Major Application / Mechanism
Curd	*Lactobacillus acidophilus* (LAB)	Bacterium	Coagulates milk casein; enriches **Vitamin B_{12}**
Bread & Alcohol	*Saccharomyces cerevisiae*	Fungus (Yeast)	Releases CO_2 (puffing) & Ethanol
Swiss Cheese	*Propionibacterium shermanii*	Bacterium	Large eye holes from rapid CO_2 generation
Roquefort Cheese	*Penicillium roqueforti*	Fungus (Mould)	Imparts specific sharp taste and texture
Citric Acid	*Aspergillus niger*	Fungus (Mould)	Food preservative, flavouring agent
Acetic Acid (Vinegar)	*Acetobacter aceti*	Bacterium	Aerobic oxidation of ethanol
Butyric Acid	*Clostridium butyricum*	Bacterium (Anaerobe)	Industrial chemical synthesis
Lactic Acid	*Lactobacillus bulgaricus*	Bacterium	Dairy, tanning, food industry
Lipases	*Candida lipolytica*	Fungus (Yeast)	Detergent formulations (dissolves oil/grease)
Pectinases & Proteases	*Aspergillus* species	Fungus (Mould)	Clarifying commercial bottled fruit juices
Streptokinase	*Streptococcus* (genetically modified)	Bacterium	**"Clot Buster"** for myocardial infarction
Cyclosporin A	*Trichoderma polysporum*	Fungus (Mould)	**Immunosuppressant** in organ transplantation
Statins	*Monascus purpureus*	Fungus (Yeast)	**Inhibits HMG-CoA reductase**; lowers blood cholesterol

2. Waste Classification Master Matrix

Material	Biodegradable?	Recyclable?	Environmental Fate / Safe Disposal Method
Wool	**Yes**	Yes	Keratin protein fibre; digested by bacteria/fungi into soil nutrients.
Cotton / Jute / Paper	**Yes**	Yes	Cellulose polymers; readily composted by microbial cellulases.
Kitchen Food Waste	**Yes**	No	Wet organic matter; converted into compost or biogas slurry.
Human Urine & Faeces	**Yes**	No	Organic waste; digested by anaerobic septic/biogas methanogens.
Plastic (Polythene, PVC)	**No**	Yes	Non-biodegradable; frays into toxic microplastics; mechanical recycling/pyrolysis.
Glass Bottles	**No**	**Yes (100%)**	Inert silicate; 100% recyclable indefinitely; completely non-biodegradable.

Material	Biodegradable?	Recyclable?	Environmental Fate / Safe Disposal Method
Aluminium Foil & Cans	**No**	**Yes (100%)**	Non-biodegradable metal; remelted and recycled with 95% energy savings.
DDT / Endosulfan / BHC	**No**	No	Persistent Organic Pollutant (POP); fat-soluble; biomagnifies in top carnivores.
Hospital Sharps & Dressings	Hazardous	Conditional	Biomedical waste; segregated in yellow/white bins; incinerated or autoclaved.

3. Shad Ritus (The Six Indian Seasons)

Ritu (Season)	Hindu Lunar Months	Gregorian Calendar Period	Meteorological Features	Ecological & Agricultural Linkages
Vasanta (Spring)	Chaitra – Vaishakha	Mid-February to Mid-April	Moderate warmth, pleasant breezes	Forest trees flower; honeybee pollination peak; ripening of Rabi crops.
Grishma (Summer)	Jyeshtha – Ashadha	Mid-April to Mid-June	Extreme heat, dry *Loo* winds	Water reservoirs shrink; cultivation of riverbed **Zaid crops** (melon, gourd).
Varsha (Rains)	Shravana – Bhadrapada	Mid-June to Mid-August	South-West Monsoon, high humidity	Sowing of **Kharif crops** (Paddy, Maize, Soybean); vector mosquito multiplication.
Sharad (Autumn)	Ashwina – Kartika	Mid-August to Mid-October	Monsoon retreat, clear skies	"October Heat"; maturation and initial harvesting of early Kharif crops.
Hemanta (Pre-Winter)	Margashirsha – Pausha	Mid-October to Mid-December	Crisp cool weather, lowering night temps	Sowing of **Rabi crops** (Wheat, Mustard, Gram, Potato); dew on grass.
Shishira (Winter)	Magha – Phalguna	Mid-December to Mid-February	Severe cold, morning frost, dense fog	Western Disturbances bring winter rains (*Mahawat*) aiding wheat; migratory wetland birds.

4. Master Matrix of Biological Interactions & Symbiosis

Interaction Type	Species A	Species B	Biological Mechanism	Classic Must-Score Examples
Mutualism	**+**	**+**	Both species derive obligate or facultative reciprocal survival/nutritional benefits.	• **Lichens**: Fungus (mycobiont) absorbs water/minerals & protects; Alga/Cyanobacterium (phycobiont) photosynthesizes sugars. • **Mycorrhiza**: Fungi colonize higher plant roots, increasing **Phosphorus/Nitrogen absorption** & pathogen protection in exchange for sugars. • **Fig & Fig Wasp**: Obligate one-to-one co-evolution; wasp pollinates fig inflorescence while laying eggs; fig seeds nourish larvae. • **Coral Reefs**: Scleractinian cnidarians provide shelter and CO_2 to symbiotic intracellular dinoflagellate algae (*Symbiodinium*). • **Root Nodules**: *Rhizobium* diazotrophs fix atmospheric N_2 inside leguminous root nodules.

Interaction Type	Species A	Species B	Biological Mechanism	Classic Must-Score Examples
Commensalism	**+**	**0**	One species benefits (commensal) while the host is neither harmed nor helped.	• **Barnacles on Whales**: Filter-feeding crustaceans get free ocean transport and access to plankton (*Phoresy*). • **Remora (Suckerfish) & Sharks**: Remora attaches to shark underside via dorsal sucker, feeding on leftover food scraps. • **Cattle Egrets & Grazing Mammals**: Egrets feed on grasshoppers/insects flushed out by grazing cattle. • **Epiphytic Orchids**: Grow perched on tall tree bark strictly for mechanical sunlight support, taking water/nutrients from air/rain.
Parasitism	**+**	**-**	Parasite derives nourishment/replication at direct physiological harm/expense of host.	• **Endoparasitism**: *Herpes simplex* virus, *Plasmodium* (malaria), and tapeworms living/replicating inside host tissue. • **Ectoparasitism**: Ticks, mites, lice, and female mosquitoes sucking blood from host skin surface.
Hyperparasitism	**+**	**-**	**Parasite within a parasite**: An organism whose host is itself a parasite or parasitoid.	• Parasitoid wasps (suborder Apocrita within Hymenoptera), Diptera flies, and Carabid beetles parasitizing other parasitic insect larvae.
Amensalism	**-**	**0**	One species is inhibited or destroyed while the other remains completely unaffected.	• *Penicillium notatum* mould secreting penicillin toxin that lyses surrounding *Staphylococcus* bacteria without benefiting from their death.

5. Master Matrix of Fungi, Lichens & Microbial Ecological Roles

Entity / Group	Kingdom / Phylum	Key Biochemical & Cellular Features	Ecological / Economic Significance & Common Traps
Yeasts	Fungi (Ascomycota / Basidiomycota)	Unicellular; cell wall of **chitin**; reproduce asexually by asymmetric **budding**; lack chlorophyll; store food as **glycogen**.	*Saccharomyces cerevisiae* (Baker's and Brewer's yeast); carries out anaerobic alcoholic fermentation.
Moulds	Fungi (Zygomycota / Ascomycota)	Filamentous multicellular networks of microscopic tubular **hyphae** forming cottony mycelia.	*Penicillium notatum* (penicillin source), *Aspergillus niger* (citric acid), *Rhizopus* (black bread mould).
Mushrooms	Fungi (mostly Basidiomycota)	Fleshy, spore-bearing macroscopic fruiting bodies (basidiocarps) growing above ground on soil or decaying wood.	Edible macrofungi (button mushroom *Agaricus*); lack chlorophyll; completely saprophytic/heterotrophic.
Gucchi (Morel)	Fungi (Ascomycota, Morchellaceae)	Genus ***Morchella***; honeycombed pileus (cap) with extensive pits and ridges; pale yellow/brown; tall hollow white stem.	**Highly priced wild edible mushroom**; grows wild in temperate **Himalayan conifer forests** (HP, UK, J&K); **cannot be commercially cultivated**.
Death Cap	Fungi (Basidiomycota, *Amanita*)	Genus ***Amanita phalloides***; greenish-white cap, prominent white gills, volva cup at stem base.	**Lethal poisonous mushroom**; contains amatoxins (\alpha-amanitin) causing irreversible liver and kidney necrosis.
Dimorphic Fungi	Fungi (Pathogenic genera)	Thermally dimorphic: exists as **mould** at ambient room temperature (25^\circ C) and transitions into **yeast** at mammalian body temperature (37^\circ C).	E.g., *Talaromyces (Penicillium) marneffei*, *Histoplasma capsulatum*; causes systemic deep mycoses in humans.

Entity / Group	Kingdom / Phylum	Key Biochemical & Cellular Features	Ecological / Economic Significance & Common Traps
Lichens	Composite Mutualistic Organism	Dual organism: **Phycobiont** (cyanobacteria/green alga) + **Mycobiont** (Ascomycete/Basidiomycete fungus).	**Pioneer species** in lithosere ecological succession on bare rocks; **exceptionally sensitive to Sulfur Dioxide (\mathbf{SO_2}) air pollution**; absent in polluted cities.
Mycorrhiza	Symbiotic Association	Mutualism between soil fungal hyphae and roots of higher vascular plants (gymnosperms/angiosperms).	Increases root absorption of **Phosphorus, Nitrogen, and minerals**; protects against root pathogens; increases drought resistance.
Oceanic Phytoplankton	Monera / Protista (**Diatoms & Cyanobacteria**)	Microscopic photosynthetic organisms floating in the photic zone; diatoms have silica cell walls (frustules).	Chief **primary producers of the oceans** (Diatoms contribute **~45%** of marine organic synthesis); sustain marine food webs.

One-Liner Revision – Microbiology, Environment and Applied Science

#	Fast Fact
1	Bacteria discovered by = **Antonie van Leeuwenhoek** (1676, using microscope)
2	Germ Theory of Disease = **Louis Pasteur** (France) and Robert Koch (Germany)
3	Koch's Postulates = rules to establish pathogen causes a specific disease
4	Penicillin = from *Penicillium notatum*; discovered by **Alexander Fleming (1928)**
5	Streptomycin = discovered by **Selman Waksman (1943)**; Nobel 1952; treats TB
6	Pasteurisation (milk) = 72°C for 15 seconds (HTST) OR 63°C for 30 min (LTST)
7	Virus = DNA OR RNA (never both) + protein coat (capsid); no metabolism outside host
8	Largest virus = Mimivirus / Pandoravirus; Smallest virus = Parvovirus
9	Prions = misfolded proteins; cause Creutzfeldt-Jakob Disease (CJD) / mad cow disease
10	Viroids = smallest known infectious agents; only RNA, no protein coat; infect plants
11	Mycology = study of fungi; Bacteriology = study of bacteria; Virology = study of viruses
12	Yeast = unicellular fungus; used in baking (CO2) and brewing (ethanol)
13	Biogas composition = **Methane (55-75%)** + CO2 (25-45%) + trace gases
14	Biogas plants = use cow dung + water; anaerobic fermentation; produces CH4
15	Composting = aerobic breakdown; humus-rich dark material; improves soil fertility
16	Vermicomposting = earthworm-based composting; best compost quality
17	Nitrogen cycle steps: Fixation -> Nitrification -> Assimilation -> Ammonification -> Denitrification
18	Denitrification = conversion of nitrates back to N2 gas; by Pseudomonas denitrificans
19	Biodiversity hotspots globally = **36** (CI criteria: >1,500 endemic plants + >70% habitat lost)
20	India's biodiversity hotspots = **4**: Western Ghats, Eastern Himalayas, Indo-Burma, Sundaland
21	IUCN Red List categories: Extinct > EW > Critical > Endangered > Vulnerable > NT > LC
22	Project Tiger (1973) = conservation of Bengal Tiger; 53 tiger reserves in India
23	Convention on Biological Diversity (CBD) = **1992**, Rio de Janeiro; India ratified 1994
24	Ramsar Convention (1971) = protection of **wetlands**; Ramsar, Iran
25	India's Ramsar sites = **75+** (largest number of wetland sites in South Asia)

#	Fast Fact
26	Biopesticide example: **Bacillus thuringiensis (Bt)** toxin against Lepidoptera larvae
27	Biofertiliser = Rhizobium, Azotobacter, Blue-Green Algae (BGA/Anabaena), Azolla
28	Eutrophication = excess N + P in water -> algae bloom -> hypoxia -> dead zone
29	BOD (Biochemical Oxygen Demand) = measures organic pollution in water; high BOD = more pollution
30	One-Health concept = human health, animal health, and environment are interconnected

Physics Topic 1 — Fundamentals, Measurement and Instruments

Consolidated — 50 Must-Score Facts

1. **Four Systems of Units**: CGS (cm, g, s), FPS (foot, pound, second — British system; 1\text{ lb} = 453.59237\text{ g}), MKS (m, kg, s), and SI (International System of Units, adopted at 11th CGPM Paris in 1960).
2. **Seven Fundamental Quantities**: Length (m), Mass (kg), Time (s), Electric Current (A), Thermodynamic Temperature (K), Luminous Intensity (cd), Amount of Substance (mol).
3. **Supplementary Units**: Plane angle (Radian, rad = m/m, dimensionless), Solid angle (Steradian, sr = m^2/m^2, dimensionless).
4. **22 SI Derived Units with Special Names**: Radian (rad), Steradian (sr), Hertz (Hz), Newton (N), Pascal (Pa), Joule (J), Watt (W), Coulomb (C), Volt (V), Farad (F), Ohm (\Omega), Siemens (S), Weber (Wb), Tesla (T), Henry (H), Degree Celsius (^\circ C), Lumen (lm), Lux (lx), Becquerel (Bq), Gray (Gy), Sievert (Sv), Katal (kat).
5. **Katal (kat)**: SI unit of catalytic activity (1\text{ kat} = 1\text{ mol/s} of substrate converted).
6. **Tesla (T)**: SI unit of magnetic flux density (B-field). 1\text{ T} = 1\text{ Wb/m}^2 = 1\text{ N}/(A\cdot m) = 1\text{ kg}\cdot s^{-2}\cdot A^{-1} = 10^4\text{ Gauss}.
7. **Magnetic Field Strength (H) vs Flux Density (B)**: Unit of H is Ampere per metre (A/m); unit of B is Tesla (T).
8. **Electrical Resistivity (\rho)**: SI unit is **Ohm-metre** (\Omega\cdot m), from R = \rho l / A \implies \rho = RA/l.
9. **Electrical Conductance (G) & Conductivity (\sigma)**: Conductance unit is Siemens (S = \Omega^{-1} or Mho); Conductivity unit is Siemens per metre (S/m or \Omega^{-1}\cdot m^{-1}).
10. **Capacitance**: SI unit is Farad (F = C/V). (Note: Faraday is a unit of electric charge: 1\text{ Faraday} \approx 96,485\text{ C}).
11. **Radioactivity Units**: SI unit is **Becquerel** (Bq = 1\text{ disintegration/s}). Historical unit is **Curie** (Ci = 3.7 \times 10^{10}\text{ Bq}). Rutherford (Rd = 10^6\text{ Bq}).
12. **Radiation Absorbed Dose vs Dose Equivalent**: Absorbed dose is **Gray** (Gy = 1\text{ J/kg} = 100\text{ rad}). Equivalent/effective biological dose is **Sievert** (Sv = 1\text{ J/kg} = 100\text{ rem}).
13. **Light-Year (ly)**: Unit of **astronomical distance** (NOT time). Distance travelled by light in vacuum in 1 Julian year (365.25\text{ days} \times 86,400\text{ s}). 1\text{ ly} = 9.46073 \times 10^{15}\text{ m} \approx 9.461 \times 10^{15}\text{ m} \approx 9.46 \times 10^{12}\text{ km}.
14. **Why Astronomical Distances are Measured in Light-Years**: The speed of light (c) is constant throughout the universe in a vacuum (unaffected by gravity or motion of medium).
15. **Astronomical Unit (AU)**: Average distance between center of Earth and center of Sun. 1\text{ AU} = 1.495978707 \times 10^{11}\text{ m} \approx 1.496 \times 10^{11}\text{ m} \approx 149.6\text{ million km}.
16. **Parsec (Parallactic Second)**: Largest practical astronomical distance unit. Distance at which average radius of Earth's orbit (1\text{ AU}) subtends an angle of 1 arcsecond. 1\text{ Parsec} = 3.0857 \times 10^{16}\text{ m} = 3.262\text{ light-years} = 206,265\text{ AU} = \frac{648,000}{\pi}\text{ AU}. Coined by Herbert Hall Turner (1913).
17. **Small Distance Units**:
    - Fermi (Femtometre, fm): 10^{-15}\text{ m} (smallest practical length unit; used for nuclear radii).
    - Angstrom (\text{\AA}): 10^{-10}\text{ m} = 0.1\text{ nm} = 10^{-8}\text{ cm} (atomic radii, X-ray & optical wavelengths).
    - Nanometre (nm): 10^{-9}\text{ m} = 10^{-7}\text{ cm} = 10\text{ \AA}. 1\text{ metre} = 10^9\text{ nm}.
    - Micron (Micrometre, \mu m): 10^{-6}\text{ m} = 10^{-4}\text{ cm} = 10^{-3}\text{ mm} = 1/1000\text{ mm} (cell dimensions).
18. **Height of a 6-foot tall person**: 1\text{ ft} = 0.3048\text{ m} \approx 30.5 \times 10^7\text{ nm} \implies 6\text{ ft} = 6 \times 30.5 \times 10^7\text{ nm} = 183 \times 10^7\text{ nm}.
19. **Mass Units Conversion**: 1\text{ kg} = 10^3\text{ g}; 1\text{ mg} = 10^{-3}\text{ g}; 1\text{ }\mu g = 10^{-6}\text{ g}; 1\text{ ng} = 10^{-9}\text{ g}; 1\text{ pg (picogram)} = 10^{-12}\text{ g}; 1\text{ fg (femtogram)} =

10^{-15}\text{ g}.

20. **Pressure SI Unit**: **Pascal** (Pa). 1\text{ Pa} = 1\text{ N/m}^2 = 1\text{ kg}\cdot m^{-1}\cdot s^{-2}.
21. **Atmospheric Pressure & Conversions**:
    - 1\text{ bar} = 10^5\text{ Pa} = 10^6\text{ dyne/cm}^2 \approx 1.02\text{ kgf/cm}^2.
    - 1\text{ atm} = 101,325\text{ Pa} = 1.01325\text{ bar} = 760\text{ mmHg} = 760\text{ torr}.
    - 1\text{ torr} = 1\text{ mmHg} = 133.322\text{ Pa}.
    - 1\text{ psi (pound per square inch)} \approx 6894.76\text{ Pa}.
    - 1\text{ kgf/cm}^2 = 98,066.5\text{ Pa} \approx 0.98\text{ bar} \approx 1.0\text{ bar}.
22. **Strain & Mach Number are Dimensionless**: Mechanical strain (\Delta l / l), Mach number (object speed / sound speed), Refractive index, Relative density, and Poisson's ratio have **NO units and NO dimensions**.
23. **Barrel of Oil**: 1\text{ barrel} = 158.9873\text{ litres} \approx 159\text{ litres} = 0.158987\text{ m}^3 = 42\text{ US gallons} = 34.9723\text{ UK gallons}.
24. **Horsepower (hp)**: 1\text{ hp} = 746\text{ Watts}. Metric horsepower \approx 735.5\text{ W}.
25. **Megawatt (MW)**: Unit of power generated at power stations. 1\text{ MW} = 10^6\text{ W} = 1,000\text{ kW}.
26. **Kilowatt-hour (kWh)**: Unit of **electrical energy** (NOT power). 1\text{ kWh} = 3.6 \times 10^6\text{ Joules} = 1\text{ B.O.T. unit}.
27. **Cusec (Cubic feet per second)**: Unit of rate of water flow in rivers/canals. 1\text{ cusec} = 1\text{ ft}^3/s = 28.317\text{ litres/second}.
28. **NTU (Nephelometric Turbidity Units)**: Measures water turbidity (cloudiness/opaqueness due to suspended particulate matter).
29. **Solar Constant**: Solar radiant energy received per unit area at top of Earth's atmosphere at 1\text{ AU} \approx 1.4\text{ kW/m}^2 (1361\text{ W/m}^2).
30. **Dobson Unit (DU)**: Measures atmospheric column ozone thickness. 1\text{ DU} = 0.01\text{ mm} thickness of pure ozone compressed to STP (0^\circ\text{C}, 1\text{ atm}). Represents 2.687 \times 10^{20}\text{ ozone molecules/m}^2. Normal ozone layer \approx 300\text{ DU}; Ozone hole <220\text{ DU}.
31. **Nautical Mile & Knot**: 1\text{ Nautical Mile} = 1.852\text{ km} = 1852\text{ m} (distance at sea). **Knot** = speed of 1 nautical mile per hour (1.852\text{ km/h} \approx 0.514\text{ m/s}).
32. **Decibel (dB)**: Logarithmic unit measuring sound intensity/level. Whispering \approx 15-20\text{ dB}, Normal speech \approx 50-60\text{ dB}, Traffic \approx 80\text{ dB}, Jet engine takeoff \approx 130-140\text{ dB} (pain threshold).
33. **Computer Processor Speed**: Measured in **Hertz (Hz) / Gigahertz (GHz)**, indicating CPU clock speed (1\text{ GHz} = 10^9\text{ cycles/second}). Byte/bit measure memory/data storage, NOT processor speed.
34. **Pyrometer (Total Radiation Pyrometer)**: Remote-sensing thermometer for extremely high temperatures (>1500^\circ C to 2000^\circ C and beyond, e.g., Sun, furnaces, molten metal). Works on **Stefan-Boltzmann Law** (E \propto T^4). Does not require physical contact.
35. **Pyrheliometer vs Pyranometer**: Pyrheliometer measures **direct beam solar irradiance** at normal incidence; Pyranometer measures **global (direct + diffuse) solar radiation** on a planar surface.
36. **Anemometer vs Wind Vane**: Anemometer measures wind **speed / velocity and force**; Wind Vane indicates wind **direction**.
37. **Barometer**: Measures atmospheric pressure. Simple barometer uses **mercury** (Hg). A sudden steep drop in mercury level indicates an approaching **storm / thunderstorm**; gradual fall indicates rain; gradual rise indicates fair/dry weather.
38. **Hygrometer vs Hydrometer**: **Hygrometer** (or Psychrometer) measures **relative humidity** of the air. **Hydrometer** measures **relative density / specific gravity of liquids**.
39. **Lactometer vs Butyrometer**: **Lactometer** checks milk purity by measuring **milk density**; **Butyrometer** measures **fat content** in milk and dairy products.

40. **SONAR (Sound Navigation and Ranging)**: Uses **ultrasonic waves** (>20\text{ kHz}) to detect, locate, and measure depth/distance of submerged submarines, icebergs, shipwrecks, and sea beds.
41. **Fathometer**: Measures **depth of the ocean/sea** using ultrasonic echo soundings (1\text{ fathom} = 6\text{ feet} = 1.8288\text{ m}).
42. **RADAR (Radio Detection and Ranging)**: Uses **radio waves / microwaves** to determine range, elevation, angle, and velocity of aircraft, ships, spacecraft, and weather formations.
43. **Polygraph**: Lie detector instrument; simultaneously records physiological parameters: arterial blood pressure, pulse rate, respiration rate, and galvanic skin resistance (electrodermal activity).
44. **EEG (Electroencephalogram)**: Records **electrical activity of the brain** via scalp electrodes. (Compare: ECG for heart, EMG for muscles, EOG for eyes).
45. **Ringelmann Scale**: Chart of graduated grid lines used to measure the **apparent density and opacity of smoke** in air pollution monitoring.
46. **Theodolite vs Tacheometer**: **Theodolite** measures precise horizontal (azimuth) and vertical (elevation) angles. **Tacheometer** is a rapid surveying theodolite that optically/electronically determines distance and height.
47. **Bolometer**: Measures the power of incident electromagnetic radiation via resistance change of a heated sensing element.
48. **Crescograph**: Invented by Sir **Jagadish Chandra Bose** in early 20th century to measure and record infinitesimal **growth in plants**.
49. **Kymograph**: Rotating cylindrical drum recording physiological changes such as muscle contraction, blood pressure variations, and vascular response.
50. **Lysimeter**: Measures **actual evapotranspiration** (combined evaporation from soil + transpiration from crops).

📋 Must-Score Drill — SI Units, Constants & Scientific Instruments

1. Master Table of Fundamental & Derived Physical Units

A. The Seven SI Base Units (2019 BIPM Anchor Constants)

Physical Quantity	SI Unit	Symbol	Base Defining Constant (2019 SI Revision)	Historical Reference Standard
Length	Metre	m	Speed of light in vacuum c = 299,792,458\text{ m/s}	Distance between two lines on Pt-Ir bar at BIPM Sèvres
Mass	Kilogram	kg	Planck constant h = 6.62607015 \times 10^{-34}\text{ kg}\cdot m^2\cdot s^{-1}	Mass of international Pt-Ir cylinder prototype (*Le Grand K*)
Time	Second	s	Caesium frequency \Delta \nu_{\text{Cs}} = 9,192,631,770\text{ Hz}	1/86,400 of a mean solar day
Electric Current	Ampere	A	Elementary charge e = 1.602176634 \times 10^{-19}\text{ C}	Force of 2 \times 10^{-7}\text{ N/m} between parallel conductors
Thermodynamic Temperature	Kelvin	K	Boltzmann constant k = 1.380649 \times 10^{-23}\text{ J/K}	1/273.16 of thermodynamic temp of triple point of water
Amount of Substance	Mole	mol	Avogadro constant N_A = 6.02214076 \times 10^{23}\text{ mol}^{-1}	Number of atoms in 0.012\text{ kg} of pure Carbon-12
Luminous Intensity	Candela	cd	Luminous efficacy K_{\text{cd}} = 683\text{ lm/W} at 540 \times 10^{12}\text{ Hz}	Radiation of candle / blackbody at Pt freezing point

B. The 22 SI Derived Units with Special Names and Symbols

Derived Quantity	Special Name	Symbol	Expressed in Terms of Other SI Units	Expressed in Terms of SI Base Units
Plane angle	radian	rad	—	m\cdot m^{-1} = 1 (dimensionless)

Derived Quantity	Special Name	Symbol	Expressed in Terms of Other SI Units	Expressed in Terms of SI Base Units
Solid angle	steradian	sr	—	m^2\cdot m^{-2} = 1 (dimensionless)
Frequency	hertz	Hz	—	s^{-1}
Force	newton	N	—	kg\cdot m\cdot s^{-2}
Pressure, stress	pascal	Pa	N/m^2	kg\cdot m^{-1}\cdot s^{-2}
Energy, work, heat	joule	J	N\cdot m	kg\cdot m^2\cdot s^{-2}
Power, radiant flux	watt	W	J/s	kg\cdot m^2\cdot s^{-3}
Electric charge	coulomb	C	—	A\cdot s
Electric potential difference, voltage	volt	V	W/A	kg\cdot m^2\cdot s^{-3}\cdot A^{-1}
Capacitance	farad	F	C/V	kg^{-1}\cdot m^{-2}\cdot s^4\cdot A^2
Electric resistance	ohm	\Omega	V/A	kg\cdot m^2\cdot s^{-3}\cdot A^{-2}
Electric conductance	siemens	S	A/V = \Omega^{-1}	kg^{-1}\cdot m^{-2}\cdot s^3\cdot A^2
Magnetic flux	weber	Wb	V\cdot s	kg\cdot m^2\cdot s^{-2}\cdot A^{-1}
Magnetic flux density (B-field)	tesla	T	Wb/m^2 = N/(A\cdot m)	kg\cdot s^{-2}\cdot A^{-1}
Inductance	henry	H	Wb/A	kg\cdot m^2\cdot s^{-2}\cdot A^{-2}
Celsius temperature	degree Celsius	^\circ C	—	K (t(^\circ\text{C}) = T(\text{K}) - 273.15)
Luminous flux	lumen	lm	cd\cdot sr	cd
Illuminance	lux	lx	lm/m^2	m^{-2}\cdot cd
Activity of radionuclide	becquerel	Bq	—	s^{-1}
Absorbed dose, kerma	gray	Gy	J/kg	m^2\cdot s^{-2}
Dose equivalent	sievert	Sv	J/kg	m^2\cdot s^{-2}
Catalytic activity	katal	kat	—	mol\cdot s^{-1}

C. Other Derived Quantities Expressed in SI Base Units

Derived Quantity	SI Unit	Dimension
Area	m^2	[M^0 L^2 T^0]
Volume	m^3	[M^0 L^3 T^0]
Speed, Velocity	m\cdot s^{-1}	[M^0 L^1 T^{-1}]
Acceleration	m\cdot s^{-2}	[M^0 L^1 T^{-2}]
Wavenumber	m^{-1}	[M^0 L^{-1} T^0]
Density, Mass Density	kg\cdot m^{-3}	[M^1 L^{-3} T^0]
Surface Density	kg\cdot m^{-2}	[M^1 L^{-2} T^0]
Specific Volume	m^3\cdot kg^{-1}	[M^{-1} L^3 T^0]
Current Density	A\cdot m^{-2}	[M^0 L^{-2} T^0 A^1]
Magnetic Field Strength (H)	A\cdot m^{-1}	[M^0 L^{-1} T^0 A^1]

Derived Quantity	SI Unit	Dimension
Molar Concentration	mol\cdot m^{-3}	[M^0 L^{-3} T^0 mol^1]
Mass Concentration	kg\cdot m^{-3}	[M^1 L^{-3} T^0]
Luminance	cd\cdot m^{-2}	[M^0 L^{-2} T^0 cd^1]

2. Astronomical, Nuclear & Practical Measurement Units

Category / Unit	Symbol	Equivalence in SI / Practical Value	Core Application & Common Trap
Parsec	pc	3.0857 \times 10^{16}\text{ m} = 3.262\text{ ly} = 206,265\text{ AU} = \frac{648,000}{\pi}\text{ AU}	Distance at which Earth's orbit (1\text{ AU}) subtends 1 arcsecond. Coined by H. H. Turner (1913).
Light-Year	ly	9.46073 \times 10^{15}\text{ m} \approx 9.461 \times 10^{15}\text{ m} \approx 9.46 \times 10^{12}\text{ km}	Distance light travels in vacuum in 1 Julian year (365.25\text{ days}). Unit of distance, NEVER time.
Astronomical Unit	AU	1.495978707 \times 10^{11}\text{ m} \approx 1.496 \times 10^{11}\text{ m}	Average distance between Earth and Sun (\approx 149.6\text{ million km}).
Angstrom	\text{\AA}	10^{-10}\text{ m} = 0.1\text{ nm} = 10^{-8}\text{ cm}	Measuring light wavelength, atomic and crystal lattices.
Nanometre	nm	10^{-9}\text{ m} = 10^{-7}\text{ cm} = 10\text{ \AA}. (1\text{ m} = 10^9\text{ nm})	Semiconductor gate lengths, CPU transistors, molecular sizes.
Micron (Micrometre)	\mu m	10^{-6}\text{ m} = 10^{-4}\text{ cm} = 10^{-3}\text{ mm} = 1/1000\text{ mm}	Biological cells, bacteria size, atmospheric particulate matter (PM_{2.5}, PM_{10}).
Fermi (Femtometre)	fm	10^{-15}\text{ m}	Nuclear dimensions and proton radii (Smallest unit of length).
Barrel (Petroleum)	bbl	158.9873\text{ litres} \approx 159\text{ litres} = 0.158987\text{ m}^3 = 42\text{ US gal}	Global standard unit of crude oil volume (34.9723\text{ UK gal}).
Horsepower	hp	746\text{ Watts} = 746\text{ J/s}	Mechanical power rating of engines and motors (1\text{ metric hp} \approx 735.5\text{ W}).
Megawatt	MW	10^6\text{ W} = 1,000\text{ kW} = 1,000,000\text{ J/s}	Power station electricity generation capacity.
Cusec	cusec	1\text{ cubic foot per second} = 28.317\text{ litres/second}	Volumetric flow rate in rivers, dams, and irrigation canals.
NTU	NTU	Nephelometric Turbidity Units	Degree of cloudiness/water turbidity caused by suspended insoluble particles.
Solar Constant	S_0	1361.6\text{ W/m}^2 \approx 1.4\text{ kW/m}^2 (1.4 \times 10^3\text{ W/m}^2)	Solar radiant energy per unit area outside Earth's atmosphere at 1\text{ AU}.
Dobson Unit	DU	0.01\text{ mm thickness of pure } O_3\text{ at STP} = 2.687 \times 10^{20}\text{ molecules/m}^2	Atmospheric column ozone layer thickness.
Nautical Mile	NM	1.852\text{ km} = 1852\text{ m}	Distance used in maritime and aeronautical navigation (1\text{ min of latitude}).
Knot	kt	1\text{ nautical mile per hour} = 1.852\text{ km/h} \approx 0.514\text{ m/s}	Speed of vessels at sea and aircraft.
Mach Number	M	Dimensionless ratio: \frac{\text{Speed of object}}{\text{Speed of sound in medium}}	Speed classification: Subsonic (<1), Supersonic (>1), Hypersonic (>5).
Bar / Atmosphere / Torr	bar, atm	1\text{ bar} = 10^5\text{ Pa} = 10^6\text{ dyn/cm}^2 \approx 1.02\text{ kgf/cm}^2; 1\text{ atm} = 760\text{ torr} = 101,325\text{ Pa}	Atmospheric pressure units (1\text{ torr} = 1\text{ mmHg} = 133.322\text{ Pa}).

3. Master Metric Prefixes Matrix

Factor	Prefix	Symbol	Factor	Prefix	Symbol
10^{30}	**Quetta**	Q	10^{-1}	**deci**	d
10^{27}	**Ronna**	R	10^{-2}	**centi**	c
10^{24}	**Yotta**	Y	10^{-3}	**milli**	m
10^{21}	**Zetta**	Z	10^{-6}	**micro**	\mu
10^{18}	**Exa**	E	10^{-9}	**nano**	n
10^{15}	**Peta**	P	10^{-10}	**angstrom**	\text{\AA}
10^{12}	**Tera**	T	10^{-12}	**pico**	p
10^{9}	**Giga**	G	10^{-15}	**femto**	f
10^{6}	**Mega**	M	10^{-18}	**atto**	a
10^{3}	**kilo**	k	10^{-21}	**zepto**	z
10^{2}	**hecto**	h	10^{-24}	**yocto**	y
10^{1}	**deca**	da	10^{-27}	**ronto**	r
—	—	—	10^{-30}	**quecto**	q

4. Master Thermometry & Temperature Scales

- **Universal Temperature Conversion Formula**: \frac{C}{5} = \frac{F - 32}{9} = \frac{K - 273.15}{5} = \frac{R}{4}
- **Key Temperature Benchmarks**:
- Absolute Zero: 0\text{ K} = -273.15^\circ\text{C} = -459.67^\circ\text{F}. Molecular kinetic energy becomes minimal.
- Celsius = Fahrenheit crossover: **-40^\circ\text{C} = -40^\circ\text{F}**.
- Normal Human Body Temperature: 37.0^\circ\text{C} = 98.6^\circ\text{F} = 310.15\text{ K}.
- Freezing Point of Pure Water: 0^\circ\text{C} = 32^\circ\text{F} = 273.15\text{ K}.
- Boiling Point of Water (at 1\text{ atm}): 100^\circ\text{C} = 212^\circ\text{F} = 373.15\text{ K}.
- Triple Point of Water: 0.01^\circ\text{C} = 273.16\text{ K} at 611.657\text{ Pa} (4.58\text{ mmHg}).

5. Comprehensive Scientific Measuring Instruments Master Directory

(Every single instrument from Ghatna Chakra and State PCS compiled alphabetically)

Instrument	Physical Quantity Measured / Operating Mechanism	Core Practical Application
Accelerometer	Acceleration, vibrations, dynamic tilt, inertial force	Airbag crash deployment, laptop drop HDD safety park, smartphone portrait/landscape tilt
Actiometer	Heating power or intensity of solar radiation	Solar energy studies, meteorological radiation monitoring
Altimeter	Altitude above sea level	Used in aircraft and high-altitude mountaineering (aneroid mechanism)
Ammeter	Magnitude of electric current in amperes (A)	Connected in series (low internal resistance, ideally zero)
Anemometer	Wind speed, velocity, force, and pressure	Meteorological stations, airport runways, wind farms
Atmometer / Evaporimeter	Water evaporation rate from a wet surface	Hydrology, agricultural irrigation scheduling
Audiometer	Hearing acuity and sound intensity threshold	Audiology clinics, hearing impairment diagnosis

Instrument	Physical Quantity Measured / Operating Mechanism	Core Practical Application
Barograph	Continuous graphical recording of atmospheric pressure	Meteorological observatories, pressure trend monitoring
Barometer	Atmospheric pressure (uses column of Mercury, Hg)	Weather forecasting; sudden drop = thunderstorm; rise = clear dry weather
Bolometer	Incident electromagnetic radiation power via resistance change of heated sensor	Infrared spectroscopy, astronomy, thermal imaging
Butyrometer	Fat content percentage in milk and dairy products	Dairy quality control (Gerber method)
Calipers (Vernier)	Internal/external dimensions, diameter, thickness	Precision engineering, machining, laboratory physics
Calorimeter	Heat of chemical reactions, phase changes, heat capacity	Thermodynamics, bomb calorimetry for food caloric values
Cardiograph	Graphic record of heart movements/cardiac cycle	Cardiology diagnostics (records mechanical heart movements)
Cathetometer	Precise vertical distances/differences in liquid levels	Liquid menisci in manometers, surface tension experiments
Chronometer	Ultra-precise timekeeping unaffected by motion or temperature	Marine navigation to determine accurate longitude at sea
Colorimeter	Absorbance of specific wavelengths of light by a solution	Chemical colorimetry, Beer-Lambert law concentration analysis
Crescograph	Infinitesimal growth rate in plants	Invented by Sir Jagadish Chandra Bose (magnifies growth up to 10,000\times)
Cryometer	Extremely low (cryogenic) temperatures	Superconductivity research, liquid nitrogen/helium physics
Cyclotron	High-energy charged particles (protons, deuterons, \alpha-particles)	Particle physics, radioisotope production in cancer therapy
Dilatometer	Thermal expansion / volume changes caused by physical/chemical changes	Metallurgy, ceramics, phase transformation studies
Dip Circle	Angle of magnetic inclination (dip angle) between Earth's magnetic field and horizon	Geomagnetic surveys; dip = 0^\circ at equator, 90^\circ at magnetic poles
Dynamo	Converts mechanical energy into direct electric current (DC)	Electromagnetic induction (Faraday's law); bicycle dynamos, generators
Dynamometer	Mechanical force, torque, and power output of rotating engines/motors	Automotive engine dynamometer testing
Electroencephalograph (EEG)	Electrical activity and brainwave patterns (\alpha, \beta, \theta, \delta) via scalp electrodes	Epilepsy, sleep disorders, encephalopathy diagnosis
Electrometer	Electrical charge or high-precision potential difference	Electrostatics research, radiation detection ionization chambers
Electroscope	Detects presence, magnitude, and polarity of electric charge	Gold-leaf electroscope for static electricity demonstration
Endoscope	Visual examination inside internal body cavities/organs	Medical diagnostics via fiber optics using Total Internal Reflection (TIR)
Eudiometer	Volume changes of gas mixtures before and after chemical/combustion reactions	Gas stoichiometry analysis, water synthesis demonstration
Fathometer	Sea depth and underwater submerged structures	Uses ultrasonic echo sounding (1\text{ fathom} = 6\text{ ft} = 1.8288\text{ m})

Instrument	Physical Quantity Measured / Operating Mechanism	Core Practical Application
Fluxmeter	Total magnetic flux (in Webers)	Magnetic materials testing, coil magnetic field characterization
Galvanometer	Detects and indicates presence of minute electric currents	Moving-coil galvanometer; converted to ammeter with low shunt in parallel
Geiger Counter (GM Counter)	Detects and counts ionizing radiation (\alpha, \beta, \gamma rays)	Nuclear radiation safety, uranium prospecting
Gravimeter	Local gravitational acceleration (g) variations	Geophysical prospecting (oil, gas, mineral reserves), geodesy
Gyroscope	Angular velocity, orientation, and maintaining rotational stability	Inertial navigation systems in spacecraft, airplanes, torpedoes, smartphones
Hydrometer	Relative density (specific gravity) of liquids based on Archimedes' principle	Testing battery acid density, specific gravity of liquid solutions
Hydrophone	Underwater sound listening and acoustic recording	Submarine detection, marine mammal acoustic monitoring
Hygrometer	Atmospheric relative humidity (moisture content of air)	Weather stations, greenhouse climate control, textile mills
Hygroscope	Indicates changes in atmospheric humidity qualitatively	Simple household humidity indicators
Hypsometer	Altitudes via boiling point of water (temperature at which water boils drops with elevation)	Calibrating thermometers, mountaineering elevation estimation
Kymograph	Graphical recording of physiological movements (BP, pulse, muscle twitches) on rotating drum	Experimental physiology, pharmacological testing
Lactometer	Purity and density of milk (based on Archimedes' principle)	Checks water adulteration in dairy industry
Luxmeter	Illuminance (incident luminous flux per unit area in Lux)	Architectural lighting design, workplace ergonomics, cinematography
Lysimeter	Actual evapotranspiration (evaporation + plant transpiration)	Agricultural hydrology, crop water requirement modeling
Magnetometer	Strength, direction, or relative variation of magnetic field	Geomagnetic mapping, geophysical surveying, planetary probes
Manometer	Differential pressure of liquids and gases in closed conduits	U-tube manometer for gas pressure measurement
Microphone	Acoustic sound waves converted into electrical audio signals	Acoustic transducer operating on electromagnetic/piezoelectric principles
Microtome	Microscopic precision cutting of ultra-thin specimen slices (sections)	Histology, pathology, optical and electron microscopy sample prep
Nephoscope	Altitude, velocity, and direction of cloud movement	Meteorological cloud drift observation
Odometer	Cumulative distance travelled by wheeled vehicles	Electromechanical/digital distance counter on automotive dashboards
Ondometer	Frequency and wavelength of high-frequency electromagnetic waves	Microwave engineering, radio frequency characterization
Periscope	Over-obstacle line-of-sight viewing using two mirrors/prisms at 45^\circ	Submarines at periscope depth, military trenches

Instrument	Physical Quantity Measured / Operating Mechanism	Core Practical Application
Phonograph	Mechanical recording and reproduction of sound	Edison's cylinder and vinyl record sound playback
Phonometer	Tests and measures the acoustic force/loudness of human voice	Speech therapy, acoustic vocal analysis
Photometer	Intensity of visible light, illuminance, and luminous flux	Optical research, photography exposure metering
Polygraph	Lie detector; simultaneously graphs BP, pulse, respiration, galvanic skin conductance	Forensic interrogation, security screening
Potentiometer	EMF and potential difference by comparison against known reference without drawing current	Null-deflection method (infinitely high effective resistance)
Pycnometer	Precision density or specific gravity of liquids and solids	Analytical laboratory standard volume vessel
Pyrheliometer	Direct-beam solar irradiance at normal incidence	Solar energy engineering, photovoltaic project site assessment
Pyrometer	Extremely high temperatures (>1500^\circ C to 2000^\circ C+) remotely via thermal radiation	Blast furnaces, molten metal, Sun/star surface temperature (Stefan's law)
Psychrometer	Relative humidity using wet-bulb and dry-bulb thermometer temperature difference	Meteorological psychrometric tables, HVAC psychrometrics
Rain Gauge (Udometer / Pluviometer / Ombrometer)	Liquid precipitation volume over a given time interval	Rainfall monitoring in meteorology and hydrology
Radiometer (Crookes)	Radiant flux of electromagnetic radiation	Solar radiometry, vane mill demonstration in partial vacuum
Refractometer	Refractive index of transparent liquids and solids	Salinity testing, sugar content (Brix) in winemaking/food
Ringelmann Scale	Apparent density and opacity of smoke emissions (0\% to 100\% in 5 shades)	Industrial chimney emissions monitoring, smoke abatement laws
Saccharimeter	Polarimetric concentration of optical active sugar (sucrose) solutions	Sugar manufacturing, sucrose purity certification
Salinometer	Electrical conductivity / density to measure water salinity	Oceanography, desalination plants, aquaculture
Seismograph / Seismometer	Detects, magnifies, and graphs seismic waves during earthquakes	Earthquake magnitude rating (Richter scale); seismic monitoring stations
Sextant	Angular distance between two visible objects (horizon and celestial body)	Celestial marine navigation to determine latitude and solar noon
Sphygmomanometer	Arterial blood pressure using inflatable cuff and mercury manometer	Medical clinical examination (Normal adult: 120/80\text{ mmHg})
Stethoscope	Acoustic medical device for listening to internal heart/lung sounds	Auscultation of cardiac lub-dub heart sounds and respiratory wheezes
Speedometer	Instantaneous linear speed of a motor vehicle in km/h or mph	Vehicle speed management
Tacheometer	Rapid surveying theodolite measuring distance, direction, and elevation differences	Topographical surveys, road alignment
Tachometer	Rotational angular speed (RPM) of an engine shaft or turbine	Automobile tachometer (rev counter), aviation engine RPM

Instrument	Physical Quantity Measured / Operating Mechanism	Core Practical Application
Theodolite	High-precision angles in horizontal (azimuth) and vertical (elevation) planes	Geodetic surveying, triangulation networks, civil construction
Thermoresistor (Thermistor)	Electronic temperature sensor whose electrical resistance varies with temperature	Precision electronic thermometers, temperature compensation circuits
Thermostat	Automatic temperature regulator maintaining a system near a preset setpoint	Refrigerators, water heaters, air conditioning systems
Viscometer	Dynamic or kinematic viscosity of fluids	Petrochemical lubricant testing, polymer fluid flow analysis
Wind Vane (Weathercock)	Direction from which the wind is blowing	Meteorological wind direction observation
SONAR	Underwater navigation, detection, and ranging using ultrasonic pulses	Locating submerged submarines, sea depth mapping (Active transmits; Passive listens)
RADAR	Range, velocity, and angle of targets using reflected radio/microwaves	Air traffic control, military air defense, weather radar

One-Liner Revision – Physics Fundamentals and Measurement

#	Fast Fact
1	SI base units = **7**: m, kg, s, A, K, mol, cd
2	Supplementary units = Radian (plane angle) + Steradian (solid angle)
3	New SI prefixes (2022) = Ronna (10^27), Quetta (10^30), Ronto (10^-27), Quecto (10^-30)
4	Light-Year = unit of **distance** (NOT time); 1 ly = 9.46 x 10^15 m
5	1 Parsec = 3.262 light-years = 206,265 AU (coined: Herbert Hall Turner, 1913)
6	1 AU = 1.496 x 10^11 m = ~150 million km (average Earth-Sun distance)
7	Smallest unit of length = **Fermi** (fm = 10^-15 m); used for nuclear radii
8	Angstrom = 10^-10 m; used for atomic radii and wavelengths
9	1 Barrel of oil = **159 litres** (~42 US gallons)
10	1 Nautical mile = **1.852 km**; Knot = 1 nautical mile/hour
11	1 Cusec = 1 ft3/s = 28.317 litres/s (river/canal flow rate)
12	Normal atmospheric pressure = **1 atm = 101,325 Pa = 760 mmHg = 760 torr**
13	1 bar = 10^5 Pa; 1 millibar = 100 Pa (used in meteorology)
14	1 Horsepower = **746 Watts**; Metric HP = 735.5 W
15	1 kWh = **3.6 x 10^6 Joules** (BOT unit; on electricity bill)
16	Dobson Unit: Normal ozone = 300 DU; Ozone hole < **220 DU**
17	Pyrometer = measures very high temperatures (>1500 degC) without contact
18	Anemometer = measures **wind speed**; Wind Vane = measures wind **direction**
19	Barometer: sudden drop = approaching storm; gradual fall = rain; rise = fair weather
20	Hygrometer = measures **humidity**; Hydrometer = measures **specific gravity of liquids**
21	Lactometer = checks **milk purity** (density); Butyrometer = measures **fat in milk**
22	SONAR = uses **ultrasonic waves** (>20 kHz); for submarine/ocean depth detection
23	RADAR = uses **radio/microwaves**; detects aircraft, ships, weather

#	Fast Fact
24	Fathometer = measures ocean depth; 1 fathom = 6 feet = 1.83 m
25	ECG = heart; EEG = brain; EMG = muscles; Polygraph = lie detector
26	1 kg defined by Planck constant h = 6.626 x 10^-34 J.s (since 2019 BIPM)
27	Cs-133 atomic clock = defines the **second** (9,192,631,770 transitions)
28	Kelvin 0 = absolute zero = -273.15 degC (theoretical minimum temperature)
29	Speed of light (c) = 299,792,458 m/s (defines the metre)
30	Mole = 6.02214076 x 10^23 entities (Avogadro's number; defines the mole)

Physics Topic 2 — Mechanics and Properties of Matter

Consolidated — 50 Must-Score Facts

1. **Brahmagupta's Precedence**: 7th-century astronomer Brahmagupta (*Brahmasphuta Siddhanta*) stated that all bodies gravitate toward Earth long before Newton.
2. **Definition of Speed**: Galileo Galilei was the first to define speed mathematically as distance per unit time (v = d/t).
3. **Mass vs Weight**: Mass (kg, scalar, constant everywhere) measures inertia. Weight (N, vector, W = mg) depends on local gravitational acceleration.
4. **Deep Breath on Sensitive Scale**: Inhaling deeply increases scale reading because 4.8\text{ L} of inhaled air adds \approx 5.88\text{ g} mass.
5. **Distance vs Displacement**: Distance is actual path length (scalar, \ge 0). Displacement is shortest straight line (vector, can be zero).
6. **Average Speed for Equal Distance Halves**: Harmonic mean v_{\text{avg}} = \frac{2 v_1 v_2}{v_1 + v_2}.
7. **Newton's First Law (Inertia)**: Passenger leans backwards when vehicle starts (inertia of rest); leans forwards on sudden braking (inertia of motion); leans left when bus turns sharp right (inertia of direction).
8. **Newton's Second Law**: F = \frac{dp}{dt} = ma. Goalkeeper pulls hands backwards to prolong contact time (t), lowering rate of change of momentum and impact force.
9. **Newton's Third Law**: Action and reaction forces are equal, opposite, and act simultaneously on **two different bodies**.
10. **Propulsion by Whistling**: On frictionless ice, whistling/blowing air forward expels momentum, generating an equal backward reaction that propels the body (conservation of momentum).
11. **Boat Jump**: Jumping at 5\text{ m/s} from a boat that recoils at 0.5\text{ m/s} implies the boat is **10 times heavier** than the person (M_{\text{boat}} = 10 M_{\text{man}}).
12. **Friction Hierarchy**: **Static Friction > Kinetic (Sliding) Friction > Rolling Friction**.
13. **Loaded Cart Principle**: Harder to start moving than to keep rolling because limiting static friction exceeds dynamic kinetic friction.
14. **Ball Bearings**: Convert sliding/static friction into **rolling friction** by drastically minimizing contact surface area.
15. **Marginal Friction at Constant Speed**: a = 0 \implies F_{\text{applied}} = f_{\text{friction}} (200\text{ N} \implies 200\text{ N}).
16. **Energy in Friction**: Moving against friction irreversibly dissipates **kinetic energy into heat energy**.
17. **Simple Machines**: Enable doing the **same amount of work with lesser effort force**; they never save total energy.
18. **Centripetal Force**: Inward radial force (F_c = mv^2/r) required for circular paths (gravity for orbits, tire friction for curves).
19. **Centrifugal Force**: Fictitious outward inertial force in rotating frames; drives cream separation from milk and washing machine spin drying.
20. **Overturning on Curves**: Unbanked vehicle moving too fast skids outwards and **overturns about its outer wheels**.
21. **Coriolis Force**: Due to Earth's rotation; deflects winds/currents to the **right in Northern Hemisphere** and **left in Southern Hemisphere**.
22. **Accelerated Liquid Tanker**: When a tanker accelerates forward, the liquid surface tilts backward, creating **larger depth at the rear end** (\tan\theta = a/g).
23. **Equal KE Stopping Distance**: Vehicles with equal KE stop over the **same distance (X = Y = Z)** under equal braking forces (W = F \cdot d = KE).

24. **de Broglie Wavelength**: \lambda = h/p. For 0.12\text{ kg} ball at 20\text{ m/s}, \lambda = 2.76 \times 10^{-34}\text{ m}.
25. **Energy Sources NOT from Sun**: **Geothermal energy** and **Nuclear energy** do NOT originate from solar radiation. Biomass and wind are solar-derived.
26. **Kepler's Laws**: 1st (elliptical orbits, Sun at focus); 2nd (areal velocity constant; fastest at perihelion, slowest at aphelion); 3rd (T^2 \propto a^3).
27. **Universal Gravitational Constant**: G = 6.67430 \times 10^{-11}\text{ N}\cdot m^2/kg^2 (measured by Henry Cavendish).
28. **Acceleration due to Gravity (g)**: g = GM_e/R_e^2. Independent of falling object's mass! Maximum at poles, minimum at equator.
29. **Free Fall in Vacuum**: Feather, rubber ball, and iron ball fall with identical acceleration g and strike ground simultaneously (v = \sqrt{2gh}).
30. **Apparent Weight in Elevator**: Upward acceleration \implies m(g+a) (heavier); Downward acceleration \implies m(g-a) (lighter); Free fall \implies 0 (weightlessness).
31. **Moon Surface Gravity**: g_{\text{moon}} = \frac{1}{6} g_{\text{earth}}. Mass of 100\text{ kg} body on Earth remains **100\text{ kg} on the Moon**, but weight is only 100/6\text{ kgf} \approx 163.3\text{ N}.
32. **No Atmosphere on Moon**: Lunar escape velocity (2.38\text{ km/s}) is **lower than the root-mean-square (RMS) thermal velocity of gas molecules**.
33. **Orbital & Escape Velocities**: V_o = \sqrt{g R_e} \approx 7.92\text{ km/s}; V_e = \sqrt{2g R_e} \approx 11.2\text{ km/s} = \sqrt{2} V_o. Projectile launched at 8\text{ km/s} falls back to Earth.
34. **Simple Pendulum Formula**: T = 2\pi \sqrt{l/g}. Independent of bob mass. Increases in summer (clocks lose time); decreases in winter (clocks gain time). Standing up on swing shortens l, making swing oscillate faster.
35. **Schuler Period & Seconds Pendulum**: Maximum pendulum period on Earth is **84.4\text{ minutes}** (l = R_e). Seconds pendulum has period of **2.0\text{ seconds}** (l \approx 1\text{ metre}).
36. **Elasticity vs Plasticity**: Elasticity is the ability to regain original shape after deforming force is removed. Perfectly plastic materials (putty, mud, wet clay) remain deformed forever.
37. **Hooke's Law**: Within proportional elastic limit, \text{Stress} \propto \text{Strain}. Modulus of elasticity E = \text{Stress}/\text{Strain}.
38. **Stress vs Strain**: Stress is internal restoring force per unit area (F/A, SI unit Pascal or N/m^2). Strain is fractional dimensional change (\Delta L/L, dimensionless and unitless).
39. **Steel is More Elastic Than Rubber**: For identical tensile strain, steel requires a much greater restoring stress than rubber (Y_{\text{steel}} \gg Y_{\text{rubber}}).
40. **Bulk Modulus of Rigid Body**: For an ideal perfectly rigid body, \Delta V = 0 \implies Bulk Modulus K = -\Delta P / 0 = \mathbf{\infty} (**Infinite**).
41. **Stretching a Spring**: Determined by **Shear Modulus (Modulus of Rigidity)**, as pulling a coil subjects the wire to shear/torsional deformation.
42. **Wire Elongation Ratio**: Extension \Delta l = \frac{F L}{A Y} = \frac{F L}{\pi r^2 Y} \propto \frac{L}{d^2}. A wire of 3\text{ m} length and 1.5\text{ mm} diameter stretches the most among equal-load wires.
43. **Surface Tension**: Elastic property of liquid surface contracting to minimum surface area (T = F/L, SI unit N/m). Drops and bubbles assume **spherical shapes** to minimize surface area and surface energy.
44. **Temperature Effect on Surface Tension**: Surface tension **decreases with rising temperature** and becomes **zero at the critical temperature**.
45. **Cohesion vs Adhesion**: Cohesion is attraction between identical molecules (e.g. water-water). Adhesion is attraction between different molecules (e.g. water-glass).
46. **Why Oil Spreads on Water**: The surface tension of water is greater than that of oil; hence water pulls oil outward into a thin film. Conversely, water on an oily surface beads up into spherical droplets.
47. **Pressure Inside a Soap Bubble**: Due to two liquid surfaces, excess pressure \Delta P = \frac{4T}{R}. Pressure inside a soap bubble is strictly **greater than atmospheric pressure**.

48. **Two Connected Soap Bubbles**: When connected by a tube, air flows from the smaller bubble (higher pressure \Delta P \propto 1/R) to the larger bubble; the **smaller bubble shrinks and the larger bubble grows**.
49. **Charging a Soap Bubble**: Placing an electric charge (positive or negative) on a soap bubble causes mutual electrostatic repulsion, causing its **radius to expand**.
50. **Capillarity (Capillary Action)**: Rise or fall of liquids in narrow tubes. Liquids that wet glass (contact angle <90^\circ, e.g. water) rise; liquids that do not wet glass (contact angle >90^\circ, e.g. mercury) show capillary depression.

One-Liner Revision – Mechanics and Properties of Matter

#	Fast Fact
1	Newton's 1st Law = Law of **Inertia** (object at rest stays at rest)
2	Newton's 2nd Law = F = ma (Force = mass x acceleration)
3	Newton's 3rd Law = For every action, there is equal and opposite **reaction**
4	1 Newton = force that accelerates 1 kg mass by 1 m/s2
5	Momentum (p) = m x v; Law of Conservation of Momentum
6	Impulse = Force x Time = change in momentum
7	g (acceleration due to gravity) = 9.8 m/s2 on Earth's surface
8	g is maximum at poles; minimum at equator (due to Earth's shape)
9	g decreases with altitude and depth (both)
10	Weight = 0 in free fall / space (apparent weightlessness)
11	Escape velocity from Earth = **11.2 km/s**
12	Orbital velocity (LEO) = ~7.9 km/s
13	Archimedes' Principle: Upthrust = weight of fluid displaced
14	A body floats if its average density <= density of fluid
15	Density of water = 1000 kg/m3 (1 g/cm3); ice floats (density ~917 kg/m3)
16	Pascal's Law: basis of hydraulic press, brakes, jacks
17	Bernoulli's Principle: explains airplane wing lift, atomizer, carburetor
18	Surface tension: water meniscus, capillarity, insects on water, dew drops
19	Viscosity decreases with temperature (liquids); increases for gases
20	Rolling friction < Sliding friction < Static friction
21	Friction is useful: walking, brakes, writing; harmful: wear of machine parts
22	Centre of Gravity = point where entire weight appears to act
23	Centripetal force = mv2/r (directed towards centre of circular motion)
24	Centrifuge separates components based on density difference (cream from milk)
25	Hooke's Law: stress = E x strain (E = Young's modulus); within elastic limit
26	Elasticity: rubber > steel? No! Steel has **higher** Young's modulus (more elastic by physics definition)
27	Pressure at depth h in liquid: P = rho x g x h
28	Ball pen works on: capillary action + gravity
29	Geostationary orbit = 35,786 km above equator; orbital period = 24 hours
30	Kepler's 3rd Law: T2 is proportional to a3 (period2 proportional to semi-major axis3)

Physics Topic 3 — Energy, Heat and Thermal Science

Consolidated — 50 Must-Score Facts

1. **Work Formula & Sign:** Work done W = \vec{F} \cdot \vec{d} = F d \cos\theta. If force and displacement are perpendicular (\theta = 90^\circ, such as a porter carrying a load horizontally or centripetal force on a satellite), work done by that force is **identically zero**.
2. **Kinetic Energy & Momentum Relation:** KE = \frac{p^2}{2m}. If the momentum p of a body is doubled, its kinetic energy becomes **4 times**. If p increases by 100\%, KE increases by 300\%. If p increases by 50\%, KE increases by 125\%.
3. **Calorific Value Leader: Hydrogen** has the highest calorific value (\approx 150\text{ kJ/g}), followed by LPG (\approx 50\text{ kJ/g}), CNG/Methane (\approx 50\text{ kJ/g}), Petrol/Diesel (\approx 45\text{ kJ/g}), Coal (25\text{–}33\text{ kJ/g}), Dry wood (17\text{ kJ/g}), and Cow-dung cake (6\text{–}8\text{ kJ/g}).
4. **Mechanical Equivalent of Heat (J):** First determined by James Prescott Joule: W = J \cdot Q, where J = 4.184\text{ J/cal} \approx 4.2\text{ J/cal}. 1\text{ calorie} is the heat required to raise the temperature of 1\text{ g} of pure water from 14.5^\circ\text{C} to 15.5^\circ\text{C} at standard atmospheric pressure.
5. **Commercial Unit of Electrical Energy:** 1\text{ Board of Trade Unit (B.O.T.U.)} = 1\text{ kilowatt-hour (kWh)} = 1000\text{ W} \times 3600\text{ s} = 3.6 \times 10^6\text{ J} = 3.6\text{ MJ}.
6. **Zeroth Law of Thermodynamics:** Formulated by Ralph H. Fowler (1931). States that if bodies A and B are each in thermal equilibrium with body C, then A and B are in thermal equilibrium with each other. This law introduces and **defines the concept of Temperature**.
7. **First Law of Thermodynamics:** \Delta Q = \Delta U + \Delta W. It is a direct manifestation of the universal **Law of Conservation of Energy**. It states that heat added to a system goes into increasing internal energy plus doing external work.
8. **Second Law of Thermodynamics:** Imposes direction on heat flow. **Clausius Statement:** Heat cannot spontaneously flow from a colder body to a hotter body without external work. **Kelvin-Planck Statement:** No engine can completely convert heat absorbed from a single reservoir into mechanical work in a cyclic process without rejecting heat.
9. **Concept of Entropy:** Clausius introduced Entropy (S, where dS = dQ_{\text{rev}}/T). In any irreversible natural process, the total entropy of the universe **always increases** (\Delta S_{\text{univ}} > 0).
10. **Absolute Zero Temperature:** The theoretical lowest possible temperature where molecular kinetic motion ceases and volume of an ideal gas drops to zero: 0\text{ K} = -273.15^\circ\text{C} = -459.67^\circ\text{F}. Negative temperatures on the Kelvin scale are physically impossible.
11. **Triple Point of Water:** The unique thermodynamic point where solid ice, liquid water, and water vapor coexist in stable dynamic equilibrium: T = 273.16\text{ K} = 0.01^\circ\text{C} at a vapor pressure of 611.65\text{ Pa} (4.58\text{ mm of Hg}).
12. **Anomalous Expansion of Water:** Pure water contracts upon heating from 0^\circ\text{C} to 4^\circ\text{C}, reaching its **maximum density (1.000\text{ g/cm}^3 = 1000\text{ kg/m}^3)** and **minimum volume** at **4^\circ\text{C} (3.98^\circ\text{C})**. Above 4^\circ\text{C}, it expands normally.
13. **Aquatic Survival in Frozen Lakes:** During severe winters, surface water cools to 4^\circ\text{C}, becomes densest, and sinks to the lake bed. Once the entire lake reaches 4^\circ\text{C}, surface water cools to 0^\circ\text{C}, expands, becomes lighter, and freezes into floating surface ice. Because ice is a poor conductor of heat, the bottom water remains liquid at **+4^\circ\text{C}**, allowing aquatic fauna and flora to survive.
14. **Bursting of Water Pipes in Freezing Weather:** When liquid water freezes into ice at 0^\circ\text{C}, its volume expands by approximately **9\%**. In rigid metal or PVC pipes, the immense hydrostatic expansion pressure ruptures the pipes.
15. **Specific Heat Capacity of Water:** Exceptionally high: c_{\text{water}} = 1\text{ cal/g}\cdot^\circ\text{C} = 4184\text{ J/kg}\cdot\text{K} \approx 4200\text{ J/kg}\cdot\text{K}. It absorbs or releases large amounts of heat

with minimal change in temperature.

16. **Why Water is Used as an Engine Coolant & in Radiators:** Because of water's high specific heat capacity, circulating water extracts massive heat energy from engine blocks without boiling rapidly.
17. **Why Water is Used in Hot Water Fermentation Bags:** Water holds thermal energy longer than any other common liquid due to its high specific heat capacity, providing prolonged, steady warmth.
18. **Maritime Climate Moderation:** Coastal areas (Mumbai, Chennai) experience mild winters and moderate summers because the adjoining sea acts as a giant thermal buffer due to water's high specific heat capacity. Conversely, continental interiors (Delhi, Nagpur) have extreme climates.
19. **Desert Temperature Extremes:** Dry silica sand has a very low specific heat capacity (\approx 800\text{ J/kg}\cdot\text{K}) and high solar absorptivity. It heats up rapidly under daytime sun and radiates heat away rapidly at night, producing dramatic diurnal temperature swings (45^\circ\text{C} day down to 10^\circ\text{C} night).
20. **Latent Heat of Fusion of Ice:** L_f = 80\text{ cal/g} = 3.34 \times 10^5\text{ J/kg} = 334\text{ kJ/kg}. Heat absorbed to convert 1\text{ g} of ice at 0^\circ\text{C} to 1\text{ g} of water at 0^\circ\text{C} without any temperature change.
21. **Latent Heat of Vaporization of Steam:** L_v = 540\text{ cal/g} = 2.26 \times 10^6\text{ J/kg} = 2260\text{ kJ/kg}. Heat absorbed to convert 1\text{ g} of water at 100^\circ\text{C} into steam at 100^\circ\text{C}.
22. **Why Steam Burns are More Severe Than Boiling Water Burns:** Both are at 100^\circ\text{C}, but each gram of steam carries an extra **540\text{ calories} (2260\text{ J})** of hidden latent heat of vaporization, which is released directly onto the skin tissues during condensation.
23. **Cooling Sensation from Ice at 0^\circ\text{C} vs Water at 0^\circ\text{C}:** Ice at 0^\circ\text{C} produces significantly more effective cooling because it absorbs an additional 80\text{ cal/g} of latent heat of fusion from the surrounding drink to melt.
24. **Pressure Cooker Physics:** Boiling point of a liquid **increases with increasing external pressure**. Inside a sealed pressure cooker, steam pressure rises to \approx 2\text{ atmospheres} (200\text{ kPa}), elevating the boiling point of water from 100^\circ\text{C} to **120^\circ\text{C}\text{–}125^\circ\text{C}**. Food cooks at this higher temperature, cutting cooking time by over 60\%.
25. **Cooking at High Altitudes (Hill Stations):** Atmospheric pressure at high altitudes (Shimla, Leh) is substantially lower than 1\text{ atm}. Consequently, water boils at a lower temperature (85^\circ\text{C}\text{–}90^\circ\text{C}). Food takes much longer to cook or remains uncooked in open pots, making pressure cookers essential on mountains.
26. **Regelation of Ice:** For substances that contract upon melting (like ice), **increasing pressure lowers the melting point**. When wire weighted by heavy loads passes through an ice block, the high pressure under the wire melts the ice below 0^\circ\text{C}. As the wire sinks, pressure above is relieved and the water refreezes immediately without splitting the block.
27. **Ice Skating Mechanism:** The sharp blade of the skate concentrates body weight over a tiny area, generating immense pressure that lowers the melting point of ice. A thin layer of liquid water forms under the blade, acting as a frictionless lubricant.
28. **Evaporative Cooling Principle:** Evaporation is an endothermic surface phenomenon where energetic molecules escape into the vapor phase, lowering the average kinetic energy (and temperature) of remaining liquid.
29. **Earthen Pitcher (Matka / Surahi) Cooling:** Porous clay walls contain millions of microscopic capillary pores. Water seeps through these pores to the outer surface and evaporates into dry ambient air, continuously extracting its latent heat of vaporization (540\text{ cal/g}) from the internal water reservoir.
30. **Desert Cooler Efficiency:** Desert coolers function via evaporative cooling. They work with **maximum cooling efficiency on hot, dry summer days** because dry air has low relative humidity, enabling rapid water evaporation. On humid monsoon days, evaporation ceases and coolers only produce muggy discomfort.
31. **Thermal Expansion Coefficients Ratio:** Linear (\alpha), Superficial/Areal (\beta), and Volumetric/Cubical (\gamma) expansion coefficients for an isotropic solid satisfy the exact ratio: \alpha : \beta : \gamma = 1 : 2 : 3.

32. **Thermal Expansion in Railway Tracks:** Small expansion gaps are deliberately left between consecutive steel rail sections, connected by fishplates with slotted bolt holes. In hot summers, rails expand linearly ($\Delta L = L_0 \alpha \Delta T$); without gaps, thermal stress causes catastrophic track buckling.
33. **Bimetallic Strip Working Principle:** Consists of two dissimilar metal strips (e.g., brass and invar/iron) riveted together. Because brass has a higher coefficient of thermal expansion ($\alpha_{\text{brass}} > \alpha_{\text{iron}}$), the strip bends into a curve with the brass on the convex outer side when heated, opening or closing electrical contacts in thermostats and fire alarms.
34. **Pendulum Clock Summer Drift:** Period of a simple pendulum is $T = 2\pi\sqrt{L/g}$. In summer, the metal rod expands ($\Delta L > 0$), lengthening T (pendulum swings slower), causing the clock to **lose time (run slow)**. In winter, contraction makes it **gain time (run fast)**.
35. **Thermal Shock in Thick Glass Tumblers:** Pouring boiling water into a thick-walled ordinary glass tumbler causes it to crack because glass is a poor thermal conductor. The inner surface expands immediately upon contact with boiling water, while the outer surface remains cold and unexpanded, creating destructive internal shearing stress. Thin glass or borosilicate (Pyrex) glass has a very low thermal expansion coefficient and withstands this.
36. **Conduction in Solids:** Heat transfers via transfer of kinetic energy between colliding adjacent atoms/molecules and through migration of free conduction electrons, without any net bulk movement of matter. Metals are excellent conductors due to abundant free electrons (Silver is the best, followed by Copper, Gold, and Aluminum).
37. **Davy's Safety Lamp:** Invented by Sir Humphry Davy for coal miners. A fine wire gauze of copper surrounds the flame. The metal gauze rapidly conducts heat away into the surrounding atmosphere, keeping the temperature outside the gauze below the ignition point of explosive methane gas (firedamp).
38. **Two Thin Blankets vs One Thick Blanket:** Two thin blankets provide significantly better thermal insulation than a single blanket of double thickness because an insulating layer of still air is trapped between the two blankets. Air is one of the worst thermal conductors ($k_{\text{air}} \approx 0.026\text{ W/m}\cdot\text{K}$).
39. **Why Birds Fluff Feathers in Winter:** Fluffing feathers traps a thick layer of dead air around their bodies, preventing body heat from escaping by conduction and convection.
40. **Convection in Fluids:** Heat transfer accompanied by the actual **macroscopic bulk movement of heated fluid matter** driven by density gradients under gravity. Warm fluid expands, becomes less dense, and ascends; cooler, denser fluid descends to replace it, establishing **convection currents**.
41. **Convection Fails in Weightlessness:** In an orbiting satellite or the International Space Station (microgravity), gravity is effectively zero ($g \approx 0$). Without buoyancy forces, convection currents cannot form. Heat can only transfer by conduction and radiation. A candle flame in space burns as a faint, spherical, blue flame and suffocates in its own CO_2.
42. **Sea Breeze vs Land Breeze:**
43. **Ventilation in Architecture:** Ventilators and exhaust windows are always placed near the ceiling because warm, exhaled air containing CO_2 is less dense and ascends to exit near the top, while fresh, cooler air enters through doors and windows below.
44. **Thermal Radiation:** Heat transfer in the form of **electromagnetic waves (primarily Infrared, $\lambda \approx 700\text{ nm to } 1\text{ mm}$)**. It requires **no material medium**, travels through vacuum at the speed of light ($c = 3 \times 10^8\text{ m/s}$), and obeys the inverse-square law.
45. **Thermos Flask (Dewar Flask) Tripartite Defense:**
46. **Kirchhoff's Law of Thermal Radiation:** At thermal equilibrium, the ratio of emissive power to absorptive power is equal for all bodies and matches the emissive power of a black body ($e/a = E_{\text{black}}$). Colloquial rule: **Good absorbers are good emitters; poor absorbers are poor emitters.** A black pot absorbs radiant heat fastest and cools fastest.
47. **Stefan-Boltzmann Law:** The total radiant energy emitted per second per unit surface area of a black body is directly proportional to the fourth power of its absolute thermodynamic temperature: $E = \sigma T^4$, where

$\sigma = 5.670 \times 10^{-8}\text{ W/m}^2\cdot\text{K}^4$. If a body's absolute temperature is doubled, its radiant emission increases by $2^4 = 16\text{ times}$.

48. **Wien's Displacement Law:** The wavelength ($\lambda_{\max}$) corresponding to the peak emission of black-body radiation is inversely proportional to absolute temperature: $\lambda_{\max} \cdot T = b$, where $b = 2.898 \times 10^{-3}\text{ m}\cdot\text{K}$. Hotter stars radiate at shorter wavelengths (blue/white), whereas cooler stars peak at longer wavelengths (red).
49. **Why Cloudy Nights are Warmer than Clear Nights:** Clouds consist of liquid water droplets and water vapor, which are strong absorbers and reflectors of terrestrial infrared radiation emitted by the Earth's surface. On a clear starry night, infrared heat escapes unobstructed into space, causing rapid radiational cooling; on cloudy nights, the cloud layer reflects this heat back, trapping warmth.
50. **Open Refrigerator Door in a Closed Room:** Leaving the door of a running refrigerator open inside a closed, insulated room **warms the room up**. A refrigerator is a heat pump that transfers heat from its interior to the exterior condenser coils while consuming electrical work: $Q_{\text{released}} = Q_{\text{absorbed}} + W_{\text{electric}}$. The heat discharged into the room exceeds the cooling effect, elevating room temperature.

📋 Must-Score Master Tables: Constants, Modes & Thermodynamic Laws

Table 1: Standard Thermal Constants of Water & Key Substances

Parameter / Constant	SI Value	Practical CGS Value	Significance in UPPCS
Freezing Point of Water	273.15 K / 0°C	32°F	Reference point for thermometric scales.
Boiling Point of Water	373.15 K / 100°C	212°F	At standard sea level pressure (760 mm Hg).
Triple Point of Water	**273.16 K / 0.01°C**	4.58 mm Hg (611.65 Pa)	Modern fundamental thermometric definition.
Maximum Density Temperature	**$3.98^\circ\text{C} \approx 4^\circ\text{C}$**	1.000 g/cm^3	Minimum volume, protects aquatic lake life.
Specific Heat Capacity of Water	**$4184\text{ J/kg}\cdot\text{K}$**	**$1.0\text{ cal/g}\cdot^\circ\text{C}$**	Used as standard; coolant and fermentation.
Specific Heat Capacity of Ice	$2090\text{ J/kg}\cdot\text{K}$	$0.5\text{ cal/g}\cdot^\circ\text{C}$	Exactly half that of liquid water.
Specific Heat Capacity of Steam	$2010\text{ J/kg}\cdot\text{K}$	$0.48\text{ cal/g}\cdot^\circ\text{C}$	Approximately half that of liquid water.
Latent Heat of Fusion of Ice (L_f)	**$3.34 \times 10^5\text{ J/kg}$**	**80 cal/g**	Ice at 0°C cools far better than water at 0°C.
Latent Heat of Vaporization (L_v)	**$2.26 \times 10^6\text{ J/kg}$**	**540 cal/g**	Steam at 100°C inflicts severe blistering burns.
Absolute Zero (0 K)	0 K	**-273.15°C / -459.67°F**	Molecular kinetic motion drops to zero.
Calorific Value of Hydrogen	**150 MJ/kg**	**150 kJ/g**	Highest of all fuels; zero carbon emissions.
Calorific Value of LPG	50 MJ/kg	50 kJ/g	Butane + Propane domestic cylinder fuel.
Calorific Value of Methane (CNG)	50 MJ/kg	50 kJ/g	Clean transport fuel (CH_4).

Table 2: Comparative Analysis of Heat Transfer Modes

Feature	Conduction	Convection	Radiation
Medium Required?	Yes (Solids primarily; liquids/gases poor)	Yes (Fluids only: Liquids & Gases)	**No** (Operates in complete vacuum)
Physical Mechanism	Atomic/molecular vibrations & free electrons	Macroscopic buoyant bulk circulation of fluid	Electromagnetic waves (Infrared photons)
Speed of Transfer	Relatively slow	Moderate	**Speed of light (3 \times 10^8\text{ m/s})**
Path Traveled	Tortuous/irregular through matter	Irregular convection currents	Straight lines (rays)
Dependence on Gravity	Independent of gravity	**Strictly depends on gravity (g > 0)**	Independent of gravity
Governing Law	Fourier's Law (Q/t = -kA\frac{dT}{dx})	Newton's Law of Cooling (Q/t = hA\Delta T)	Stefan-Boltzmann Law (E = \sigma T^4)
Everyday Example	Metal spoon heating in hot soup; Davy's lamp	Sea/land breezes; room ventilators; geysers	Solar warmth reaching Earth; campfire glow

Table 3: Summary of Thermodynamic Laws & Radiation Principles

Law / Principle	Core Formula / Expression	Physical Meaning & Key Takeaway	PCS Application / Example
Zeroth Law	If A = B and B = C, then A = C	Establishes temperature as a fundamental state property.	Basis of all thermometers.
First Law	\Delta Q = \Delta U + \Delta W	Conservation of energy: Heat added equals change in internal energy plus work done.	Adiabatic tire burst causes gas cooling.
Second Law (Clausius)	Spontaneous heat flow: Hot \to Cold	Heat will never transfer spontaneously from cooler to hotter body without external work.	Refrigerator requires compressor motor power.
Second Law (Kelvin-Planck)	\eta = W / Q_1 < 1	100\% conversion of heat into mechanical work in a cyclic engine is impossible.	Thermal power plants must reject waste heat.
Entropy Principle	\Delta S_{\text{universe}} \ge 0	Natural spontaneous processes increase disorder; time has a thermodynamic arrow.	Irreversible degradation of energy.
Third Law (Nernst)	\lim_{T \to 0} S = 0	Entropy of a pure crystalline solid approaches zero at absolute zero (0\text{ K}).	Absolute zero cannot be attained in finite steps.
Stefan-Boltzmann Law	E = \sigma T^4	Total radiant emission per unit area is proportional to fourth power of absolute temperature.	If Sun's surface temperature doubles, radiant flux increases 16\times.
Wien's Law	\lambda_{\max} \cdot T = b	Peak radiant wavelength is inversely proportional to absolute temperature.	Red stars are cooler; blue stars are hotter.
Kirchhoff's Law	e_\lambda / a_\lambda = E_{\text{black}}	Good absorbers are good emitters; poor absorbers are poor emitters.	Black painted solar collectors; dark spots on Sun.
Newton's Law of Cooling	\frac{dT}{dt} = -k(T - T_0)	Rate of cooling of a hot body is proportional to excess temperature over surroundings.	Hot tea cools faster initially than when near room temperature.

Table 4: Everyday Thermal Phenomena & Scientific Rationale

Phenomenon	Direct Scientific Cause & Mechanism	High-Yield Common Trap
Pressure Cooker Fast Cooking	High steam pressure elevates boiling point to 120^\circ\text{C}; food cooks at higher temperature.	Cooking speed depends on **temperature of boiling water**, not boiling rate.

Phenomenon	Direct Scientific Cause & Mechanism	High-Yield Common Trap
Mountain Cooking Difficulty	Low atmospheric pressure lowers boiling point to 85^\circ\text{C}; water evaporates before cooking food.	Water boils earlier, but at a **lower temperature**, undercooking pulses.
Matka / Pitcher Water Cooling	Seepage of water through porous clay capillaries; continuous surface evaporation extracts 540\text{ cal/g}.	Works only if relative humidity is low; fails in humid monsoon.
Desert Air Cooler Cooling	Fan forces dry ambient air across wet pads, driving evaporative cooling.	Highly effective in hot dry May/June; ineffective during July/August rains.
Sweating / Perspiration	Sweat evaporates from skin surface, drawing latent heat of vaporization from the body.	High humidity prevents sweat evaporation, making humans feel sticky and hot.
Why Steam Burns Worse Than Water	Steam at 100^\circ\text{C} contains an extra 540\text{ cal/g} (2260\text{ J/g}) latent heat of vaporization.	Both are at 100^\circ\text{C}, but steam releases latent heat on condensing.
Cloudy Nights Are Warmer	Clouds absorb and reflect terrestrial infrared radiation back to the ground.	Clear starry nights permit unrestricted radiative heat escape into space.
Two Blankets vs One Double-Thick	Still air layer trapped between two blankets acts as an exceptional thermal insulator.	Air has extremely low thermal conductivity (k = 0.026\text{ W/m}\cdot\text{K}).
Open Refrigerator in Closed Room	Compressor electrical work is dissipated as heat at condenser coils (Q_H = Q_C + W).	The room warms up; it does **not** cool down.
Shattering of Thick Glass Tumbler	Unequal thermal expansion: inner surface expands instantly while outer surface remains cold.	Borosilicate glass resists shattering due to very low thermal expansion.

One-Liner Revision – Energy, Heat and Thermal Physics

#	Fast Fact
1	1 Joule = 1 Newton x 1 metre (work done)
2	1 Calorie = **4.186 Joules** (heat to raise 1 g water by 1 degC)
3	1 kWh = 3.6 x 10^6 J = 3.6 MJ (unit on electricity bill)
4	KE = 1/2 mv2; PE = mgh; Total ME = KE + PE = constant (no friction)
5	Power = work done / time = Energy / time; SI unit = Watt (W)
6	1 Watt = 1 Joule/second; 1 kW = 1000 W; 1 MW = 10^6 W
7	1 Horsepower = **746 Watts**; British HP used in motors/vehicles
8	Water has the HIGHEST specific heat (4200 J/kg/K); best coolant and heat reservoir
9	Specific heat of water = 4200 J/kg/K; iron = 450 J/kg/K; lead = 128 J/kg/K
10	Latent heat of vaporisation of water = 2260 kJ/kg (much higher than fusion = 336 kJ/kg)
11	0 degC = 273.15 K; 100 degC = 373.15 K; -273.15 degC = 0 K (absolute zero)
12	Celsius to Fahrenheit: F = (9/5)C + 32; at -40 degC = -40 degF
13	Heat transfer: Conduction (solids) > Convection (fluids) > Radiation (vacuum/space)
14	Metals = good conductors; Wood, glass, air = insulators; vacuum = no conduction/convection
15	Thermos flask = prevents heat loss by all 3 methods (vacuum, silvered walls, stopper)
16	Black surfaces = absorb AND emit heat best (best radiator and absorber)
17	Stefan-Boltzmann Law: E = sigma x T^4 (radiation proportional to 4th power of temp)
18	Newton's Law of Cooling: rate of cooling proportional to (T_object - T_surrounding)
19	Greenhouses: glass transmits visible light but blocks infrared (re-radiated heat)
20	Greenhouse gases: CO2, CH4, N2O, H2O vapour, O3, CFCs

#	Fast Fact
21	Most potent GHG (per molecule) = **SF6** (sulfur hexafluoride); most volume-significant = CO2
22	Global warming potential (GWP): CH4 = 25x CO2; N2O = 298x CO2 (100-yr)
23	1st Law of Thermodynamics = Conservation of Energy (dU = Q - W)
24	2nd Law of Thermodynamics = entropy of universe always increases; heat flows hot to cold
25	3rd Law of Thermodynamics = entropy of perfect crystal = 0 at absolute zero
26	Zeroth Law = if A and B are in thermal equilibrium with C, then A and B are in equilibrium
27	Thermal expansion: gases > liquids > solids (rail gaps, bimetallic strips)
28	Bimetallic strip = two metals with different expansion rates; used in thermostats
29	Regelation = ice melts under pressure and refreezes when pressure released (ice skating)
30	Clinical thermometer range = 35 degC to 42 degC; mercury (now digital)

Physics Topic 4 — Light, Optics and Laser Technology

Consolidated — 50 Must-Score Facts

1. **Luminous vs Non-Luminous Bodies**: A luminous object emits its own light (Sun, candle flame, electric lamp). The **Moon is non-luminous**; it is visible solely because it reflects incident sunlight.
2. **Speed of Light Hierarchy**: In vacuum, speed of light is maximum: c = 3.0 \times 10^8\text{ m/s}. In water: 2.25 \times 10^8\text{ m/s}; in glass: 2.0 \times 10^8\text{ m/s}; in diamond: 1.24 \times 10^8\text{ m/s}. Speed is minimum in diamond (highest refractive index \approx 2.42).
3. **Speed of Light in Water vs Vacuum**: \frac{v_{\text{water}}}{c} = \frac{2.25 \times 10^8}{3.0 \times 10^8} = \mathbf{0.75} (three-fourths), NOT 0.95.
4. **Sunlight Travel Time**: Sunlight takes approximately **499\text{ seconds} \approx 8.3\text{ minutes} (or 8.5\text{ minutes})** to reach Earth across 1\text{ AU}.
5. **Invariance of Frequency in Refraction**: When light passes from one medium to another (e.g. air to glass), its **frequency (f) remains completely unchanged**, while its velocity (v) and wavelength (\lambda) change proportionally (v = f \lambda).
6. **Refraction Direction Rules**: Light entering an optically denser medium (higher refractive index) slows down and **bends towards the normal**. Light entering a rarer medium speeds up and **bends away from the normal**.
7. **Apparent Depth in Water**: A swimming pool or pond appears shallower than its actual depth due to refraction (light rays from bottom bend away from normal on exiting water). Real depth d = \mu \times \text{apparent depth}.
8. **Bent Stick & Raised Coin**: A pencil or stick partially immersed in water appears bent at the interface; a coin in a bucket appears raised due to refraction.
9. **Underwater Observer Looking Up**: An underwater diver looking at an aerial object sees the object at a **higher level than it actually is** (light travels from rarer air into denser water, bending toward the normal).
10. **Twinkling of Stars**: Caused by continuous **atmospheric refraction** through moving air layers of varying temperatures and optical densities.
11. **Early Sunrise and Delayed Sunset**: Due to atmospheric refraction, the Sun is visible **\approx 2\text{ minutes} before actual geometric sunrise** and **\approx 2\text{ minutes} after actual sunset**, lengthening the day by \approx 4\text{ minutes}.
12. **Elliptical Sun and Moon at Horizon**: Near the horizon, unequal atmospheric refraction across the upper and lower limbs flattens the vertical diameter, making them appear elliptical.
13. **Total Internal Reflection (TIR) Conditions**: (1) Light must travel from an optically **denser to a rarer medium**, and (2) Angle of incidence must exceed the **critical angle (i > i_c)**.
14. **Critical Angles for Common Media**: Water-air: i_c \approx 48.75^\circ; Glass-air: i_c \approx 41.14^\circ; Diamond-air: i_c \approx \mathbf{24.4^\circ} (extremely small, causing repeated internal reflections).
15. **Sparkle of Diamond**: A cut diamond's brilliant sparkle is due to **Total Internal Reflection** enabled by its very high refractive index (2.42) and tiny critical angle (24.4^\circ).
16. **Applications of TIR**:
17. **Soap Film & Bubble Colours vs Spherical Droplets**: The shimmering rainbow colors seen on soap bubbles and oil films in daylight are produced by the **Interference of light** reflecting from front and back surfaces. *(Cross-Concept Note: Liquid droplets and soap bubbles assume an inherently spherical shape due to Surface Tension, which minimizes surface area for a given volume; their optical brilliance and colors are due to thin-film light interference)*.
18. **Compact Disc (CD) Iridescence**: A CD viewed in sunlight displays brilliant rainbow colors due to **Reflection and Diffraction** of light from microscopic pits and tracks behaving as a diffraction grating.
19. **Dispersion in a Prism**: White light splits into 7 constituent colors (VIBGYOR). **Violet light has the shortest wavelength, highest frequency, travels slowest in glass, has highest refractive index, and deviates the**

most. **Red light has the longest wavelength, lowest frequency, travels fastest in glass, has lowest refractive index, and deviates the least**.

20. **Rainbow Formation Physics**: Rainbows are formed by the combined action of **refraction, dispersion, and total internal reflection** within falling raindrops.
21. **Rainbow Colors and Position**:
22. **Rayleigh's Law of Scattering**: The intensity of scattered light is inversely proportional to the fourth power of its wavelength: I \propto \frac{1}{\lambda^4}. Shorter wavelengths (blue/violet) scatter \approx 16 times more than red.
23. **Blue Sky & Red Sunset**:
24. **Sky Appearance to Astronauts & High-Altitude Flyers**: At very high altitudes or in outer space where there is no atmosphere to scatter light, the sky appears **completely black**.
25. **Red Danger Signals**: Red light has the longest visible wavelength and suffers the **least atmospheric scattering**, allowing it to penetrate fog and smoke and remain visible from the greatest distance.
26. **Green Grass Appearance**: Grass appears green because its chlorophyll **reflects green light** while absorbing red, blue, and other wavelengths.
27. **Doppler Redshift in Astronomy**: Light from a receding celestial source or rocketship undergoes a Doppler shift towards longer wavelengths (redshift). A blue light emitted on a rocket moving away at 0.2c appears **yellow-orange** to an observer on Earth.
28. **Plane Mirror Image Properties**: Virtual, erect, laterally inverted, same size as object (m = +1), and located at the same distance behind the mirror as the object is in front.
29. **Plane Mirror Rotation**: If a plane mirror is rotated by an angle \theta, the reflected ray rotates by **2\theta**.
30. **Minimum Height of Plane Mirror**: To view one's full standing height (H), the minimum vertical height of a plane mirror must be **H/2 (half the person's height)**.
31. **Images Between Parallel Mirrors**: When an object is placed between two parallel plane mirrors (\theta = 0^\circ), the number of images formed is **infinite (\infty)**.
32. **Formula for Number of Images**: If two mirrors are inclined at angle \theta: n = \frac{360^\circ}{\theta} - 1 (if 360/\theta is even).
33. **Concave Mirror (Converging Mirror)**:
34. **Convex Mirror (Diverging Mirror)**:
35. **Convex Lens (Converging Lens)**:
36. **Concave Lens (Diverging Lens)**:
37. **Air Bubble in Water**: An air bubble trapped inside water has convex physical surfaces, but because the refractive index of water (1.33) is greater than air (1.0), it behaves optically as a **concave (divergent) lens**.
38. **Convex + Concave Lens in Contact**: If a convex lens and concave lens of identical focal length are placed in contact, their combined focal length is 1/F = 1/f - 1/f = 0 \implies F = \infty. The combination acts as a **flat plane glass sheet**.
39. **Power of Lens (P)**: Defined as the reciprocal of focal length in metres: P = \frac{1}{f(\text{m})}. Measured in **Dioptres (D)**. Converging (convex) lens has positive power (+D); diverging (concave) lens has negative power (-D).
40. **Lens Power Calculation**: A +2\text{ D} lens has focal length f = \frac{1}{2}\text{ m} = \mathbf{50\text{ cm}}. Sunglasses with zero curvature power have **0\text{ Dioptres}**.
41. **Least Distance of Distinct Vision (D)**: For a normal, healthy human eye, the minimum clear focusing distance is **25\text{ cm}**.
42. **Human Eye Retina as Photographic Film**: The retina acts like photographic film in a camera; light is transduced into electrical impulses by rods and cones and sent via the optic nerve to the brain. Images formed on retina are **real, inverted, and diminished**.
43. **Accommodation of the Eye**: Focusing on objects at different distances is achieved by changing the curvature (convexity) and focal length of the eye lens via the **ciliary muscles**.

44. **Dark Room Adaptation & Rhodopsin**: Moving from bright sunlight into a dark room causes temporary blindness because pupils need time to dilate and bleached rod visual pigments (**rhodopsin**) require time to regenerate.
45. **Stereoscopic Vision (Two Eyes)**: Having two forward-facing eyes provides binocular depth perception (stereopsis), allowing accurate judgment of distance and three-dimensionality.
46. **Solar Eclipse Retinal Burns**: Looking directly at a solar eclipse causes irreversible retinal thermal burns caused primarily by invisible **ultraviolet (UV) radiation**.
47. **Star Color and Temperature**: The color of a star indicates its surface temperature: **Blue/White stars are youngest and hottest**; **Red stars are coolest and oldest**.
48. **Primary Colors of Light (Additive)**: **Red, Green, and Blue (RGB)**. Mixing them in equal intensity yields **White light**. Color TV screens use triads of RGB phosphors.
49. **Primary Pigment Colors (Subtractive)**: **Cyan, Magenta, and Yellow (CMY)**. Magenta + Yellow = Red; Cyan + Magenta = Blue; Cyan + Yellow = Green.
50. **Raman Effect**: Inelastic scattering of light photons when traversing any transparent medium (solid, liquid, or gas), discovered by Sir C.V. Raman (Nobel Prize 1930).

One-Liner Revision – Light, Optics and Laser

#	Fast Fact
1	Speed of light in vacuum (c) = **3 x 10^8 m/s** (299,792,458 m/s exactly)
2	Light is both wave (interference, diffraction) and particle (photoelectric effect)
3	Visible spectrum = 400-700 nm; violet (400nm) to red (700nm)
4	VIBGYOR spectrum: Violet has shortest wavelength; Red has longest wavelength
5	Speed of light = lowest in **diamond** (n=2.42; highest refractive index)
6	Laws of reflection: angle of incidence = angle of reflection; in same plane
7	Concave mirror = **converging** mirror; used in torches, telescopes, solar furnaces
8	Convex mirror = **diverging** mirror; used in rear-view mirrors, security mirrors
9	Convex lens = **converging**; used in magnifying glass, camera, eye
10	Concave lens = **diverging**; corrects myopia (short-sightedness)
11	Myopia = cannot see far; eyeball too long; corrected by **concave lens**
12	Hypermetropia = cannot see near; eyeball too short; corrected by **convex lens**
13	Presbyopia = old-age vision; bifocals (convex for near, concave for far)
14	Total Internal Reflection occurs when light goes from **denser to rarer** medium above critical angle
15	Diamond sparkles due to **total internal reflection** (critical angle = 24.4 deg)
16	Optical fibre uses TIR to transmit light signals; glass/silica core
17	Sky is blue = **Rayleigh scattering**; blue light scattered more (shorter wavelength)
18	Sunset/sunrise red = blue scattered away; long-wavelength red remains
19	Mirages = due to total internal reflection of light in hot air layers near ground
20	Rainbow = formed by reflection + refraction + dispersion in water droplets
21	Dispersion: violet light bends most; red light bends least (through prism)
22	Primary colours of light = Red, Green, Blue (RGB) – additive mixing
23	Primary colours of pigment = Red, Yellow, Blue (RYB) – subtractive mixing
24	LASER = monochromatic (one wavelength) + coherent + collimated beam

#	Fast Fact
25	First working laser = **Theodore Maiman (1960)**; used ruby crystal
26	LASER used in: surgery, CD/DVD, barcode scanner, laser cutting, LIDAR
27	Holography = 3D image using laser; invented by **Dennis Gabor (1948)**; Nobel 1971
28	Photoelectric effect = Einstein (1905); Nobel 1921; basis of solar cells, photomultipliers
29	Endoscopy = uses optical fibres to view internal organs without surgery
30	Periscope = uses two plane mirrors at 45 deg; used in submarines

Physics Topic 5 — Sound and Wave Motion

Consolidated — 50 Must-Score Facts

1. A wave transports **energy and momentum** from one point to another without any net transport of matter.
2. Mechanical waves require an elastic material medium possessing **inertia and elasticity (restoring force)** for propagation.
3. Transverse mechanical waves can propagate only in **solids** (which sustain shear stress) and on liquid surfaces; they **cannot propagate through the bulk of fluids (liquids and gases)**.
4. Ocean surface waves are a hybrid: water particles execute **both vertical and horizontal orbital motions** (elliptical trajectories).
5. Longitudinal waves propagate through **all three states of matter (solids, liquids, and gases)** as alternating compressions and rarefactions.
6. Sound waves in air and all fluid media are strictly **longitudinal mechanical waves**.
7. In solids (such as rocks and metals), sound can propagate as **both longitudinal waves and transverse (shear) waves**.
8. In an earthquake, **Primary (P) waves are longitudinal** (compressional, travel through solid, liquid, gas, recorded first on seismograph), while **Secondary (S) waves are transverse** (shear, travel only through solids, recorded later).
9. Plucking the string of a Sitar produces **transverse stationary (standing) waves** along the string, which induce longitudinal pressure sound waves in the surrounding air.
10. Electromagnetic (EM) waves do not require any material medium and travel through a vacuum at the speed of light (c \approx 3.0 \times 10^8\text{ m/s}).
11. In a vacuum, **all electromagnetic waves travel at the exact same velocity** (c), regardless of their wavelength or frequency.
12. EM waves are self-sustaining oscillating electric (\vec{E}) and magnetic (\vec{B}) fields perpendicular to each other and to the direction of propagation (**transverse nature**).
13. Electromagnetic spectrum in decreasing order of frequency (increasing order of wavelength): **Gamma rays > X-rays > Ultraviolet > Visible light > Infrared > Microwaves > Radio waves**.
14. **Gamma rays (\gamma)** have the shortest wavelength (< 10^{-12}\text{ m}), highest frequency, highest energy per photon (E = h\nu), and maximum penetrating power; used in radiotherapy to destroy cancerous cells.
15. **Wilhelm Conrad Röntgen** discovered X-rays (1\text{ nm} - 10^{-3}\text{ nm}); X-rays are not deflected by electric or magnetic fields because they carry no electric charge.
16. X-rays are used in **crystallography (crystal structure analysis by diffraction)**, medical radiography, CT scans (computed tomography), and diagnosing intestinal disorders.
17. **Cosmic rays are NOT electromagnetic waves**; they are streams of extremely high-energy charged subatomic particles (chiefly protons and alpha particles) arriving from outer space.
18. Ultraviolet radiation is divided into UV-A (320-400\text{ nm}), UV-B (280-320\text{ nm}), and UV-C (100-280\text{ nm}). UV-C is lethal to cells but is completely absorbed by the stratospheric ozone layer.
19. Ozone layer depletion exposes Earth to UV radiation of minimum wavelength around **100\text{ nm} = 10^{-7}\text{ m}**.
20. Infrared radiation (1\text{ mm} - 700\text{ nm}) was discovered by **William Herschel**; it is the physical agent responsible for radiant thermal heating and is used in night-vision devices and TV remotes.
21. Common TV remote controls primarily emit **Infrared (IR) rays** (some smart remotes use radio waves/Bluetooth).
22. Microwaves (0.1\text{ m} - 1\text{ mm}, 300\text{ MHz} - 300\text{ GHz}) were pioneered by Sir J.C. Bose and Heinrich Hertz; widely used in Radar, satellite communication, and microwave ovens.

23. The carrier frequency of a 30\text{ m} radio band is: \nu = c/\lambda = (3 \times 10^8\text{ m/s}) / 30\text{ m} = 10\text{ MHz}.
24. **Sky wave propagation** depends on the reflection of medium- and shortwave radio waves from the electrically ionized gas layers of the **Ionosphere**.
25. Satellite dish antennas experience signal dropout during heavy rain because **raindrops absorb and scatter/disperse the radio wave energy**, attenuating high-frequency satellite downlinks.
26. In television broadcasting, **picture signals are transmitted by Amplitude Modulation (AM)**, while **audio signals are transmitted by Frequency Modulation (FM)**.
27. **Edwin Howard Armstrong** invented wideband Frequency Modulation (FM) in 1935; FM operates at higher frequency (88-108\text{ MHz}) and wider bandwidth, offering complete immunity from amplitude noise and static.
28. Medium-speed data communication (1200 - 2400\text{ bps}) conventionally employs **Frequency Modulation (Frequency Shift Keying)**.
29. Speed of sound in various media at standard temperature: **Steel (5960\text{ m/s}) > Iron (5950\text{ m/s}) > Seawater (1531\text{ m/s}) > Fresh Water (1498\text{ m/s}) > Hydrogen (1284\text{ m/s}) > Air at 22^\circ\text{C} (344\text{ m/s}) > Air at 0^\circ\text{C} (331\text{ m/s})**.
30. The speed of sound in steel is approximately **17 times** greater than in air.
31. When sound moves from one medium to another, its **frequency remains completely unchanged**, while its velocity and wavelength change proportionally (v = f\lambda).
32. Speed of sound in a gas is directly proportional to the square root of absolute temperature (v \propto \sqrt{T}); speed of sound is higher in summer than in winter.
33. Speed of sound in air is **completely independent of atmospheric pressure changes** at constant temperature.
34. Speed of sound is greater in **humid air than in dry air** because the molecular weight of water vapour (18\text{ g/mol}) is lower than that of dry air (\sim 29\text{ g/mol}), reducing density \rho.
35. The audible frequency range for human ears is **20\text{ Hz} - 20,000\text{ Hz} (20\text{ kHz})**.
36. **Infrasonic waves** have frequencies below 20\text{ Hz}; generated by earthquakes, volcanoes, ocean swells, avalanches, and elephants, whales, and rhinos.
37. Prior to the arrival of destructive main earthquake shockwaves, **infrasound waves** are generated, alerting animals.
38. The human heart vibrates at **infrasonic frequencies**, picked up by the acoustic diaphragm of a stethoscope.
39. **Ultrasonic waves** have frequencies exceeding 20,000\text{ Hz} (20\text{ kHz}); bats, dolphins, and porpoises use ultrasound for echolocation.
40. High-frequency ultrasonic vibrations are used in **lithotripsy** to pulverize kidney stones into fine gravel that passes harmlessly in urine.
41. **Echocardiography (Echo test)** utilizes reflected ultrasound beams to produce dynamic real-time images of cardiac valves and chambers.
42. **Galton Whistle** emits ultrasonic frequencies (> 20\text{ kHz}), completely inaudible to human ears but heard clearly by dogs for training.
43. Stethoscopes operate on the principle of **multiple reflection of sound waves** within the hollow tubing.
44. The persistence of hearing in the human brain lasts for **0.1\text{ second} (1/10\text{ s})**.
45. To hear a distinct, clear echo, the reflecting barrier must be situated at a minimum distance of **17.2\text{ metres}** (approx 17\text{ m} or 16.5\text{ m}) at 22^\circ\text{C} (d = vt/2 = 344 \times 0.1 / 2).
46. Reverberation in large halls and auditoriums is caused by **multiple consecutive reflections**; music hall walls are lined with porous, acoustic-absorbing materials.
47. **Pitch** is governed strictly by **frequency** (and inversely by wavelength); high pitch produces a sharp, shrill voice (females, children), while low pitch produces a deep, grave voice.
48. **Loudness** is governed by the **amplitude** of wave vibration (I \propto A^2); perceived loudness is measured in decibels (\text{dB}).

49. **Timbre (Quality)** is governed by the waveform and harmonic overtones, allowing the human ear to distinguish between a Sitar and a Flute playing the identical musical note at identical loudness.
50. **Mach Number** is the ratio of object velocity to local speed of sound; Supersonic jets (M > 1) flying through the stratosphere emit nitrogen oxides (\text{NO}_x) that catalyze ozone layer depletion.

📋 Must-Score Master Tables

Table 1: Speed of Sound Across Common Media (Standard Conditions)

Medium	State	Approximate Speed (v) in \text{m/s}	Must-Score Notes
Aluminium	Solid Metal	6,420\text{ m/s}	Extremely high speed due to high Young's modulus
Steel	Solid Alloy	5,960\text{ m/s}	\approx 17 times faster than air; highest common civil question
Iron	Solid Metal	5,950\text{ m/s}	Tested in UPPCS / BPSC rail-track questions
Granite / Rock	Solid Mineral	\sim 6,000\text{ m/s}	Transmits both P (longitudinal) and S (transverse) waves
Glass	Solid Amorphous	\sim 4,000 - 5,000\text{ m/s}	Lower than aluminium and steel
Seawater (25^\circ\text{C})	Liquid	1,531\text{ m/s}	Higher than fresh water due to salinity and density
Distilled Water (25^\circ\text{C})	Liquid	1,498\text{ m/s}	Standard reference liquid speed
Hydrogen Gas (25^\circ\text{C})	Gas	1,284\text{ m/s}	Fastest gas speed due to very low molecular weight (M=2)
Helium Gas (25^\circ\text{C})	Gas	965\text{ m/s}	High speed in light inert gas
Air (22^\circ\text{C})	Gas Mixture	344\text{ m/s}	Standard room temperature speed (used for 17.2\text{ m} echo)
Air (0^\circ\text{C})	Gas Mixture	331\text{ m/s}	Zero degrees standard speed
Vacuum	Free Space	**0\text{ m/s}**	**Cannot propagate**; astronauts on Moon cannot hear speech

Table 2: The Decibel Scale & Physiological Impact

Sound Source / Environment	Sound Intensity Level (\text{dB})	Physiological / Regulatory Impact
Threshold of Hearing	0\text{ dB}	Faintest audible sound at 1000\text{ Hz}
Rustling of Tree Leaves	**20\text{ dB}**	Standard quiet nature level (UPPCS Pre 2018)
Whisper (at 1\text{ m})	15 - 20\text{ dB}	Very quiet acoustic level
Quiet Library / Soft Music	30\text{ dB}	Comfortable reading level
Normal Conversation	**30 - 60\text{ dB}**	Optimal human communication level
WHO Safe Urban Level	**45\text{ dB}**	WHO recommended daytime city ambient limit
Average Street Traffic	60 - 70\text{ dB}	Moderate noise
Anger / Heated Conversation	70 - 80\text{ dB}	Annoying level
Heavy Truck / Motorcycle	90 - 95\text{ dB}	Hazardous under prolonged exposure
Machine Shop / Factory Floor	**100 - 110\text{ dB}**	Permanent hearing loss risk over 8 hours

Sound Source / Environment	Sound Intensity Level (\text{dB})	Physiological / Regulatory Impact
Orchestra / Rock Concert	110 - 120\text{ dB}	Extreme acoustic intensity
Moradabad Peak Noise (UNEP)	**114\text{ dB}**	Ranked 2nd noisiest city globally (2022)
Threshold of Pain	**120 - 130\text{ dB}**	Physical pain in ear canal
Jet Aircraft Takeoff (close)	140 - 150\text{ dB}	Immediate acoustic trauma / eardrum rupture

CPCB Ambient Noise Standards (Noise Pollution Rules, 2000)

Category of Zone	Day Time (6\text{ AM} - 10\text{ PM})	Night Time (10\text{ PM} - 6\text{ AM})
Industrial Area	75\text{ dB}	70\text{ dB}
Commercial Area	65\text{ dB}	55\text{ dB}
Residential Area	**55\text{ dB}**	**45\text{ dB}**
Silence Zone (Hospital, School \le 100\text{ m})	**50\text{ dB}**	**40\text{ dB}**

Table 3: Complete Electromagnetic Spectrum Breakdown

Wave Type	Discoverer	Wavelength Range (\lambda)	Frequency Range (\nu)	Key Applications & High-Yield Facts
Gamma Rays (\gamma)	Paul Villard (identified), Henri Becquerel (named)	< 10^{-3}\text{ nm} (< 10^{-12}\text{ m})	> 3 \times 10^{20}\text{ Hz}	Maximum penetrating power; radiotherapy for cancer; nuclear emissions
X-Rays	Wilhelm Conrad Röntgen (1895)	1\text{ nm} - 10^{-3}\text{ nm} (10^{-9} - 10^{-12}\text{ m})	3 \times 10^{17} - 3 \times 10^{20}\text{ Hz}	Crystal structure diffraction; CT scans; intestinal disease diagnosis
Ultraviolet (UV)	Johann Wilhelm Ritter (1801)	400\text{ nm} - 1\text{ nm} (4 \times 10^{-7} - 10^{-9}\text{ m})	7.5 \times 10^{14} - 3 \times 10^{17}\text{ Hz}	Water & surgical sterilization; Vitamin D synthesis; absorbed by stratospheric ozone
Visible Light	Isaac Newton (1666)	700\text{ nm} - 400\text{ nm} (7 \times 10^{-7} - 4 \times 10^{-7}\text{ m})	4.3 \times 10^{14} - 7.5 \times 10^{14}\text{ Hz}	Human vision (VIBGYOR); peak sensitivity at 555\text{ nm} (yellowish-green)
Infrared (IR)	William Herschel (1800)	1\text{ mm} - 700\text{ nm} (10^{-3} - 7 \times 10^{-7}\text{ m})	3 \times 10^{11} - 4.3 \times 10^{14}\text{ Hz}	Thermal radiation; greenhouse effect; night-vision goggles; TV remote controllers
Microwaves	J.C. Bose, Heinrich Hertz	0.1\text{ m} - 1\text{ mm} (10^{-1} - 10^{-3}\text{ m})	300\text{ MHz} - 300\text{ GHz}	Radar systems; microwave ovens; point-to-point satellite & cellular telecommunication
Radio Waves	Heinrich Hertz (proof), Guglielmo Marconi	> 0.1\text{ m} (> 10^{-1}\text{ m} to 10^4\text{ m})	< 300\text{ MHz}	Radio and television broadcast; reflected by ionosphere (sky waves)

Table 4: Diatonic Scale Musical Octave

Indian Note	Western Note	Frequency Ratio to Key Note	Frequency (if \text{Sa} = 256\text{ Hz})
Sa	C	1	256\text{ Hz}
Re	D	9/8	288\text{ Hz}
Ga	E	5/4	320\text{ Hz}
Ma	F	4/3	341.3\text{ Hz}
Pa	G	3/2	384\text{ Hz}

Indian Note	Western Note	Frequency Ratio to Key Note	Frequency (if \text{Sa} = 256\text{ Hz})
Dha	A	5/3	426.7\text{ Hz}
Ni	B	15/8	**480\text{ Hz}**
\text{Sa}_1 (Octave)	\text{C}_1	2	**512\text{ Hz}**

One-Liner Revision – Sound and Wave Motion

#	Fast Fact
1	Sound is a **mechanical longitudinal** wave (particles vibrate parallel to wave direction)
2	Sound needs a **medium** to travel; cannot travel in vacuum
3	Speed of sound: 343 m/s in air (20 degC); 1500 m/s in water; 5100 m/s in steel
4	Speed of sound increases with temperature and humidity
5	Sound travels fastest in **solids**, slowest in **gases**
6	Human audible range = **20 Hz to 20,000 Hz**
7	Infrasound (< 20 Hz) = elephants, whales, earthquakes; long-range communication
8	Ultrasound (> 20,000 Hz) = bats (echolocation), dolphins; SONAR; medical USG
9	SONAR uses ultrasound to detect submarines, fish, ocean depth
10	Medical ultrasonography = uses 1-20 MHz ultrasound to image internal organs
11	Echo = minimum distance = **17 metres** (at 20 degC); time gap > 0.1 sec
12	Reverberation = multiple reflections; undesirable in auditoriums; curtains absorb
13	Resonance = when forced frequency = natural frequency; Tacoma Narrows Bridge collapse (1940)
14	Doppler Effect: source approaching = frequency increases (blue shift); receding = decreases (red shift)
15	Doppler Effect used in: radar speed guns, echocardiography, astronomy (red shift)
16	Mach number = object speed / speed of sound; Mach 1 = speed of sound
17	Sonic boom = shock wave when object crosses Mach 1 (supersonic)
18	dB levels: Whisper = 20 dB; Conversation = 60 dB; Rock concert = 110-120 dB
19	Safe noise level = <85 dB (prolonged); >140 dB = pain threshold
20	Noise pollution limit (India): day = 55 dB (residential), night = 45 dB
21	Beats = alternating loud/soft sound when two slightly different frequencies combine
22	Beat frequency =
23	Musical instruments: String (sitar), Wind (flute, trumpet), Percussion (tabla)
24	Stethoscope = uses sound conduction (not amplification) to hear heartbeats
25	Thunder heard after lightning = because light (3x10^8 m/s) >> sound (343 m/s)
26	Sound wavelength (audible range): 1.7 cm (20 kHz) to 17 m (20 Hz)
27	Intensity of sound: I = P/(4*pi*r^2); decreases as 1/r^2 from point source
28	Transverse waves: light, EM waves; Longitudinal waves: sound, seismic P-waves
29	Seismic P-waves = longitudinal; S-waves = transverse (can't pass through liquid outer core)
30	Infrasound from earthquakes detected by animals before humans = early warning

Physics Topic 6 — Electricity, Magnetism and Electromagnetism

Consolidated — 50 Must-Score Facts

1. Electric current is defined as the rate of flow of electric charge: I = Q/t; its SI unit is the **Ampere (1\text{ A} = 1\text{ C/s})**.
2. By international convention, the direction of electric current is the direction of flow of positive charge, which is **opposite to the drift of electrons**.
3. **Current density (\vec{j} = I/A) is a VECTOR quantity** (pointing in the direction of positive charge motion), whereas electric current (I) and electric potential (V) are **scalars**.
4. The **dielectric constant** (relative permittivity \varepsilon_r) is a dimensionless quantity (M^0 L^0 T^0).
5. The kinetic energy acquired by an electron of charge e falling through potential difference V is **E = eV\text{ Joules} = 1.602 \times 10^{-19}\text{ J}** for 1\text{ volt}.
6. The commercial unit of electrical energy is the **kilowatt-hour (kWh)**, commonly called one **'unit'**: 1\text{ kWh} = 1000\text{ W} \times 3600\text{ s} = 3.6 \times 10^6\text{ Joules}.
7. Household electrical consumption of 250\text{ units} = 250 \times 3.6 \times 10^6 = \mathbf{9 \times 10^8\text{ Joules}} (71st BPSC 2025).
8. Electric power equations in simple ohmic circuits: P = VI = I^2 R = V^2 / R. The expression **IR^2 DOES NOT represent electric power** (71st BPSC 2025).
9. Resistance of a conductor: R = \rho L / A, where \rho is specific resistance (electrical resistivity), measured in **Ohm-metre (\Omega\cdot\text{m})**.
10. If a wire is stretched so that its length doubles (L' = 2L) at constant volume, its area halves (A' = A/2), causing its resistance to quadruple (R' = 4R).
11. Of two domestic light bulbs connected to the same voltage supply, **the dimmer bulb has a LARGER electrical resistance** (P = V^2 / R).
12. Connected across the mains in parallel, two identical bulbs glow significantly brighter than when connected in series (P_{\text{parallel}} = 2V^2/R vs P_{\text{series}} = V^2/2R).
13. Resistor colour code memory key **BBROYGBVGW**: Brown = 1, Green = 5, Black = 0 (multiplier 10^0 = 1). A resistor with bands Brown-Green-Black has resistance (15) \times 10^0 = 15\,\Omega; under 15\text{ V}, it draws **1\text{ Ampere}** (UPPCS RO/ARO 2023).
14. Capacitive reactance is inversely proportional to frequency: X_C = \frac{1}{2\pi f C} \propto \frac{1}{f}.
15. Inductive reactance is directly proportional to frequency: X_L = 2\pi f L \propto f.
16. A **dynamo (DC generator)** converts mechanical energy into direct electrical energy using a **commutator (split rings)**.
17. An **AC generator (alternator)** converts mechanical energy into alternating current using **continuous slip rings**.
18. An **electric motor** converts electrical energy into mechanical rotational energy based on the Lorentz force on a current-carrying loop.
19. Electric motors operating at severely depressed voltages burn out because they **draw excessively high current** (I = P/V) to deliver rated mechanical power, generating excessive Joule heating (I^2 R).
20. The **Faraday constant (F \approx 96,485\text{ C/mol})** represents the magnitude of electric charge per mole of electrons and is a universal physical constant.
21. Motor car lead-acid batteries have six 2\text{ V} cells connected in series giving a nominal **12\text{ V}**; capacity is rated in **ampere-hours (Ah)**; electrolyte is approx 35\% sulphuric acid (\text{H}_2\text{SO}_4) in water; positive electrode is lead dioxide (\text{PbO}_2) and negative electrode is spongy lead (\text{Pb}).
22. **Nickel-Cadmium (Ni-Cd)** rechargeable batteries use nickel oxide hydroxide and metallic cadmium electrodes with potassium hydroxide (\text{KOH}) alkaline electrolyte; used in calculators, cordless tools, and transistors.

23. The filament of an incandescent light bulb is made of **Tungsten** (Wolfram, W), discovered by Juan José and Fausto Elhuyar; melting point \approx 3422^\circ\text{C}, operating temperature 1500 - 2500^\circ\text{C}.
24. Ordinary incandescent bulbs have short lifespans because: (i) filament thickness is non-uniform, (ii) complete vacuum cannot be maintained, (iii) support wires degrade. They are filled with inert **Argon and Nitrogen** to inhibit tungsten evaporation.
25. Incandescent bulbs convert only **5\% - 10\%** of electrical energy into visible light; the remaining 90\% - 95\% is dissipated as infrared heat.
26. **Lightning** contains tremendous electrical energy (10^8 - 10^9\text{ Joules}), sufficient to vaporize moisture inside a tree and cause explosive combustion.
27. A **lightning conductor** is a copper rod mounted at the apex of a building, bonded to ground through thick conductor wire, directing atmospheric surge harmlessly into Earth.
28. In a **three-pin electrical plug, the Earth pin is LONGER and THICKER** than the live and neutral pins so that the chassis is grounded first upon insertion and disconnected last upon removal.
29. Electric current **cannot flow** between two connected conductors if both are maintained at the **SAME electrical potential** (I = \Delta V / R = 0).
30. An **electric fuse** is a critical safety device connected in **SERIES** with the live line; it operates on the **heating effect of electric current** (H = I^2 R t).
31. A fuse wire must possess **HIGH RESISTIVITY and a LOW MELTING POINT**.
32. Domestic fuse wires are made of an alloy of **63\% Tin (Sn) and 37\% Lead (Pb)**, with a eutectic melting point of only **183^\circ\text{C}** (vastly lower than copper at 1085^\circ\text{C} or aluminium at 660^\circ\text{C}).
33. For high-current industrial circuits (> 15\text{ A}), tin-coated copper fuse wires are standard.
34. During a **short circuit**, line resistance drops to near zero, causing electric current to **increase drastically / surge massively** (71st BPSC 2025).
35. The filament of a **halogen lamp** is an alloy of **Tungsten and Sodium**, producing intense yellowish-white light.
36. When electric current flows through a metal wire, temperature rises because moving drift electrons collide continuously with vibrating lattice atoms, transferring kinetic energy into thermal vibrations.
37. Domestic electrical distribution wiring is connected in **PARALLEL**; every appliance receives the full rated voltage (220\text{ V}), and turning off one appliance does not disrupt others.
38. Standard Indian household insulation colour coding: **Red/Brown = Live**, **Black/Blue = Neutral**, **Green/Yellow-Green = Earth**.
39. The electrical resistance of a dry human body is on the order of **10^4\,\Omega (10,000\,\Omega)** to 10^5\,\Omega; wet or broken skin drops resistance to below 1000\,\Omega.
40. **CFL (Compact Fluorescent Lamp)** contains mercury vapour and argon; electric discharge excites mercury atoms to emit ultraviolet radiation, which strikes inner phosphor coating to fluoresce visible white light (22\% energy efficiency; lasts 6,000 - 15,000\text{ hours}).
41. **Light Emitting Diode (LED)** operates on the principle of **ELECTROLUMINESCENCE**; electrons recombine with electron holes in a direct-bandgap semiconductor (e.g. Gallium Arsenide/Phosphide), releasing energy directly as photons.
42. LEDs consume \sim 85\% less power than incandescent bulbs, emit directional light (180^\circ), and have an enormous operational lifespan of **50,000 - 100,000\text{ hours}**.
43. **Sodium vapour street lamps** are omnidirectional (360^\circ) and emit monochromatic yellow light (589\text{ nm}), producing shadowy-black colour rendering compared to high-CRI multi-spectrum LEDs.
44. The **choke coil** in a fluorescent tubelight acts as an inductor that produces a high-voltage inductive kick to ignite gas discharge, and subsequently limits and stabilizes alternating current through the tube.
45. If potential difference (V) and current (I) across a load are each doubled, power increases **fourfold** (P' = 2V \times 2I = 4P).
46. If n small identical spherical drops charged to potential V coalesce into a single large drop, the potential of the composite drop becomes **V' = n^{2/3} V** (UPPCS Mains).

47. Magnetic field at the centre of a flat circular coil of N turns carrying current I: B = \frac{\mu_0 N I}{2r}. For N = 100, r = 0.1\text{ m}, I = 1\text{ A} \implies B = \mathbf{6.28 \times 10^{-4}\text{ Tesla}} (UKPCS 2024).
48. A **transformer** functions exclusively on alternating current (AC) via **mutual electromagnetic induction**; it cannot operate on steady direct current (DC).
49. A **mobile phone charger** incorporates a **step-down transformer** (stepping down 220\text{ V AC} \to 5-10\text{ V AC}) followed by a **rectifier** (converting AC to DC).
50. Earth's geomagnetic field is generated by convective electric currents in the molten, iron-rich outer liquid core (**geodynamo effect**).

📋 Must-Score Master Tables

Table 1: Electrical & Magnetic Units and Dimensions

Physical Quantity	SI Unit	Dimension	Scalar / Vector
Electric Charge (Q)	Coulomb (\text{C} = \text{A}\cdot\text{s})	[A T]	Scalar
Electric Current (I)	Ampere (\text{A})	[A]	Scalar
Current Density (\vec{j})	\text{A/m}^2	[A L^{-2}]	**Vector**
Electric Potential / EMF (V)	Volt (\text{V} = \text{J/C})	[M L^2 T^{-3} A^{-1}]	Scalar
Electrical Resistance (R)	Ohm (\Omega = \text{V/A})	[M L^2 T^{-3} A^{-2}]	Scalar
Electrical Resistivity (\rho)	Ohm-metre (\Omega\cdot\text{m})	[M L^3 T^{-3} A^{-2}]	Scalar
Electrical Conductivity (\sigma)	Siemens/metre (\text{S/m})	[M^{-1} L^{-3} T^3 A^2]	Scalar
Capacitance (C)	Farad (\text{F} = \text{C/V})	[M^{-1} L^{-2} T^4 A^2]	Scalar
Magnetic Flux (\Phi_B)	Weber (\text{Wb})	[M L^2 T^{-2} A^{-1}]	Scalar
Magnetic Field (B)	Tesla (\text{T} = \text{Wb/m}^2)	[M T^{-2} A^{-1}]	**Vector**
Dielectric Constant (\varepsilon_r)	Dimensionless	[M^0 L^0 T^0]	Scalar

Table 2: Comparative Analysis of Lighting Technologies

Parameter	Incandescent Bulb	Fluorescent Tube	CFL	LED
Light Generation	Thermal incandescence of tungsten	Gas discharge of mercury vapour	Gas discharge in compact tube	Solid-state electroluminescence
Efficiency	Very low (5 - 10\%)	Moderate (\sim 22\%)	High (\sim 25\%)	Superior (\sim 85 - 90\%)
Operating Temp	1500 - 2500^\circ\text{C}	Low (\sim 40^\circ\text{C})	Low (\sim 60^\circ\text{C})	Very low (cool running)
Average Lifespan	\sim 1,000\text{ hours}	\sim 8,000\text{ hours}	6,000 - 15,000\text{ h}	**50,000 - 100,000\text{ h}**
Environmental Hazard	Minimal (inert gas)	Contains toxic mercury	Contains toxic mercury	**Zero toxic mercury**
Radiation Pattern	Omnidirectional (360^\circ)	Cylindrical	Omnidirectional	Directional (180^\circ)

One-Liner Revision – Electricity and Magnetism

#	Fast Fact
1	Charge on electron = -1.6 x 10^-19 C; proton = +1.6 x 10^-19 C
2	1 Coulomb = charge of 6.24 x 10^18 electrons
3	Ohm's Law: V = IR (Voltage = Current x Resistance)

#	Fast Fact
4	Resistance of wire: R = rho x L / A (longer and thinner = higher resistance)
5	Series circuit: same current; resistances add (R_total = R1 + R2)
6	Parallel circuit: same voltage; conductances add (1/R_total = 1/R1 + 1/R2)
7	Power consumed: P = VI = I^2 R = V^2/R
8	Household supply (India) = 220V, 50Hz AC
9	Fuse wire = tin-lead alloy; low melting point; melts to break circuit on overload
10	MCB (Miniature Circuit Breaker) = modern fuse; can be reset; trips on overload
11	Earth wire = green/yellow; safety; carries fault current to earth
12	Neutral wire = blue/black; Live/Phase wire = red/brown
13	Short circuit = live and neutral wires touch directly; large current; fire risk
14	Fleming's Left Hand Rule = direction of force on current-carrying conductor in magnetic field
15	Fleming's Right Hand Rule = direction of induced EMF in generator
16	Faraday's Law: induced EMF proportional to rate of change of magnetic flux
17	Lenz's Law: induced current opposes the change causing it (conservation of energy)
18	Transformer: step-up (V2>V1, N2>N1); step-down (V2<V1, N2<N1)
19	Electric motor: converts electrical energy to mechanical energy
20	Generator/Dynamo: converts mechanical energy to electrical energy
21	AC generator uses **slip rings**; DC generator uses **split-ring commutator**
22	Magnetic poles: like poles repel, unlike poles attract
23	Earth's magnetic pole (north seeking) = geographic south pole of Earth
24	Magnetic declination = angle between geographic north and magnetic north
25	Solenoid with current = acts as bar magnet; strength increases with turns and current
26	Electromagnet = soft iron core + coil; temporary; used in cranes, MRI, doorbells
27	MRI (Magnetic Resonance Imaging) = uses strong magnetic field + radio waves
28	Lightning conductor = iron rod on tall buildings; safely channels lightning to earth
29	Electric bell works on: **electromagnetic induction + make-and-break circuit**
30	Galvanometer = detects small currents; can be converted to voltmeter (series R) or ammeter (parallel R)

Physics Topic 7 — Electronics, Semiconductors and Computers

Consolidated — 50 Must-Score Facts

1. A computer executes sequences of arithmetic or logical operations automatically; raw unorganized facts and figures are **Data**, while organized, interpreted, and meaningful processed data is **Information**.
2. **Bit (Binary Digit)** is the smallest unit of digital data (0 or 1); a **Byte** comprises **8\text{ bits}**, and a **Nibble** comprises **4\text{ bits}**.
3. **Charles Babbage** is celebrated as the "Father of Computing" for designing the **Analytical Engine (1834/1837)**, which pioneered the ALU, integrated memory, and flow control.
4. **Alan Turing** formulated the theoretical foundation of modern computer science and artificial intelligence (Turing Machine, 1936; Turing Test / Imitation Game, 1949).
5. The first electronic digital computer was **Colossus** (designed by Tommy Flowers and Alan Turing in Britain, 1943–1945 for deciphering the Lorenz cipher) and **ENIAC** (1946, USA).
6. The first commercial digital computer built with integrated circuit (IC) chips was the **IBM System/360** (1964).
7. The physical circuit board housing the CPU, memory sockets, and chipset is the **Motherboard**.
8. The computer mouse was invented in **1964 by Dr. Douglas Engelbart**; the original prototype body was carved out of **wood** with two metal wheels.
9. **Primary (Internal) Memory** is semiconductor memory directly addressed by the CPU: volatile **RAM** (Random Access Memory) and non-volatile **ROM** (Read-Only Memory).
10. **ROM** stores system firmware, specifically the **BIOS (Basic Input/Output System) / UEFI**, which initializes hardware during the Power-On Self-Test (POST).
11. **Cache Memory** is an extremely high-speed, small SRAM memory situated between the CPU core and main RAM to accelerate data access and eliminate latency.
12. **Virtual Memory** is a memory management technique where the operating system uses allocated space on the secondary hard drive (pagefile/swap) to supplement physical RAM.
13. Storage capacity equivalence ladder:
    - 1\text{ Byte} = 8\text{ bits}
    - 1\text{ KB} = 1024\text{ Bytes} = 8192\text{ bits}
    - 1\text{ MB} = 1024\text{ KB} = 1024 \times 1024 = \mathbf{1,048,576\text{ Bytes}} (approximately 1\text{ million bytes})
    - 1\text{ GB} = 1024\text{ MB} = 2^{30}\text{ Bytes} (approximately 1\text{ billion bytes})
    - 1\text{ TB} = 1024\text{ GB} = 2^{40}\text{ Bytes}
    - 1\text{ PB (Petabyte)} = 1024\text{ TB} = 2^{50}\text{ Bytes}
14. A standard single-layer **CD-ROM holds \sim 650 - 700\text{ MB}**; a standard single-layer **DVD holds 4.7\text{ GB}** (approx. 7 times a CD); a single-layer **Blu-ray Disc holds 25\text{ GB}** (50\text{ GB} dual-layer).
15. Blu-ray Discs utilize a **blue-violet laser (405\text{ nm})** with shorter wavelength than the red laser (650\text{ nm}) used in DVDs, allowing significantly denser data pit packing.
16. Binary arithmetic addition rule: **1 + 1 = 0 with a carry of 1** (which represents 2 in decimal, binary 10).
17. A **self-complementing code** has the mathematical property that replacing 1 with 0 and 0 with 1 yields the 9's complement of the decimal digit; examples include **2421 and 5211 codes** (sum of weights equals 9). The standard 8421 BCD code is NOT self-complementing.
18. **MAC Address (Media Access Control):** A permanent physical hardware identifier burned into the Network Interface Card (NIC) at **Layer 2 (Data Link Layer)**, measuring **48\text{ bits}** in length, expressed as six pairs of hexadecimal digits (e.g., `00:1A:2B:3C:4D:5E`).
19. **IP Address:** A logical software address assigned at **Layer 3 (Network Layer)** of the OSI model. **IPv4 is 32\text{-bit}** (2^{32} \approx 4.3\text{ billion} addresses); **IPv6 is 128\text{-bit}** (2^{128} \approx 3.4 \times 10^{38}

addresses).

20. **Domain Name System (DNS):** A hierarchical distributed database that translates human-friendly domain names (e.g. `upsc.gov.in`) into numerical IP addresses.
21. The **World Wide Web (WWW)** was invented in **1989 by British scientist Sir Tim Berners-Lee** at CERN, Geneva, using HyperText Markup Language (HTML) over HTTP.
22. The **World Wide Web Consortium (W3C)**, founded in 1994 by Tim Berners-Lee, is the primary international standards organization for the Web.
23. **Email** network addressing using the **"@"** symbol was introduced by **Raymond (Ray) Tomlinson** in 1971 on ARPANET.
24. An email address consists of two parts separated by "@": **username@domainname** (e.g. in `mark.sttol@ITdesk.info`, `ITdesk.info` is the domain name).
25. **Bluetooth** (invented 1994 by Jaap Haartsen) operates in the **2.4\text{ GHz} ISM radio band** over short distances (10\text{ m}); **Wi-Fi** operates in both **2.4\text{ GHz} and 5\text{ GHz}** bands over local areas; **Li-Fi (Light Fidelity)** transmits wireless data using **visible light spectrum (375 - 780\text{ nm})** via high-speed pulsing LEDs, achieving speeds up to **10\text{ Gbps}** with zero electromagnetic interference.
26. **Near Field Communication (NFC):** Contactless short-range wireless communication using electromagnetic radio fields operating at **\le 4\text{ cm}** with maximum transfer speed of **424\text{ kbps}**.
27. A **Virtual Private Network (VPN)** extends an encrypted private network tunnel across the public Internet, ensuring end-to-end security, confidentiality, and data integrity.
28. **Modem (Modulator/Demodulator):** A hardware device that converts digital computer signals into analog signals for transmission over telephone lines and demodulates incoming analog signals back into digital data.
29. A **Network Switch** connects devices within a LAN and directs data packets specifically to the destination port; a **Router** directs data packets between distinct interconnected computer networks at the Network Layer.
30. A **Repeater** operates at the Physical Layer to regenerate and amplify electrical or optical signals to overcome cable attenuation without loss of signal strength.
31. The Internet operates fundamentally on **packet switching** (not circuit switching), breaking messages into discrete packets routed independently using TCP/IP.
32. In the **OSI 7-Layer Model**, the layers from 1 to 7 are: **1. Physical \to 2. Data Link \to 3. Network \to 4. Transport \to 5. Session \to 6. Presentation \to 7. Application**. The Data Link Layer sits precisely between Physical and Network layers.
33. **Search Engines:** Google, Bing (Microsoft, 2009), Baidu (China), DuckDuckGo (privacy-focused, no personal profiling), Kurrently (social media real-time search engine). **Orkut was a social networking site, NOT a search engine.**
34. A **Web Crawler (Spider / Spiderbot):** An automated bot that systematically traverses hyperlinks across the World Wide Web for indexing and search engine optimization.
35. **VIRUS** stands for **Vital Information Resources Under Siege**; an intentionally destructive computer program that infects executable files and replicates itself across systems.
36. **Worm:** A standalone self-replicating malware program that spreads across computer networks automatically without requiring a host file or human intervention.
37. **Trojan Horse:** A malware disguised as legitimate, harmless software (e.g. games or utilities) that tricks the user into installing it; **Trojans do NOT replicate or inject themselves into other files**.
38. **Ransomware:** Malicious software that encrypts user data and demands cryptocurrency payment for decryption keys; major global strains include **WannaCry, Petya, and EternalBlue** (exploiting Windows SMB vulnerabilities).
39. **Pegasus Spyware:** Highly invasive surveillance spyware developed by the **NSO Group (Israel)**, capable of zero-click infections on iOS and Android devices to access messages, microphones, and cameras.
40. **Stuxnet:** A sophisticated cyberwarfare worm detected in 2010 that targeted industrial supervisory control and data acquisition (SCADA) systems at the **Bushehr nuclear enrichment facility in Iran**, physically destroying uranium gas centrifuges.

41. A **Firewall** monitors and filters incoming and outgoing network traffic based on predetermined security access rules, preventing unauthorized external access to private networks.
42. **Denial of Service (DoS) / DDoS:** A cyberattack that intentionally floods a server with millions of spoofed requests from distributed botnets, exhausting bandwidth and denying legitimate user access.
43. **Black Hat Hackers (Crackers)** breach computer security unlawfully for malicious intent, data theft, or financial fraud; **White Hat Hackers (Ethical Hackers)** uncover system vulnerabilities to assist organizations in strengthening defensive security.
44. **MS-DOS** was released by Microsoft in **August 1981** as a non-graphical **Command Line Interface (CLI)** operating system.
45. **UNIX** was developed in **1969 at AT&T Bell Labs** by Ken Thompson, Dennis Ritchie, and others; its central core is the **Kernel**, which controls memory allocation, process scheduling, and hardware access.
46. **GNU Project:** Initiated in **1983 by Richard Stallman** at MIT to build a completely free, open-source Unix-compatible operating system ("GNU's Not Unix").
47. Programming Language Generations: **1GL = Machine Code (0s and 1s)**; **2GL = Assembly Language (Mnemonics)**; **3GL = High-Level Procedural (C, C++, Java, Python, FORTRAN, COBOL, BASIC)**; **4GL = Declarative / Query (SQL, 4GL tools)**; **5GL = AI / Logic Programming (Prolog, OPS5, Mercury)**.
48. **John McCarthy** coined the term **Artificial Intelligence** in 1955 and organized the 1956 Dartmouth Conference; recognized globally as the **"Father of Artificial Intelligence"**.
49. The **Turing Test**, originally proposed as the **"Imitation Game"** by Alan Turing in 1949, evaluates whether a machine can exhibit conversational behaviour indistinguishable from a human.
50. **Blockchain** is a decentralized, distributed, cryptographically secured public ledger; under **Web 3.0**, blockchain enables user data sovereignty, decentralized finance (DeFi), and censorship-resistant social networks.

📋 Must-Score Master Tables

Table 1: Computer Generations Summary Matrix

Generation	Era	Primary Hardware Switching Logic	Primary Memory / Storage	Programming Languages	Representative Systems
I	1940–1956	**Vacuum Tubes (Thermionic Valves)**	Magnetic Drums, Mercury Delay Lines	Machine Language, Binary code	ENIAC, EDVAC, EDSAC, UNIVAC-I
II	1956–1963	**Discrete Transistors**	Magnetic Core Memory, Magnetic Tapes	Early High-Level: FORTRAN, COBOL	IBM 1620, IBM 7094, UNIVAC 1107
III	1964–1971	**Integrated Circuits (SSI & MSI ICs)**	Semiconductor Memory, Magnetic Disks	Operating Systems, BASIC, Pascal	IBM System/360, PDP-8 Minicomputer
IV	1971–Present	**Microprocessors (LSI & VLSI Chips)**	Semiconductor RAM/ROM, Hard Disks	High-Level, C, C++, SQL, GUI OS	Personal Computers (PCs), Apple Macintosh
V	Present & Beyond	**ULSI, Parallel Processing, Quantum Chips**	Optical Storage, High-Speed SSDs, Cloud	AI Languages, Python, Java, Prolog	Supercomputers, Notebooks, AI Systems

Table 2: Programming Language Generations

Generation	Taxonomy Category	Operational Characteristics	High-Yield Must-Score Examples
1GL	Machine Language	Directly executed by CPU; pure binary strings (0s and 1s); architecture-dependent	Native machine code
2GL	Assembly Language	Low-level symbolic code using human-readable mnemonics; translated by an Assembler	Assembly Language (x86, ARM)

Generation	Taxonomy Category	Operational Characteristics	High-Yield Must-Score Examples
3GL	High-Level Languages	English-like procedural / object-oriented syntax; translated by Compilers or Interpreters	FORTRAN, COBOL, BASIC, C, C++, Java, Python
4GL	Non-Procedural Languages	Declarative syntax stating "what" result is desired rather than "how" to compute it	SQL, Report Generators, QBE
5GL	AI / Logic Programming	Constraint- and logic-based rule programming used for artificial intelligence	**Prolog, OPS5, Mercury**

Table 3: Digital Memory Conversion Ladder

Storage Unit	Equivalent in Bits / Bytes	Exact Calculation	Common Applications
Bit (b)	0 or 1	Single binary digit	Elementary unit of data
Nibble	4\text{ bits}	Half a Byte	BCD representation
Byte (B)	8\text{ bits}	1\text{ Byte}	Base memory addressing unit
Kilobyte (KB)	1,024\text{ Bytes}	2^{10}\text{ Bytes} = 8,192\text{ bits}	Small text documents
Megabyte (MB)	1,024\text{ KB}	2^{20}\text{ Bytes} = \mathbf{1,048,576\text{ Bytes}}	MP3 audio files (\sim 5\text{ MB})
Gigabyte (GB)	1,024\text{ MB}	2^{30}\text{ Bytes} \approx 1.07 \times 10^9\text{ Bytes}	System RAM, High-Def movies
Terabyte (TB)	1,024\text{ GB}	2^{40}\text{ Bytes} \approx 1.10 \times 10^{12}\text{ Bytes}	Secondary Hard Drives (HDDs/SSDs)
Petabyte (PB)	1,024\text{ TB}	2^{50}\text{ Bytes}	Enterprise Data Centres, Cloud Storage
Exabyte (EB)	1,024\text{ PB}	2^{60}\text{ Bytes}	Global Internet monthly traffic
Zettabyte (ZB)	1,024\text{ EB}	2^{70}\text{ Bytes}	Worldwide aggregate digital data
Yottabyte (YB)	1,024\text{ ZB}	2^{80}\text{ Bytes}	Global data projections

Table 4: OSI 7-Layer Reference Model vs TCP/IP

Layer #	OSI Layer Name	Primary Function	Data Unit	Protocols & Hardware Devices
7	**Application**	User-facing network services & interfaces	Data	HTTP, HTTPS, FTP, SMTP, DNS, Telnet
6	**Presentation**	Data formatting, encryption, and compression	Data	SSL/TLS, JPEG, MPEG, ASCII
5	**Session**	Establishes, manages, and terminates dialogs	Data	NetBIOS, RPC, Sockets
4	**Transport**	End-to-end reliable data delivery & flow control	Segment	TCP (Reliable), UDP (Fast/Streaming)
3	**Network**	Logical addressing and shortest path routing	**Packet**	**IP (IPv4, IPv6)**, ICMP; **Routers**
2	**Data Link**	Physical frame transmission & error detection	**Frame**	**MAC Address (48\text{-bit})**, Ethernet; **Switches, Bridges**
1	**Physical**	Raw bit stream transmission over physical medium	**Bits**	Cables, Connectors, **Repeaters, Hubs**

Table 5: India's Supercomputing Installations (NSM & Legacy)

Supercomputer	Host Institution	Computing Power	Key Applications
PowerEdge XE 9680	Shakti Cloud, Yotta Data Services, Mumbai	**84.31\text{ Petaflops}**	Commercial Cloud AI workloads (Fastest installed in India)
AIRAWAT-PSAI	C-DAC, Pune	**8.50\text{ Petaflops}**	National AI compute platform (Fastest indigenously developed)
PARAM Pravega	IISc, Bengaluru	3.3\text{ Petaflops}	High-performance academic research simulations
PARAM Kamrupa	IIT-Guwahati	838\text{ TFLOPS}	Weather modeling, bioinformatics, high-power microwaves
PARAM Ganga	IIT-Roorkee	1.66\text{ Petaflops}	Computational fluid dynamics, civil engineering modeling
PARAM Shivay	IIT-BHU	833\text{ TFLOPS}	First supercomputer installed under NSM
Pratyush & Mihir	IITM Pune & NCMRWF Noida	6.8\text{ Petaflops} (Combined)	Monsoon forecasting, cyclone prediction, climate research
ANUPAM Series	Bhabha Atomic Research Centre (BARC)	Multi-node parallel systems	Nuclear reactor design, molecular dynamics, atomic research

One-Liner Revision – Electronics, Semiconductors and Computers

#	Fast Fact
1	Silicon (Si) = most widely used semiconductor; Germanium (Ge) = used in early devices
2	N-type semiconductor = doped with phosphorus/arsenic (extra electrons)
3	P-type semiconductor = doped with boron/indium (holes = positive charge carriers)
4	Diode = PN junction; allows current in forward bias only; used in rectifiers, LEDs
5	LED = Light Emitting Diode; converts electrical energy to light; more efficient than bulb
6	Solar cell = photovoltaic diode; converts light to electricity
7	Transistor invented = **1947** (Bell Labs; Bardeen, Brattain, Shockley); Nobel 1956
8	Integrated Circuit (IC) invented = **1958** (Jack Kilby at TI; Nobel 2000)
9	IC generations: SSI, MSI, LSI, VLSI, ULSI (Ultra-large scale integration)
10	Moore's Law: transistor count on chip doubles every ~2 years (Gordon Moore, 1965)
11	1 byte = 8 bits; 1 KB = 1024 bytes; 1 MB = 1024 KB; 1 GB = 1024 MB
12	Data sizes: bit < nibble(4 bits) < byte(8) < KB < MB < GB < TB < PB < EB
13	Binary: 0,1; Octal: 0-7 (base 8); Decimal: 0-9 (base 10); Hexadecimal: 0-9,A-F (base 16)
14	CPU = ALU (Arithmetic Logic Unit) + CU (Control Unit) + Registers
15	RAM = volatile (lost on power off); ROM = non-volatile (permanent)
16	Cache memory = fastest memory; SRAM; between CPU and RAM
17	First computer (ENIAC) = 1945; used vacuum tubes; weighed 30 tons
18	1st generation = vacuum tubes; 2nd = transistors; 3rd = ICs; 4th = microprocessors; 5th = AI
19	Microprocessor invented = **1971** (Intel 4004; Federico Faggin)
20	Internet began as **ARPANET** (1969, USA; Defense department)
21	WWW (World Wide Web) invented = **Tim Berners-Lee (1989)**; CERN, Switzerland

#	Fast Fact
22	.com domain = commercial; .org = organization; .gov = government; .edu = education
23	GPS = Global Positioning System; 24+ satellites; US DoD; free civilian use
24	Nanotechnology = 1-100 nm scale; applications: drug delivery, sensors, materials
25	Quantum bits (qubits) = can be 0, 1, or both simultaneously (superposition)
26	5G frequency bands = sub-6 GHz + mmWave (above 24 GHz)
27	Optical fibre internet = uses light pulses through silica glass; faster than copper
28	Blockchain = distributed ledger; each block cryptographically linked; basis of cryptocurrency
29	Deep Blue (IBM, 1997) = first chess computer to defeat world champion (Kasparov)
30	ChatGPT (OpenAI, 2022) = large language model AI; generative AI milestone

Physics Topic 8 — Nuclear and Atomic Physics

Consolidated — 50 Must-Score Facts

1. **Nuclear fission** is the splitting of a heavy atomic nucleus (e.g. U-235, Pu-239) into two medium-sized fragments of comparable mass upon absorbing a thermal (slow) neutron.
2. In each fission of a Uranium-235 nucleus, approximately **200\text{ MeV}** of energy is liberated, along with 2 to 3 fast neutrons and energetic gamma rays.
3. Fission of 1\text{ gram} of Uranium-235 yields approximately **5 \times 10^{23}\text{ MeV} of energy**, equivalent to detonating **20\text{ tonnes of TNT}**.
4. Most of the energy released during nuclear fission appears as the **kinetic energy (thermal heat) of the fission fragments**.
5. When a neutron is captured by a heavy nucleus without causing fission, it forms a heavier isotope via **radiative neutron capture**, not fission.
6. An **Atomic Bomb** utilizes an **uncontrolled chain reaction** of fast neutrons in highly enriched U-235 or Pu-239.
7. **Critical Mass** is the minimum quantity of fissile material required to sustain a self-propagating nuclear chain reaction.
8. The atomic bomb dropped on Hiroshima ("Little Boy", Aug 6, 1945) used **enriched Uranium-235** in a gun-type assembly; the bomb dropped on Nagasaki ("Fat Man", Aug 9, 1945) used **Plutonium-239** in an implosion assembly.
9. A **Nuclear Reactor (Atomic Pile / Molecular Furnace)** sustains a **controlled, self-propagating chain reaction** for peaceful power generation and radioisotope production.
10. The world's first artificial nuclear reactor was built in 1942 at the University of Chicago by **Enrico Fermi** ("Chicago Pile-1").
11. **Nuclear Reactor Fuel:** Fissile materials such as **Uranium-235 (^{235}_{92}\text{U})**, **Plutonium-239 (^{239}_{94}\text{Pu})**, or synthetic **Uranium-233 (^{233}_{92}\text{U})**.
12. **Moderator:** Slows down energetic fast fission neutrons (2\text{ MeV}) to thermal speeds (0.025\text{ eV}) through elastic collisions so they can induce subsequent fission.
13. Common moderators: **Light water (\text{H}_2\text{O}, most common globally)**, **Heavy water (\text{D}_2\text{O})**, **High-purity Graphite**, and **Beryllium oxide (BeO)**.
14. Heavy water (\text{D}_2\text{O}) is the superior moderator because deuterium has an exceptionally **low neutron absorption cross-section**, permitting the use of un-enriched **natural uranium** as fuel (as in Indian PHWRs).
15. **Control Rods:** Absorb surplus slow-moving thermal neutrons to regulate or halt the fission rate. Inserting control rods dampens the chain reaction; withdrawing them increases reactivity.
16. Control rods are fabricated from neutron-absorbing materials: **Cadmium (Cd) or Boron (B)** (often as boron steel or boron carbide).
17. **Cadmium is NOT a nuclear fuel**; it is strictly a neutron absorber used in control rods and radiation shielding.
18. **Coolant:** Extracts intense heat generated in the reactor core; typical coolants include light water, heavy water, liquid sodium metal (in fast breeder reactors), \text{CO}_2 gas, and helium.
19. **Nuclear Fusion:** The thermonuclear coalescence of two or more light nuclei at extreme velocity/temperature to form a heavier, more tightly bound nucleus, with a mass deficit converted to energy (E = \Delta m c^2).
20. The energy released per unit mass in **nuclear fusion is vastly greater than in nuclear fission**.
21. Nuclear fusion powers the **Sun and all stars**, primarily through the proton-proton chain and CNO cycle at core temperatures exceeding **1.5 \times 10^7\text{ K}**.
22. The theoretical mechanism of stellar thermonuclear energy generation was formulated in 1938 by **Hans Albrecht Bethe** (Nobel Prize 1967).

23. The **Hydrogen Bomb (Thermonuclear Weapon)** operates on the principle of **uncontrolled nuclear fusion** of heavy hydrogen isotopes (deuterium and tritium).
24. Because nuclear fusion requires activation temperatures of millions of degrees, an **atomic fission bomb serves as the trigger (igniter)** inside a hydrogen bomb.
25. The "Father of the Hydrogen Bomb" is Hungarian-American physicist **Edward Teller**; the first full-scale thermonuclear test ("Ivy Mike") was detonated by the USA in 1952 at Eniwetok Atoll.
26. Controlled thermonuclear fusion research utilizes **Tokamaks**—toroidal magnetic confinement chambers that trap superheated plasma using helical magnetic fields (e.g. ITER project in Cadarache, France; SST-1 and ADITYA at IPR Gandhinagar, India).
27. **Mesons** were theoretically predicted in 1935 by Japanese physicist **Hideki Yukawa** as carriers of the strong nuclear force binding nucleons together.
28. The **Positron (e^+)**, the antiparticle of the electron, was discovered in cosmic rays in 1932 by **Carl D. Anderson and Victor Hess** (Nobel Prize 1936).
29. **Transuranic elements** (elements with atomic number Z > 92, beyond Uranium) were synthesized primarily by **Glenn T. Seaborg** and his team at UC Berkeley (Neptunium, Plutonium, Americium, Curium), NOT by Enrico Fermi.
30. A **Cyclotron** is a particle accelerator invented by **Ernest O. Lawrence** that accelerates **charged particles (protons, deuterons, alpha particles)** to high kinetic energies using alternating electric fields and a static magnetic field.
31. A cyclotron **cannot accelerate neutrons** because neutrons are electrically neutral and experience no Lorentz force.
32. A **Scintillation Counter** is an optical radiation detector that utilizes the luminescent flashes produced when ionizing radiation strikes a scintillator crystal (e.g. NaI); used to detect **gamma-emitting radioisotopes in drinking water**.
33. **Ernest Rutherford** famously expressed skepticism before the Royal Society that atomic energy would ever be harnessed industrially, misjudging the commercial viability of mass-energy conversion.
34. Two German physicists (Gunter Nimtz and Alfons Stahlhofen) claimed that microwave photons exhibited superluminal tunneling velocities, attempting to rebut Einstein's speed-of-light postulate.
35. **Radioactivity** is an intrinsic nuclear property of unstable isotopes that undergo spontaneous decay by emitting alpha particles (^4_2\text{He}), beta particles (electrons/positrons), and gamma rays.
36. **Monazite sand** is an anhydrous phosphate of thorium and rare-earth elements, representing India's primary nuclear fuel resource (13.15\text{ million tonnes}).
37. **Thorium (Th-232)** is fertile, not fissile; it cannot sustain a chain reaction directly. It must capture a neutron in a breeder reactor to transmute into fissile **Uranium-233 (^{233}_{92}\text{U})**.
38. **Advantages of Thorium over Uranium:**
    - Thorium is \sim 3 times more abundant in Earth's crust than uranium.
    - Yields more energy per unit mass of mined ore.
    - Operates in thermal breeder cycles without generating long-lived transuranic actinides (like Plutonium, Americium, Curium).
    - Highly proliferation-resistant (produces no weapon-usable plutonium).
39. Non-nuclear fuel elements commonly tested in PSCs: **Cadmium (absorber), Chromium, Calcium, Lead, Helium (inert gas)**. None of these can be used as nuclear fuels.
40. **Enriched Uranium** refers to natural uranium that has undergone isotopic separation (via gas centrifuges or gaseous diffusion) to artificially elevate the concentration of fissile U-235 above its natural 0.7\%.
41. Nuclear power reactors operating in India: **Tarapur (Maharashtra), Rawatbhata (Rajasthan), Kalpakkam (Tamil Nadu), Narora (Uttar Pradesh), Kakrapar (Gujarat), Kaiga (Karnataka), Kudankulam (Tamil Nadu)**.
42. **Kudankulam Nuclear Power Plant** utilizes Russian-designed **VVER-1000** light-water pressurized water reactors (PWRs), using enriched uranium and light water.

43. **Alpha decay (\alpha):** Atomic number decreases by 2 (Z \to Z - 2) and mass number decreases by 4 (A \to A - 4).
44. **Beta-minus decay (\beta^-):** A nuclear neutron converts into a proton, emitting an electron and an antineutrino; atomic number increases by 1 (Z \to Z + 1), mass number A remains constant (produces **isobars**).
45. **Gamma emission (\gamma):** High-energy electromagnetic photon emitted when an excited nucleus de-excites to ground state; Z and A remain completely unchanged.
46. Penetrating power sequence: **\gamma (Gamma) > \beta (Beta) > \alpha (Alpha)**. Ionizing power sequence: **\alpha > \beta > \gamma**.
47. **Radioactive Half-Life (T_{1/2}):** Time required for half of the radioactive nuclei in a sample to disintegrate: T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}.
48. **Carbon-14 Dating:** Used to date organic archaeological specimens up to \sim 50,000\text{ years}; half-life of C-14 is **5,730\text{ years}**.
49. **Uranium-Lead Dating:** Used to determine the absolute age of geological rocks, meteorites, and the Earth (\sim 4.54\text{ billion years}).
50. Medical radioisotopes: **Cobalt-60** (cancer radiotherapy), **Iodine-131** (thyroid disorders and cancer), **Sodium-24** (circulatory blood clots), **Phosphorus-32** (leukemia and blood disorders), **Technetium-99m** (diagnostic imaging).

📋 Must-Score Master Tables

Table 1: Operating Nuclear Power Stations in India

Power Station	Location & State	Reactor Type	Installed Capacity / Units	Critical Must-Score Notes
Narora (NAPS)	Bulandshahr, **Uttar Pradesh**	PHWR	2 \times 220\text{ MWe} = 440\text{ MWe}	**Only commercial nuclear plant in UP**; banks of Ganga
Tarapur (TAPS)	Palghar, **Maharashtra**	BWR & PHWR	2 \times 160 + 2 \times 540 = 1,400\text{ MWe}	**Oldest nuclear power plant in India** (1969; US collaboration)
Rawatbhata (RAPS)	Chittorgarh, **Rajasthan**	PHWR	6\text{ units} = 1,180\text{ MWe}	Established with Canadian collaboration (CANDU design)
Kalpakkam (MAPS)	Chengalpattu, **Tamil Nadu**	PHWR & PFBR	2 \times 220 + 500\text{ (PFBR)} = 940\text{ MWe}	Home to **Indira Gandhi Centre for Atomic Research (IGCAR)** & PFBR
Kakrapar (KAPS)	Surat, **Gujarat**	PHWR	2 \times 220 + 2 \times 700 = 1,840\text{ MWe}	Unit-3 & 4 are India's first indigenous **700\text{ MWe} PHWRs**
Kaiga (KGS)	Uttara Kannada, **Karnataka**	PHWR	4 \times 220\text{ MWe} = 880\text{ MWe}	Holds world record for continuous operation (962\text{ days})
Kudankulam (KKNPP)	Tirunelveli, **Tamil Nadu**	VVER (PWR)	2 \times 1000\text{ MWe} = 2,000\text{ MWe}	**Largest operating nuclear station in India**; Russian VVER-1000
Gorakhpur (GHAVP)	Fatehabad, **Haryana**	PHWR	4 \times 700\text{ MWe} (Under construction)	Key upcoming northern nuclear capacity addition

Table 2: Nuclear Discoveries & Pioneers Matrix

Phenomenon / Particle	Discoverer / Theorist	Year	High-Yield Civil Must-Score Significance
Radioactivity	Henri Becquerel	1896	Spontaneous emission of rays from uranium salts
Polonium and Radium	Marie Curie & Pierre Curie	1898	Isolated highly radioactive elements from pitchblende
Nuclear Fission	Otto Hahn & Fritz Strassmann	1938	Splitting of uranium by thermal neutrons (interpreted by Lise Meitner)
First Nuclear Reactor	Enrico Fermi	1942	Built Chicago Pile-1; proved sustained controlled fission

Phenomenon / Particle	Discoverer / Theorist	Year	High-Yield Civil Must-Score Significance
Meson Theory	**Hideki Yukawa**	1935	Strong nuclear force carrier in atomic nucleus
Positron (e^+)	**Carl D. Anderson & Victor Hess**	1932	First discovered antiparticle (antimatter)
Stellar Energy Theory	**Hans Albrecht Bethe**	1938	Proven thermonuclear fusion reactions powering Sun
Transuranic Elements	**Glenn T. Seaborg**	1940+	Synthesized elements beyond uranium (Z > 92, Plutonium)
Cyclotron	**Ernest O. Lawrence**	1932	Circular particle accelerator for positive ions
Hydrogen Bomb	**Edward Teller**	1952	Thermonuclear weapon utilizing fusion of deuterium/tritium

Table 3: Comparative Properties of Nuclear Radiations

Property	Alpha Particles (\alpha)	Beta Particles (\beta)	Gamma Rays (\gamma)
Physical Nature	Helium-4 Nucleus (^4_2\text{He}^{2+})	Fast electrons (e^-) or positrons (e^+)	High-energy EM Photons
Charge	+2e	-1e (or +1e)	**Zero (0)**
Rest Mass	4\text{ amu} (6.64 \times 10^{-27}\text{ kg})	1/1836\text{ amu} (9.11 \times 10^{-31}\text{ kg})	**Zero (0)**
Velocity	\sim 0.05 - 0.1\, c	Up to 0.99\, c	**Exactly c (3 \times 10^8\text{ m/s})**
Ionizing Power	**Highest (10,000)**	Moderate (100)	Lowest (1)
Penetrating Power	**Lowest (stopped by paper / skin)**	Moderate (stopped by few mm aluminium)	**Highest (requires thick lead/concrete)**
Deflection in Fields	Deflected slightly towards negative plate	Deflected strongly towards positive plate	**No deflection in electric/magnetic fields**

One-Liner Revision – Nuclear and Atomic Physics

#	Fast Fact
1	Radioactivity discovered by = **Henri Becquerel (1896)**; Nobel 1903 (with Curies)
2	Marie Curie = discovered Polonium and Radium; 2 Nobel Prizes (1903 Physics, 1911 Chemistry)
3	Rutherford's nuclear model = nucleus-centred atom; alpha scattering experiment (1909)
4	Bohr's model = electrons in fixed orbits (energy shells); explains hydrogen spectrum
5	Electron discovered = **J.J. Thomson (1897)**; cathode ray tube experiment
6	Proton discovered = **Ernest Rutherford (1917)**; Nobel 1908
7	Neutron discovered = **James Chadwick (1932)**; Nobel 1935
8	Positron (anti-electron) discovered = **Carl Anderson (1932)**; Nobel 1936
9	Atomic number (Z) = number of **protons**; unique to each element
10	Mass number (A) = protons + neutrons; isotopes have same Z, different A
11	Alpha particles: stopped by **paper / skin**; most ionising; dangerous if inhaled
12	Beta particles: stopped by **aluminium sheet** (few mm)
13	Gamma rays: stopped by **thick lead** (several cm) or thick concrete
14	Half-life of C-14 = **5730 years** (radiocarbon dating of organic matter)
15	Half-life of U-235 = 700 million years; U-238 = 4.5 billion years

#	Fast Fact
16	Nuclear fission: U-235 + neutron -> Ba + Kr + 3 neutrons + 200 MeV
17	Critical mass = minimum fissile material for self-sustaining chain reaction
18	Atom bomb (Little Boy) = U-235; (Fat Man) = Pu-239 (Nagasaki/Hiroshima, 1945)
19	Nuclear fusion: D + T -> He-4 + neutron + 17.6 MeV; requires >10^7 K
20	Hydrogen bomb (thermonuclear bomb) = fusion bomb; triggered by fission bomb
21	Einstein's mass-energy equivalence: E = mc^2
22	Nuclear reactor: controlled fission; moderator (heavy water/graphite) slows neutrons
23	India's 1st nuclear test = **Pokhran (1974)**; "Smiling Buddha"
24	India's 2nd nuclear test = **Pokhran-II (1998)**; "Operation Shakti"
25	India's nuclear doctrine = **No First Use (NFU)** + credible minimum deterrence
26	Chernobyl disaster (1986) = RBMK reactor explosion; Ukraine (Soviet Union)
27	Fukushima disaster (2011) = tsunami-induced meltdown; Japan
28	ITER project = International Thermonuclear Experimental Reactor; France; fusion experiment
29	Radiocarbon dating = measures C-14 decay; used for dating up to ~50,000 years
30	X-rays discovered = **Wilhelm Roentgen (1895)**; Nobel 1901 (first ever)

Physics Topic 9 — Scientists, Discoveries, Space, Defence & Applied Science

Consolidated — 50 Must-Score Facts

1. **DRDO Genesis & Motto:** Established in **1958** by amalgamating Technical Development Establishment (TDEs), DTDP, and Defence Science Organisation (DSO). Motto: ***"Balasya Mulam Vigyanam"*** ("The source of strength is science").
2. **Integrated Guided Missile Development Programme (IGMDP):** Conceived by **Dr. A.P.J. Abdul Kalam** in 1983 and sanctioned on 26 July 1983; completed in March 2012. Developed five indigenous missile systems: **P-A-T-A-N** (Prithvi, Agni, Trishul, Akash, Nag).
3. **Prithvi Missile Family:** Surface-to-surface, single-stage liquid-propellant ballistic missiles.
4. **Agni Missile Hierarchy:** Surface-to-surface ballistic missile family:
5. **Trishul:** Short-range (9 km), quick-reaction, supersonic surface-to-air missile developed under IGMDP; officially closed in 2008 due to guidance limitations.
6. **Akash:** Medium-range (25–30 km) mobile surface-to-air missile (SAM) system using an air-breathing **ramjet rocket** propulsion system; guided by indigenous **Rajendra 3D phased array radar**.
7. **Nag & HELINA / Dhruvastra:** Third-generation, 'fire-and-forget' anti-tank guided missile (ATGM). Employs Imaging Infrared (IIR) passive seeker. Land range: **500 m to 4 km**. Air-launched helicopter version: **HELINA / Dhruvastra** (range **7–8 km**).
8. **Amogha ATGM Series:** Developed by Bharat Dynamics Limited (BDL), Hyderabad. Amogha-III is a 3rd-gen man-portable fire-and-forget ATGM with a tandem HEAT warhead (range 200 m to 2.5 km).
9. **BrahMos Supersonic Cruise Missile:** Joint venture between India (DRDO) and Russia (NPO Mashinostroyeniya), named after **Brahmaputra and Moskva** rivers.
10. **Nirbhay Cruise Missile:** Indigenously designed long-range, all-weather, **subsonic cruise missile** (Mach 0.7, range **1,000 km**); uses a solid booster and a small turbofan engine.
11. **Pralay Missile:** Battlefield quasi-ballistic solid-propellant missile (range **150–500 km**) capable of performing evasive maneuvers mid-course to defeat anti-ballistic missile systems.
12. **K-Missile Family (Submarine Launched Ballistic Missiles - SLBMs):** Named in honor of Dr. A.P.J. Abdul Kalam.
13. **Indian Ballistic Missile Defence (BMD) Program:** Double-tiered shield:
14. **Astra Missile:** India's first indigenous Beyond-Visual-Range Air-to-Air Missile (**BVRAAM**), developed by DRDO. Range: 80–110 km (Astra Mk-1); integrated on Su-30 MKI and LCA Tejas.
15. **RudraM Anti-Radiation Missiles:**
16. **LCA Tejas:** Indigenous 4.5th generation, single-engine, compound delta-wing tailless multirole supersonic light combat aircraft; designed by Aeronautical Development Agency (ADA) and manufactured by HAL.
17. **Fighter Jet Generations:**
18. **Nuclear Submarine Fleet:**
19. **Scorpene-Class Conventional Submarines (Project-75):** Built by Mazagon Dock Shipbuilders Limited (MDL) under technology transfer from Naval Group, France:
20. **Aircraft Carriers of Indian Navy:**
21. **Main Battle Tanks (MBT):**
22. **Unmanned Aerial Vehicles (UAVs):**
23. **Radars & Surveillance Systems:**
24. **Foreign Air Defence Systems:**
25. **Multilateral Export Regimes:**
26. **Disarmament & Non-Proliferation Treaties:**

27. **New START Treaty:** Signed in Prague (2010) between USA and Russia; limited deployed strategic warheads to 1,550 and launchers to 700. Expired on **5 February 2026**.
28. **NATO (North Atlantic Treaty Organization):** Formed on 4 April 1949 (Washington Treaty). Collective defence enshrined in **Article 5**. Headquartered in **Brussels, Belgium**. 32 members (Finland joined 2023, Sweden joined March 2024).
29. **Father & Son Nobel Physics Prize:** Sir **William Henry Bragg and William Lawrence Bragg** shared the 1915 Nobel Prize in Physics for crystal analysis using X-ray spectrometry.
30. **Einstein's Miraculous Year (Annus Mirabilis, 1905):** Published 4 papers: Photoelectric Effect (light quanta), Brownian Motion (atomic proof), Special Relativity (speed of light constancy), and Mass-Energy Equivalence (E = mc^2).
31. **Einstein's Nobel Prize (1921):** Awarded specifically for the **Law of the Photoelectric Effect**, *not* for Relativity.
32. **Sir C.V. Raman:** Discovered the **Raman Effect on 28 February 1928** (inelastic scattering of light by molecules); 1930 Nobel Prize in Physics; Bharat Ratna in 1954.
33. **National Days in Science & Technology:**
34. **Acharya J.C. Bose:** Generated millimeter microwaves (60\text{ GHz}) in 1895; invented the **Crescograph** measuring plant physiological responses to stimuli.
35. **Satyendra Nath Bose:** Formulated **Bose-Einstein Statistics** (1924) for integer-spin particles (**Bosons**, named by Dirac); predicted the **Bose-Einstein Condensate (BEC)**.
36. **Meghnad Saha:** Formulated the **Saha Ionization Equation** (1920) linking stellar spectra to temperature and pressure; chaired Calendar Reform Committee (1952) establishing the Saka National Calendar.
37. **Subrahmanyan Chandrasekhar:** Formulated the **Chandrasekhar Limit (1.44 M_\odot)** in 1930; white dwarf stars exceeding 1.44 M_\odot collapse into neutron stars or black holes. 1983 Nobel Prize in Physics.
38. **Dr. Homi J. Bhabha:** Father of Indian Nuclear Program; founded TIFR (1945) and BARC (1954); formulated India's 3-stage nuclear power strategy.
39. **Dr. Vikram Sarabhai:** Father of Indian Space Program; founded PRL (1947), TERLS (1963), and ISRO (1969).
40. **Dr. Raja Ramanna:** Nuclear physicist who directed India's first nuclear test ('Smiling Buddha', 18 May 1974) at Pokhran.
41. **John Bardeen:** Only person to win **two Nobel Prizes in Physics**: 1956 for the **Transistor** (with Brattain and Shockley) and 1972 for the **BCS Theory of Superconductivity**.
42. **Landmark Inventions & Pioneers:**
43. **Black Hole Physics:** Celestial body with escape velocity exceeding light (v_{\text{esc}} > c). Boundary is the **Event Horizon**; central density is the **Singularity**.
44. **Gravitational Waves:** Spacetime ripples predicted by Einstein (1916); first directly observed by **LIGO on 14 September 2015** from a binary black hole merger.
45. **Rocket Propulsion Dynamics:** Governed by **Newton's Third Law** and the **Law of Conservation of Linear Momentum** (m \Delta v = -v_e \Delta m).
46. **Indian Launch Vehicle Fleet:**
47. **Cryogenics Chemistry:** Involves temperatures below **-150^\circ\text{C} (120\text{ K})**. Rocket propellant combination: **Liquid Hydrogen (LH_2, fuel, boiling point -252.9^\circ\text{C})** and **Liquid Oxygen (LOX, oxidizer, boiling point -183.0^\circ\text{C})**.
48. **Ramjet vs Scramjet:** Ramjet slows airflow to **subsonic** speeds inside the burner (Mach 2–5). Scramjet maintains **supersonic** airflow throughout the combustor (velocities **> Mach 6**).
49. **Geostationary vs Geosynchronous Orbits:** Both have a 24-hour orbital period at **35,786\text{ km} (22,236\text{ miles})**. Geostationary is circular and strictly on the equatorial plane (stationary relative to Earth). Geosynchronous is inclined (figure-8 path).
50. **NavIC (IRNSS):** Regional navigation satellite constellation covering India and **1,500 km** beyond its borders, using **7 satellites** (3 in Geostationary Orbit + 4 in Geosynchronous Orbit).

📋 Must-Score Master Tables

Table 1: India's Missile Arsenal & Strategic Delivery Fleet

Missile System	Type / Category	Propulsion & Stages	Operational Range	Warhead Capacity	Strategic Highlights & Induction Status
Prithvi-II	Short-Range Ballistic Missile (SRBM)	Single-stage liquid propellant	250–350 km	500–1,000 kg	Inducted by Strategic Forces Command (SFC); high-accuracy tactical surface-to-surface strike.
Agni-I	Short-Range Ballistic Missile (SRBM)	Single-stage solid propellant	700–1,250 km	1,000 kg	Road/rail-mobile, operational with SFC.
Agni-II	Medium-Range Ballistic Missile (MRBM)	Two-stage solid propellant	2,000–2,500 km	1,000 kg	Advanced navigation, operational with SFC.
Agni-III	Intermediate-Range Ballistic Missile (IRBM)	Two-stage solid propellant	3,000–3,500 km	1,500 kg	High circular error probable (CEP), operational.
Agni-IV	Intermediate-Range Ballistic Missile (IRBM)	Two-stage solid propellant	4,000 km	1,000 kg	Equipped with composite rocket motor and modern avionics.
Agni-V	Intercontinental Ballistic Missile (ICBM)	Three-stage solid propellant	5,000+ km	1,500 kg	Canisterised; tested with **MIRV technology** under **Mission Divyastra (March 2024)**.
Agni P (Prime)	New-Gen Canisterised Ballistic Missile	Two-stage solid propellant	1,000–2,000 km	1,000 kg	Maneuverable Re-entry Vehicle (MaRV); rail-mobile launch tested Sep 2025.
Pralay	Surface-to-Surface Quasi-Ballistic	Single-stage solid propellant	150–500 km	350–700 kg	Battlefield conventional weapon; salvo launch tested Dec 2025; evades interceptor radars.
BrahMos	Supersonic Cruise Missile	Two-stage: Solid booster + liquid ramjet	290 km (original); 450–500 km (ER)	200–300 kg	Speed **Mach 2.8–3.0**; land/sea/air launch; exported to the **Philippines** (2024–2026).
Nirbhay	Subsonic Cruise Missile	Solid booster + small turbofan engine	1,000 km	300 kg	Speed **Mach 0.7**; terrain-hugging, loitering capability.
Astra Mk-1	Beyond-Visual-Range Air-to-Air (BVRAAM)	Solid-fuel smokeless motor	80–110 km	15 kg HE pre-fragmented	Indigenous active radar homing; deployed on Su-30 MKI and LCA Tejas.
Akash	Mobile Surface-to-Air Missile (SAM)	Solid booster + ramjet rocket motor	25–30 km	60 kg	Guided by Rajendra 3D radar; multi-target engagement capability.
Nag	3rd-Gen Anti-Tank Guided Missile (ATGM)	Solid propellant motor	500 m – 4 km	8 kg tandem HEAT	'Fire-and-forget' IIR seeker; NAMICA carrier vehicle. Air-launched: **HELINA / Dhruvastra** (7–8 km).
Sagarika (K-15)	Submarine-Launched Ballistic Missile	Two-stage solid propellant	700–750 km	1,000 kg nuclear/conventional	Deployed on **INS Arihant** SSBN; solidifies India's nuclear triad second-strike.
K-4	Intermediate-Range SLBM	Two-stage solid propellant	3,500 km	2,000 kg nuclear	Developed for Arihant-class submarines.

Missile System	Type / Category	Propulsion & Stages	Operational Range	Warhead Capacity	Strategic Highlights & Induction Status
RudraM-I / II	Anti-Radiation Air-to-Surface Missile	Solid propellant rocket motor	100–300 km	Varies	SEAD weapon to destroy enemy air-defence radars and communication nodes.

Table 2: Multilateral Export Control Regimes & Disarmament Treaties

Regime / Treaty	Established	Core Mandate & Control Focus	Total Members	India's Status & Crucial Key Points
MTCR (Missile Technology Control Regime)	1987 (by G-7)	Non-proliferation of delivery systems for WMD carrying \ge 500\text{ kg} payload over \ge 300\text{ km}.	35 partners	**India joined on 27 June 2016** as 35th partner, enabling BrahMos range extension and predator drone imports.
HCoC (Hague Code of Conduct)	2002	Multilateral transparency code against ballistic missile proliferation; pre-launch notification.	145 subscribing states	**India joined on 1 June 2016**. Qatar became 145th member in 2024.
Australia Group	1985	Harmonisation of export controls to prevent proliferation of **chemical and biological weapons**.	43 members	**India joined on 19 January 2018** as 43rd member.
Wassenaar Arrangement	1996	Transparency and export controls on **conventional arms and dual-use goods/technologies**.	42 participating states	**India joined on 8 December 2017** as 42nd participating state.
CWC (Chemical Weapons Convention)	1993 / 1997	Total prohibition of development, production, and stockpiling of chemical weapons.	193 states parties	India signed 1993, ratified 1996, enacted CWC Act 2000; completely destroyed declared stockpiles.
CCW (Certain Conventional Weapons)	1980 / 1983	Restricts excessively injurious weapons (landmines, blinding lasers, incendiaries).	126 parties	India ratified in 1984 and is committed to **all five Protocols**.
CTBT (Comprehensive Test Ban Treaty)	1996	Bans all nuclear test explosions everywhere.	Not in force	India **did not sign**, citing discriminatory nature and lack of time-bound universal disarmament.
New START	2010 / 2011	Bilateral US–Russia limits on deployed strategic nuclear warheads (1,550) and ICBMs/SLBMs (700).	USA & Russia	**Officially expired on 5 February 2026**; Russia suspended participation in 2023.
NATO (North Atlantic Alliance)	1949	Collective military defence (**Article 5**: attack against one is attack against all).	32 member states	Headquarters: **Brussels, Belgium**. **Sweden joined as 32nd member on 7 March 2024**; Finland 31st (2023).

Table 3: Landmark Bilateral & Multilateral Military Exercises

Exercise Name	Partner Nation(s)	Service Branch	Typical Locations & Must-Score Significance
Surya Kiran	**India – Nepal**	Army	Alternates between **Pithoragarh (Uttarakhand)** and Saljhandi (Nepal); counter-insurgency and jungle warfare.
Yudh Abhyas	**India – USA**	Army	Alternates; conducted at high-altitude training area in **Auli (Chamoli, Uttarakhand)** and Alaska/Washington.
Mitra Shakti	**India – Sri Lanka**	Army	Counter-terrorism operations; 9th edition Aundh (Pune, 2023), 11th edition Belagavi (Karnataka, Nov 2025).

Exercise Name	Partner Nation(s)	Service Branch	Typical Locations & Must-Score Significance
Khanjar	**India – Kyrgyzstan**	Special Forces	High-altitude counter-terrorism; 12th edition Kyrgyzstan (March 2025), 13th edition Missamari (Assam, Feb 2026).
Dustlik	**India – Uzbekistan**	Army	Semi-urban/counter-terror operations; 7th edition Gurumsaray (Namangan, Uzbekistan, April 2026).
Harimau Shakti	**India – Malaysia**	Army	Jungle warfare; 2025 edition conducted at Mahajan Field Firing Range (Rajasthan).
Maitree	**India – Thailand**	Army	Counter-insurgency drills; 14th edition Umroi (Meghalaya, Sep 2025).
Malabar	**QUAD (India, USA, Japan, Australia)**	Navy	Multinational maritime warfare; 29th edition conducted off **Guam (Nov 2025)**.
JIMEX	**India – Japan**	Navy	Maritime interoperability; 8th edition Yokosuka (Japan, June 2024), 9th edition Yokosuka (Oct 2025).
SLINEX	**India – Sri Lanka**	Navy	Fleet exercise; 12th edition Colombo (Sri Lanka, Aug 2025).
Milan	Multilateral (70+ navies)	Navy	Biennial mega naval exercise hosted by Indian Navy at **Visakhapatnam (13th edition, Feb 2026)**.
AFINDEX	**India – 25 African Nations**	Army	Field training exercise in counter-terror operations; held at Aundh (Pune).
CINBAX	**India – Cambodia**	Army	1st Joint Table Top Exercise conducted at Foreign Training Node, Pune (Dec 2024).

Table 4: Key Inventions, Inventors, Years & Operating Principles

Invention	Inventor(s)	Year / Origin	Underlying Scientific Principle / Operating Law	Key Must-Score Angle
X-Ray Machine	Wilhelm Conrad Röntgen	1895 (Germany)	Deceleration of high-energy electrons colliding with a heavy metal target (Bremsstrahlung)	First-ever Nobel Prize in Physics (1901).
Radio / Wireless Telegraph	Guglielmo Marconi	1901 (Italy/UK)	Propagation and reception of electromagnetic radio waves	Shared 1909 Nobel Prize with Karl Ferdinand Braun.
Telephone	Alexander Graham Bell	1876 (USA)	Conversion of sound wave pressure fluctuations into varying electrical currents	Patent granted March 1876.
Electric Incandescent Lamp	Thomas Alva Edison	1879 (USA)	Joule heating (H = I^2Rt) of carbonized/tungsten filament in an evacuated glass bulb	First commercial electric lighting system (1882).
Steam Engine	James Watt	1769 (Scotland)	Transformation of heat energy into mechanical work (First Law of Thermodynamics)	Significant improvements over Newcomen/Savery engines.
Television (Mechanical)	John Logie Baird	1926 (UK)	Mechanical scanning of optical images using Nipkow discs	First working television system.
Laser	Theodore H. Maiman	1960 (USA)	Stimulated emission of radiation inside an optically pumped ruby crystal rod	Emitted coherent red light at \lambda = 694.3\text{ nm}.
Penicillin	Sir Alexander Fleming	1928 (Scotland)	Biosynthesis of antibacterial metabolite by fungus *Penicillium notatum*	Discovered accidentally; 1945 Nobel Prize in Medicine.
Dynamite	Alfred Nobel	1867 (Sweden)	Stabilization of liquid nitroglycerin absorbed into porous diatomaceous earth (kieselguhr)	Founded the Nobel Foundation in his will (1900).
Transistor	Bardeen, Brattain, Shockley	1947 (USA)	Electronic amplification and switching across semiconductor p-n junctions	1956 Nobel Prize in Physics; replaced vacuum tubes.

Invention	Inventor(s)	Year / Origin	Underlying Scientific Principle / Operating Law	Key Must-Score Angle
Integrated Circuit (Microchip)	Jack Kilby & Robert Noyce	1958/59 (USA)	Fabrication of entire circuits (transistors, resistors, capacitors) on a single silicon substrate	Kilby won 2000 Nobel Prize in Physics.
Diode Valve	John Ambrose Fleming	1904 (UK)	Thermionic emission of electrons from heated cathode to anode in a vacuum tube	First vacuum-tube electronic detector.
Crescograph	Acharya J.C. Bose	Early 1900s (India)	Electromechanical clockwork gear train magnifying plant tissue movements 10,000\times	Proved plants respond to electrical, chemical stimuli.
Machine Gun (Puckle Gun)	James Puckle	1718 (England)	Manually rotated cylinder mechanism firing multiple rounds ('Defence Gun')	First patented mechanical machine gun.
Radar	Robert Watson-Watt	1935 (Scotland)	Reflection of radio waves from metal targets measuring range and bearing via Doppler shift	Decisive technology for Royal Air Force in WWII.
Cyclotron	Ernest O. Lawrence	1932 (USA)	Acceleration of charged particles by high-frequency alternating electric field in magnetic spiral	1939 Nobel Prize in Physics.

Table 5: India's Polar & Deep Ocean Exploration Stations & Missions

Station / Mission	Region / Sea	Year Commissioned	Coordinating Body / Location	Key Mandate & Features
Dakshin Gangotri	Antarctica (Ice Shelf)	1983–84	NCPOR, Goa	India's 1st Antarctic research base; decommissioned & ice-buried in 1990.
Maitri	Antarctica (Schirmacher Oasis)	1989	NCPOR, Goa	2nd permanent Antarctic base on rocky ice-free ground; fresh-water lake Priyadarshini.
Bharati	Antarctica (Larsemann Hills)	2012	NCPOR, Goa	3rd & newest base; constructed of 134 shipping containers overlooking Prydz Bay.
Himadri	Arctic (Ny-Ålesund, Norway)	2008	NCPOR, Goa	India's 1st Arctic research station; glaciology, atmospheric & microbial research.
IndARC	Arctic (Kongsfjorden, Svalbard)	2014	NCPOR, Goa	India's 1st moored underwater observatory; Arctic teleconnections with Indian monsoons.
Samudrayaan (MATSYA 6000)	Deep Indian Ocean	2026 (target)	NIOT, Chennai / MoES	India's 1st manned deep-ocean submersible; carries 3 scientists to 6,000 m depth.
CANI OFC Project	Bay of Bengal / Andaman Sea	2020	BSNL / MoC	2,312 km undersea optical fibre linking Chennai to Port Blair & 7 islands.

Table 6: Nanotechnology Dimensions, Confinements & 4 Generations

Generation / Category	Dimensional Confinement	Examples & Materials	Core Mechanical / Physical Feature
Quantum Well	1D Confinement (2D Free)	Thin semiconductor films, 2D quantum sheets	Quantized in thickness; free electron motion in plane.
Quantum Wire	2D Confinement (1D Free)	Nanowires, Carbon Nanotubes (CNTs)	Quantized in two spatial directions; 1D electrical transport.
Quantum Dot (QD)	3D Confinement (0D Free)	CdSe, InP nanocrystals (1.5–10.0 nm)	Completely discrete atom-like energy levels; tunable fluorescence.
Generation I (Passive)	Macroscopic dispersion	Aerosols, colloids, ceramics, polymer coatings	High surface area-to-volume ratio; friction dominates forces.

Generation / Category	Dimensional Confinement	Examples & Materials	Core Mechanical / Physical Feature
Generation II (Active)	Targeted bio-physicochemical	Targeted drug delivery, 3D transistors, amplifiers	Bio-conjugation; cellular penetration without metabolic breakdown.
Generation III (Systems)	Self-assembling networks	Nanorobotics, 3D molecular networking	Guided multi-component autonomous functionality.
Generation IV (Molecular)	Atom-by-atom assembly	Molecular manufacturing, nanofactories	Mechanosynthesis; atomically precise engineering.

Table 7: Premier National Research Institutions, Locations & Mandates

Institution Name	Acronym	Headquarters / Location	Founding Architect / Key Milestone	Primary Mandate
National Physical Laboratory	CSIR-NPL	New Delhi	CSIR (1947)	National metrology; custodian of Indian standard measurements & time.
National Chemical Laboratory	CSIR-NCL	Pune, Maharashtra	CSIR (Est. 1950)	Chemical sciences, polymer chemistry, catalysis, process engineering.
Tata Institute of Fundamental Research	TIFR	Mumbai, Maharashtra	Dr. Homi J. Bhabha & Dorabji Tata (1945)	Fundamental physics, mathematics, cosmic rays, nuclear science.
Central Electrochemical Research Institute	CSIR-CECRI	Karaikudi, Tamil Nadu	Alagappa Chettiar, S.S. Bhatnagar, Nehru (1953)	Electrochemistry, batteries, corrosion, South Asia's largest electrochem lab.
Indian Institute of Science	IISc	Bengaluru, Karnataka	Jamshedji Tata & Swami Vivekananda (1909)	Premier institute for advanced scientific and engineering education & research.
National Centre for Earth Science Studies	NCESS	Thiruvananthapuram, Kerala	MoES (Est. 1978 as CESS)	Earth system sciences, geodynamics, coastal and atmospheric processes.
National Institute of Oceanography	CSIR-NIO	Dona Paula, Goa	CSIR (Est. 1966)	Oceanography, marine resources, physical/chemical/biological ocean studies.
National Centre for Polar & Ocean Research	NCPOR	Vasco da Gama, Goa	MoES (Est. 1998)	Nodal agency executing Antarctic, Arctic, Himalayan, & Southern Ocean missions.
Central Institute of Indian Languages	CIIL	Mysuru, Karnataka	Ministry of Education (1969)	Promotion and development of Indian languages.
Central Drug Research Institute	CSIR-CDRI	Lucknow, Uttar Pradesh	CSIR (Est. 1951)	Drug research, biomedical discoveries (developed Centchroman/Saheli).
Aryabhatta Research Inst. of Obs. Sciences	ARIES	Nainital, Uttarakhand	Autonomous body under DST	Observational astronomy, astrophysics, Devasthal 3.6m optical telescope.
Doppler Weather Radar (Mukteshwar)	IMD-DWR	Mukteshwar, Nainital, UK	MoES / IMD (Jan 2021)	Uttarakhand's 1st DWR; severe weather, storm & heavy precipitation monitoring.

Table 8: Interdisciplinary Scientific Disciplines & Core Principles

Discipline	Defining Pioneer / Year	Core Subject Matter & Domain	Must-Score Importance / Real-World Application
Cybernetics	Norbert Wiener (1948)	Scientific study of control and communication in animals and machines.	Foundation of robotics, automated feedback loops, neural networks.
Geodesy	Classical Greek / Modern Geophysics	Measurement of Earth's geometric shape, orientation in space, and gravity field.	Satellite navigation, GPS datum, tectonic deformation monitoring.
Horology	Classical Astronomy & Horologists	Science and art of measuring time and constructing timekeeping devices.	Mechanical clocks, quartz crystal oscillators, atomic clocks.

Discipline	Defining Pioneer / Year	Core Subject Matter & Domain	Must-Score Importance / Real-World Application
Tribology	Peter Jost (1966)	Science of interacting surfaces in relative motion: friction, wear, and lubrication.	Engine efficiency, bearings, industrial lubricants, nanomaterials friction.
Plasma Physics	Irving Langmuir (1928)	4th state of matter: ionized gas containing free ions and electrons.	Fusion reactors (Tokamak), stellar physics (Sun & stars), interstellar medium.
Degenerate Astronomy	R.H. Fowler / S. Chandrasekhar	White dwarfs supported by electron degeneracy pressure (M < 1.44 M_\odot).	Final evolutionary stage of low-to-intermediate mass stars.

One-Liner Revision – Key Scientists and Their Discoveries

#	Scientist	Discovery/Invention	Year
1	**Isaac Newton**	Laws of Motion + Law of Gravitation	1687
2	**Galileo Galilei**	Telescope (improved); heliocentrism; laws of falling bodies	1609
3	**Charles Darwin**	Theory of Evolution by Natural Selection	1859
4	**Wilhelm Roentgen**	X-rays (Nobel 1901)	1895
5	**Henri Becquerel**	Radioactivity (Nobel 1903)	1896
6	**Marie Curie**	Polonium + Radium; 2 Nobel prizes (1903, 1911)	1898
7	**J.J. Thomson**	Electron (Nobel 1906); plum pudding model	1897
8	**Ernest Rutherford**	Nuclear model of atom; proton (Nobel 1908)	1911
9	**Niels Bohr**	Atomic model with electron shells (Nobel 1922)	1913
10	**Albert Einstein**	Special Relativity (E=mc2); Photoelectric effect (Nobel 1921)	1905
11	**Dmitri Mendeleev**	Periodic Table of Elements (1869)	1869
12	**James Chadwick**	Neutron (Nobel 1935)	1932
13	**Enrico Fermi**	First nuclear reactor (Chicago Pile-1, 1942)	1942
14	**Alexander Fleming**	Penicillin (Nobel 1945 with Chain and Florey)	1928
15	**Watson and Crick**	DNA double helix structure (Nobel 1962)	1953
16	**Frederick Sanger**	Insulin protein sequence (Nobel 1958); DNA sequencing (Nobel 1980)	1953/80
17	**Kary Mullis**	PCR (Polymerase Chain Reaction) (Nobel 1993)	1983
18	**Tim Berners-Lee**	World Wide Web (WWW)	1989
19	**C.V. Raman**	Raman Effect (Nobel 1930); first Asian Nobel in Science	1928
20	**S. Chandrasekhar**	Chandrasekhar limit for white dwarfs (Nobel 1983)	1930
21	**Homi J. Bhabha**	Founded India's nuclear programme; TIFR, BARC	1940s-60s
22	**Vikram Sarabhai**	Father of Indian Space Programme; ISRO founder	1969
23	**A.P.J. Abdul Kalam**	IGMDP; Agni/Prithvi missiles; "Missile Man of India"	1980s-90s
24	**M.S. Swaminathan**	Green Revolution in India; HYV seeds; "Father of Green Revolution India"	1960s
25	**Har Gobind Khorana**	Genetic code cracking (Nobel 1968, Medicine)	1968
26	**Yellapragada SubbaRow**	Folic acid, methotrexate, aureomycin	1930s-40s
27	**Edward Jenner**	Smallpox vaccine (first vaccine)	1796
28	**Louis Pasteur**	Germ Theory; pasteurisation; rabies vaccine	1860s-80s
29	**Robert Koch**	Identified TB (*Mycobacterium tuberculosis*) (Nobel 1905)	1882

#	Scientist	Discovery/Invention	Year
30	**Gregor Mendel**	Laws of Heredity (“Father of Genetics”); pea plant experiments	1865

Chemistry Topic 1 — Atomic Structure and Periodic Table

Consolidated — 50 Must-Score Facts

1. **Subatomic Mass & Charge Hierarchy:** - **Electron:** Discovered by **J.J. Thomson (1897)** via cathode ray discharge tubes. Charge = -1.602 \times 10^{-19}\text{ C}; Mass = 9.109 \times 10^{-31}\text{ kg} = 0.000548\text{ u} (approx. **1/1836\text{th} of proton mass**). Charge-to-mass ratio (e/m) = 1.7588 \times 10^{11}\text{ C/kg}. - **Proton:** Observed as canal rays/anode rays by Eugen Goldstein (1886) and identified as the hydrogen nucleus by **Ernest Rutherford (1919)**. Charge = +1.602 \times 10^{-19}\text{ C}; Mass = 1.6726 \times 10^{-27}\text{ kg} = 1.00727\text{ u}. - **Neutron:** Discovered by **James Chadwick in 1932** by bombarding beryllium foils with alpha particles (^9_4Be + ^4_2He \to ^{12}_6C + ^1_0n). Charge = 0 (neutral); Mass = 1.6749 \times 10^{-27}\text{ kg} = 1.00866\text{ u}. **The neutron is the heaviest subatomic particle in an ordinary atom.**
2. **Rutherford's Gold Foil \alpha-Scattering Experiment (1911):** Bombarded a thin gold foil (100\text{ nm} thick) with fast \alpha-particles: - Most \alpha-particles passed straight through undeflected \implies **most of the atomic volume is empty space**. - A few deflected at large angles, and roughly 1 in 20,000 bounced back (180^\circ) \implies all positive charge and nearly all atomic mass is concentrated in a tiny central core called the **Nucleus**. - Nuclear diameter is \approx 10^{-15}\text{ m} (1\text{ Fermi}), whereas atomic diameter is \approx 10^{-10}\text{ m} (1\text{ \AA}). The atom is 100,000\text{ times larger} than its nucleus.
3. **Bohr's Quantum Postulates (1913):** - Electrons revolve around the nucleus only in certain discrete, non-radiating circular orbits called **stationary states**. - Angular momentum of revolving electron is an integral multiple of h / 2\pi: L = mvr = \frac{nh}{2\pi} \quad (n = 1, 2, 3, \dots) - Radiation is emitted or absorbed only when an electron jumps from one stationary orbit to another: \Delta E = E_2 - E_1 = h\nu.
4. **Four Quantum Numbers:** - **Principal Quantum Number (n = 1, 2, 3, \dots / K, L, M, N):** Dictates main energy level, electron shell, and average distance from nucleus. Max electrons in shell = 2n^2. - **Azimuthal / Orbital Angular Momentum (l = 0\text{ to } n-1):** Dictates orbital shape (l=0 \to s spherical; l=1 \to p dumbbell; l=2 \to d double-dumbbell; l=3 \to f complex). - **Magnetic Quantum Number (m_l = -l\text{ to } +l):** Dictates spatial orientation of orbitals in 3D space. Number of orbitals in a subshell = 2l + 1. - **Spin Quantum Number (m_s = +1/2\text{ or } -1/2):** Dictates intrinsic electron spin orientation (clockwise or counter-clockwise).
5. **Aufbau Principle:** Electrons occupy orbitals of the **lowest available energy level first** before occupying higher levels. Governed by the **(n + l) rule**: - Orbitals with lower (n + l) value fill first (e.g., 4s with n+l = 4+0=4 fills before 3d with n+l = 3+2=5). - If two orbitals share the same (n + l) value, the one with lower n fills first (3d fills before 4p).
6. **Pauli's Exclusion Principle (1925): No two electrons in the same atom can possess the exact same set of all four quantum numbers.** Consequently, an orbital can hold a maximum of **two electrons**, and they must have **opposite (antiparallel) spins** (\uparrow\downarrow).
7. **Hund's Rule of Maximum Multiplicity:** Electron pairing in degenerate orbitals belonging to the same subshell (p, d, f) **cannot occur until each orbital is singly occupied with parallel spins** (\uparrow, \uparrow, \uparrow).
8. **Anomalous Electronic Configurations of Chromium and Copper:** - **Chromium (Z=24):** Expected [Ar] 3d^4 4s^2; Actual = [Ar] 3d^5 4s^1. - **Copper (Z=29):** Expected [Ar] 3d^9 4s^2; Actual = [Ar] 3d^{10} 4s^1. - *Reason:* **Completely filled (d^{10}) and exactly half-filled (d^5) subshells possess extra thermodynamic stability** due to symmetrical electron distribution and maximum quantum exchange energy.
9. **Isotopes:** Atoms of the same element having the **same atomic number (Z) but different mass numbers (A)**. Identical chemical properties, different nuclear stability. - Hydrogen Isotopes: Protium (^1_1H, 0 neutrons), Deuterium (^2_1H, 1 neutron), Tritium (^3_1H, 2 neutrons, radioactive).
10. **Isobars:** Atoms of different chemical elements having the **same mass number (A) but different atomic numbers (Z)** (e.g., ^{40}*{18}Ar and ^{40}*{20}Ca). Different chemical and physical properties.
11. **Isotones:** Atoms having the **exact same number of neutrons (N = A - Z)** but different atomic numbers (e.g., ^{14}_6C and ^{16}_8O, both having 8 neutrons).

12. **Isoelectronic Species:** Atoms, molecules, or ions containing the **exact same total number of electrons** (e.g., N^{3-}, O^{2-}, F^-, Ne, Na^+, Mg^{2+}, Al^{3+}, each having 10 electrons).
    - *Ionic Radius Trend:* Among isoelectronic ions, **ionic radius decreases as positive nuclear charge (Z) increases**: N^{3-} (1.71\text{ \AA}) > O^{2-} (1.40\text{ \AA}) > F^- (1.33\text{ \AA}) > Na^+ (1.02\text{ \AA}) > Mg^{2+} (0.72\text{ \AA}) > Al^{3+} (0.53\text{ \AA})
13. **Dobereiner's Law of Triads (1829):** Arranged elements in groups of three with similar chemical properties. The atomic weight of the middle element is approximately the arithmetic mean of the other two:
    - Triad 1: Lithium (7), Sodium (23), Potassium (39) \to (7+39)/2 = 23.
    - Triad 2: Calcium (40), Strontium (88), Barium (137).
    - Triad 3: Chlorine (35.5), Bromine (80), Iodine (127).
14. **Newlands' Law of Octaves (1864):** English chemist John Newlands arranged elements in order of increasing atomic weight; every **eighth element showed properties similar to the first**, analogous to musical musical notes (Sa, Re, Ga, Ma...). **Valid only up to Calcium (Z=20)**.
15. **Mendeleev's Periodic Law (1869):** Russian chemist Dmitri Mendeleev stated: *"The physical and chemical properties of elements are periodic functions of their Atomic Weights."*
    - Contained 63 elements arranged in 8 Groups and 6 Periods.
    - Correctly predicted undiscovered elements and left gaps: **Eka-Boron (Scandium)**, **Eka-Aluminum (Gallium)**, and **Eka-Silicon (Germanium)**.
    - Flaws: Inverted atomic weights (Te before I, Co before Ni); no fixed position for Hydrogen; anomalous placement of isotopes.
16. **Modern Periodic Law (Henry Moseley, 1913):** English physicist Henry Moseley measured characteristic X-ray frequencies (\nu) emitted by elements under electron bombardment, discovering \sqrt{\nu} = a(Z - b). Proved that **Atomic Number (Z) is more fundamental than atomic weight**: *"The physical and chemical properties of elements are periodic functions of their Atomic Numbers."*
17. **Architecture of the Modern Long Form Periodic Table:** Based on Bohr's electronic configurations:
    - **18 Vertical Columns** called **Groups**.
    - **7 Horizontal Rows** called **Periods**.
    - Divided into 4 blocks (s, p, d, f) based on the subshell receiving the differentiating valence electron.
18. **Period Breakdown:**
    - Period 1: Very short (2 elements: H, He).
    - Period 2 & 3: Short (8 elements each: Li to Ne; Na to Ar).
    - Period 4 & 5: Long (18 elements each: K to Kr; Rb to Xe).
    - Period 6: Very long (32 elements: Cs to Rn; includes Lanthanides Z=57\text{–}71).
    - Period 7: Incomplete/complete (32 elements: Fr to Og; includes Actinides Z=89\text{–}103).
19. **Magic Numbers in the Periodic Table:** The electron capacities of successive shells that confer exceptional nuclear and electronic stability: **2, 8, 8, 18, 18, 32**.
20. **Groups and Chemical Families:**
    - Group 1: **Alkali Metals** (Li, Na, K, Rb, Cs, Fr) — configuration ns^1; highly reactive, soft, low melting points, store in kerosene.
    - Group 2: **Alkaline Earth Metals** (Be, Mg, Ca, Sr, Ba, Ra) — configuration ns^2.
    - Groups 3 to 12: **Transition Metals (d-block)** — partially filled d-orbitals; exhibit variable oxidation states, form colored catalytic compounds.
    - Group 15: **Pnictogens** (N, P, As, Sb, Bi).
    - Group 16: **Chalcogens** (O, S, Se, Te, Po) — 'ore-forming' elements.
    - Group 17: **Halogens** (F, Cl, Br, I, At) — 'salt-producers'; ns^2 np^5; highly electronegative.

- Group 18: **Noble Gases / Inert Gases** (He, Ne, Ar, Kr, Xe, Rn, Og) — completely filled ns^2 np^6 (except He 1s^2); zero valency, highest ionization energies.

21. **Inner Transition Elements (f-block):** Placed in two separate rows at the bottom:
    - **Lanthanides (4f series):** 14 elements from Cerium (Z=58) to Lutetium (Z=71), following Lanthanum (Z=57).
    - **Actinides (5f series):** 14 elements from Thorium (Z=90) to Lawrencium (Z=103), following Actinium (Z=89). **All actinides are radioactive.**
22. **Lanthanide Contraction:** The steady, progressive decrease in atomic and ionic radii of Lanthanide elements (La^{3+} \to Lu^{3+}) with increasing atomic number.
    - *Cause:* Extremely poor shielding effect of diffused inner 4f electrons against increasing nuclear charge (Z_{\text{eff}}), pulling the outer 6s shell inward.
    - *Consequence:* Elements of the 2nd and 3rd transition series have virtually identical atomic radii: **Zirconium (Zr, 1.60\text{ \AA}) and Hafnium (Hf, 1.59\text{ \AA})**, and **Niobium (Nb) and Tantalum (Ta)**, making their chemical separation difficult.
23. **Atomic Radius Periodic Trend:**
    - **Across a Period (Left to Right): Decreases**. Principal shell n remains constant while nuclear charge (Z) increases, pulling electrons closer.
    - **Down a Group (Top to Bottom): Increases**. New electron shells are added, increasing screening/shielding.
    - Smallest neutral atom: **Helium (He)** / Hydrogen.
    - Largest neutral atom: **Cesium (Cs)** / Francium.
24. **Ionization Energy / Enthalpy (IE):** Minimum energy required to remove the most loosely bound valence electron from an isolated neutral gaseous atom in its ground state: X(g) + IE_1 \longrightarrow X^+(g) + e^-
    - **Across a Period: Increases** (higher Z_{\text{eff}}, smaller radius).
    - **Down a Group: Decreases** (larger radius, valence electron further away).
    - Highest IE_1 in the Periodic Table: **Helium (He, 2372\text{ kJ/mol})**.
    - Lowest IE_1 in the Periodic Table: **Cesium (Cs, 376\text{ kJ/mol})**.
25. **Ionization Energy Anomalies (High-Yield UPPCS Traps):**
    - **IE_1(\text{Beryllium}) > IE_1(\text{Boron}):** Be (1s^2 2s^2) has a fully filled stable 2s subshell, requiring more energy to remove an electron than B (1s^2 2s^2 2p^1), where removing the single 2p electron leaves a stable 2s^2 core.
    - **IE_1(\text{Nitrogen}) > IE_1(\text{Oxygen}):** N (1s^2 2s^2 2p^3) has an exactly half-filled 2p subshell with extra quantum stability, making its first ionization energy higher than O (1s^2 2s^2 2p^4).
26. **Electron Gain Enthalpy / Electron Affinity (EA):** Energy released when an electron is added to an isolated neutral gaseous atom: X(g) + e^- \longrightarrow X^-(g) + \text{Energy}
    - **Across a Period:** Generally becomes more negative / **increases**.
    - **Down a Group:** Generally **decreases**.
    - **The Fluorine vs Chlorine Trap: Chlorine (Cl) has a HIGHER electron affinity than Fluorine (F)** (Cl = -349\text{ kJ/mol} vs F = -328\text{ kJ/mol}). Fluorine's tiny 2p subshell has extremely high electron density; incoming electrons suffer severe inter-electronic repulsion, lowering affinity. Sequence: **Cl > F > Br > I**.
27. **Electronegativity (EN):** The relative tendency of an atom in a chemical molecule to attract the shared pair of bonding electrons toward itself. Dimensionless property measured on the **Pauling Scale**.
    - **Across a Period: Increases** from alkali metals to halogens.
    - **Down a Group: Decreases**.
    - **Most Electronegative Element: Fluorine (EN = 4.0)**, followed by **Oxygen (3.5)**, **Nitrogen (3.0)**, and **Chlorine (3.0)**.
    - Least Electronegative / Most Electropositive: **Cesium / Francium (0.7)**.

28. **Valency Across a Period:** Valency with respect to hydrogen increases from **1\text{ to } 4 and then decreases back to 0** (NaH=1, MgH_2=2, AlH_3=3, SiH_4=4, PH_3=3, H_2S=2, HCl=1, Ar=0). Valency with respect to oxygen increases from 1\text{ to } 7 (Na_2O \to Cl_2O_7). Valency remains constant down a group.
29. **Metallic vs Non-Metallic Character:**
    - Metallic character (electropositivity, ability to lose electrons) **decreases across a period** and **increases down a group**.
    - Non-metallic character (electronegativity, ability to gain electrons) **increases across a period** and **decreases down a group**.
    - Nature of Oxides: Metallic oxides are **basic** (Na_2O, CaO); non-metallic oxides are **acidic** (SO_2, CO_2, P_2O_5); metalloid/intermediate oxides are **amphoteric** (Al_2O_3, ZnO, PbO).
30. **Diagonal Relationship:** Light elements of the second period share surprising chemical similarities with diagonally adjacent elements of the third period due to nearly identical charge-to-size ratios (ionic potential):
    - **Lithium (Li) resembles Magnesium (Mg)**: Both form normal oxides, decompose carbonates on heating, and have insoluble fluorides.
    - **Beryllium (Be) resembles Aluminum (Al)**: Both form amphoteric oxides and covalently bonded chlorides that sublime.
    - **Boron (B) resembles Silicon (Si)**: Both form covalent polymers, acidic oxides, and hydrides that spontaneously ignite.
31. **Metalloids (Semimetals):** Elements with border-line properties between metals and non-metals, lining the staircase divide: **Boron (B), Silicon (Si), Germanium (Ge), Arsenic (As), Antimony (Sb), Tellurium (Te), and Polonium (Po)**.
32. **Physical State of Elements at Room Temperature (25^\circ\text{C}):**
    - **Liquids (Only Two Elements): Mercury (Hg, metal)** and **Bromine (Br_2, non-metal)**. Elements melting slightly above room temperature: Gallium (29.7^\circ\text{C}), Cesium (28.4^\circ\text{C}), and Francium (27^\circ\text{C}).
    - **Gases (11 Elements):** Hydrogen (H_2), Nitrogen (N_2), Oxygen (O_2), Fluorine (F_2), Chlorine (Cl_2), and all six Noble Gases (He, Ne, Ar, Kr, Xe, Rn).
    - **Solids:** All remaining \approx 105\text{ elements}.
33. **Most Abundant Elements in the Earth's Crust (by Weight):** \text{Oxygen } (46.6\%) > \text{Silicon } (27.7\%) > \text{Aluminum } (8.1\%) > \text{Iron } (5.0\%) > \text{Calcium } (3.6\%) > \text{Sodium } (2.8\%) > \text{Potassium } (2.6\%) > \text{Magnesium } (2.1\%)
    - **Most abundant element in crust: Oxygen (46.6\%, non-metal)**.
    - **Most abundant metalloid in crust: Silicon (27.7\%)**.
    - **Most abundant metal in crust: Aluminum (8.1\%)**, followed by Iron (5.0\%).
34. **Most Abundant Elements in the Human Body (by Weight):** \text{Oxygen } (65\%) > \text{Carbon } (18.5\%) > \text{Hydrogen } (9.5\%) > \text{Nitrogen } (3.2\%) > \text{Calcium } (1.5\%) > \text{Phosphorus } (1.0\%)
    - **Most abundant element in human body: Oxygen (65\%)**.
    - **Most abundant metal in human body: Calcium (1.5\%)**, located primarily in bones and teeth as hydroxyapatite.
35. **Most Abundant Element in the Entire Universe: Hydrogen (\approx 75\%)**, followed by **Helium (\approx 24\%)**.
36. **Most Abundant Gas in the Earth's Atmosphere: Nitrogen (78.08\%)**, followed by **Oxygen (20.95\%)**, **Argon (0.93\%, most abundant noble gas)**, and **Carbon Dioxide (0.04\%)**.
37. **Lightest and Densest Elements:**
    - **Lightest element & gas: Hydrogen (H_2)**.
    - **Lightest metal: Lithium (Li, density 0.534\text{ g/cm}^3)**, floats on water and petroleum.

- **Densest element / metal: Osmium (Os, density 22.59\text{ g/cm}^3)**, closely followed by Iridium (Ir, 22.56\text{ g/cm}^3).

38. **Hardest Natural vs Synthetic Substances:**
    - Hardest natural substance: **Diamond** (allotrope of pure Carbon, Mohs hardness 10).
    - Hardest metal: **Tungsten / Chromium**.
    - Highest melting point of all elements: **Carbon (sublimes at \approx 3825^\circ\text{C})**. Highest melting point among metals: **Tungsten (3422^\circ\text{C})**.
39. **Lowest Melting Point Element: Helium (-272.2^\circ\text{C} = 0.95\text{ K} at 25\text{ atm} pressure)**; cannot freeze into solid at standard atmospheric pressure even at absolute zero.
40. **Coinage Metals:** Group 11 transition elements used historically for minting currency: **Copper (Cu), Silver (Ag), and Gold (Au)**.
41. **Noble Metals:** Metals that resist oxidation, corrosion, and acid attack: **Gold (Au), Platinum (Pt), Silver (Ag), and Rhodium (Rh)**. Dissolve only in **Aqua Regia** (3:1 concentrated HCl and HNO_3).
42. **Strategic Metals:** Metals essential for national defense, aerospace, and energy independence: **Titanium (Ti, 'Metal of the Future'), Zirconium (Zr), Chromium (Cr), Manganese (Mn)**.
43. **Liquid Metal at Room Temperature: Mercury (Hg, atomic number 80)**. Has low melting point (-38.83^\circ\text{C}) due to relativistic contraction of the filled 6s^2 shell resisting metallic bonding.
44. **Best Electrical Conductors: Silver (Ag) > Copper (Cu) > Gold (Au) > Aluminum (Al)**. Silver is not used for domestic power lines solely due to high cost.
45. **Poor Conductors Among Metals: Lead (Pb)** and **Mercury (Hg)** are the poorest conductors of heat and electricity among metals.
46. **Alkali Metals Stored in Liquid:** Sodium (Na) and Potassium (K) react violently with atmospheric moisture and oxygen (2Na + 2H_2O \to 2NaOH + H_2 \uparrow with explosive heat), catching fire. They are preserved by immersing in **kerosene oil**.
47. **White Phosphorus Storage:** White phosphorus (P_4) has a very low ignition temperature (30^\circ\text{C}) and undergoes spontaneous combustion in air, emitting dense white smoke. It is **stored submerged under water** (in which it is insoluble).
48. **Liquid Non-Metal: Bromine (Br_2, atomic number 35)** is the only non-metallic element that exists as a fuming reddish-brown liquid at room temperature (25^\circ\text{C}).
49. **Non-Metal Exhibiting Metallic Luster: Iodine (I_2)** forms dark violet-black lustrous crystals that sublime upon gentle heating into violet vapors. **Diamond** and **Graphite** also exhibit distinctive luster.
50. **Non-Metal That Conducts Electricity: Graphite** (allotrope of Carbon). Each carbon atom is sp^2 hybridized, bonded to three neighbors in planar hexagonal sheets, leaving one **delocalized free \pi-electron** per atom to drift freely under electric potential.

📋 Must-Score Master Tables: Periodic Reference & Abundance

Table 1: Fundamental Subatomic Particles & Atomic Dimensions

Subatomic Particle	Discoverer & Year	Absolute Electric Charge	Relative Charge	Rest Mass (kg)	Rest Mass (u)	Location in Atom
Electron (e^-)	**J.J. Thomson (1897)**	**-1.602 \times 10^{-19}\text{ C}**	-1	**9.109 \times 10^{-31}\text{ kg}**	0.000548\text{ u}	Extra-nuclear orbitals
Proton (p^+)	**Goldstein / Rutherford (1919)**	**+1.602 \times 10^{-19}\text{ C}**	+1	**1.6726 \times 10^{-27}\text{ kg}**	1.00727\text{ u}	Inside nucleus

Subatomic Particle	Discoverer & Year	Absolute Electric Charge	Relative Charge	Rest Mass (kg)	Rest Mass (u)	Location in Atom
Neutron (n^0)	**James Chadwick (1932)**	**0\text{ C} (Neutral)**	0	**1.6749 \times 10^{-27}\text{ kg}**	1.00866\text{ u}	Inside nucleus
Positron (e^+)	Carl Anderson (1932)	+1.602 \times 10^{-19}\text{ C}	+1	9.109 \times 10^{-31}\text{ kg}	0.000548\text{ u}	Antiparticle of electron
Atomic Radius	—	—	—	\approx 10^{-10}\text{ m} = 1\text{ \AA} = 0.1\text{ nm}	Angstrom scale	
Nuclear Radius	Rutherford (1911)	—	—	\approx 10^{-15}\text{ m} = 1\text{ Fermi (fm)}	Fermi scale (10^5\times smaller)	

Table 2: Periodic Table Groups, Periods & Chemical Families

Group Number	Family Common Name	Valence Configuration	Typical Oxidation State	Key Characteristic Chemical Properties
Group 1	**Alkali Metals**	ns^1	+1	Soft, highly reactive, tarnish in air, stored under kerosene.
Group 2	**Alkaline Earth Metals**	ns^2	+2	Basic oxides, flame test colors (Ca brick red, Sr crimson, Ba apple green).
Groups 3–12	**Transition Elements**	(n-1)d^{1-10} ns^{1-2}	Variable (+2\text{ to } +7)	High melting points, paramagnetic, form colored catalytic compounds.
Group 13	Boron Family	ns^2 np^1	+3, +1	Boron is a metalloid; Aluminum is most abundant crust metal.
Group 14	Carbon Family	ns^2 np^2	+4, +2	Catenation ability; Silicon and Germanium are key semiconductors.
Group 15	**Pnictogens**	ns^2 np^3	-3, +3, +5	Nitrogen is inert diatomic gas; half-filled stable configuration.
Group 16	**Chalcogens**	ns^2 np^4	-2, +4, +6	Ore-forming non-metals (oxides and sulfides).
Group 17	**Halogens**	ns^2 np^5	-1	Salt formers; highest electronegativity (F); highest affinity (Cl).
Group 18	**Noble / Inert Gases**	ns^2 np^6	0	Chemically unreactive monoatomic gases; highest ionization energy.

Table 2B: Alpha (\alpha), Beta (\beta) and Gamma (\gamma) Radiation Master Comparison

Parameter / Property	\alpha-Particles	\beta-Particles	\gamma-Rays
Nature	Helium nuclei (^4_2He^{2+}, \alpha^{2+})	High-speed electrons (\beta^-) or positrons (\beta^+)	High-energy electromagnetic photons
Charge	+2 units (+3.204 \times 10^{-19}\text{ C})	-1 unit (for \beta^-) or +1 unit (for \beta^+)	Neutral (0 charge)
Rest Mass	\approx 4\text{ amu} (6.64 \times 10^{-27}\text{ kg})	0.0005486\text{ amu} (9.11 \times 10^{-31}\text{ kg})	Zero rest mass (m_0 = 0)
Velocity	\approx 1/10\text{th} velocity of light (c/10)	Up to \approx 99\% velocity of light (\sim c)	Exactly velocity of light (c = 3 \times 10^8\text{ m/s})

Parameter / Property	\alpha-Particles	\beta-Particles	\gamma-Rays
Penetrating Power	Very low (stopped by 0.002\text{ cm} Al or sheet of paper)	Moderate (can penetrate \approx 0.5\text{ cm} Al sheet)	Extremely high (penetrates > 5\text{ cm} lead or Al sheet)
Ionizing Power	Extremely high (maximum ionization)	Moderate (100\times less than \alpha-rays)	Very weak (100\times less than \beta, 10,000\times less than \alpha)
Deflection in Fields	Deflected towards negative electric plate & in magnetic field	Deflected towards positive plate (\beta^-) & in magnetic field	Completely undeflected in electric & magnetic fields
Detection Method	Geiger-Müller counter, Wilson cloud chamber	Geiger counter, scintillation counter	Scintillation counter, GM counter, semiconductor detector

[!IMPORTANT] **Radioactivity Discovery & Historical Anchors:** - Discovered in **1896 by Henri Becquerel** while researching phosphorescent uranium salts. - **Marie Curie and Pierre Curie (1898):** Discovered **Polonium** (Po) and **Radium** (Ra) isolated from uranium ores such as **Pitchblende** (principally U_3O_8 and variable UO_2). - Radioactivity is strictly a **nuclear phenomenon**, not a chemical reaction. - **Mass Number (A):** Derived from the German term *Atomgewicht* (atomic weight); represents total nucleons (p + n).

Table 2C: Comprehensive Milestones in the Classification of Elements

Milestone / Scientist	Year	Core Principle & Features	Key Examples / Anomalies
Prout's Hypothesis	1815–16	William Prout hypothesized that all atomic weights are exact whole-number multiples of hydrogen, considering hydrogen as the only fundamental unit (**protyle**).	Rejected as atomic weights of chlorine (35.5), etc., are fractional, though anticipates isotope mass numbers.
Dobereiner's Law of Triads	1829	Johann Wolfgang Döbereiner arranged chemically similar elements in groups of three (**Triads**); the atomic weight of the middle element was approximately the arithmetic mean of the terminal two.	- Li (7), Na (23), K (39) \to (7+39)/2 = 23 - Ca (40), Sr (88), Ba (137) \to (40+137)/2 = 88.5 - Cl (35.5), Br (80), I (127)
Newlands' Law of Octaves	1865	John Newlands arranged elements by increasing atomic weight; every 8th element exhibited properties similar to the 1st, resembling musical octaves (Sa, Re, Ga, Ma, Pa, Dha, Ni).	Valid only up to **Calcium (Z=20)**; broke down for transition elements.
Lothar Meyer's Curve	1870	Lothar Meyer plotted physical properties (specifically **atomic volume** = molar mass / solid density) against atomic weight; chemically similar elements occupied identical positions on the curve (alkali metals at peaks, halogens on ascending slopes, alkaline earth metals on descending slopes).	Contemporary with Mendeleev, but focused purely on physical curves rather than predictive tabular laws.
Mendeleev's Periodic Law	1869/1871	Dmitri Mendeleev stated: *"Physical and chemical properties of elements are periodic functions of their atomic weights."* Arranged 63 known elements into **6 Periods** and **8 Groups** (subgroups A & B). Left predictive gaps: **Eka-Boron (Scandium)**, **Eka-Aluminium (Gallium)**, **Eka-Silicon (Germanium)**.	Anomalous pairs (deliberately inverted heavier before lighter): **Co (58.9) before Ni (58.7)**, and **Te (127.6) before I (126.9)** to keep chemical families aligned.
Modern Periodic Table (Moseley)	1913	Henry Moseley established that **Atomic Number (Z)** is the true fundamental property using X-ray emission frequencies: \sqrt{\nu} = a(Z - b). Periodic Law: *"Properties of elements are periodic functions of their atomic numbers."*	18 Groups (Arabic 1–18), 7 Periods. Resolved isotope positioning and anomalous atomic weight pairs.

Table 2D: Superheavy Synthetic Elements (Atomic Numbers 101 to 118)

Z	IUPAC Name	Symbol	Z	IUPAC Name	Symbol	Z	IUPAC Name	Symbol
101	Mendelevium	Md	**107**	Bohrium	Bh	**113**	Nihonium	Nh
102	Nobelium	No	**108**	Hassium	Hs	**114**	Flerovium	Fl
103	Lawrencium	Lr	**109**	Meitnerium	Mt	**115**	Moscovium	Mc
104	Rutherfordium	Rf	**110**	Darmstadtium	Ds	**116**	Livermorium	Lv
105	Dubnium	Db	**111**	Roentgenium	Rg	**117**	Tennessine	Ts
106	Seaborgium	Sg	**112**	Copernicium	Cn	**118**	Oganesson	Og

Table 3: Summary of Periodic Trends Across Periods and Down Groups

Periodic Property	Across a Period (Left to Right \to)	Down a Group (Top to Bottom \downarrow)	Physical Reason / Governing Mechanism
Atomic Radius	**Decreases**	**Increases**	Z_{\text{eff}} pulls shell closer across period; new shells added down group.
Ionization Energy (IE)	**Increases** (with Be/B & N/O exceptions)	**Decreases**	Valence electrons held more tightly across; shielded further down.
Electron Affinity (EA)	**Increases** (Cl > F)	**Decreases**	Atom becomes smaller and attracts added electrons more strongly.
Electronegativity (EN)	**Increases** (Fluorine is 4.0)	**Decreases**	Smaller atomic radius exerts stronger pull on shared bonding pairs.
Metallic Character	**Decreases**	**Increases**	Ease of losing valence electrons drops across and rises down.
Non-Metallic Character	**Increases**	**Decreases**	Tendency to gain electrons rises across and drops down.
Basic Nature of Oxides	**Decreases** (Basic \to Acidic)	**Increases**	Electropositive metals form basic oxides; non-metals form acidic.

Table 4: Elemental Abundance in Earth's Crust vs Human Body

Rank	Earth's Crust Element	Crust Percentage	Human Body Element	Body Percentage	Must-Score Takeaway
1	**Oxygen (O)**	**46.6\%**	**Oxygen (O)**	**65.0\%**	**Most abundant element in both Crust and Body**.
2	**Silicon (Si)**	**27.7\%**	**Carbon (C)**	**18.5\%**	Silicon dominates mineral crust; Carbon dominates organic life.
3	**Aluminum (Al)**	**8.1\%**	**Hydrogen (H)**	**9.5\%**	**Aluminum is the #1 most abundant metal in Earth's crust**.
4	**Iron (Fe)**	**5.0\%**	**Nitrogen (N)**	**3.2\%**	Iron is the #2 metal in crust; Nitrogen forms amino acids.
5	**Calcium (Ca)**	**3.6\%**	**Calcium (Ca)**	**1.5\%**	**Calcium is the #1 most abundant metal in the Human Body**.
6	Sodium (Na)	2.8\%	Phosphorus (P)	1.0\%	ATP and bone mineral phosphates.
7	Potassium (K)	2.6\%	Potassium (K)	0.4\%	Major intracellular cation.
8	Magnesium (Mg)	2.1\%	Sulfur (S)	0.3\%	Disulfide bridges in protein keratin.

One-Liner Revision – Atomic Structure and Periodic Table

#	Fast Fact
1	Mendeleev's Periodic Table (1869) = arranged by **atomic mass**; predicted missing elements
2	Modern Periodic Table = arranged by **atomic number** (Moseley, 1913)
3	Periodic table: **7 periods** (horizontal rows) + **18 groups** (vertical columns)
4	Group 1 = Alkali metals (H, Li, Na, K, Rb, Cs, Fr); highly reactive
5	Group 17 = Halogens (F, Cl, Br, I, At); 7 valence electrons; most electronegative
6	Group 18 = Noble gases (He, Ne, Ar, Kr, Xe, Rn); inert; completely filled shells
7	Most electronegative element = **Fluorine (F)**
8	Largest atom = Cesium (Cs) / Francium (Fr) [by atomic radius]
9	Smallest atom = Helium (He) / Hydrogen (H)
10	Hardest natural substance = **Diamond** (form of Carbon)
11	Lightest element = **Hydrogen (H)**; atomic number 1
12	Lightest metal = **Lithium (Li)**; also most reactive metal
13	Heaviest naturally occurring element = **Uranium (U)**; atomic number 92
14	Most abundant element in universe = **Hydrogen**
15	Most abundant element in Earth's crust = **Oxygen** (46%)
16	Most abundant metal in Earth's crust = **Aluminium** (Al; 8%)
17	Most abundant gas in atmosphere = **Nitrogen** (78%)
18	Electronic configuration rule: 2n^2 (max electrons per shell: 2, 8, 18, 32...)
19	Octet rule: atoms tend to gain/lose electrons to have 8 valence electrons
20	Isotopes = same Z, different A; e.g., H-1 (protium), H-2 (deuterium), H-3 (tritium)
21	Isobars = same A, different Z; e.g., Ar-40 and Ca-40
22	Isotones = same number of neutrons, different Z; e.g., C-13 and N-14
23	Radioactive elements: all elements with Z > 83 (Bismuth) are naturally radioactive
24	Lanthanides = f-block elements 57-71; Actinides = f-block elements 89-103
25	Transition metals = d-block (Groups 3-12); variable oxidation states; coloured ions
26	Metalloids (semiconductors): B, Si, Ge, As, Sb, Te, At
27	Allotropy = element existing in multiple forms; Carbon (diamond, graphite, fullerene)
28	Diamond = hardest; Graphite = soft, good conductor; both are pure carbon
29	Polonium = only element with all naturally occurring isotopes being radioactive
30	Element 118 (Oganesson) = heaviest known element; all artificial/synthetic

Chemistry Topic 2 — Matter, Solutions and Purification

Consolidated — 50 Must-Score Facts

1. **Matter** is defined as anything possessing mass, inertia, and occupying three-dimensional physical space.
2. The five recognized physical states of matter are: **Solid**, **Liquid**, **Gas**, **Plasma**, and **Bose-Einstein Condensate (BEC)**.
3. In **solids**, intermolecular forces are strongest, intermolecular space is minimal, kinetic energy of particles is lowest, and particles vibrate strictly around fixed mean positions; solids possess definite shape and fixed volume.
4. In **liquids**, intermolecular forces are moderate, particles possess translational freedom to slip over one another, having fixed volume but indefinite shape (adapting to their container).
5. In **gases**, intermolecular attraction is negligible, intermolecular space is maximum, mean free path and particle kinetic energy are highest; gases exhibit neither fixed shape nor fixed volume and expand indefinitely to fill any container.
6. **Plasma** (the fourth state) consists of superheated, highly ionized gas containing free electrons and positive ions; it conducts electricity, generates magnetic fields, and makes up >99\% of the visible universe (stars, Sun, auroras, lightning, neon sign tubes).
7. **Bose-Einstein Condensate (BEC)** (the fifth state) forms when integer-spin boson gases of extremely low density (about one-hundred-thousandth the density of normal air) are cooled to ultra-low temperatures near absolute zero (0\text{ K} or -273.15^\circ\text{C}).
8. **Phase Transitions**: Melting (Solid \to Liquid), Freezing (Liquid \to Solid), Vaporization/Boiling (Liquid \to Gas), Condensation (Gas \to Liquid), Sublimation (Solid \to Gas directly), Deposition (Gas \to Solid directly).
9. During any phase change, **temperature remains strictly constant** because absorbed/released thermal energy functions entirely as **Latent Heat** (breaking or establishing intermolecular lattice bonds without increasing particle kinetic energy).
10. **Sublimation** occurs because the substance's vapor pressure exceeds atmospheric pressure before reaching its melting point. Examples: **Camphor, Naphthalene, Anthracene, Ammonium Chloride (NH_4Cl), Iodine (I_2), Dry Ice (solid CO_2)**.
11. **Triple Point of Water** is the unique thermodynamic coordinate of temperature and pressure at which solid ice, liquid water, and water vapor coexist in stable thermodynamic equilibrium: strictly 273.16\text{ K} (0.01^\circ\text{C}) and 4.58\text{ mmHg} (611.65\text{ Pa} or 0.006\text{ atm}).
12. **Critical Temperature (T_c)** is the temperature above which a gas cannot be liquefied, regardless of the magnitude of pressure applied. For CO_2, $T_c = 31.1^\circ\text{C}$.
13. A **Physical Change** alters physical properties (shape, size, state, volume) without changing chemical composition or molecular identity; it produces no new substances and is typically reversible (e.g., melting of ice, glowing of an electric bulb, crystallization, dissolution of salt in water).
14. A **Chemical Change** alters molecular structure, breaking and reforming chemical bonds to generate entirely new substances with distinct chemical properties; it is accompanied by substantial energy changes and is generally irreversible (e.g., rusting of iron, curdling of milk, digestion of food, burning of coal).
15. **Burning of a candle** is uniquely **both physical and chemical**: melting and vaporization of paraffin wax is a physical change, while the combustion of wax vapor into CO_2, carbon soot, and H_2O is a chemical change.
16. **Rusting of Iron**: When iron reacts with atmospheric oxygen and water vapor, hydrated ferric oxide ($Fe_2O_3 \cdot xH_2O$) is formed. Rusting is an oxidation chemical change that results in an **increase in the total mass of the iron object** due to the addition of bonded oxygen and water molecules.
17. An **Element** is a pure chemical substance consisting of only one species of atom that cannot be decomposed into simpler substances by ordinary chemical methods (e.g., Au, Fe, O_2, N_2).
18. A **Compound** is formed by two or more different elements chemically combined in a **strictly fixed ratio by mass** (Law of Definite Proportions by Joseph Proust); its properties are radically distinct from its constituent

elements (e.g., water H_2O, ratio of H:O = 1:8 by mass).

19. A **Mixture** contains two or more elements or compounds combined physically in variable proportions without chemical bonding; constituents retain their individual properties and can be separated by physical techniques (e.g., air, brass, crude oil, muddy water).
20. A **Homogeneous Mixture (True Solution)** has uniform composition and identical properties throughout its entire mass; its constituent particles are mixed at molecular/atomic dimensions ($< 1\text{ nm}$ or $< 10^{-9}\text{ m}$) and do not scatter visible light (e.g., sugar dissolved in water, air, aqueous NaCl, alloys like brass and bronze).
21. A **Heterogeneous Mixture** does not possess uniform composition; it exhibits distinct phase boundaries and visible interfaces (e.g., sand in water, oil-water mixture, suspension, colloid).
22. **Solubility** of most solid solutes in liquid solvents increases with increasing temperature (endothermic dissolution), but decreases for exothermic dissolutions (e.g., $Ce_2(SO_4)_3$, CaO).
23. **Henry's Law** states that the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas above the liquid ($m = k_H \cdot P$):
    - Packaging carbonated soft drinks under high CO_2 pressure increases effervescence.
    - Deep-sea divers breathe helium-diluted oxygen ($11.7\%\text{ He}$, $56.2\%\text{ }N_2$, $32.1\%\text{ }O_2$) to prevent painful and lethal nitrogen bubbles in the bloodstream ("bends" or decompression sickness).
    - High-altitude dwellers suffer from **anoxia** (impaired cognitive function) due to lower partial pressure of atmospheric oxygen.
24. **Molarity (M)** = moles of solute per liter of solution (mol/L). Because liquid volume expands or contracts with temperature, Molarity is **temperature-dependent**.
25. **Molality (m)** = moles of solute per kilogram of pure solvent (mol/kg). Because mass is independent of temperature, Molality is **strictly temperature-independent** and preferred in high-precision thermodynamic and colligative calculations.
26. **Normality (N)** = gram equivalents of solute per liter of solution ($N = M \times \text{Valence factor} / \text{n-factor}$).
27. **Parts per Million (ppm)** is used for ultra-trace concentrations (pollutants in air, fluoride/arsenic in drinking water): $\text{ppm} = \frac{\text{Mass of component}}{\text{Total mass}} \times 10^6$.
28. **Colligative Properties** are physical properties of dilute solutions that depend **strictly on the total number of solute particles** (ions or molecules) present, and are completely independent of the chemical nature, size, or identity of the solute.
29. The four fundamental colligative properties are:
    - **Relative Lowering of Vapor Pressure** (Raoult's Law)
    - **Elevation of Boiling Point** ($\Delta T_b = i \cdot K_b \cdot m$)
    - **Depression of Freezing Point** ($\Delta T_f = i \cdot K_f \cdot m$)
    - **Osmotic Pressure** ($\Pi = i \cdot C \cdot R \cdot T$)
30. The **van 't Hoff factor (i)** accounts for solute ionization or association: i = 1 for non-electrolytes (glucose, urea, sucrose); i = 2 for NaCl; i = 3 for $CaCl_2$ or Na_2SO_4; i = 4 for $AlCl_3$. For equal molal concentrations, $AlCl_3$ produces the greatest boiling point elevation and freezing point depression.
31. Adding common salt (NaCl) or ethylene glycol to water **elevates its boiling point** and **depresses its freezing point**.
32. **Automobile Antifreeze**: In sub-zero polar/winter conditions, **Ethylene glycol ($C_2H_6O_2$)** is added to vehicle radiator coolant water; it depresses the freezing point well below 0°C to prevent engine block cracking and elevates the boiling point to prevent summer overheating.
33. **De-icing of Frozen Highways**: Rock salt (NaCl) or calcium chloride ($CaCl_2$) is broadcast over icy roads. $CaCl_2$ is superior because it dissociates into 3 ions (i=3) and dissolves exothermically, melting snow down to -25°C.
34. **Pressure Cooker Mechanics**: By trapping steam, the internal chamber pressure rises to ~2 atmospheres (~15 psi gauge pressure), which **elevates the boiling point of water from 100°C to approximately**

120^\circ\text{C}–122^\circ\text{C}. Food cooks substantially faster because it absorbs heat at a significantly higher temperature, not because of the pressure itself.

35. At **high altitudes** (e.g., Leh, Mount Everest), atmospheric pressure drops significantly; consequently, water boils at a temperature below 100^\circ\text{C} (at 70^\circ\text{C}–80^\circ\text{C}), which fails to supply adequate cooking heat and makes boiling pulses/meat very slow without a pressure cooker.
36. **Osmosis** is the spontaneous net movement of solvent molecules from a region of lower solute concentration (higher chemical potential of solvent) to higher solute concentration across a **semi-permeable membrane (SPM)**.
37. **Osmotic Pressure (\Pi)** is the exact excess hydrostatic pressure that must be applied to the solution side to completely halt the osmotic influx of pure solvent.
38. **Reverse Osmosis (RO)**: When an external hydrostatic pressure greater than the osmotic pressure (P > \Pi) is applied to the concentrated solution side, pure solvent molecules are forced backwards through the semi-permeable membrane into the pure water reservoir. This is the cornerstone technology for **desalination of seawater** and domestic water purifiers.
39. **Isotonic Solutions** possess identical osmotic pressure across a semi-permeable boundary. A 0.9\%\text{ (w/v) } NaCl solution ("normal saline") is strictly isotonic with human red blood cells (RBCs).
40. If RBCs are placed in a **hypotonic solution** (distilled water), water enters by endosmosis, causing cells to swell and burst (**hemolysis**). In a **hypertonic solution** (concentrated brine), water exits by exosmosis, causing cells to shrink (**plasmolysis/crenation**).
41. **Colloidal Solution**: A heterogeneous system wherein the dispersed phase particle diameter lies between **1\text{ nm} (10^{-9}\text{ m}) and 1000\text{ nm} (1\text{ }\mu\text{m} or 10^{-6}\text{ m})**. Colloidal particles pass through ordinary filter paper but are arrested by animal/parchment membranes.
42. **Tyndall Effect**: The scattering of a beam of visible light by colloidal particles suspended in a medium, illuminating the path of light (e.g., sunlight entering a dusty room through a slit, headlights in dense fog, blue appearance of the sky). True solutions do NOT exhibit the Tyndall effect.
43. **Brownian Motion**: The continuous, zig-zag, random kinetic bombardment of colloidal particles by solvent molecules (first observed by botanist Robert Brown in 1827). Brownian motion opposes gravity and confers physical stability upon colloidal sols.
44. **Electrophoresis (Cataphoresis)**: The migration of electrically charged colloidal particles toward the oppositely charged electrode under the influence of an applied external electric field; it definitively proves that **colloidal particles carry electric charge**.
45. **Coagulation (Flocculation)**: The process of neutralizing the electric charge of colloidal particles, causing them to aggregate into larger macro-particles and settle out as a precipitate.
46. **Hardy-Schulze Rule**: The coagulating (flocculating) power of an active electrolyte ion is directly proportional to the fourth power of its valence. For negatively charged colloidal particles (e.g., clay particles in muddy water, As_2S_3 sol), the coagulation power follows: **Al^{3+} > Ba^{2+} > Na^+**.
47. **Purification of Muddy Water by Potash Alum**: River and pond mud consists of negatively charged colloidal clay suspended in water. When potash alum (KAl(SO_4)_2 \cdot 12H_2O) dissolves, trivalent aluminium cations (Al^{3+}) neutralize the negative surface charges of clay particles, precipitating them rapidly through coagulation.
48. **Delta Formation**: River water carries negatively charged colloidal clay and silt. When river water enters the ocean, the abundant multivalent metal cations (Na^+, Mg^{2+}, Ca^{2+}) in seawater immediately coagulate and deposit the silt at the river mouth, forming triangular geological deltas.
49. **Arresting Bleeding (Hemostasis) by Alum / Ferric Chloride (FeCl_3)**: Blood is a negatively charged colloidal suspension of albuminous matter. Applying alum (Al^{3+}) or styptic pencil (Fe^{3+}) instantly coagulates the blood proteins, forming a clot that seals the ruptured capillary.
50. **Classification of Colloids by Physical State**:
    - **Emulsion**: Liquid dispersed in Liquid (Milk, Mayonnaise, Cod liver oil, Cold cream).

- **Solid Aerosol**: Solid dispersed in Gas (Smoke, Automobile exhaust, Airborne dust).
- **Liquid Aerosol**: Liquid dispersed in Gas (Fog, Mist, Cloud, Insecticide sprays).
- **Sol**: Solid dispersed in Liquid (Paints, Inks, Cell fluids, Blood, Starch sol).
- **Gel**: Liquid dispersed in Solid (Jelly, Cheese, Butter, Gelatin, Curd).
- **Solid Sol**: Solid dispersed in Solid (Ruby glass, Colored gemstones, Pearls).
- **Foam**: Gas dispersed in Liquid (Soap froth, Whipped cream, Shaving lather).
- **Solid Foam**: Gas dispersed in Solid (Pumice stone, Foam rubber, Bread).

One-Liner Revision – Matter, Solutions and Purification

#	Fast Fact
1	States of matter: Solid, Liquid, Gas, Plasma (4th state in stars/lightning)
2	Sublimation = solid directly to gas (dry ice/CO2, iodine, naphthalene, camphor)
3	Deposition = gas directly to solid (frost formation)
4	Water: boiling point = 100 degC; freezing = 0 degC; maximum density at **4 degC**
5	Universal solvent = **Water** (H2O)
6	Solution components: solute (minor, dissolved) + solvent (major, dissolving)
7	Mixture types: homogeneous (uniform) = solution; heterogeneous (non-uniform) = suspension
8	Colloid particle size = 1-1000 nm; shows Tyndall effect; e.g., milk, fog, smoke
9	Tyndall effect = scattering of light by colloidal particles (explains why beam visible in dusty room)
10	Suspension = particles > 1000 nm; settles on standing; filtered by filter paper
11	Distillation = separates liquids with different boiling points
12	Fractional distillation = separates crude oil into petrol, diesel, kerosene fractions
13	Chromatography = separates pigments, amino acids (paper, column, gas, HPLC)
14	Centrifugation = separates components by density using centrifugal force; cream from milk
15	Crystallisation = purification method; produces pure crystals; for alum, salt
16	Miscible liquids = dissolve in each other in all proportions; e.g., alcohol + water
17	Immiscible liquids = do not mix; separated by separating funnel; e.g., oil + water
18	Henry's Law: solubility of gas in liquid increases with pressure (soda carbonation)
19	Molarity (M) = moles of solute per litre of solution
20	Molality (m) = moles of solute per kg of solvent
21	Saturated solution = maximum solute dissolved at given temp; undissolved solute remains
22	Supersaturated = more solute than normally possible; unstable; seeding causes crystallisation
23	Hard water = contains Ca2+ and Mg2+ salts; does not form lather easily
24	Temporary hardness = Ca/Mg bicarbonates; removed by boiling or lime
25	Permanent hardness = Ca/Mg sulphates/chlorides; removed by washing soda or ion exchange
26	Alloy = mixture of metals (or metal + non-metal); e.g., brass = Cu+Zn, steel = Fe+C
27	Amalgam = alloy with mercury; dental fillings (Ag-Sn-Hg amalgam)
28	Electrolysis = chemical decomposition using electricity; used to refine metals, electroplate
29	Osmosis = water moves from low to high solute concentration through semipermeable membrane
30	Reverse osmosis (RO) = applied pressure forces water against osmosis; water purification

Chemistry Topic 3 — Metals and Chemical Reactions

Consolidated — 50 Must-Score Facts

1. **Metals** are electropositive elements that readily donate valence electrons to form positive cations (M \to M^{n+} + n e^-).
2. **Gold (Au)** is the most **malleable** and most **ductile** metal known: a single gram of gold can be drawn into a microscopic wire over 2\text{ kilometers} long or beaten into gold leaf only 0.00001\text{ mm} thick.
3. **Silver (Ag)** has the **highest electrical and thermal conductivity** of all metals, followed in sequence by Copper (Cu), Gold (Au), and Aluminium (Al).
4. **Lead (Pb)** and **Mercury (Hg)** are the **poorest conductors of heat** among common commercial metals.
5. **Mercury (Hg, Hydrargyrum, Z=80)** is the only elemental metal that exists in the **liquid state** at standard room temperature (25^\circ\text{C}); its freezing point is -38.83^\circ\text{C} and normal boiling point is 356.7^\circ\text{C}.
6. **Gallium (Ga, m.p. 29.76^\circ\text{C})** and **Caesium (Cs, m.p. 28.44^\circ\text{C})** are low-melting metals that liquefy directly in the palm of a human hand.
7. **Osmium (Os, density 22.59\text{ g/cm}^3)** is the **densest naturally occurring element**, closely followed by Iridium (Ir, 22.56\text{ g/cm}^3) and Platinum (Pt, 21.45\text{ g/cm}^3).
8. **Lithium (Li, density 0.534\text{ g/cm}^3)** is the **lightest metal** and lowest-density solid element; it floats easily on water and paraffin oil.
9. **Tungsten (W, Wolfram, Z=74)** has the **highest melting point of all metals** (3422^\circ\text{C}); its high tensile strength and high electrical resistance make it the standard incandescent light bulb filament.
10. **Alkali metals (Sodium Na, Potassium K)** are exceptionally soft metals that can be **easily sliced with an ordinary table knife**; they have low densities, low melting points, and react vigorously with atmospheric moisture and oxygen, requiring storage submerged in **kerosene oil**.
11. **Phosphorus (P_4)** is an exceptionally reactive non-metal that spontaneously catches fire in moist air (yielding dense white fumes of P_4O_{10}); it is stored safely submerged under **water**.
12. **Bromine (Br_2, Z=35)** is the **only non-metal that is liquid** at room temperature; it is a dense, volatile, reddish-brown liquid with fuming, corrosive vapors.
13. **Iodine (I_2, Z=53)** is a non-metal that anomalously displays a **bright metallic-like luster** and sublimes into deep violet vapors upon gentle heating.
14. **Diamond** is an allotrope of carbon where each carbon atom is bonded tetrahedrally to 4 other carbons via rigid sp^3 covalent bonds; it is the **hardest known naturally occurring mineral** (10 on Mohs scale), an electrical insulator, but an exceptional conductor of heat (better than copper).
15. **Graphite** is an allotrope of carbon consisting of planar hexagonal sheets bonded via sp^2 hybridization with delocalized \pi electrons; it is soft, slippery (used as a solid dry lubricant), and **conducts electricity**.
16. Thermodynamically, **Graphite is the most stable allotrope of carbon** under standard ambient conditions (298\text{ K}, 1\text{ atm}); diamond is metastable and converts to graphite over geological timescales.
17. **Fullerenes (C_{60}, Buckminsterfullerene)** consist of 60 carbon atoms arranged in a truncated icosahedron containing 20 hexagons and 12 pentagons, resembling a hollow soccer ball (“buckyball”).
18. **Graphene** is a single two-dimensional planar sheet of carbon atoms packed in a honeycomb crystal lattice; it is nearly transparent, conducts electricity with zero effective mass electrons, and is >100 times stronger than structural steel.
19. Most metal oxides are **basic** (e.g., Na_2O, CaO, MgO), forming hydroxides with water. Most non-metal oxides are **acidic** (e.g., SO_2, SO_3, CO_2, P_4O_{10}), forming oxyacids with water.
20. **Amphoteric Oxides** react with both acids and strong bases to yield salt and water. The two classic examples are **Aluminium oxide (Al_2O_3)** and **Zinc oxide (ZnO)**: ZnO + 2HCl \to ZnCl_2 + H_2O, \quad ZnO + 2NaOH \to Na_2ZnO_2\text{ (Sodium zincate)} + H_2O

21. **Neutral Oxides** exhibit neither acidic nor basic properties: Nitrous oxide (N_2O, laughing gas), Nitric oxide (NO), Carbon monoxide (CO), and Water (H_2O).
22. **The Reactivity Series of Metals** in descending order: K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Pt > Au
23. **Potassium (K) and Sodium (Na)** react violently with cold water, evolving hydrogen gas which catches fire spontaneously with an explosive, yellow/lilac flame.
24. **Calcium (Ca)** reacts moderately with cold water; the bubbles of liberated H_2 gas stick to the metal surface, causing the calcium lumps to **float to the top of the water**.
25. **Magnesium (Mg)** does not react with cold water; it reacts slowly with hot water and vigorously with steam to form magnesium oxide (MgO) and H_2.
26. **Aluminium (Al), Zinc (Zn), and Iron (Fe)** react neither with cold nor hot water, but react exclusively with **superheated steam**: 3Fe + 4H_2O\text{ (steam)} \to Fe_3O_4\text{ (Magnetic iron oxide)} + 4H_2 \uparrow
27. **Lead, Copper, Silver, Platinum, and Gold** do not react with water, hot water, or steam at any temperature.
28. Metals situated **above hydrogen** in the reactivity series displace hydrogen gas (H_2 \uparrow) from dilute mineral acids (HCl, H_2SO_4). Metals situated **below hydrogen** (Cu, Ag, Au, Pt) cannot displace H_2 from dilute acids.
29. **Nitric Acid (HNO_3)** is a powerful oxidizing agent. When metals react with HNO_3, hydrogen gas is generally NOT evolved; instead, HNO_3 oxidizes H_2 to H_2O and gets reduced to nitrogen oxides (NO_2, NO, N_2O).
30. The only two metals that react with **very dilute (\approx 1\%) cold nitric acid** to evolve hydrogen gas (H_2) are **Magnesium (Mg)** and **Manganese (Mn)**.
31. **Aqua Regia ("Royal Water")** is a freshly prepared fuming mixture of **3\text{ parts concentrated } HCl and 1\text{ part concentrated } HNO_3** (strict volumetric ratio 3:1). It dissolves noble metals like Gold (Au) and Platinum (Pt) by producing nascent chlorine and nitrosyl chloride (NOCl), forming soluble tetrachloroaurate complexes ([AuCl_4]^-).
32. **Bauxite (Al_2O_3 \cdot 2H_2O)** is the principal commercial ore of aluminium; it is purified by Bayer's chemical leaching and reduced electrolytically in the **Hall-Héroult process** using molten cryolite (Na_3AlF_6) and fluorspar (CaF_2) to lower the operating melting point from 2050^\circ\text{C} to \sim 950^\circ\text{C}.
33. **Iron Ores**: Haematite (Fe_2O_3, red ore, >65\% Fe), Magnetite (Fe_3O_4, black ore, magnetic, up to 72\% Fe), Limonite (2Fe_2O_3 \cdot 3H_2O), Siderite (FeCO_3).
34. **Iron Pyrites (FeS_2)** displays a brassy golden metallic luster and is popularly termed **"Fool's Gold"**; it is an ore of sulphur, not iron extraction.
35. In the metallurgical blast furnace, iron extraction yields **Pig Iron** (\sim 4\% C, highly brittle), which is remelted into **Cast Iron** (\sim 3\% C). **Wrought Iron** (0.1–0.25\% C) is the **purest commercial form of iron**, possessing high malleability and fibrous toughness.
36. **Cinnabar (HgS)** is the primary ore of Mercury; heating cinnabar in air roasts it directly into volatile mercury vapor, which condenses into liquid mercury (HgS + O_2 \to Hg + SO_2).
37. **Galena (PbS)** is the primary ore of Lead; **Pitchblende (U_3O_8)** is the primary ore of Uranium (from which Marie Curie discovered Radium and Polonium).
38. **Monazite sand**, abundant in the placer beach sands of Kerala, is the primary global resource for **Thorium (Th)** and rare earth elements (cerium, lanthanum).
39. **Froth Flotation Process** (invented by Mrs. Carrie Everson) is used selectively to concentrate **sulphide ores** (e.g., ZnS, PbS, CuFeS_2) using pine oil as a frothing collector and xanthates to make ore particles hydrophobic.
40. **Calcination vs. Roasting**: Calcination heats carbonate/hydrated ores in the **absence or limited supply of air** below melting point (CaCO_3 \xrightarrow{\Delta} CaO + CO_2). Roasting heats sulphide ores in a **regular, excess supply of air** (2ZnS + 3O_2 \to 2ZnO + 2SO_2).
41. **Thermite Process (Goldschmidt Reaction)**: Highly exothermic reduction of metal oxides using aluminium powder (Fe_2O_3 + 2Al \to Al_2O_3 + 2Fe\text{ (molten)} + \text{immense heat}). The generated molten iron reaches >2500^\circ\text{C} and is used on-site for welding broken railway tracks and heavy machinery cracks.

42. **Brass** is an alloy of **Copper (60–70\%) and Zinc (30–40\%)**; it is corrosion-resistant, acoustic, and used for musical instruments, cartridge casings, and decorative hardware.
43. **Bronze** is an alloy of **Copper (88–90\%) and Tin (10–12\%)**; it is harder than brass and used for statues, medals, ship propellers, and coins.
44. **German Silver** consists of **Copper (50–60\%), Zinc (20–30\%), and Nickel (10–20\%)**; it contains **ZERO percent silver** and is named solely for its silvery appearance.
45. **Gun Metal** is an alloy of **Copper (88\%), Tin (10\%), and Zinc (2\%)**; used for heavy-duty ordnance, gears, and high-pressure steam valves.
46. **Solder (Tanka)** is an alloy of **Lead (50\%) and Tin (50\%)**; it possesses a sharp, low melting point (\sim 183^\circ\text{C}) ideal for joining electrical circuits and plumbing joints.
47. **Stainless Steel** consists of **Iron (\sim 73\%), Chromium (18\%), Nickel (8–10\%), and Carbon (0.1–0.2\%)**:
    - **Chromium** forms a self-healing, microscopic passive layer of chromium oxide (Cr_2O_3) that prevents corrosion and rust.
    - **Nickel** imparts mechanical toughness, ductility, and glossy luster.
    - **Carbon** imparts high tensile hardness.
48. **Nichrome** is an alloy of **Nickel (80\%) and Chromium (20\%)**; it exhibits high electrical resistivity, high melting point (1400^\circ\text{C}), and does not oxidize at red heat, making it the standard heating element in domestic toasters, geysers, and room heaters.
49. **Amalgams** are alloys formed by any metal with **Mercury (Hg)**. Notable exception: **Iron (Fe), Cobalt, Nickel, and Platinum do NOT form amalgams**; hence, mercury is stored and transported industrially in sealed **iron flasks**.
50. **Galvanization** is the metallurgical process of coating iron or steel sheets with a thin protective layer of **Zinc (Zn)** by dipping them into molten zinc at \sim 450^\circ\text{C}. Zinc acts as a sacrificial anode because Zn is more electropositive than Fe (E^\circ_{Zn^{2+}/Zn} = -0.76\text{ V} vs E^\circ_{Fe^{2+}/Fe} = -0.44\text{ V}); even if the zinc surface is scratched, zinc corrodes preferentially, shielding the underlying iron from rust.

One-Liner Revision – Metals and Chemical Reactions

#	Fast Fact
1	Most reactive metal = **Potassium (K)**; reacts violently with water
2	Least reactive metal = **Platinum (Pt)** / Gold (Au) = noble metals
3	Most malleable metal = **Gold (Au)** (1g can be beaten into 1m2 sheet)
4	Best conductor of electricity = **Silver (Ag)**; then Cu, then Au, then Al
5	Hardest metal = **Chromium (Cr)**; hardest element = Carbon (diamond)
6	Lightest metal = **Lithium (Li)**; floats on water and oil
7	Mercury (Hg) = only metal that is liquid at room temperature
8	Gallium (Ga) = melts in palm (melting point 29.76 degC)
9	Stainless steel = Fe + Cr (10.5%+) + Ni; rust-resistant
10	Brass = **Cu + Zn**; Bronze = Cu + Sn; Gunmetal = Cu + Sn + Zn
11	German silver / Nickel silver = Cu + Zn + Ni (no silver!)
12	Solder = **Pb + Sn** (used in electronics for joining); being replaced by Sn-Ag-Cu
13	Bell metal = **Cu + Sn** (high tin content); used for bells and cymbals
14	Duralumin = **Al + Cu + Mg + Mn**; lightweight strong alloy; aircraft construction
15	Rusting requires both **water** and **oxygen**; prevented by galvanisation, painting, oiling

#	Fast Fact
16	Galvanisation = coating iron with **zinc (Zn)**; zinc acts as sacrificial anode
17	Thermite reaction: 2Al + Fe2O3 -> 2Fe + Al2O3 + heat (~2500 degC); railway welding
18	Reactivity series: metals above H displace H from acids; K, Na, Ca react with cold water
19	Gold does not corrode; found native (in free state) in nature
20	Bauxite (Al2O3.2H2O) = ore of aluminium; Haematite (Fe2O3) = ore of iron
21	Bayer Process = extraction of alumina from bauxite; then Hall-Heroult for Al electrolysis
22	Blast furnace = extraction of iron from haematite using coke + limestone
23	Pig iron = crude iron from blast furnace; high carbon (4-5%); brittle
24	Wrought iron = nearly pure iron; very low carbon (<0.02%); most malleable
25	Steel = iron + carbon (0.2-2%); iron + Cr = stainless; iron + Mn = manganese steel
26	Amalgamation = extraction of Ag/Au using Hg (forms amalgam)
27	Electrolytic refining: crude metal at anode (-) dissolves; pure at cathode (-) deposits
28	Patina = green coating on copper = basic copper carbonate [Cu(OH)2.CuCO3]
29	Verdigris (patina on Statue of Liberty) = green due to copper oxidation
30	Chemical change = new substance formed (rust); Physical change = no new substance (melting)

Chemistry Topic 4 — Acids, Bases, Salts and Sucrose

Consolidated — 50 Must-Score Facts

1. **Arrhenius Theory**: An **acid** is a chemical substance that dissociates in aqueous solution to yield hydrogen ions (H^+ or hydronium ions H_3O^+); a **base** dissociates to yield hydroxide ions (OH^-).
2. **Brønsted-Lowry Theory**: An **acid is a proton (H^+) donor**, and a **base is a proton (H^+) acceptor**. Every acid has a corresponding conjugate base, and every base has a conjugate acid.
3. **Lewis Theory**: An **acid is an electron-pair acceptor** (electrophile, e.g., BF_3, AlCl_3, H^+); a **base is an electron-pair donor** (nucleophile, e.g., NH_3, H_2O, OH^-).
4. The **pH scale** was devised in 1909 by Danish biochemist **S.P.L. Sørensen** at the Carlsberg Laboratory: \text{pH} = -\log_{10}[H^+] = \log_{10}\left(\frac{1}{[H^+]}\right)
5. In pure neutral water at standard room temperature (25^\circ\text{C} or 298\text{ K}), [H^+] = [OH^-] = 10^{-7}\text{ M}, yielding: \text{pH} = 7.0 \quad (\text{Strictly Neutral})
6. **Acidic solutions** possess [H^+] > 10^{-7}\text{ M}, corresponding to **\text{pH} < 7**. The lower the pH value, the stronger the acidity.
7. **Basic (alkaline) solutions** possess [H^+] < 10^{-7}\text{ M} and [OH^-] > 10^{-7}\text{ M}, corresponding to **\text{pH} > 7**. The higher the pH value, the stronger the alkalinity.
8. At 25^\circ\text{C}, the ionic product of water is K_w = [H^+][OH^-] = 10^{-14}\text{ mol}^2/\text{L}^2, which dictates: \text{pH} + \text{pOH} = 14
9. **Temperature Effect on pH**: Because auto-ionization of water (H_2O \rightleftharpoons H^+ + OH^-) is endothermic, increasing temperature shifts equilibrium to the right, increasing K_w. Consequently, at 50^\circ\text{C} the pH of pure neutral water drops to \approx 6.63, but the water **remains completely neutral** because [H^+] remains exactly equal to [OH^-].
10. A change of **1 pH unit** represents a **tenfold (10\times) change in hydrogen ion concentration**. A solution of pH 2 is 100 times more acidic than a solution of pH 4 (10^{4-2} = 10^2).
11. **Human Blood pH**: Strictly maintained within the narrow, slightly alkaline window of **7.35\text{ to }7.45** (mean \approx 7.40). A blood pH drop below 7.0 (acidosis) or rise above 7.8 (alkalosis) is fatal.
12. The primary biological buffer maintaining human blood pH is the **Carbonic acid–Bicarbonate buffer system**: H_2CO_3 \rightleftharpoons H^+ + HCO_3^- regulated cooperatively by the lungs (eliminating gaseous CO_2) and the kidneys (reabsorbing or excreting HCO_3^-).
13. **Human Gastric Juice** contains hydrochloric acid (HCl, \sim 0.5\%), with a highly acidic pH of **1.0\text{ to }1.5**, which activates the protein-digesting enzyme pepsinogen into pepsin and sterilizes ingested pathogens.
14. Normal **Human Saliva** has a pH range of **6.5\text{ to }7.5**. When oral bacteria ferment dietary sugars, dropping saliva pH below **5.5**, tooth enamel (calcium hydroxyapatite, Ca_{10}(PO_4)_6(OH)_2) begins to demineralize, initiating dental decay.
15. **Acid Rain** is defined as precipitation with a **\text{pH} < 5.6**, caused primarily by atmospheric emissions of sulphur dioxide (SO_2) and nitrogen oxides (NO_x) reacting with rainwater to form sulphuric acid (H_2SO_4) and nitric acid (HNO_3).
16. **Natural Organic Acids & Food Sources**:
    - **Acetic Acid (Ethanoic Acid, CH_3COOH)**: Vinegar (4–8\% aqueous solution).
    - **Formic Acid (Methanoic Acid, HCOOH)**: Stings of red ants, bees, wasps, and stinging nettle plant hairs.
    - **Citric Acid (C_6H_8O_7)**: Citrus fruits (lemons, oranges, limes, sweet limes).
    - **Tartaric Acid (C_4H_6O_6)**: Tamarind (imli), unripe grapes, cream of tartar; essential component of baking powder.
    - **Lactic Acid (CH_3CH(OH)COOH)**: Sour milk, curd (dahi), yogurt; muscle accumulation during anaerobic respiration causes muscular fatigue and cramps.

- **Oxalic Acid (H_2C_2O_4)**: Tomatoes, spinach, sorrel; used industrially to remove stubborn rust/ink stains from textiles.
- **Malic Acid (C_4H_6O_5)**: Apples (especially green/unripe apples), cherries.
- **Ascorbic Acid (Vitamin C, C_6H_8O_6)**: Amla (Indian gooseberry), citrus fruits, guava.
- **Tannic Acid**: Tea leaves (causes astringent taste).
- **Butyric Acid**: Rancid, oxidized butter.
- **Carbonic Acid (H_2CO_3)**: Carbonated soft drinks, club soda.

17. **Kidney Stones (Renal Calculi)**: Approximately 80\% of all human kidney stones consist of insoluble crystals of **Calcium Oxalate (CaC_2O_4)**, formed when high concentrations of dietary oxalic acid bind with urinary calcium.
18. **Acid-Base Indicators**:
    - **Litmus**: Natural purple dye extracted from **Lichens** (Roccella tinctoria). Turns **Red in acidic solutions** and **Blue in basic/alkaline solutions**.
    - **Phenolphthalein**: Synthetic organic indicator. Remains **Colorless in acidic and neutral solutions**, but turns **Intense Magenta/Pink in basic solutions** (pH transition interval 8.2–10.0).
    - **Methyl Orange**: Synthetic azo indicator. Turns **Red/Pink in acidic solutions** (pH < 3.1) and **Yellow in basic/neutral solutions** (pH > 4.4).
    - **Turmeric (Haldi)**: Natural yellow indicator. Remains yellow in acids/neutrals, but turns **Reddish-Brown in alkaline solutions** (explaining why yellow curry stains turn deep red when scrubbed with alkaline laundry soap).
    - **Red Cabbage Extract**: Natural anthocyanin indicator. Turns red/pink in acids, purple at neutral pH, and green-to-yellow in strong bases.
19. **Olfactory Indicators** are substances whose characteristic odor changes depending on whether they are in an acidic or basic medium. Examples: **Vanilla extract, Onion juice, and Clove oil** completely lose their smell in alkaline solutions (NaOH), but retain their scent in acids.
20. **Neutralization Reaction**: The exothermic chemical reaction between an acid and a base to produce a salt and water: \text{Acid} + \text{Base} \to \text{Salt} + \text{Water} + \text{Heat} For any strong acid reacting with a strong base in dilute aqueous solution, the **standard enthalpy of neutralization is constant**: \Delta H_{\text{neut}}^\circ = -57.1\text{ kJ/mol} \quad (-13.7\text{ kcal/mol}) because the net reaction is always: H^+\text{(aq)} + OH^-\text{(aq)} \to H_2O\text{(l)}.
21. **Salt Hydrolysis and Solution pH**:
    - *Salt of Strong Acid + Strong Base* (NaCl, KNO_3): Neutral solution (\text{pH} = 7.0).
    - *Salt of Strong Acid + Weak Base* (NH_4Cl, CuSO_4): Acidic solution (\text{pH} < 7.0).
    - *Salt of Weak Acid + Strong Base* (CH_3COONa, Na_2CO_3): Basic/alkaline solution (\text{pH} > 7.0).
    - *Salt of Weak Acid + Weak Base* (CH_3COONH_4): pH depends on relative K_a and K_b values.
22. **Baking Soda**: Chemically **Sodium Hydrogen Carbonate / Sodium Bicarbonate (NaHCO_3)**:
    - Prepared industrially by the **Solvay Process** using brine (NaCl), ammonia (NH_3), and carbon dioxide (CO_2).
    - Mild, non-corrosive basic salt.
    - Used in medicine as an **antacid** to neutralize hyperacidity in the stomach (NaHCO_3 + HCl \to NaCl + H_2O + CO_2 \uparrow).
    - Used in **soda-acid fire extinguishers**: when inverted, concentrated sulphuric acid mixes with NaHCO_3 solution, generating high-pressure CO_2 gas that smothers flames.
23. **Baking Powder**: A dry commercial blend of **Baking Soda (NaHCO_3)** and a mild edible solid acid like **Tartaric Acid** or cream of tartar (potassium bitartrate).
    - *Why Tartaric Acid is Added*: When heated or mixed with water, NaHCO_3 releases CO_2 gas which causes dough to rise and become soft/spongy. However, the residual sodium carbonate (Na_2CO_3) would impart an

unpleasant, bitter alkaline taste. The tartaric acid neutralizes Na_2CO_3 into sodium tartrate, completely eliminating the bitter taste.

24. **Washing Soda**: Chemically **Sodium Carbonate Decahydrate ($Na_2CO_3 \cdot 10H_2O$)**:
    - Contains **10 molecules of water of crystallization**.
    - Undergoes **efflorescence**: when exposed to dry atmospheric air, it loses 9 molecules of water of crystallization to form a white powdery monohydrate: $Na_2CO_3 \cdot 10H_2O \xrightarrow{\text{air}} Na_2CO_3 \cdot H_2O\text{ (Monohydrate)} + 9H_2O \uparrow$
    - Anhydrous sodium carbonate (Na_2CO_3) is known commercially as **Soda Ash**.
    - Widely used in laundry detergents, glass, soap, and paper manufacturing, and for **removing permanent hardness** of water caused by dissolved $CaCl_2$ and $MgSO_4$.
25. **Bleaching Powder**: Chemically **Calcium Oxychloride / Calcium Hypochlorite ($CaOCl_2$)**:
    - Manufactured by passing chlorine gas over dry slaked lime at 40°C: $Ca(OH)_2 + Cl_2 \to CaOCl_2 + H_2O$
    - Gives off a strong odor of chlorine in air because it reacts with atmospheric CO_2: $CaOCl_2 + CO_2 \to CaCO_3 + Cl_2 \uparrow$
    - Functions as a powerful **oxidizing agent, textile bleach, and bactericide/disinfectant** for municipal drinking water sterilization.
26. **Plaster of Paris (PoP)**: Chemically **Calcium Sulphate Hemihydrate ($CaSO_4 \cdot \frac{1}{2}H_2O$ or $(CaSO_4)_2 \cdot H_2O$)**:
    - Manufactured by carefully heating mineral Gypsum ($CaSO_4 \cdot 2H_2O$) in a rotary kiln to exactly **373 K (100°C)**: $CaSO_4 \cdot 2H_2O \xrightarrow{373\text{ K}} CaSO_4 \cdot \frac{1}{2}H_2O + 1\frac{1}{2}H_2O \uparrow$
    - *Crucial Thermal Trap*: If gypsum is heated above **473 K (200°C)**, it loses all water of crystallization, yielding anhydrous calcium sulphate ($CaSO_4$) known as **"Dead Burnt Plaster"**, which completely loses the property of setting with water.
    - *Setting Action*: When mixed with water, PoP rehydrates back into a hard crystalline mass of gypsum within 10–15 minutes, undergoing a **slight volume expansion ($\approx 1\%$)**, which makes it ideal for medical orthopedic casts (immobilizing fractured bones), architectural false ceilings, dental moulds, and statue casting.
27. **Gypsum**: Chemically **Calcium Sulphate Dihydrate ($CaSO_4 \cdot 2H_2O$)**:
    - Naturally occurring mineral containing **2 molecules of water of crystallization**.
    - Added in small amounts (2–3%) to **Portland cement clinker** during final grinding to retard the initial flash setting time of cement, providing masons sufficient working time.
    - Used in agriculture as a soil conditioner to reclaim alkaline/sodic soils and supply calcium and sulphate nutrients.
28. **Blue Vitriol**: Chemically **Copper Sulphate Pentahydrate ($CuSO_4 \cdot 5H_2O$)**:
    - Distinctive deep blue crystalline solid containing 5 water molecules (4 coordinated to Cu^{2+}, 1 hydrogen-bonded).
    - Heating above 250°C drives off all water, turning it into white anhydrous $CuSO_4$.
    - Key ingredient in **Bordeaux Mixture** ($CuSO_4 + Ca(OH)_2$ in water), an essential agricultural fungicide sprayed on grapevines and fruit orchards.
29. **Green Vitriol**: Chemically **Ferrous Sulphate Heptahydrate ($FeSO_4 \cdot 7H_2O$)**:
    - Pale green crystals used in ink manufacturing, iron supplement medicine (iron deficiency anaemia), and water purification coagulants.
30. **White Vitriol**: Chemically **Zinc Sulphate Heptahydrate ($ZnSO_4 \cdot 7H_2O$)**:

- Used in the manufacture of lithopone pigment, rayon spinning baths, and as an ophthalmic astringent eye-drop.

31. **Oil of Vitriol**: Historical alchemical name for concentrated **Sulphuric Acid (H_2SO_4)**, so named because it was originally synthesized by distilling green vitriol crystals.
32. **Epsom Salt**: Chemically **Magnesium Sulphate Heptahydrate (MgSO_4 \cdot 7H_2O)**:
    - Used medically as a saline osmotic laxative and in therapeutic soothing bath salts.
33. **Potash Alum (Fitkari)**: Chemically **Potassium Aluminium Sulphate Dodecahydrate (K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O)**:
    - A classic double salt containing 24 molecules of water of crystallization.
    - Used as a hemostatic agent (stops shaving bleeding by coagulating blood proteins) and for clarifying turbid water.
34. **Caustic Soda**: Sodium Hydroxide (NaOH). Strong base used in soap making, rayon, petroleum refining.
35. **Caustic Potash**: Potassium Hydroxide (KOH). Used in luxury soft shaving soaps and alkaline storage batteries.
36. **Slaked Lime**: Calcium Hydroxide (Ca(OH)_2). Clear aqueous solution is called **Lime Water**. Reacts with CO_2 to turn milky due to insoluble CaCO_3 precipitation.
37. **Quicklime**: Calcium Oxide (CaO). Prepared by thermal calcination of limestone. Highly exothermic reaction with water yields slaked lime: CaO + H_2O \to Ca(OH)_2 + \text{heat}.
38. **Carbohydrates**: Polyhydroxy aldehydes or polyhydroxy ketones (or compounds that yield them upon hydrolysis), having the general empirical formula C_x(H_2O)*y (though exceptions like rhamnose C_6H*{12}O_5 and acetic acid C_2H_4O_2 exist).
39. **Monosaccharides**: Simplest carbohydrate monomers that cannot be hydrolyzed into smaller carbohydrate units.
    - **Glucose (C_6H_{12}O_6)**: An aldohexose known as **Dextrose, Grape Sugar, or Blood Sugar**. Chief physiological energy fuel of the human body (4\text{ kcal/g}).
    - **Fructose (C_6H_{12}O_6)**: A ketohexose known as **Levulose or Fruit Sugar**.
    - **Galactose (C_6H_{12}O_6)**: Component of milk sugar.
40. **Fructose Sweetness**: Fructose is the **sweetest naturally occurring carbohydrate**, with a relative sweetness rating of **\sim 170** (relative to sucrose standardized at 100). Present abundantly in **honey (\approx 38\%)** and sweet ripe fruits.
41. **Disaccharides**: Carbohydrates that yield two monosaccharide molecules upon acid or enzymatic hydrolysis:
    - **Sucrose (C_{12}H_{22}O_{11})**: Cane sugar or table sugar. Formed by \alpha\text{-D-glucose} + \beta\text{-D-fructose} linked via an \alpha-1,\beta-2 glycosidic bond.
    - **Lactose (C_{12}H_{22}O_{11})**: Milk sugar (\approx 5\% in cow milk, \approx 7\% in human milk). Formed by \beta\text{-D-galactose} + \beta\text{-D-glucose} via a \beta-1,4 glycosidic bond. Least sweet natural sugar (rating \approx 16).
    - **Maltose (C_{12}H_{22}O_{11})**: Malt sugar. Formed by two \alpha\text{-D-glucose} units linked via an \alpha-1,4 glycosidic bond. Produced during starch digestion by amylase.
42. **Reducing vs. Non-Reducing Sugars**:
    - **Reducing Sugars**: Contain free hemiacetal or hemiketal groups; reduce **Fehling's solution** (blue cupric \to red cuprous oxide Cu_2O\downarrow) and **Tollens' reagent** (ammoniacal silver nitrate \to silver mirror).
    - **All Monosaccharides** (Glucose, Fructose, Galactose) and **Maltose & Lactose** are reducing sugars!
    - **Sucrose is a NON-REDUCING sugar** because both of its anomeric carbons (C-1 of glucose and C-2 of fructose) are mutually locked in the glycosidic bond, leaving no free functional group to reduce Tollens' or Fehling's reagents.
43. **Invert Sugar**:
    - Sucrose is **dextrorotatory** ([\alpha]_D = +66.5^\circ).

- When sucrose undergoes hydrolysis by dilute acid or the enzyme **invertase**: C_{12}H_{22}O_{11}\text{ (Sucrose)} + H_2O \to C_6H_{12}O_6\text{ (D-Glucose)} + C_6H_{12}O_6\text{ (D-Fructose)}
- D-Glucose is dextrorotatory (+52.7^\circ), but D-Fructose is strongly levorotatory (-92.4^\circ).
- The resulting equimolar mixture has a net negative (levorotatory) specific rotation of **-19.8^\circ**.
- Because the sign of optical rotation **inverts from dextro (+) to levo (-)**, the hydrolyzed product is called **Invert Sugar** (present naturally in honey).

44. **Artificial Sweetening Agents (Non-Nutritive Sweeteners)**:
    - **Saccharin (o-Sulphobenzimide)**: The first popular artificial sweetener (discovered in 1879 by Remsen and Fahlberg); approximately **550\text{ times} sweeter than sucrose**. Excreted unchanged in urine; provides zero calories.
    - **Aspartame**: A methyl ester of phenylalanine/aspartic acid dipeptide; approximately **100–200\text{ times} sweeter than sucrose**. **Thermally unstable at cooking temperatures**, so restricted exclusively to cold foods and soft drinks. Unsafe for individuals suffering from phenylketonuria (PKU).
    - **Sucralose**: Trichloro derivative of sucrose; approximately **600\text{ times} sweeter than sucrose**. Stable at cooking and baking temperatures; non-caloric.
    - **Alitame**: High-potency aspartic acid dipeptide; approximately **2000\text{ times} sweeter than sucrose**. Extremely sweet, making precise sweetness control difficult.
45. **Relative Sweetness Scale Matrix**: \text{Alitame (2000)} > \text{Sucralose (600)} > \text{Saccharin (550)} > \text{Aspartame (100–200)} > \text{Fructose (170)} > \text{Sucrose (100)} > \text{Glucose (74)} > \text{Maltose (32)} > \text{Lactose (16)}
46. **Aqua Fortis**: Alchemical name for **Nitric Acid (HNO_3)**.
47. **Muriatic Acid**: Commercial trade name for industrial **Hydrochloric Acid (HCl)**.
48. **Battery Acid**: Moderately concentrated **Sulphuric Acid (H_2SO_4, \sim 33–38\%)** used as the electrolyte in automotive lead-acid storage batteries (specific gravity 1.28).
49. **Antacids**: Weakly basic compounds that neutralize excess gastric HCl without irritating the stomach lining. Classic examples: **Milk of Magnesia (Mg(OH)_2)**, Aluminium Hydroxide gel (Al(OH)_3), and Sodium Bicarbonate (NaHCO_3).
50. **Aqua Regia**: The royal solvent consisting of **3\text{ parts concentrated } HCl + 1\text{ part concentrated } HNO_3** by volume (3:1 ratio).

One-Liner Revision – Acids, Bases, Salts

#	Fast Fact
1	pH scale = 0-14; pH < 7 = acid; pH 7 = neutral; pH > 7 = base/alkaline
2	Strongest acid = HClO4 (perchloric); in chemistry; practically H2SO4 is “king”
3	Common acid in stomach (gastric juice) = **HCl** (hydrochloric acid)
4	Formic acid (HCOOH) = in ant sting; methanoic acid; simplest organic acid
5	Acetic acid (CH3COOH) = vinegar; ethanoic acid
6	Citric acid = lemons, oranges; ascorbic acid = Vitamin C
7	Lactic acid = sour milk (curd); produced by Lactobacillus
8	Oxalic acid = in spinach/tomato leaves; kidney stone-causing
9	Sulphuric acid (H2SO4) = “King of Chemicals”; most produced industrial chemical
10	Aqua Regia = 3 parts HCl + 1 part HNO3; dissolves gold and platinum
11	Litmus = turns red in acid, blue in base; from lichen (Rocella tinctoria)
12	Phenolphthalein = colourless in acid; pink/red in base

#	Fast Fact
13	Methyl orange = red in acid; yellow in base
14	NaOH (Caustic soda) = used in soap-making, paper, textile
15	Ca(OH)2 (slaked lime / calcium hydroxide) = whitewash; water treatment; mortar
16	Baking soda (NaHCO3) = antacid for acidity; releases CO2 in baking
17	Washing soda (Na2CO3.10H2O) = removes permanent hardness; laundry
18	Plaster of Paris (CaSO4.1/2H2O) = made from gypsum (CaSO4.2H2O) by heating
19	Bleaching powder Ca(OCl)Cl = disinfects water; bleaches cloth/paper
20	Common salt (NaCl) = table salt; from sea water by evaporation or rock salt mines
21	Blood pH = 7.4 (slightly basic); maintained by bicarbonate buffer
22	Acid rain pH < 5.6 (H2SO4 + HNO3 from SO2 + NOx emissions)
23	Antacids = neutralise excess HCl in stomach; Mg(OH)2, Al(OH)3, NaHCO3
24	Hard water + soap = scum (Ca/Mg stearate); soft water lathers well
25	Soap is sodium salt of fatty acid; detergent = synthetic sulphonate/sulphate
26	Saponification = hydrolysis of fats/oils with NaOH/KOH to make soap + glycerol
27	Sugar (sucrose) = C12H22O11; invert sugar = glucose + fructose (after acid hydrolysis)
28	Glucose = C6H12O6; dextrose; simplest monosaccharide; brain's primary fuel
29	Fructose = fruit sugar; sweetest naturally occurring sugar
30	Lactose = milk sugar; disaccharide (glucose + galactose); lactose intolerance = deficiency of lactase

Chemistry Topic 5 — Carbon, Organic Chemistry and Polymers

Consolidated — 50 Must-Score Facts

1. **Carbon (Z=6)** exhibits exceptional chemical versatility due to two foundational properties: **Catenation** (the unique ability of carbon atoms to link together via strong covalent bonds to form long open chains, branched structures, or rings) and **Tetravalency** (sharing four valence electrons to form 4 covalent bonds).
2. The C-C single bond dissociation enthalpy is exceptionally high (**348\text{ kJ/mol}**), imparting extraordinary thermodynamic stability to organic chains (unlike Silicon, whose Si-Si bond energy is only 222\text{ kJ/mol}).
3. **Hybridization States of Carbon**: - sp^3 (Tetrahedral, bond angle 109^\circ 28′, all 4 are single \sigma-bonds, e.g., Methane, Diamond, Alkanes). - sp^2 (Trigonal planar, bond angle 120^\circ, one double bond containing 1\sigma + 1\pi, e.g., Ethene, Graphite, Graphene, Benzene). - sp (Linear, bond angle 180^\circ, one triple bond containing 1\sigma + 2\pi, e.g., Ethyne/Acetylene).
4. **Alkanes** (C_n H_{2n+2}): Saturated hydrocarbons with single bonds. Formerly termed **paraffins** (Latin *parum affinis* = little affinity) due to their chemical inertness toward acids and bases under ambient conditions.
5. **Methane (CH_4)**: The simplest alkane; known historically as **Marsh Gas** because it is generated by anaerobic methanogenic archaebacteria in stagnant swamps, waterlogged paddy fields, ruminant animal digestive tracts (cattle cud-chewing), and decomposing municipal landfill sites. It is the primary constituent of Natural Gas (\approx 85–95\%) and Biogas/Gobar Gas (\approx 55–70\%).
6. **Alkenes** (C_n H_{2n}): Unsaturated hydrocarbons containing at least one carbon-carbon double bond (C=C). Known as **olefins** (oil-forming).
7. **Ethylene / Ethene (C_2H_4)**: The only gaseous **plant hormone** that naturally promotes fruit ripening and leaf abscission; also serves as the chemical monomer for manufacturing polyethylene/polythene.
8. **Alkynes** (C_n H_{2n-2}): Unsaturated hydrocarbons containing at least one carbon-carbon triple bond (C \equiv C).
9. **Acetylene / Ethyne (C_2H_2)**: Burns in pure oxygen to produce an intensely hot **Oxy-acetylene flame (\approx 3200^\circ\text{C})** used for industrial cutting and welding of steel.
10. **Artificial Fruit Ripening**: Industrial ripening often uses **Calcium Carbide (CaC_2)**, which reacts with atmospheric moisture to release acetylene gas ($CaC_2 + 2H_2O \to Ca(OH)_2 + C_2H_2 \uparrow$). *Must-Score*: Commercial calcium carbide typically contains toxic trace impurities of arsenic hydride (arsine, AsH_3) and phosphorus hydride (phosphine, PH_3), leading FSSAI to ban its use in food trade.
11. **Benzene (C_6H_6)**: The archetypal aromatic hydrocarbon. Proposed by August Kekulé in 1865 as a planar hexagonal ring of 6 sp^2 carbon atoms with alternating double bonds. Thermodynamically stabilized by resonance with a delocalized cloud of 6\pi electrons complying with **Hückel's Rule (4n+2 \pi-electrons, where n=1)**.
12. **Allotropes of Carbon**: Pure elemental carbon exists as crystalline allotropes (**Diamond, Graphite, Fullerenes, Carbon Nanotubes, Graphene**) and amorphous forms (Coal, Charcoal, Lampblack/Carbon Black, Coke).
13. **Diamond**: Each carbon is sp^3 hybridized and bonded tetrahedrally to 4 other carbons in a rigid 3D framework (C-C bond length 1.54\text{ \AA}). It is the hardest known natural substance (Mohs 10), an electrical insulator, but an exceptional thermal conductor.
14. **Graphite**: Each carbon is sp^2 hybridized and bonded to 3 others in planar hexagonal rings (C-C bond length 1.42\text{ \AA}). Layers are held by weak van der Waals dispersion forces (3.35\text{ \AA} gap). Delocalized \pi electrons allow it to **conduct electricity**. It is soft, slippery, and acts as a **solid dry lubricant**.
15. **Thermodynamic Stability of Carbon**: **Graphite is the thermodynamically most stable allotrope of carbon** under standard ambient conditions (298.15\text{ K}, 1\text{ bar}); its standard enthalpy of formation is defined as zero ($\Delta_f H^\circ = 0$). Diamond is metastable ($\Delta_f H^\circ = +1.9\text{ kJ/mol}$).
16. **Fullerenes (C_{60}, Buckminsterfullerene)**: Discovered in 1985 by Harold Kroto, Robert Curl, and Richard Smalley. Spherical cage resembling a soccer ball ("buckyball"), composed of **20 hexagonal rings and 12**

pentagonal rings.

17. **Carbon Nanotubes (CNTs)**: Discovered by Sumio Iijima in 1991. Cylindrical tubules of rolled graphene sheets with diameters on the nanometer scale; possess tensile strength >50 times that of steel and high thermal/electrical conductivity.
18. **Graphene**: Isolated in 2004 by Andre Geim and Konstantin Novoselov at Manchester University using adhesive tape exfoliation of graphite (Nobel Prize in Physics, 2010). A single atom-thick, two-dimensional honeycomb sheet of sp^2 carbon; nearly transparent (97.7\% light transmission), ballistic electron conductor, and over 100\text{ times} stronger than structural steel.
19. **Black Gold**: Synthesized in 2019 by scientists at the Tata Institute of Fundamental Research (TIFR, Mumbai) by rearranging the inter-nanoparticle spacing of gold nanoparticles; absorbs visible and near-infrared solar light, acting as a solar water desalination absorber and CO_2 conversion catalyst.
20. **Chronology of Landmark Advanced Materials**: \text{Kevlar (1965)} \to \text{Fullerenes (1985)} \to \text{Carbon Nanotubes (1991)} \to \text{Graphene (2004)} \to \text{Black Gold (2019)}
21. **Polymer**: A high-molecular-weight macromolecule formed by the repetitive chemical linkage of small, reactive monomer molecules ($M_w \approx 10^3\text{ to }10^7\text{ u}$).
22. **Addition (Chain-Growth) Polymers**: Formed by the repeated addition of unsaturated monomer units (containing C=C or C \equiv C) without the elimination of any small byproduct molecules (e.g., Polyethylene, Polypropylene, PVC, Polystyrene, Teflon).
23. **Condensation (Step-Growth) Polymers**: Formed by repeated condensation reactions between bi-functional or poly-functional monomers, accompanied by the **elimination of small byproduct molecules** such as water (H_2O), alcohol (CH_3OH), or ammonia (NH_3) (e.g., Nylon-6,6, Terylene/Dacron, Bakelite, Melamine).
24. **Classification Based on Intermolecular Forces**:
    - **Elastomers**: Possess the **weakest intermolecular forces** (van der Waals forces). Polymer chains are randomly coiled and held together by occasional cross-links, allowing them to stretch under tension and snap back to original shape (e.g., Natural rubber, Vulcanized rubber, Buna-S, Buna-N, Neoprene).
    - **Fibres**: Possess the **strongest intermolecular forces** (extensive intermolecular hydrogen bonding or strong dipole-dipole interactions). Long, linear chains pack tightly into crystalline arrays, providing high tensile strength and high melting points (e.g., Nylon-6,6, Terylene, Kevlar, Silk, Wool).
    - **Thermoplastics**: Possess **intermediate intermolecular forces** between elastomers and fibres. Linear or slightly branched polymers that **soften on heating and harden upon cooling reversibly**; can be repeatedly melted, remoulded, and recycled (e.g., Polyethylene, PVC, Polystyrene, Teflon).
    - **Thermosetting Plastics**: Form extensive **three-dimensional cross-linked network polymers**. During moulding, heating triggers irreversible chemical cross-linking. Once set, they **cannot be remelted, softened, or reshaped**; excessive heating causes degradation. They are completely non-recyclable (e.g., Bakelite, Melamine-formaldehyde, Urea-formaldehyde).
25. **Polyethylene / Polythene**:
    - **Low-Density Polyethylene (LDPE)**: Polymerized under extreme pressure (1000–2000\text{ atm}) and temperature (200–350^\circ\text{C}) via a free-radical mechanism. Possesses extensive branched chains that prevent close packing, yielding low density and high flexibility; used for squeeze bottles, packaging films, and flexible tubing.
    - **High-Density Polyethylene (HDPE)**: Polymerized under low pressure (6–7\text{ atm}) and moderate temperature (60–70^\circ\text{C}) in the presence of a **Ziegler-Natta Catalyst** (triethylaluminium and titanium tetrachloride, $Al(C_2H_5)_3 + TiCl_4$). Consists of linear chains that pack tightly into crystalline structures, providing high density, rigidity, and chemical resistance; used for heavy buckets, dustbins, pipes, and milk crates.
26. **Polyvinyl Chloride (PVC)**: Formed by the addition polymerization of **Vinyl Chloride ($CH_2=CHCl$)**. An electrical insulator used for drainage pipes, electrical wire coatings, raincoats, and vinyl flooring.

27. **Teflon (PTFE, Polytetrafluoroethylene)**: Formed by the addition polymerization of **Tetrafluoroethylene (CF_2=CF_2)** under high pressure with persulphate catalyst: n(CF_2=CF_2) \to [-CF_2-CF_2-]_n Possesses high thermal stability (operating range up to 260^\circ\text{C}), chemical inertness against boiling aqua regia and strong alkalis, and an exceptionally low coefficient of friction; used as a **non-stick coating on kitchen cookware**, low-friction machinery bearings, and chemical valve gaskets.
28. **Bakelite (Phenol-Formaldehyde Resin)**: The first fully synthetic commercial plastic, invented in 1907 by Belgian-American chemist **Leo Baekeland**:
    - A **thermosetting cross-linked polymer** formed by the condensation of **Phenol (C_6H_5OH)** with **Formaldehyde (HCHO)** in the presence of an acid or base catalyst.
    - The initial linear intermediate polymer is **Novolac** (used in paints and wood lacquers). Heating Novolac with excess formaldehyde produces the rigid, cross-linked infusible solid Bakelite.
    - An electrical insulator resistant to heat and chemicals; standard material for **electrical switches, plugs, sockets, distributor caps**, and heat-insulated handles of frying pans and pressure cookers.
29. **Melamine-Formaldehyde Resin**: A thermosetting polymer formed by the condensation of Melamine (2,4,6-triamino-1,3,5-triazine) with Formaldehyde. Highly durable and impact-resistant; used to manufacture **unbreakable household crockery, dinnerware**, and decorative laminate countertops.
30. **Polystyrene**: Addition polymer of **Styrene (phenylethene, C_6H_5-CH=CH_2)**. In its foamed expanded form, it is commercially known as **Thermocol**; used as thermal insulation in refrigerators and packaging material for fragile electronics.
31. **Rayon (“Artificial Silk”)**:
    - A **semi-synthetic regenerated cellulose fibre**, NOT a synthetic petrochemical plastic or modified starch!
    - **Manufacturing**: Natural cellulose (from wood pulp or cotton linters) is dissolved in caustic soda (NaOH) and carbon disulphide (CS_2) to form a viscous cellulose xanthate solution (“viscose”), which is extruded through fine spinnerets into a bath of dilute sulphuric acid (H_2SO_4). The acid regenerates pure cellulose as continuous filaments.
    - Absorbs moisture, drapes like silk, and blends well with cotton or wool.
32. **Nylon-6,6**:
    - The first fully synthetic commercial textile fibre, synthesized in 1935 by **Wallace Carothers** at DuPont.
    - A **condensation polyamide** formed by the reaction of **Adipic Acid** (6\text{ carbons}, HOOC-(CH_2)_4-COOH) with **Hexamethylenediamine** (6\text{ carbons}, H_2N-(CH_2)_6-NH_2) at 280^\circ\text{C} under pressure: n HOOC(CH_2)_4COOH + n H_2N(CH_2)_6NH_2 \to [-CO(CH_2)_4CONH(CH_2)_6NH-]_n + 2n H_2O
    - Joined by peptide-like **amide linkages (-CO-NH-)**. High tensile strength, elasticity, and abrasion resistance; used for toothbrush bristles, tyre cords, ropes, climbing gear, and parachutes.
33. **Nylon-6 (Perlon)**: Formed by the ring-opening polymerization of the cyclic monomer **Caprolactam** (a 7-membered ring containing 6 carbon atoms) when heated with water at 260^\circ\text{C}.
34. **Terylene / Dacron (Polyester)**:
    - Formed by the condensation polymerization of **Ethylene Glycol (CH_2OH-CH_2OH)** with **Terephthalic Acid (p-HOOC-C_6H_4-COOH)** at 200^\circ\text{C} with zinc acetate–antimony trioxide catalyst.
    - Joined by **ester linkages (-COO-)**. Crease-resistant, quick-drying, and resistant to moths and chemical attack; blended with cotton to make **terycot** and with wool to make **terywool**.
35. **Polyethylene Terephthalate (PET)**: The chemical equivalent of Dacron used for blowing rigid, transparent, recyclable beverage bottles, food jars, and synthetic fleece jackets.
36. **Kevlar (Poly-para-phenylene terephthalamide)**:
    - An aromatic polyamide (**aramid fibre**) synthesized in 1965 by American chemist **Stephanie Kwolek** at DuPont.
    - Monomers: **1,4-phenylenediamine** and **terephthaloyl chloride**.

- Liquid crystalline polymer chains align parallel, held together by dense networks of intermolecular hydrogen bonds between amide groups and \pi-\pi aromatic stacking.
- Possesses an extraordinary tensile strength-to-weight ratio (**\approx 5\text{ times} stronger than structural steel on an equal weight basis**); widely used for **bulletproof tactical vests, combat helmets, military armor plating**, cut-resistant gloves, and aerospace structural composites.

37. **Lexan / Polycarbonate**: A high-impact, optically transparent thermoplastic polyester formed from bisphenol A and phosgene; used for **bulletproof transparent security windows**, riot shields, and astronaut helmet visors.
38. **Polymethyl Methacrylate (PMMA / Plexiglass / Lucite / Perspex)**: Formed by the addition polymerization of methyl methacrylate; a transparent, shatterproof thermoplastic substitute for glass used in airplane cockpit canopies, illuminated signs, and dental prosthetics.
39. **Natural Rubber**:
    - A natural addition polymer extracted from the milky latex of the rubber tree (*Hevea brasiliensis*).
    - Chemically, it is strictly **cis-1,4-polyisoprene**: n(CH_2=C(CH_3)-CH=CH_2) \to [-CH_2-C(CH_3)=CH-CH_2-]_n
    - Monomer: **Isoprene (2-methyl-1,3-butadiene)**.
    - Because of its all-*cis* stereochemical configuration, polymer chains cannot pack closely and remain coiled, conferring high elasticity. In contrast, the all-*trans* isomer is **Gutta-percha**, which is non-elastic, hard, and crystalline.
40. **Vulcanization of Rubber**:
    - Discovered serendipitously in 1839 by **Charles Goodyear**.
    - Raw natural rubber is sticky and soft at warm temperatures (>33^\circ\text{C}), brittle at low temperatures (<10^\circ\text{C}), and soluble in non-polar organic solvents (benzene, petrol).
    - **Mechanism**: Heating raw rubber with **3–5\% elemental Sulphur** and basic activators (ZnO) at 140^\circ\text{C}.
    - Sulphur atoms form stable **disulphide cross-links (-S-S-)** across the reactive double bonds of adjacent polyisoprene chains, locking them into a three-dimensional elastic network.
    - Vulcanized rubber resists abrasion, remains elastic across -40^\circ\text{C} to 100^\circ\text{C}, does not absorb water, and is used for **heavy-duty automobile tyres**. If sulphur content is increased to 30–35\%, extremely hard, non-elastic **Ebonite (vulcanite)** is formed.
41. **Neoprene (Polychloroprene)**:
    - The first synthetic commercial elastomer (invented at DuPont, 1930).
    - Monomer: **Chloroprene (2-chloro-1,3-butadiene)**: n(CH_2=C(Cl)-CH=CH_2) \to [-CH_2-C(Cl)=CH-CH_2-]_n
    - Highly resistant to petroleum oils, greases, gasoline, and atmospheric ozone oxidation; widely used for manufacturing fuel delivery hoses, automotive engine gaskets, conveyor belts, and diving wetsuits.
42. **Buna-S (SBR, Styrene-Butadiene Rubber)**:
    - A synthetic copolymer formed by the addition polymerization of **75% 1,3-Butadiene and 25% Styrene** in the presence of sodium catalyst ("Bu" = Butadiene, "Na" = Natrium/Sodium catalyst, "S" = Styrene).
    - High abrasion resistance and mechanical toughness; primarily used for automotive road tyres.
43. **Buna-N (NBR, Nitrile Rubber)**:
    - A synthetic copolymer of **1,3-Butadiene and Acrylonitrile (CH_2=CH-CN)**.
    - Resistant to the swelling action of petrol, oils, and organic solvents; used for oil seals, fuel tank linings, and laboratory safety gloves.
44. **Natural Protein Fibres**:
    - **Silk**: Natural structural animal protein fibre secreted by silkworms (*Bombyx mori*). Composed of **Fibroin** (\approx 75\%, rich in glycine, alanine, and serine forming pleated \beta-sheets) bound together by an outer gummy protein sheath called **sericin** (\approx 25\%).

- **Wool**: Natural protein fibre harvested from sheep fleece. Composed of **\alpha-Keratin**, a fibrous protein with an \alpha-helical secondary structure stabilized by hydrogen bonds and extensive **cystine disulphide bridges (-S-S-)**; burning wool releases a characteristic pungent odor of burning sulphur.

45. **Cellulose**: The most abundant organic polymer on Earth. A linear homopolysaccharide composed of **\beta-D-glucose** units joined by **\beta-1,4 glycosidic bonds**. Humans cannot digest cellulose because our digestive tract lacks the enzyme cellulase.
46. **Starch**: Plant storage carbohydrate consisting of **\alpha-Amylose** (water-soluble, linear \alpha-1,4 polymer) and **Amylopectin** (water-insoluble, branched \alpha-1,4 and \alpha-1,6 polymer). Gives a characteristic deep blue-black color with iodine solution.
47. **PHBV (Poly-\beta-hydroxybutyrate-co-\beta-hydroxyvalerate)**: A commercially significant **biodegradable aliphatic polyester** synthesized by bacterial fermentation (*Ralstonia eutropha*) from glucose and propionic acid; degrades completely in soil and landfill environments via microbial enzymes.
48. **Glyptal**: A cross-linked polyester resin formed by the condensation of Ethylene Glycol with Phthalic Acid; used in manufacturing heavy-duty industrial paints, lacquers, and electrical building primers.
49. **Urea-Formaldehyde Resin**: Thermosetting adhesive resin used for manufacturing particle boards, decorative laminated sheets, and unbreakable plastic bottle closures.
50. **Plasticizer**: Low-molecular-weight organic liquids (e.g., dioctyl phthalate, DOP) added to rigid polymers like PVC to lower glass transition temperature (T_g) and impart mechanical softness and flexibility (used for garden hoses, synthetic leather).

One-Liner Revision – Carbon, Organic Chemistry and Polymers

#	Fast Fact
1	Carbon = tetravalent; can form 4 bonds; catenation (chain-forming) ability
2	Allotropes of carbon: Diamond (hardest), Graphite (soft, conductor), Fullerene (C60)
3	Diamond = 3D tetrahedral; hardest; electrical insulator
4	Graphite = layered hexagonal; soft; good electrical conductor; used as lubricant and in pencils
5	Fullerene (Buckminsterfullerene C60) = 60 carbon atoms in soccer-ball shape; discovered 1985
6	Methane (CH4) = simplest alkane; natural gas; greenhouse gas
7	Ethylene (C2H4) = simplest alkene; plant hormone (fruit ripening); makes polyethylene
8	Acetylene (C2H2) = simplest alkyne; used in oxyacetylene welding; very hot flame
9	Benzene (C6H6) = simplest aromatic compound; carcinogen; aromatic ring
10	Nylon = first synthetic fibre (1935, Du Pont); polyamide; used in parachutes, stockings
11	Polyester (PET) = from ethylene glycol + terephthalic acid; soft-drink bottles, clothing
12	Bakelite = first synthetic plastic (1907, Leo Baekeland); thermosetting; phenol + formaldehyde
13	PVC (polyvinyl chloride) = monomer = vinyl chloride; pipes, wires, credit cards
14	Polythene (polyethylene) = monomer = ethylene; plastic bags, bottles
15	Teflon (PTFE) = polytetrafluoroethylene; non-stick cookware; very low friction
16	Natural rubber = polymer of isoprene (C5H8)n; from Hevea brasiliensis
17	Vulcanisation = rubber + sulphur at 140 degC; makes rubber stronger and elastic (Goodyear, 1839)
18	Cellulose = polymer of beta-glucose; plant cell wall; not digestible by humans
19	Starch = polymer of alpha-glucose; energy storage in plants; digested by amylase
20	Proteins = polymer of amino acids; peptide bonds; 20 standard amino acids
21	DNA = polymer of nucleotides (deoxyribose + phosphate + base)

#	Fast Fact
22	Soap = sodium/potassium salt of fatty acids; amphipathic molecule (hydrophilic + hydrophobic end)
23	Biodegradable plastics = break down by microorganisms; e.g., PLA (polylactic acid)
24	Micro-plastics = plastic particles < 5 mm; accumulate in food chain; major environmental threat
25	Rayon = regenerated cellulose fibre; “artificial silk”; from wood pulp
26	LPG = mainly propane (C3H8) + butane (C4H10); ethyl mercaptan added as odorant
27	CNG = compressed natural gas (mainly methane CH4); cleaner than diesel
28	Ethanol (C2H5OH) = ethyl alcohol; from fermentation; used as fuel additive (E10 = 10% ethanol)
29	Methanol (CH3OH) = wood alcohol; poisonous; causes blindness and death
30	Chloroform (CHCl3) = trichloromethane; used as anaesthetic historically; now solvent

Chemistry Topic 6 — Gases, Fuels and Energy

Consolidated — 50 Must-Score Facts

1. A **Fuel** is any combustible substance containing chemical energy that undergoes controlled exothermal reaction with oxygen to generate useful thermal energy.
2. An **Ideal Fuel** possesses: high calorific value, moderate ignition temperature (neither too low nor too high), low moisture content, burns without producing poisonous gases or ash, and is safe and economical to transport.
3. **Calorific Value** is the total quantity of heat liberated by the complete combustion of a unit mass (1\text{ kg}) or unit volume (1\text{ m}^3) of a fuel, expressed in **\text{kJ/kg}** or \text{kcal/kg}.
4. **Hydrogen (H_2)** has the **highest calorific value of all known fuels (150,000\text{ kJ/kg})**, roughly 3 times higher than petrol or diesel.
5. Despite its immense calorific value, hydrogen is not yet used as a common domestic fuel because of its high flammability, wide explosion limits (4–75\% in air), low ignition energy, and technical difficulty in liquefying and storing it under high pressures.
6. **Master Hierarchy of Calorific Values (\text{kJ/kg})**: \text{Hydrogen (150,000)} > \text{Methane (55,000)} > \text{LPG (50,000)} > \text{CNG (50,000)} > \text{Petrol/Diesel (45,000)} > \text{Kerosene (45,000)} > \text{Biogas (35,000–40,000)} > \text{Coal (25,000–33,000)} > \text{Wood (17,000)} > \text{Cow dung cake (6,000–8,000)}
7. **LPG (Liquefied Petroleum Gas)**: - Sourced as a byproduct of petroleum crude fractional distillation and natural gas processing. - Primarily composed of **n-Butane and Isobutane (\approx 60\%)** with **Propane (\approx 40\%)**. In standard state PCS questions where single choices are given, **Butane** is the primary constituent. - Liquefies under moderate pressure (6–8\text{ bar}) inside steel cylinders at room temperature, expanding \approx 250 times into gas upon release through the pressure regulator.
8. **Ethyl Mercaptan (Ethanethiol, C_2H_5SH)**: Natural alkane gases (butane, propane, methane) are completely colorless and odorless. To detect accidental leaks instantly and prevent catastrophic household explosions, a minute concentration of foul, pungent **Ethyl Mercaptan** (resembling the odor of rotten cabbages or sulfur) is deliberately blended into commercial LPG cylinders.
9. **Natural Gas**: Naturally occurring gaseous fossil fuel extracted from underground geological reservoirs (often associated with crude petroleum deposits). Composed predominantly of **Methane (CH_4, 85–95\%)**, with minor fractions of ethane (C_2H_6), propane, butane, and traces of nitrogen and CO_2.
10. **CNG (Compressed Natural Gas)**:
    - Chemically identical to Natural Gas; composed primarily of **Methane (CH_4, >85\%)**.
    - Compressed to a high pressure of **200\text{ to }250\text{ bar}** (20–25\text{ MPa}) inside high-strength composite vehicle cylinders.
    - Used as a clean alternative automotive fuel for buses, autos, and cars; emits significantly lower carbon monoxide (CO), nitrogen oxides (NO_x), and zero lead or sulfur oxides compared to petrol/diesel.
11. **LNG (Liquefied Natural Gas)**:
    - Natural gas chilled to cryogenic temperatures of **-162^\circ\text{C} (-260^\circ\text{F})** at atmospheric pressure.
    - Liquefaction reduces the physical gas volume by approximately **600\text{ times}**, enabling economical overseas shipping in double-hulled, insulated cryogenic carrier vessels.
12. **Biogas (Gobar Gas)**:
    - Produced by the **anaerobic microbial digestion** of animal manure (cow dung), human sewage, and wet agricultural biomass by specialized methanogenic archaebacteria (*Methanococcus*, *Methanobacterium*).
    - Composition: **Methane (CH_4, 55–70\%)**, **Carbon Dioxide (CO_2, 30–45\%)**, with traces of Hydrogen (H_2, 1–5\%), Nitrogen (N_2, 1–2\%), and Hydrogen Sulphide (H_2S, trace).
    - Methane is the sole combustible, heat-generating component in biogas.

- The spent digested slurry remaining in the digester tank is exceptionally rich in **Nitrogen (\approx 1.5–2\%) and Phosphorus (\approx 1\%)**, serving as high-grade organic bio-fertilizer.

13. **Water Gas (Synthesis Gas / Syngas)**:
    - An equimolar gaseous fuel mixture composed of **Carbon Monoxide and Hydrogen (CO + H_2)**: C\text{ (red-hot coke)} + H_2O\text{ (superheated steam)} \xrightarrow{1000^\circ\text{C}} CO + H_2 \quad (\Delta H = +131\text{ kJ/mol, Endothermic})
    - Burns with an intensely hot, non-luminous blue flame (historically termed "Blue Water Gas").
    - Key industrial feedstock for the catalytic synthesis of methanol (CH_3OH) and liquid hydrocarbons (Fischer-Tropsch process).
14. **Producer Gas**:
    - A gaseous fuel mixture composed primarily of **Carbon Monoxide and Nitrogen (CO + N_2)**: 2C\text{ (coke)} + O_2 + 4N_2\text{ (air)} \xrightarrow{1100^\circ\text{C}} 2CO + 4N_2 \quad (\Delta H = -221\text{ kJ/mol, Exothermic})
    - Typical volumetric composition: **\sim 30\% CO, \sim 60\% N_2**, and small fractions of H_2 and CO_2.
    - Because inert atmospheric nitrogen makes up nearly two-thirds of the volume, producer gas has a relatively low calorific value (\approx 5000\text{ kJ/m}^3); widely used for firing industrial open-hearth steel furnaces, glass-melting tanks, and ceramic kilns.
15. **Coal Gas**:
    - Manufactured by the destructive distillation (pyrolysis) of bituminous coal in sealed retorts in the absence of air at 1000^\circ\text{C}.
    - Composition: **Hydrogen (H_2, \approx 50\%)**, **Methane (CH_4, \approx 30–35\%)**, **Carbon Monoxide (CO, \approx 8–10\%)**, with traces of ethylene and nitrogen.
    - Historically used for municipal street lighting in 19th-century London and Paris before the advent of electricity.
16. **Coal Classification & Ranks** (by increasing geological age, carbon content, and calorific value): \text{Peat} \to \text{Lignite} \to \text{Bituminous} \to \text{Anthracite}
17. **Peat**: Earliest geological stage of coal formation; brown fibrous mass containing **<60\% Carbon**, high moisture (>70\%), burns with heavy smoke and low calorific value (10,000–15,000\text{ kJ/kg}).
18. **Lignite ("Brown Coal")**: Soft, crumbly coal containing **60–70\% Carbon** and high residual moisture; lowest commercial rank of coal. In India, massive deposits are mined at **Neyveli (Tamil Nadu)** for thermal power generation.
19. **Bituminous Coal ("Soft Coal" / Commercial Coal)**: The most abundant and widely used commercial coal; black, dense, contains **75–85\% Carbon** and low moisture; standard fuel for thermal electricity generation and metallurgical coking in blast furnaces.
20. **Anthracite Coal ("Hard Coal")**: The highest rank and geologically oldest coal; black, lustrous, contains **>90–95\% Carbon**; burns slowly with a smokeless, pale blue flame, minimal ash, and the highest calorific value (\approx 33,000\text{ kJ/kg}). In India, found exclusively in small deposits in **Jammu and Kashmir (Kalakot, Reasi)**.
21. **Petroleum Fractions by Boiling Point**:
    - Petroleum Gas (LPG): b.p. < 20^\circ\text{C} (C_1–C_4)
    - Petrol / Gasoline: b.p. 40–120^\circ\text{C} (C_5–C_{10})
    - Kerosene: b.p. 150–250^\circ\text{C} (C_{10–14})
    - Diesel Oil: b.p. 250–350^\circ\text{C} (C_{15–18})
    - Lubricating Oil & Paraffin Wax: b.p. > 350^\circ\text{C} (C_{18–25})
    - Bitumen / Asphalt: Non-volatile residue (C_{30+}) used for road surfacing.

22. **Engine Knocking**: In an internal combustion engine, if the air-fuel mixture ignites prematurely (pre-ignition or detonation) before the spark plug fires, shock waves hit the cylinder walls, producing an audible metallic rattling noise known as **knocking**, which reduces power output and causes engine wear.
23. **Octane Rating (Octane Number)**:
    - Measures the anti-knock resistance of **Petrol (Gasoline)** in spark-ignition engines.
    - Standardized scale:
    - **n-Heptane (C_7H_{16})** knocks violently and is assigned an **Octane Number = 0**.
    - **Iso-octane (2,2,4-trimethylpentane, C_8H_{18})** burns smoothly and is assigned an **Octane Number = 100**.
    - If a petrol sample has an octane rating of 91, it exhibits the exact anti-knock combustion characteristics of a mixture of 91\% isooctane and 9\% n-heptane by volume.
24. **Anti-Knock Additives**:
    - Historically, **Tetraethyl Lead (TEL, Pb(C_2H_5)_4)** was added to petrol to boost octane rating. Banned globally because lead emissions cause neurological toxicity in children and permanently deactivate automotive catalytic converters.
    - Modern unleaded petrol uses oxygenates such as **MTBE (Methyl tert-butyl ether)** and **Ethanol**.
25. **Cetane Rating (Cetane Number)**:
    - Measures the ignition delay and combustion quality of **Diesel Fuel** in compression-ignition engines.
    - Standardized scale:
    - **n-Hexadecane (Cetane, C_{16}H_{34})** has a very short ignition delay and is assigned **Cetane Number = 100**.
    - **\alpha-Methylnaphthalene** has a long ignition delay and is assigned **Cetane Number = 0**.
    - Higher cetane numbers indicate quicker cold starting and smoother diesel engine operation.
26. **Generations of Biofuels**:
    - **1G (First Generation)**: Produced from edible human food crops (sugarcane juice, corn starch, wheat, edible vegetable oils). Creates food-versus-fuel conflict.
    - **2G (Second Generation)**: Produced from non-food agricultural lignocellulosic residues (rice straw, wheat stalk, sugarcane bagasse, corn cobs, bamboo). India's commercial 2G bio-ethanol refinery is operational at **Panipat (Haryana)**.
    - **3G (Third Generation)**: Extracted from aquatic algal biomass (microalgae) grown in sewage or saline wastewater ponds; rapid growth rates, zero arable land needed.
    - **4G (Fourth Generation)**: Utilizes genetically modified photosynthetic microorganisms and solar fuels that capture and convert atmospheric CO_2 directly into biofuels.
27. **Ethanol Blending in India**:
    - **Ethanol (C_2H_5OH)**: Produced by yeast (*Saccharomyces cerevisiae*) fermentation of molasses (byproduct of sugar manufacturing), broken grains (damaged rice, maize), and agricultural cellulose.
    - **Blending Ratio**: E10 (10\% ethanol, 90\% petrol); E20 (20\% ethanol, 80\% petrol).
    - Ethanol contains **34.7\% oxygen by mass**, functioning as an effective oxygenate that promotes complete fuel combustion, reducing carbon monoxide (CO) tailpipe emissions by up to 30\%, while boosting the fuel's octane rating.
28. **Biodiesel**:
    - A clean-burning renewable diesel substitute produced by the **transesterification** of non-edible vegetable seed oils.
    - Primary Indian feedstock: Seeds of **Jatropha curcas (Ratanjyot)** and **Pongamia pinnata (Karanja)**, as well as Used Cooking Oil (UCO).

- **Transesterification Reaction**: Vegetable oil triglycerides react with methanol (CH_3OH) in the presence of a base catalyst (NaOH or KOH) to produce **Fatty Acid Methyl Esters (FAME, Biodiesel)** and a commercially valuable byproduct, **Glycerol (Glycerine)**.

29. **Combustion**: An exothermic redox chemical reaction between a fuel and an oxidant (usually atmospheric oxygen) accompanied by the evolution of heat and light.
    - *Complete Combustion*: Occurs in excess oxygen; hydrocarbons burn with a clean, non-luminous blue flame, yielding exclusively CO_2 and H_2O: CH_4 + 2O_2 \to CO_2 + 2H_2O + 890\text{ kJ/mol}
    - *Incomplete Combustion*: Occurs in restricted/deficient oxygen supply; burns with a yellow, smoky, sooty flame, producing toxic **Carbon Monoxide (CO)** and unburnt carbon particulate soot.
30. **Carbon Monoxide (CO) Toxicity**: Known as the **"Silent Killer"**; a colorless, odorless, non-irritating toxic gas. When inhaled, CO binds to human hemoglobin with an affinity **\approx 200–250\text{ times} greater than oxygen**, forming stable **Carboxyhemoglobin (CO-Hb)**. This halts systemic oxygen transport to brain and heart tissues, causing headache, dizziness, unconsciousness, and asphyxiation within minutes (often occurring when charcoal braziers/angithis are burned inside closed, unventilated bedrooms during winter).
31. **Ignition Temperature**: The minimum temperature to which a combustible substance must be heated in air before it catches fire and sustains combustion. A substance cannot catch fire if its temperature is below its ignition temperature.
32. **Inflammable Substances**: Substances with exceptionally low ignition temperatures that easily catch fire with a flame (e.g., Petrol, Alcohol, LPG, Acetone).
33. **Structure of a Candle Flame (Three Concentric Zones)**:
    - **Innermost Dark Zone**: Surrounds the wick; contains unburnt vaporized paraffin wax; dark black; lowest temperature (coolest zone); no combustion occurs due to lack of oxygen.
    - **Middle Luminous Zone**: Region of incomplete combustion; yellow, glowing flame; carbon particles glow incandescently, emitting light; moderately hot.
    - **Outermost Non-Luminous Zone**: Region of complete combustion; blue flame; excess oxygen supply; **hottest part of the flame**. Goldsmiths blow air into this outermost blue zone with a metallic blowpipe to melt gold and silver.
34. **Fire Extinguisher Physics & Chemistry**:
    - Combustion requires three simultaneous elements (the **Fire Triangle**): **Fuel**, **Oxygen**, and **Heat (above ignition temperature)**. Eliminating any one component extinguishes the fire.
    - **Water**: Extinguishes wood and paper fires by cooling the burning material below its ignition temperature, while steam envelopes the fuel, displacing oxygen.
    - *Why Water Cannot Extinguish Oil/Petrol Fires*: Petrol and oil are immiscible with water and have a lower density (0.7–0.8\text{ g/cm}^3); water sinks to the bottom, while the burning petrol continues to float and spread across the water surface.
    - *Why Water Cannot Extinguish Electrical Fires*: Impure tap water conducts electricity due to dissolved mineral ions, creating an electric shock hazard for emergency responders.
    - **Carbon Dioxide (CO_2) Extinguishers**: Ideal for electrical and flammable liquid fires. CO_2 is \approx 1.5\text{ times} denser than air; it blankets the fire like a heavy gas curtain, cutting off atmospheric oxygen supply, while its rapid expansion from high-pressure cylinders cools the fuel below ignition temperature.
35. **Boyle's Law (Robert Boyle, 1662)**: At constant temperature, the volume (V) of a given mass of dry gas is inversely proportional to its pressure (P): P \propto \frac{1}{V} \implies P_1 V_1 = P_2 V_2 \quad (\text{at constant } T, n)
36. **Charles's Law (Jacques Charles, 1787)**: At constant pressure, the volume (V) of a given mass of dry gas is directly proportional to its absolute thermodynamic temperature (T in Kelvin): V \propto T \implies \frac{V_1}{T_1} = \frac{V_2}{T_2} \quad (\text{at constant } P, n)

37. **Gay-Lussac's Law (Pressure-Temperature Law)**: At constant volume, the pressure (P) of a given mass of gas is directly proportional to its absolute temperature (T): \frac{P_1}{T_1} = \frac{P_2}{T_2} \quad (\text{at constant } V, n) Explains why vehicle tyre pressure increases noticeably during long, hot summer highway drives.
38. **Avogadro's Law (Amedeo Avogadro, 1811)**: Equal volumes of all gases under identical conditions of temperature and pressure contain an equal number of molecules (V \propto n).
39. **Molar Volume of Gas at STP**: One mole of any ideal gas occupies strictly **22.414\text{ Liters} (22.4\text{ dm}^3)** at Standard Temperature and Pressure (0^\circ\text{C} or 273.15\text{ K}, and 1\text{ atm} pressure).
40. **Ideal Gas Equation**: Combining Boyle's, Charles's, and Avogadro's laws yields: P V = n R T where R is the Universal Gas Constant: R = 8.314\text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1} = 0.0821\text{ L}\cdot\text{atm}\cdot\text{mol}^{-1}\cdot\text{K}^{-1} \approx 2.0\text{ cal}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}
41. **Real Gases vs. Ideal Gases**: Real gases deviate from ideal behavior at **high pressures and low temperatures** because intermolecular attractive forces become significant and the volume occupied by gas molecules is no longer negligible. A real gas behaves most ideally under conditions of **low pressure and high temperature**.
42. **van der Waals Equation of State**: \left(P + \frac{a n^2}{V^2}\right)(V - n b) = n R T where a corrects for intermolecular attractive forces and b (co-volume) corrects for the finite physical volume of gas molecules.
43. **Graham's Law of Diffusion (Thomas Graham, 1829)**: At constant temperature and pressure, the rate of diffusion or effusion of a gas (r) is inversely proportional to the square root of its molar mass (M) or vapor density (d): \frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{d_2}{d_1}} Because Hydrogen (M=2) is lighter than Oxygen (M=32), hydrogen diffuses \sqrt{32/2} = \sqrt{16} = 4\text{ times faster} than oxygen.
44. **Dalton's Law of Partial Pressures (John Dalton, 1801)**: The total pressure exerted by a mixture of non-reacting gases is equal to the sum of the partial pressures that each individual gas would exert if it alone occupied the entire volume: P_{\text{total}} = P_1 + P_2 + P_3 + \dots
45. **Joule-Thomson Effect**: When a real gas under high pressure is allowed to expand adiabatically through a porous plug or narrow throttle valve into a region of low pressure, its temperature falls (cooling effect). This phenomenon is the engineering foundation for the **liquefaction of gases (Linde and Claude processes)** and household compression refrigeration cycles. (Exceptions: Hydrogen and Helium warm up slightly at room temperature because their inversion temperatures are well below 0^\circ\text{C}).
46. **Laughing Gas**: **Nitrous Oxide (N_2O)**; used as an anesthetic in dentistry and as an oxidizer in racing engines.
47. **Tear Gas**: Chloropicrin (CCl_3NO_2) or CS gas (2-chlorobenzalmalononitrile); causes severe lachrymation and temporary incapacitation.
48. **Phosgene (COCl_2)**: Carbonyl chloride; an extremely toxic chemical warfare asphyxiant gas formed when chloroform (CHCl_3) slowly oxidizes in air and sunlight: 2CHCl_3 + O_2 \xrightarrow{\text{light}} 2COCl_2 + 2HCl Hence, medical chloroform is stored in completely filled, dark amber-colored bottles with 1\% ethanol added to neutralize trace phosgene.
49. **Mustard Gas (Dichlorodiethyl sulphide)**: Vesicant chemical warfare blister agent deployed in World War I, causing painful chemical burns on skin and lungs.
50. **Bhopal Gas Tragedy (December 2–3, 1984)**: The world's worst industrial disaster, caused by the accidental release of over 40 tonnes of lethal **Methyl Isocyanate (MIC, CH_3NCO)** gas from the Union Carbide pesticide plant in Bhopal, Madhya Pradesh, reacting with water in a runaway exothermic reaction.

One-Liner Revision – Gases, Fuels and Energy Chemistry

#	Fast Fact
1	Calorific values (approx): Hydrogen = 142 MJ/kg; CNG/methane = 55 MJ/kg; Petrol = 47 MJ/kg
2	Highest calorific value fuel = **Hydrogen** (142 MJ/kg; "fuel of the future")
3	LPG composition: Propane (C3H8) + Butane (C4H10); stored as liquid; ethyl mercaptan = smell agent

#	Fast Fact
4	CNG = mainly **methane (CH4)**; stored at 200-250 bar; cleanest fossil fuel
5	Coal types by carbon %: Peat (60%) < Lignite (70%) < Bituminous (80%) < Anthracite (90-95%)
6	Best quality coal = **Anthracite** (max carbon, max calorific value)
7	Petroleum (crude oil) = fractional distillation: gas > petrol > kerosene > diesel > fuel oil > bitumen
8	Octane number 87 = regular petrol; 91-93 = premium; higher = less knocking
9	Knocking in engine = premature ignition; prevented by anti-knock additives (earlier: TEL; now: MTBE)
10	Producer gas (CO + N2) = made by blowing air through burning coke
11	Water gas (CO + H2) = made by steam over hot coke; also called “synthesis gas”
12	Biogas = mainly methane; from cow dung + water; anaerobic digestion
13	Hydrogen fuel cell = H2 + O2 -> H2O + electricity; zero emission vehicle
14	Ozone (O3) layer = 15-35 km altitude (stratosphere); absorbs UV-B radiation
15	CFCs (chlorofluorocarbons) = destroy ozone; one Cl atom destroys 100,000 ozone molecules
16	Montreal Protocol (1987) = international treaty to phase out ozone-depleting substances
17	CO2 in atmosphere = ~420 ppm (2024); pre-industrial = ~280 ppm; rising due to fossil fuels
18	Methane GWP = **25x CO2** (100-year); N2O = 298x CO2; SF6 = 22,800x CO2
19	Solar panels = photovoltaic cells; convert sunlight to DC electricity; silicon-based
20	Wind energy = kinetic energy of wind -> electricity via turbine generator
21	Geothermal energy = heat from Earth’s interior; Iceland, Philippines, India (Puga Valley, Ladakh)
22	Tidal energy = from rise and fall of ocean tides; predictable; coastal locations
23	Nuclear energy: fission (U-235, Pu-239) = current reactors; fusion = future
24	Hydroelectric power = 3rd largest renewable in India; HP, Uttarakhand, NE India
25	India’s largest solar project = Bhadla Solar Park (Rajasthan; 2.2 GW)
26	India’s target: 500 GW renewable energy by 2030 (Paris Agreement commitment)
27	Ethanol blending in India: E20 target = 20% ethanol in petrol by 2025
28	Hydrogen Mission (2021) = Green Hydrogen from electrolysis using renewable electricity
29	Coal India Ltd (CIL) = world’s largest coal mining company; state-owned
30	India’s coal reserves = mainly in Jharkhand, Odisha, Chhattisgarh, WB

Chemistry Topic 7 — Chemistry in Daily Life, Industry and Agriculture

Consolidated — 50 Must-Score Facts

1. **Soaps** are sodium or potassium salts of higher fatty acids (containing 12 to 18 carbon atoms) such as **Stearic acid (C_{17}H_{35}COOH)**, **Palmitic acid (C_{15}H_{31}COOH)**, and **Oleic acid (C_{17}H_{33}COOH)**.
2. **Saponification**: The chemical alkaline hydrolysis of animal fats or vegetable oils (triglycerides) with aqueous sodium hydroxide (NaOH) or potassium hydroxide (KOH), producing soap and a commercially valuable byproduct, **Glycerol (Glycerine, CH_2OH-CHOH-CH_2OH)**.
3. **Hard Soap vs. Soft Soap**: Soaps made with **Sodium hydroxide (NaOH)** are hard soaps (laundry washing bars); soaps made with **Potassium hydroxide (KOH)** are soft soaps (toilet luxury soaps, shaving creams, liquid handwashes).
4. **Salting Out of Soap**: Common salt (**NaCl**) is added to the saponification mixture to decrease the solubility product of soap, precipitating it out as a curdled solid through the common-ion effect.
5. **Micelle Cleansing Action**: A soap molecule consists of a long non-polar hydrophobic (water-repelling / oil-soluble) hydrocarbon tail and a polar hydrophilic (water-loving) ionic carboxylate head (-COO^-Na^+). In water, soap molecules assemble into spherical aggregates called **micelles**: the hydrophobic hydrocarbon tails point inward to trap non-polar oil/grease dirt droplets, while the hydrophilic polar heads face outward into water. Agitation forms an emulsion that is rinsed away.
6. **Why Soap Fails in Hard Water**: Hard water contains dissolved calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions. These cations react with soluble sodium soap molecules to precipitate an insoluble, sticky white curd called **Scum** (calcium/magnesium stearate): 2C_{17}H_{35}COONa + Ca^{2+} \to (C_{17}H_{35}COO)_2Ca\downarrow\text{ (Insoluble Scum)} + 2Na^+ Scum adheres to fabrics, dulls garments, and wastes large amounts of soap before lathering can begin.
7. **Synthetic Detergents (Syndets)**: Sodium salts of long-chain alkyl hydrogen sulphates or alkylbenzene sulphonic acids (e.g., **Sodium Lauryl Sulphate**, **Sodium Dodecylbenzene Sulphonate**).
8. **Why Detergents Work in Hard Water**: The calcium and magnesium salts of alkyl sulphates and sulphonates are **completely soluble in water**; hence, synthetic detergents form abundant rich lather in both soft and hard water without producing insoluble scum.
9. **Analgesics**: Pharmaceutical drugs that relieve or eliminate bodily pain without causing loss of consciousness. Classified into: - *Non-narcotic (non-opioid) analgesics*: **Aspirin** (Acetylsalicylic acid) and **Paracetamol** (Acetaminophen). Aspirin also inhibits platelet blood clotting (anti-thrombotic) and is administered to prevent heart attacks. - *Narcotic (opioid) analgesics*: **Morphine**, **Codeine**, and **Heroin**; derived from opium poppy (*Papaver somniferum*); relieve severe post-operative and cancer pain, but induce addiction and respiratory depression.
10. **Antipyretics**: Drugs used to reduce elevated body temperature in fever (e.g., **Paracetamol**, **Aspirin**, **Phenacetin**).
11. **Antiseptics**: Chemical antimicrobial agents applied to **living human and animal tissues** (wounds, cuts, skin ulcers, mouth infections) to kill or inhibit microbial growth without damaging tissue cells.
12. **Dettol Composition**: The commercial household antiseptic Dettol is a blend of **Chloroxylenol** (\approx 4.8\%) and **\alpha-Terpineol** dissolved in pine oil and alcohol.
13. **Bithionol**: An organosulfur chlorinated compound added to toilet soaps to impart long-lasting deodorant and antiseptic properties.
14. **Tincture of Iodine**: An effective antiseptic solution consisting of **2–3\%\text{ elemental Iodine } (I_2)** dissolved in an alcohol-water mixture.
15. **Iodoform (CHI_3)**: A yellow crystalline solid with a medicinal odor, historically used as an antiseptic dusting powder for septic surgical wounds due to the slow liberation of free iodine.
16. **Boric Acid (H_3BO_3)**: A very mild, non-irritating antiseptic used as an ophthalmic soothing **eye-wash**.

17. **Disinfectants**: Chemical antimicrobial agents applied strictly to **non-living inanimate surfaces** (floors, toilets, drainage pipes, surgical instruments).
18. **The Phenol Concentration Rule**: A **0.2\% aqueous solution of Phenol (C_6H_5OH) acts as an Antiseptic**, whereas a **1.0\% aqueous solution of Phenol acts as a Disinfectant**!
19. **Chlorine Disinfection**: Chlorine gas at concentrations of **0.2\text{ to }0.4\text{ parts per million (ppm)}** is standardly used to disinfect municipal drinking water supplies and swimming pools.
20. **Antibiotics**: Chemical metabolic substances produced by microorganisms (fungi, bacteria) that inhibit the growth of or destroy pathogenic bacteria at low concentrations.
21. **Discovery of Penicillin**: The world's first true antibiotic, discovered serendipitously in **1928 by Sir Alexander Fleming** at St. Mary's Hospital, London, from the green mold **Penicillium notatum**. Purified as a clinical drug by Howard Florey and Ernst Chain (Nobel Prize in Medicine, 1945).
22. **Broad-Spectrum Antibiotics**: Effective against a wide range of both Gram-positive and Gram-negative bacteria (e.g., **Chloramphenicol**, **Ampicillin**, **Amoxicillin**, **Tetracycline**). Chloramphenicol is effective in treating typhoid, meningitis, and pneumonia.
23. **Antacids**: Weakly basic compounds administered orally to neutralize excess hydrochloric acid in gastric juice (HCl), elevating stomach pH above 3.5. Classic examples: **Milk of Magnesia (Mg(OH)_2)**, Aluminium Hydroxide gel (Al(OH)_3), and Sodium Bicarbonate (NaHCO_3). Modern antacids like **Ranitidine** and **Omeprazole** act by blocking histamine H_2-receptors or proton pumps (H^+/K^+ ATPase).
24. **Food Preservatives**: Chemical additives that prevent food spoilage caused by microbial fermentation, mold growth, or oxidative rancidity:
    - **Sodium Benzoate (C_6H_5COONa)**: The most widely utilized chemical preservative in commercial packaged foods (fruit squashes, tomato ketchup, jams, soft drinks). It metabolizes safely in the human liver into hippuric acid and is excreted in urine.
    - **Sodium Metabisulphite / Potassium Metabisulphite (K_2S_2O_5)**: Releases sulfur dioxide gas (SO_2), which inhibits bacterial and fungal enzymes in wines, fruit juices, and cut dried fruits.
    - **Table Salt (NaCl), Sugar Syrup, and Vinegar (CH_3COOH)**: Natural osmotic and acidic food preservatives used in traditional pickling and fruit preservation.
25. **Food Antioxidants**: Sacrificial reducing agents added to fats, oils, and fried snacks (potato chips) to retard oxidative rancidity: **BHA (Butylated Hydroxyanisole)** and **BHT (Butylated Hydroxytoluene)**. Flushing snack bags with **Nitrogen gas (N_2)** prevents oxygen from contacting edible oils.
26. **Glass**: An amorphous, non-crystalline, transparent supercooled inorganic liquid of high viscosity, manufactured by fusing silica (sand, SiO_2) with metal carbonates and cooling the melt rapidly without crystallization.
27. **Soda-Lime Glass (Soft Glass)**: Approximate composition Na_2O \cdot CaO \cdot 6SiO_2. Low melting point, inexpensive; used for window panes, beverage bottles, mirror plates, and cheap glassware.
28. **Potash Glass (Hard Glass)**: Made using potassium carbonate (K_2CO_3), calcium carbonate, and silica (K_2O \cdot CaO \cdot 6SiO_2). Highly resistant to chemical reagents and acids; used for high-temperature chemical laboratory equipment (flasks, combustion tubes).
29. **Pyrex Glass (Borosilicate Glass)**: Formed by substituting boron trioxide (B_2O_3, \approx 13\%) and alumina for alkali. Possesses an exceptionally **low thermal expansion coefficient** and high thermal shock resistance; does not crack upon rapid heating/cooling; used for standard laboratory glassware (beakers, burettes) and microwave oven cookware.
30. **Flint Glass (Optical / Lead Crystal Glass)**: Contains **Lead Oxide (PbO, 20–30\%)**. Possesses a high density, high refractive index, and high optical dispersion; used for manufacturing achromatic camera lenses, optical prisms, telescope objectives, and brilliant crystal chandeliers.
31. **Crookes Glass**: A specialty glass containing **Cerium Oxide (CeO_2)**. Cerium atoms possess electronic energy levels that selectively and sharply **absorb harmful ultraviolet (UV) rays** while transmitting visible light; used for high-protection sunglasses and optical UV shields.

32. **Photochromic Glass**: Specialty glass containing microscopic dispersed crystals of **Silver Chloride (AgCl) or Silver Bromide (AgBr)**. When exposed to sunlight (UV photons), silver halide decomposes into tiny clusters of metallic silver (AgCl \xrightarrow{h\nu} Ag + Cl), turning the lens dark. In dim indoor light, copper sensitizers catalyze recombination, restoring optical clarity.
33. **Annealing of Glass**: The process of slowly, uniformly cooling newly shaped hot glass articles inside a temperature-controlled lehr tunnel oven. Rapid cooling produces severe internal structural stress, causing glass to shatter spontaneously upon minor contact. Annealing removes internal stress.
34. **Glass Colorants**:
    - **Cobalt Oxide (CoO)**: Deep Royal Blue
    - **Ferrous Oxide (FeO) / Chromium Oxide (Cr_2O_3)**: Emerald Green
    - **Ferric Oxide (Fe_2O_3)**: Yellowish-Brown / Amber
    - **Cuprous Oxide (Cu_2O) / Gold Chloride**: Rich Ruby Red
    - **Manganese Dioxide (MnO_2)**: Purple / Amethyst
    - **Cadmium Sulphide (CdS)**: Brilliant Lemon Yellow
35. **Portland Cement**:
    - Patented in 1824 by British bricklayer **Joseph Aspdin**, named for its visual resemblance to Portland limestone building stone in Dorset, England.
    - Raw materials: Calcareous materials (Limestone, chalk, CaCO_3 \to \mathbf{CaO, 60–65\%}) and Argillaceous materials (Clay, shale, \mathbf{SiO_2, 20–25\%}; \mathbf{Al_2O_3, 5–10\%}; \mathbf{Fe_2O_3, 2–3\%}).
36. **Role of Gypsum (CaSO_4 \cdot 2H_2O) in Cement**:
    - Tricalcium aluminate (C_3A, 3CaO \cdot Al_2O_3) reacts rapidly with water, causing premature flash setting within minutes.
    - Intergrinding **2–3\%\text{ Gypsum}** with clinker reacts with C_3A to form an insoluble coating of calcium sulphoaluminate (ettringite), which **retards the initial setting time of cement** to 30–45 minutes, providing masons sufficient working time.
37. **Essential Plant Nutrients**: Higher plants require 17 essential chemical elements:
    - Structural elements (from air/water): C, H, O.
    - Primary Macronutrients: **Nitrogen (N)**, **Phosphorus (P)**, **Potassium (K)**.
    - Secondary Macronutrients: Calcium (Ca), Magnesium (Mg), Sulphur (S).
    - Micronutrients: Zinc (Zn), Iron (Fe), Manganese (Mn), Copper (Cu), Boron (B), Molybdenum (Mo), Chlorine (Cl), Nickel (Ni).
38. **Urea (NH_2CONH_2)**:
    - The most widely utilized solid nitrogenous fertilizer in India.
    - Contains **46\%\text{ Nitrogen (N)}** by weight (the highest nitrogen concentration among all solid nitrogen fertilizers).
    - Synthesized by reacting anhydrous ammonia with carbon dioxide under high pressure (2NH_3 + CO_2 \to NH_2CONH_2 + H_2O).
    - In soil, urea is hydrolyzed by the enzyme **urease** into ammonium carbamate, which converts into ammonium (NH_4^+) and nitrate (NO_3^-) ions that are absorbed by plant roots.
39. **Neem-Coated Urea (NCU)**:
    - Mandatory in India since 2015. Urea prills are coated with a thin film of natural **Neem oil (0.05\%)**.
    - **Nitrification Inhibition**: Active triterpenoids in neem (nimbin, azadirachtin) inhibit soil nitrifying bacteria (*Nitrosomonas*, *Nitrobacter*), slowing the conversion of ammonium to nitrate.
    - **Benefits**: Reduces nitrogen leaching into groundwater, prevents nitrous oxide greenhouse gas emissions, boosts Nitrogen Use Efficiency (NUE) by 10–15\%, and completely halts the illegal diversion of subsidized

agricultural urea to chemical resin factories.

40. **Phosphatic Fertilizers**:
    - **Single Superphosphate (SSP)**: Contains **16\%\text{ available } P_2O_5**, plus 11\% Sulphur and 19\% Calcium; manufactured by treating rock phosphate with sulphuric acid (Ca_3(PO_4)_2 + 2H_2SO_4 \to Ca(H_2PO_4)_2 + 2CaSO_4).
    - **Triple Superphosphate (TSP)**: Contains **46–48\%\text{ } P_2O_5**; made by treating rock phosphate with phosphoric acid.
    - **Diammonium Phosphate (DAP)**: High-analysis complex fertilizer containing **18\%\text{ Nitrogen (N)}** and **46\%\text{ Phosphorus (P}_2\text{O}_5\text{)}**.
41. **Potassic Fertilizers**:
    - **Muriate of Potash (MOP)**: Potassium chloride (KCl); contains **60\%\text{ } K_2O**. India lacks indigenous commercial potash mineral reserves and **imports 100\% of its MOP consumption**.
    - **Sulphate of Potash (SOP)**: Potassium sulphate (K_2SO_4, 50\% K_2O); preferred for chloride-sensitive crops (tobacco, potato, grapes).
42. **Soil pH Remediation**:
    - **Acidic Soils (\text{pH} < 6.0)**: Contain excess H^+ and toxic soluble Al^{3+} and Mn^{2+}. Corrected by **Liming**: applying agricultural **Slaked Lime (Ca(OH)_2)**, Quicklime (CaO), or powdered **Limestone (CaCO_3)**, which neutralizes soil acidity.
    - **Alkaline / Sodic / Usar Soils (\text{pH} > 8.5)**: Contain high exchangeable sodium (Na^+) that deflocculates soil structure. Corrected by applying **Gypsum (CaSO_4 \cdot 2H_2O)** or **Iron Pyrites (FeS_2)**: 2\text{Na-Clay} + CaSO_4 \to \text{Ca-Clay} + Na_2SO_4\text{ (leached out by irrigation)}
43. **DDT (Dichlorodiphenyltrichloroethane)**:
    - A chlorinated organic insecticide synthesized in 1874 by Othmar Zeidler; its insecticidal properties were discovered in 1939 by Swiss chemist Paul Hermann Müller (Nobel Prize in Medicine, 1948).
    - Extremely persistent, lipophilic, and **non-biodegradable**.
    - Undergoes **Biomagnification**: accumulates in the fatty tissues of organisms and concentrates across ascending trophic levels of food webs (phytoplankton \to zooplankton \to fish \to fish-eating raptors). Causes reproductive failure in birds by disrupting calcium metabolism, leading to fragile, eggshell thinning. Banned globally under the UN Stockholm Convention on Persistent Organic Pollutants (POPs).
44. **BHC / Gammaxene / Lindane**:
    - Benzene Hexachloride / Hexachlorocyclohexane (C_6H_6Cl_6); formed by the addition of chlorine to benzene under UV light. The \gamma-isomer (**Lindane**) is an active agricultural and anti-lice insecticide.
45. **Herbicides / Weedicides**: Chemical agents used to selectively or non-selectively destroy unwanted invasive weeds:
    - **2,4-D (2,4-Dichlorophenoxyacetic acid)**: A synthetic plant auxin; selectively kills broad-leaved dicotyledonous weeds without harming narrow-leaved monocot cereal crops (wheat, rice).
    - **Sodium Chlorate (NaClO_3)**: A powerful, non-selective chemical herbicide that desiccates and destroys all plant vegetation.
    - **Glyphosate**: Widely used systemic non-selective broad-spectrum herbicide that inhibits the plant EPSPS shikimate enzyme pathway.
46. **Rodenticides (Rat Poisons)**: Chemical poisons deployed to eliminate rodent agricultural pests:
    - **Zinc Phosphide (Zn_3P_2)**: Ingested rat poison; reacts with gastric hydrochloric acid in the rodent's stomach to liberate lethal, toxic **Phosphine gas (PH_3)**, causing heart failure and organ collapse: Zn_3P_2 + 6HCl \to 3ZnCl_2 + 2PH_3 \uparrow
    - **Warfarin**: Anticoagulant rodenticide; inhibits vitamin K epoxide reductase, triggering internal hemorrhaging.

47. **Artificial Rain (Cloud Seeding)**:
    - Discovered in 1946 by Bernard Vonnegut and Vincent Schaefer.
    - Seeded into subcooled moisture clouds using aircraft or ground generators.
    - Primary chemical agent: **Silver Iodide (AgI)**, which possesses a hexagonal crystal lattice almost identical to natural ice crystals, serving as an effective ice-nucleating template that triggers rapid ice crystal growth and rainfall.
    - Auxiliary seeding agents: **Dry Ice (solid CO_2)** and hygroscopic common salt (NaCl).
48. **Photographic Chemistry**:
    - **Light-Sensitive Salt**: Photographic film is coated with a gelatin emulsion containing microscopic crystals of **Silver Bromide (AgBr)** (and some AgI). When exposed to light, photons reduce trace silver ions to metallic silver nuclei, creating a latent image (Ag^+ + e^- \xrightarrow{h\nu} Ag).
    - **Developer**: Hydroquinone (quinol) reduces light-exposed silver bromide grains into black metallic silver.
    - **Fixer (Hypo Solution)**: Sodium Thiosulphate Pentahydrate (**Na_2S_2O_3 \cdot 5H_2O**). Dissolves unexposed, unreduced AgBr crystals as a soluble thiosulphato-argentate complex, preventing the film from darkening further in sunlight: AgBr + 2Na_2S_2O_3 \to Na_3[Ag(S_2O_3)_2]\text{ (Soluble complex)} + NaBr
49. **Aspartame Alert**: The artificial sweetener Aspartame decomposes into phenylalanine, aspartic acid, and methanol; products containing aspartame must carry a mandatory warning label for individuals with the rare genetic disorder **Phenylketonuria (PKU)**.
50. **Liquid Bleach**: An aqueous solution of **Sodium Hypochlorite (NaOCl, \approx 5\%)**; acts as a powerful disinfectant and textile stain remover via oxidation.

One-Liner Revision – Chemistry in Daily Life and Agriculture

#	Fast Fact
1	Urea (CO(NH2)2) = most widely used nitrogen fertiliser; 46% N; water-soluble
2	DAP = Diammonium Phosphate; provides both N and P to crops
3	NPK fertilisers = provide Nitrogen, Phosphorus, Potassium (potash)
4	Bordeaux mixture = CuSO4 + Ca(OH)2; first effective fungicide; grape vines, potato blight
5	DDT = banned insecticide (persistent organic pollutant); biomagnification in food chain
6	Biomagnification = increasing concentration of persistent toxin up the food chain
7	BHC/BHC (Benzene Hexachloride) = lindane; insecticide; restricted use
8	Common antiseptics: Dettol (chloroxylenol), Savlon (chlorhexidine), Iodine (tincture)
9	Common disinfectants: bleaching powder, phenol (carbolic acid), formaldehyde
10	Paracetamol = analgesic (pain reliever) + antipyretic (fever reducer); most common OTC drug
11	Aspirin (acetylsalicylic acid) = analgesic, antipyretic, anti-inflammatory; anti-platelet
12	Antacids neutralise excess HCl: Mg(OH)2 (milk of magnesia), Al(OH)3, NaHCO3
13	Tranquilizers = CNS depressants; reduce anxiety; e.g., diazepam (Valium), barbiturates
14	Antibiotics kill bacteria; NOT effective against viruses
15	Antihistamines = counter histamine (allergic reaction); e.g., cetirizine, diphenhydramine
16	Soap = biodegradable; forms scum in hard water (Ca/Mg stearate precipitate)
17	Detergent = synthetic (non-soap); does NOT form scum in hard water
18	Food preservatives: salt (NaCl), sugar, vinegar (acetic acid), sodium benzoate, SO2
19	Food colours: Tartrazine (E102, yellow), Carmoisine (red) are synthetic; saffron, turmeric are natural

#	Fast Fact
20	Baking powder = baking soda (NaHCO3) + cream of tartar (tartaric acid); releases CO2
21	Vinegar = 4-8% acetic acid (CH3COOH); food preservative and condiment
22	Chlorination of water = Cl2 or bleaching powder kills pathogens; residual Cl = 0.2-0.5 ppm
23	Fluoridation of water = NaF added; 0.7-1.2 ppm F- prevents tooth decay
24	Iodised salt = NaCl + KIO3 (potassium iodate); prevents iodine deficiency/goitre
25	Organic farming = no synthetic pesticides/fertilisers; uses compost, biopesticides
26	Green Chemistry = design of chemical products that minimise environmental hazards
27	Food packaging: BPA (Bisphenol A) in polycarbonate plastics = endocrine disruptor
28	MSG (Monosodium glutamate) = flavour enhancer; “Umami” taste
29	Aspartame = artificial sweetener; ~200x sweeter than sugar; 0 calories
30	Saccharin = oldest artificial sweetener (1879); 300-500x sweeter; no calories

Chemistry Topic 8 — Radiation, Nuclear Chemistry, Explosives and Environmental Chemistry

Consolidated — 50 Must-Score Facts

1. **Radioactivity** is the spontaneous, uncontrolled nuclear disintegration of unstable heavy atomic nuclei accompanied by the emission of ionizing particles or electromagnetic radiation.
2. Radioactivity was discovered in **1896 by French physicist Henri Becquerel** while investigating phosphorescence in uranium potassium sulphate crystals wrapped in black paper.
3. In 1898, **Marie Curie and Pierre Curie** isolated two new radioactive elements from pitchblende ore (U_3O_8): **Polonium (Po, named after Poland)** and **Radium (Ra)**. Marie Curie is the only scientist to win Nobel Prizes in two distinct sciences (Physics 1903, Chemistry 1911).
4. **Nuclear Property**: Radioactivity is governed exclusively by nuclear stability (neutron-to-proton ratio, N/Z); it is completely **independent of temperature, pressure, chemical combination, or physical state**.
5. **Alpha (α) Particles**: Helium nuclei carrying a double positive charge (**$^4_2He^{2+}$**). Possess the highest mass (4 u), **highest ionizing power ($\approx 10,000\times \gamma$)**, but the **lowest penetrating power**; completely stopped by a single sheet of paper or the dead outer keratin layer of human skin.
6. **Beta (β^-) Particles**: Fast-moving electrons (**$^0_{-1}e$**) originating from the nucleus when a neutron decays into a proton, electron, and antineutrino ($n \to p + e^- + \bar{\nu}$). Moderate ionizing power and moderate penetrating power; stopped by a few millimeters of aluminium foil.
7. **Gamma (γ) Rays**: High-energy, short-wavelength electromagnetic photons emitted by excited nuclei transitioning to the ground state. Carry zero charge and zero rest mass; travel at the speed of light ($3 \times 10^8\text{ m/s}$). Possess the **lowest ionizing power**, but the **highest penetrating power** among nuclear emissions; require thick lead blocks or dense reinforced concrete barriers for shielding.
8. **Soddy-Fajans Group Displacement Law**: - Emission of an **α-particle** reduces the atomic number (Z) by 2 units and the mass number (A) by 4 units: $^A_Z X \xrightarrow{\alpha} {}^{A-4}_{Z-2} Y + {}^4_2He$ *(The daughter element shifts 2 groups to the left in the Periodic Table). - Emission of a* ***β^--particle*** *increases the atomic number (Z) by 1 unit while leaving the mass number (A) unchanged: $^A_Z X \xrightarrow{\beta^-} {}^{A}_{Z+1} Y + {}^0_{-1}e$* (The daughter element shifts 1 group to the right; forms an **isobar** of the parent). - Emission of a **γ-ray** causes no change in atomic number or mass number, merely dissipating excess nuclear excitation energy.
9. **Radioactive Decay Kinetics & Half-Life ($t_{1/2}$)**: - Radioactive decay strictly follows **first-order reaction kinetics**: $-\frac{dN}{dt} = \lambda N \implies N(t) = N_0 e^{-\lambda t}$ - The Half-Life ($t_{1/2}$) is the time required for half the radioactive nuclei in a sample to disintegrate: $t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$ - After n half-lives, the remaining radioactive fraction is: $N = N_0 \left(\frac{1}{2}\right)^n$.
10. **Units of Radioactivity**:
    - **Becquerel (Bq)**: SI unit; $1\text{ Bq} = 1\text{ nuclear disintegration per second}$.
    - **Curie (Ci)**: Traditional historical unit; $1\text{ Ci} = 3.7 \times 10^{10}\text{ disintegrations/sec}$ (the activity of 1 g of pure Radium-226).
    - **Rutherford (Rd)**: $1\text{ Rd} = 10^6\text{ disintegrations/sec}$.
11. **Radiometric Dating Techniques**:
    - **Radiocarbon Dating (^{14}C)**: Devised by Willard Libby (1949, Nobel Prize 1960). Based on the decay of cosmogenic ^{14}C ($t_{1/2} = 5,730\text{ years}$) into ^{14}N. Used for dating once-living biological archaeological artifacts (wood, bone, charcoal, fossils) up to $\approx 50,000\text{ years}$ old.
    - **Uranium-Lead Dating ($^{238}U \to {}^{206}Pb$)**: Half-life $\approx 4.5\text{ billion years}$; used to calculate the geological age of the Earth ($\approx 4.54\text{ billion years}$), ancient continental cratons, and lunar meteorites.

- **Potassium-Argon Dating ($^{40}K \to {}^{40}Ar$)**: Half-life $\approx 1.25\text{ billion years}$; used to date ancient volcanic igneous rocks and early hominid fossil strata.

12. **Nuclear Fission**:
    - Discovered in 1938 by German chemists **Otto Hahn and Fritz Strassmann**; theoretical mechanism explained by Lise Meitner and Otto Frisch.
    - The splitting of a heavy, fissile atomic nucleus (such as **Uranium-235** or **Plutonium-239**) upon absorbing a slow-moving **thermal neutron (0.025 eV)** into two medium-sized daughter nuclei, releasing $2\text{ to }3$ fast neutrons and an enormous quantity of energy ($\approx 200\text{ MeV}$ per fission event): $^{235}{92}U + {}^1_0n \to {}^{236}{92}U^* \to {}^{141}{56}Ba + {}^{92}{36}Kr + 3{}^1_0n + 200\text{ MeV}$
    - The liberated energy arises from the conversion of missing mass into energy according to Einstein's mass-energy equivalence equation ($E = \Delta m \cdot c^2$).

13. **Atomic Bomb (Fission Weapon)**:
    - Operates on an **uncontrolled, supercritical nuclear chain reaction**.
    - If the mass of fissile material exceeds the **Critical Mass** ($k > 1$), neutron multiplication is exponential ($1 \to 3 \to 9 \to 27 \dots$), causing a catastrophic thermonuclear explosion within microseconds.
    - World's first nuclear detonations: "Little Boy" (Uranium-235, Hiroshima, August 6, 1945) and "Fat Man" (Plutonium-239, Nagasaki, August 9, 1945).

14. **Nuclear Power Reactor (Controlled Fission)**:
    - Maintains a steady critical chain reaction ($k = 1.0$) to generate commercial steam for electricity.
    - **Moderator**: Material with light atomic mass used to slow down energetic fast neutrons (2 MeV) to thermal energy (0.025 eV) via elastic collisions without absorbing them. Standard moderators: **Heavy Water (D_2O, deuterium oxide)** and high-purity **Graphite**.
    - **Control Rods**: Neutron-absorbing materials inserted into the core to regulate reaction rate or shut down the reactor: **Cadmium (Cd)** and **Boron (B)** (have exceptionally high neutron capture cross-sections).
    - **Coolant**: Fluid circulated through the core to extract fission heat: **Light water (H_2O)**, Heavy water (D_2O), or liquid metallic Sodium (Na).

15. **Nuclear Fusion**:
    - The process where two light atomic nuclei combine to form a heavier, more stable nucleus at extremely high kinetic energies, liberating vast energy per unit mass: $^2_1H\text{ (Deuterium)} + {}^3_1H\text{ (Tritium)} \xrightarrow{\text{extreme } T, P} {}^4_2He + {}^1_0n + 17.6\text{ MeV}$
    - Nuclear fusion powers the **Sun and all celestial stars** (via the proton-proton chain and CNO cycle).
    - Fusion requires overcoming immense electrostatic Coulomb repulsion between positively charged nuclei, requiring **temperatures exceeding 10^7 K (100 million $^\circ\text{C}$)** and massive pressures to form a thermonuclear plasma.

16. **Hydrogen Bomb (Thermonuclear Bomb)**:
    - Operates primarily on **uncontrolled Nuclear Fusion**.
    - Conceived by American physicists **Edward Teller and Stanislaw Ulam** (first detonated in 1952, "Ivy Mike").
    - *Must-Score Fact*: Because thermonuclear fusion requires stellar ignition temperatures, **a hydrogen bomb incorporates an internal atomic (fission) bomb as its primary trigger**. The fission detonation generates the extreme heat ($>10^7\text{ K}$) and radiation pressure needed to ignite the secondary fusion fuel (lithium deuteride).
    - Yields are measured in **Megatons of TNT** (thousands of times more destructive than atomic fission bombs) and produce less long-lived direct radioactive fission fallout per megaton.

17. **Dynamite**:
    - Invented and patented in **1867 by Swedish chemist Alfred Nobel**.

- Composed of liquid **Nitroglycerin (Glyceryl Trinitrate, GTN)** absorbed into an inert, porous siliceous earth called **Kieselguhr (diatomaceous earth)** or wood pulp, mixed with sodium nitrate.
- Nitroglycerin is shock-sensitive; absorbing it into Kieselguhr created the first safe, transportable, shock-stable high explosive, revolutionizing civil engineering (tunnels, canals, railways) and establishing the fortune that funds the **Nobel Prizes**.

18. **TNT (Trinitrotoluene)**:
    - Chemically **2,4,6-Trinitrotoluene (C_7H_5N_3O_6)**; formed by the stepwise nitration of toluene with mixed acid (HNO_3 + H_2SO_4).
    - A yellow crystalline military high explosive; insensitive to friction and mechanical shock; can be safely melted at 80^\circ\text{C} and poured directly into artillery shells.
19. **RDX (Research Department Explosive / Cyclonite / Hexogen / T4)**:
    - Chemically **Cyclotrimethylenetrinitramine (C_3H_6N_6O_6)**; white crystalline solid.
    - High velocity of detonation (\approx 8750\text{ m/s}) and brisance. When blended with motor oil and plasticizers (polyisobutylene), it forms malleable **C-4 Plastic Explosive**, favored in military demolition and counter-terrorism scenarios.
    - Known as **Hexogen** in Germany, **T4** in Italy, and **RDX** in the USA and Commonwealth nations.
20. **PETN (Pentaerythritol Tetranitrate)**:
    - One of the most powerful known secondary high explosives; highly shock-sensitive; used in military detonating cords (Primacord) and booster charges.
21. **ANFO (Ammonium Nitrate Fuel Oil)**:
    - Industrial blasting explosive consisting of **94\%\text{ porous prilled Ammonium Nitrate } (NH_4NO_3)** mixed with **6\%\text{ No. 2 Fuel Oil}**.
    - Low sensitivity; requires a high-explosive booster charge (TNT or dynamite) to detonate; widely used in commercial open-cast coal mining and limestone quarrying.
22. **Gunpowder (Black Powder)**:
    - The oldest known chemical explosive, invented in 9th-century China.
    - A mechanical mixture of: **Potassium Nitrate (Saltpetre, KNO_3, \approx 75\%)**, **Charcoal (Carbon fuel, \approx 15\%)**, and **Sulphur (\approx 10\%)**. Burns rapidly via deflagration to generate high volumes of hot N_2, CO_2, and potassium salts.
23. **Picric Acid (2,4,6-Trinitrophenol)**: A strong, yellow crystalline organic acid and explosive; historically used under the names **Lyddite** and **Melinite** as an artillery shell filling in the late 19th and early 20th centuries.
24. **Acid Rain (Acid Precipitation)**:
    - Defined as any precipitation (rain, snow, fog) having a **\text{pH} < 5.6**.
    - *Natural Background Rain*: Normal unpolluted rainwater is naturally slightly acidic (\text{pH} \approx 5.6) due to atmospheric CO_2 dissolving to form weak carbonic acid: CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-
    - *Anthropogenic Acid Rain*: Caused primarily by atmospheric emissions of **Sulphur Dioxide (SO_2)** from coal-fired thermal power plants and petroleum refineries, and **Nitrogen Oxides (NO_x)** from high-temperature vehicular exhaust.
25. **Acid Rain Reactions**:
    - Sulphuric acid formation accounts for **\approx 65–70\% of total acidity**: 2SO_2 + O_2 + 2H_2O \xrightarrow{\text{catalytic dust}} 2H_2SO_4
    - Nitric acid formation accounts for **\approx 30–35\% of total acidity**: 4NO_2 + O_2 + 2H_2O \to 4HNO_3
26. **Marble Cancer (Stone Leprosy)**:

- Sulphuric acid in acid rain attacks calcium carbonate (CaCO_3) in historical white marble monuments (notably the **Taj Mahal in Agra**, threatened by emissions from the nearby Mathura Oil Refinery and Firozabad glass foundries): CaCO_3\text{ (white marble)} + H_2SO_4 \to CaSO_4 \cdot 2H_2O\text{ (Gypsum)} + CO_2 \uparrow
- The formation of soluble, crumbly gypsum causes the marble surface to blister, turn yellowish, flake away, and suffer irreversible pitting corrosion.

27. **Ecological Damage of Acid Rain**:
    - Drops lake and river water pH below 5.0, triggering the leaching of toxic free **Aluminium (Al^{3+}) ions** from surrounding soils into waterways, precipitating mucus on fish gills and causing asphyxiation.
    - Strips essential magnesium and calcium nutrients from forest soils, causing dieback in coniferous forests.
28. **Classical Smog (London-type Smog)**:
    - First documented extensively during the catastrophic London Great Smog of December 1952.
    - Occurs in **cool, humid winter climates**.
    - Chemically, it is a **Reducing Smog** consisting of a dense mixture of coal smoke, particulate soot, atmospheric fog, and high concentrations of **Sulphur Dioxide (SO_2)**.
29. **Photochemical Smog (Los Angeles-type Smog)**:
    - Occurs in **warm, dry, sunny climates** with heavy automotive vehicular traffic.
    - Chemically, it is an **Oxidizing Smog** formed by complex solar ultraviolet photolytic reactions between **Nitrogen Oxides (NO_x)** and **Volatile Organic Compounds (Hydrocarbons)**.
    - **Key Ingredients & Byproducts**:
    - **Ozone (O_3)**: Formed at ground level via photolysis of NO_2: NO_2 \xrightarrow{h\nu} NO + O^\bullet, \quad O^\bullet + O_2 \to O_3
    - **PAN (Peroxyacetyl Nitrate, CH_3COOONO_2)**: Potent eye and lung irritant, causing severe tearing (lachrymation) and plant leaf bronzing.
    - **Acrolein (CH_2=CH-CHO) and Formaldehyde (HCHO)**.
30. **Stratospheric Ozone Layer**:
    - Concentrated in the lower stratosphere between **15\text{ and }35\text{ kilometers}** altitude ("Ozonosphere").
    - Formed and maintained naturally via the dynamic **Chapman Cycle**: O_2 \xrightarrow{\text{solar UV-C } (<240\text{ nm})} O + O O + O_2 + M \to O_3 + M O_3 \xrightarrow{\text{UV-B } (240–310\text{ nm})} O_2 + O
    - Shields terrestrial life by absorbing >99\% of solar UV-C and biological harmful UV-B radiation (280–315\text{ nm}), which causes human skin melanoma, cataract blindness, and immune suppression.
31. **Dobson Unit (DU)**:
    - The standard physical unit used to measure total column atmospheric ozone thickness.
    - 1\text{ DU} is equivalent to a pure ozone layer thickness of **0.01\text{ mm} (10\text{ }\mu\text{m})** at standard temperature and pressure (0^\circ\text{C}, 1\text{ atm}).
    - Normal atmospheric ozone levels range from **300\text{ to }350\text{ DU}**. An **"Ozone Hole"** is technically defined when total column ozone falls below **220\text{ Dobson Units}**.
32. **Catalytic Ozone Depletion by Chlorofluorocarbons (CFCs)**:
    - Discovered in 1974 by **Mario Molina and F. Sherwood Rowland** (Nobel Prize in Chemistry, 1995).
    - CFCs (Freons, such as CF_2Cl_2 / CFC-12) are non-flammable, non-toxic refrigerants and aerosol propellants that do not break down in the troposphere.
    - Upon diffusing into the stratosphere, high-energy UV radiation photolytically cleaves C-Cl bonds, releasing reactive **free chlorine radicals (Cl^\bullet)**: CF_2Cl_2 \xrightarrow{h\nu} CF_2Cl^\bullet + Cl^\bullet

Cl^\bullet + O_3 \to ClO^\bullet + O_2 ClO^\bullet + O \to Cl^\bullet + O_2

- *Catalytic Cycle*: The chlorine radical is continually regenerated; **a single chlorine atom can catalytically destroy over 100,000\text{ ozone molecules}** before being removed by reservoir species (HCl, ClONO_2).

33. **The Antarctic Ozone Hole**:
    - Discovered in 1985 by British Antarctic Survey scientists (Joe Farman, Brian Gardiner, and Jonathan Shanklin) over Halley Bay.
    - Develops intensely each austral spring (September–October) due to unique meteorological conditions: the **Polar Vortex** isolates Antarctic air, chilling temperatures below -80^\circ\text{C} and forming **Polar Stratospheric Clouds (PSCs / nacreous clouds)** of nitric acid trihydrate. PSC surfaces catalyze the conversion of inactive chlorine reservoirs (HCl, ClONO_2) into active molecular chlorine (Cl_2), which is photolyzed by returning spring sunlight into massive bursts of destructive Cl^\bullet radicals.
34. **Montreal Protocol**:
    - Landmark international environmental treaty adopted on **September 16, 1987** (celebrated annually as **World Ozone Day**); successfully phased out the global production and consumption of ozone-depleting substances (CFCs, halons, carbon tetrachloride).
    - **Kigali Amendment (2016)**: Mandates the phased reduction of Hydrofluorocarbons (HFCs), which do not deplete ozone but are potent greenhouse gases.
35. **Greenhouse Effect**:
    - The natural thermodynamic phenomenon discovered by Joseph Fourier (1824) and quantified by Svante Arrhenius (1896).
    - Earth absorbs short-wavelength solar radiation (visible and UV) and re-radiates it toward space as **long-wavelength infrared (thermal IR) radiation (\lambda \approx 4–50\text{ }\mu\text{m})**.
    - Polyatomic greenhouse gas molecules absorb outgoing thermal IR photons via molecular bond stretching and bending vibrations, trapping heat within the lower troposphere and raising Earth's mean surface temperature from -18^\circ\text{C} (inhospitable ice world) to a habitable **+15^\circ\text{C}**.
36. **Master Greenhouse Gases (GHGs)**:
    - **Water Vapor (H_2O)**: The largest natural contributor to the natural greenhouse effect (\approx 60\%).
    - **Carbon Dioxide (CO_2)**: Responsible for \approx 26\% of overall warming; baseline reference gas (\text{GWP} = 1.0).
    - **Methane (CH_4)**: Responsible for \approx 9\%; \text{GWP}_{100} \approx 28–36.
    - **Nitrous Oxide (N_2O)**: \text{GWP}_{100} \approx 265–298; emitted from nitrogenous fertilizers.
    - **Fluorinated Gases**: Hydrofluorocarbons (HFCs), Perfluorocarbons (PFCs), and **Sulphur Hexafluoride (SF_6, highest GWP \approx 23,500)**.
37. **Greenhouse Non-Absorbers (State PCS Distractor Trap)**:
    - **Argon (Ar)**, **Nitrogen (N_2)**, and **Oxygen (O_2)** are **NOT greenhouse gases**!
    - Homonuclear diatomic molecules (N_2, O_2) and noble gases (Ar, He) possess no permanent dipole moment, and symmetrical vibrations produce zero dynamic dipole change; therefore, they are transparent to infrared radiation and cannot absorb thermal IR.
38. **Minamata Disease (Methylmercury Toxicity)**:
    - Severe neurological poisoning caused by chronic ingestion of fish contaminated with **Methylmercury (CH_3Hg^+)**, first identified in 1956 around Minamata Bay, Kumamoto Prefecture, Japan.
    - Industrial effluent from the Chisso chemical factory discharged inorganic mercury into the bay, where anaerobic sulfate-reducing bacteria methylated it into lipid-soluble methylmercury.
    - Biomagnified through marine food webs into human consumers, crossing the blood-brain barrier to cause ataxia, peripheral numbness, tunnel vision, muscle weakness, and congenital neurological deformities.

39. **Itai-Itai ("Ouch-Ouch") Disease (Cadmium Toxicity)**:
    - Mass chronic **Cadmium (Cd) poisoning** documented along the Jinzū River basin in Toyama Prefecture, Japan, in 1912.
    - Upstream mining runoff contaminated river water used to irrigate downstream paddy fields.
    - Cadmium replaced calcium in the human skeletal matrix, causing severe **osteomalacia**, bone softening, multiple pathological fractures, and severe renal tubular failure accompanied by excruciating skeletal pain ("itai-itai").
40. **Blackfoot Disease (Chronic Arsenic Toxicity)**:
    - Caused by prolonged consumption of drinking water contaminated with high levels of inorganic **Arsenic (As > 0.05\text{ mg/L})**, endemic in the Bengal delta basin (West Bengal and Bangladesh).
    - Arsenic binds to cellular sulfhydryl groups, disrupting cellular respiration and inducing peripheral vascular gangrene (causing blackened, necrotic feet that require amputation), hyperkeratosis of palms/soles, and skin cancers.
41. **Blue Baby Syndrome (Methemoglobinemia / Nitrate Poisoning)**:
    - Occurs in human infants under 6 months of age who consume infant formula prepared with well water containing excess **Nitrate (NO_3^- > 45\text{ mg/L})** from agricultural fertilizer runoff.
    - Intestinal bacteria reduce nitrate to nitrite (NO_2^-), which oxidizes the ferrous iron (Fe^{2+}) in normal hemoglobin to ferric iron (Fe^{3+}), generating **Methemoglobin**.
    - Methemoglobin cannot bind oxygen, causing systemic tissue hypoxia and characteristic infant cyanosis (turning skin a bluish-slate color), which can be fatal without methylene blue therapy.
42. **Skeletal & Dental Fluorosis**:
    - Caused by excessive fluoride concentration in drinking water (**>1.5\text{ mg/L} or ppm**); widely prevalent across Rajasthan, Gujarat, and Uttar Pradesh.
    - At optimal concentrations (0.7–1.0\text{ ppm}), fluoride converts dental hydroxyapatite to acid-resistant fluorapatite (Ca_5(PO_4)_3F), preventing cavities.
    - Above 1.5\text{ ppm}, it causes **Dental Fluorosis** (permanent brownish-yellow mottling of tooth enamel).
    - Above 4.0\text{ ppm}, it deposits in bones, causing **Skeletal Fluorosis** (calcification of spinal ligaments, severe joint stiffness, and debilitating "knock-knee" deformity / genu valgum).
43. **Lead (Pb) Toxicity (Plumbism / Saturnism)**:
    - Toxic heavy metal that mimics calcium; crosses the blood-brain barrier, causing cognitive impairment, lowered IQ, behavioral deficits in children, and microcytic anemia by inhibiting the heme-synthesis enzyme delta-aminolevulinic acid dehydratase (ALAD). Characterized by a dark blue line along the gums (**Burton's Line**).
44. **Biochemical Oxygen Demand (BOD)**:
    - The amount of dissolved oxygen (\text{mg/L} or \text{ppm}) required by aerobic microorganisms to biologically oxidize and decompose organic matter present in a unit volume of water over 5 days at 20^\circ\text{C} (\text{BOD}_5).
    - Pure drinking water has a \text{BOD} < 1\text{ ppm}; highly polluted domestic sewage has a \text{BOD} > 100–400\text{ ppm}. High BOD indicates severe organic pollution and depletion of dissolved oxygen (DO), causing fish kills.
45. **Chemical Oxygen Demand (COD)**:
    - The amount of oxygen consumed to chemically oxidize both biodegradable and non-biodegradable organic matter in water using a powerful chemical oxidizing agent (**Potassium Dichromate, K_2Cr_2O_7**, in acidic medium). For any polluted wastewater, **\text{COD} is always greater than \text{BOD}** (\text{COD} > \text{BOD}).

46. **Eutrophication**: The accelerated aging of an aquatic ecosystem (lakes, ponds) caused by nutrient enrichment with agricultural **Nitrates (NO_3^-) and Phosphates (PO_4^{3-})** from fertilizers and detergents, triggering dense **algal blooms** that deplete dissolved oxygen and suffocate aquatic life.
47. **Tear Gas Chemistry**: Chemical riot-control agents: **CS Gas** (2-chlorobenzalmalononitrile), **CN Gas** (chloroacetophenone), and **Chloropicrin (CCl_3NO_2)**; react with moisture on mucous membranes to trigger intense eye burning and lachrymation.
48. **Mustard Gas (Dichlorodiethyl Sulphide)**: An oily, volatile blister agent (vesicant) chemical weapon; causes severe, slow-healing chemical burns and ulcerations on skin, eyes, and lungs.
49. **Lewisite**: An arsenic-based blistering chemical warfare agent (ClCH=CH-AsCl_2); antidote is British Anti-Lewisite (BAL / Dimercaprol).
50. **Bhopal Gas Tragedy (MIC Chemistry)**: On December 2–3, 1984, in Bhopal, water accidentally entered a storage tank containing over 40 tonnes of liquid **Methyl Isocyanate (CH_3NCO)**. The resulting exothermic hydrolysis reaction caused temperatures to exceed 200^\circ\text{C}, bursting the relief valve and releasing toxic MIC gas across the city, resulting in thousands of fatalities.

One-Liner Revision – Radiation, Nuclear and Environmental Chemistry

#	Fast Fact
1	Ionising radiation: alpha, beta, gamma, X-rays, cosmic rays – can cause cancer
2	Non-ionising: radio, microwave, visible, UV-A – generally safer (UV-B intermediate)
3	UV-B (280-315 nm) = causes sunburn, skin cancer; blocked by ozone layer
4	UV-A (315-400 nm) = causes tanning; glass blocks UV; used in sterilisation
5	Microwave oven = 2.45 GHz; water molecules absorb and vibrate -> heat food
6	C-14 half-life = **5730 years**; used for dating organic material
7	Radioactive decay: alpha (weakest penetration), beta, gamma (strongest penetration)
8	Radiation doses: mSv (millisievert); background dose ~2-3 mSv/year
9	Radiation therapy = high-dose gamma/X-ray to kill cancer cells
10	Chernobyl (1986) = RBMK reactor explosion; released Cs-137 and I-131
11	Acid rain: SO2 + H2O -> H2SO3; SO3 + H2O -> H2SO4; causes damage to Taj Mahal marble
12	Acid rain pH < 5.6 (normal rain = 5.6 due to natural CO2)
13	London smog = industrial smog (SO2 + soot + fog); cold; 1952 Great Smog killed 4,000+
14	LA smog = photochemical smog (NOx + VOC + sunlight -> O3 + PAN); warm, sunny
15	Ozone (O3) in troposphere = pollutant (causes respiratory problems); in stratosphere = protective
16	PM2.5 = particulate matter < 2.5 micron; deepest lung penetration; most harmful
17	PM10 = particles < 10 micron; cause respiratory problems
18	BOD: high BOD = more organic pollution; clean water BOD < 3 mg/L
19	COD (Chemical Oxygen Demand) = total oxygen needed to chemically oxidise all organics
20	Eutrophication: excess N and P (from fertiliser runoff) -> algal bloom -> dead zone
21	Biomagnification = persistent pollutants (DDT, Hg, PCBs) concentrate up the food chain
22	Mercury (Hg) poisoning = Minamata disease (Japan, 1950s); methylmercury in fish
23	Cadmium poisoning = Itai-itai disease (Japan); bone pain; from mine runoff
24	Lead (Pb) poisoning = cognitive damage in children; banned from petrol (TEL) and paint
25	Arsenic contamination = groundwater in Bangladesh, West Bengal; skin lesions, cancer

#	Fast Fact
26	Fluorosis = excess fluoride (>1.5 ppm) = mottled teeth (dental); skeletal damage (skeletal)
27	e-waste = toxic heavy metals; India = 3rd largest e-waste generator globally
28	Basel Convention (1989) = controls transboundary movement of hazardous wastes
29	Stockholm Convention (2001) = bans persistent organic pollutants (POPs) like DDT, PCBs
30	CPCB = Central Pollution Control Board (India); SPCB = State Pollution Control Board

Chemistry Topic 9 — Discoveries, Discoverers and Miscellaneous Chemistry

Consolidated — 50 Must-Score Facts

1. **Antoine Laurent Lavoisier (1743–1794)**: Revered as the **"Father of Modern Chemistry"**. Formulated the fundamental **Law of Conservation of Mass** (1789), proved that combustion and respiration are chemical combinations with oxygen, disproved the ancient phlogiston theory, named the elements Oxygen and Hydrogen, and published the first modern list of chemical elements.
2. **John Dalton (1766–1844)**: Formulated the modern **Atomic Theory (1808)** (matter consists of indivisible atoms); established the **Law of Multiple Proportions** and **Dalton's Law of Partial Pressures**.
3. **Joseph Priestley (1774)** and **Carl Wilhelm Scheele (1772)**: Independently discovered **Oxygen gas (O_2)**. Priestley termed it "dephlogisticated air" by heating mercuric oxide; Lavoisier recognized it as an elemental gas and named it oxygen (meaning "acid-former").
4. **Henry Cavendish (1766)**: Discovered **Hydrogen gas (H_2)**, calling it "inflammable air" by reacting zinc with dilute sulphuric acid; later proved that burning hydrogen in air produces water, demonstrating that water is a compound, not an element.
5. **Humphry Davy (1778–1829)**: Pioneer of electrochemistry; used electric battery currents to isolate elemental **Sodium (Na), Potassium (K), Calcium (Ca), Magnesium (Mg), Strontium (Sr), and Barium (Ba)**; established that chlorine is an element; invented the **Davy Miner's Safety Lamp** to prevent firedamp explosions in coal mines; discovered the anesthetic properties of nitrous oxide.
6. **Friedrich Wöhler (1828)**: Synthesized **Urea (NH_2CONH_2)** by heating inorganic ammonium cyanate (NH_4CNO): NH_4CNO \xrightarrow{\Delta} NH_2CONH_2 This landmark synthesis disproved Jöns Jacob Berzelius's **"Vital Force Theory" (Vis Vitalis)**, establishing that organic compounds can be synthesized from purely inorganic laboratory reagents.
7. **Jöns Jacob Berzelius (1779–1848)**: Introduced the modern system of **chemical symbols using Latin letters** (H, O, C, Fe, Au); determined accurate atomic weights; discovered the elements Cerium, Selenium, Silicon, and Thorium; coined the terms *catalysis*, *isomerism*, *allotropy*, and *protein*.
8. **August Kekulé (1865)**: Proposed the cyclic hexagonal ring structure of **Benzene (C_6H_6)** with alternating double bonds, inspired by a famous daydream of a snake biting its own tail (Ouroboros); established the tetravalency of carbon.
9. **Harold Clayton Urey (1931)**: Discovered **Deuterium (^2H or D)**, the heavy isotope of hydrogen, by fractional distillation of liquid hydrogen (Nobel Prize in Chemistry, 1934).
10. **Linus Pauling (1901–1994)**: Formulated the quantum concepts of **orbital hybridization (sp, sp^2, sp^3)**, the nature of the chemical bond, the **Pauling Electronegativity Scale** (Fluorine = 4.0), and the \alpha-helical secondary structure of proteins (Nobel Prize in Chemistry, 1954; Nobel Peace Prize, 1962).
11. **Acharya Prafulla Chandra Ray (P. C. Ray, 1861–1944)**:
    - Revered as the **"Father of Indian Chemistry"**.
    - Discovered and synthesized **Mercurous Nitrite (Hg_2(NO_2)_2)** in 1896, earning global acclaim as the "Master of Nitrites".
    - Founded **Bengal Chemical & Pharmaceutical Works (1901)** in Calcutta, India's first modern indigenous chemical and pharmaceutical enterprise.
    - Authored the comprehensive historical treatise *A History of Hindu Chemistry* (1902).
12. **Sir Shanti Swaroop Bhatnagar (1894–1955)**:
    - Pioneer of magneto-chemistry; invented the high-sensitivity **Bhatnagar-Mathur Magnetic Interference Balance**.

- First Director-General of the **Council of Scientific and Industrial Research (CSIR)**; spearheaded the establishment of national research laboratories across India. The prestigious **SSB Prize** is named in his honor.

13. **Chintamani Nagesa Ramachandra Rao (C. N. R. Rao, b. 1934)**:
    - Internationally recognized solid-state and structural chemist; authority on transition metal oxides, high-temperature superconductivity, and carbon nanomaterials (graphene, nanotubes). Awarded India's highest civilian honor, the **Bharat Ratna (2014)**.
14. **Asima Chatterjee (1917–2006)**:
    - Renowned Indian organic chemist; first woman to receive a Doctor of Science (D.Sc.) degree from an Indian university (Calcutta University, 1944).
    - Pioneered research on vinca alkaloids from *Catharanthus roseus*, developed the anti-epileptic formulation **Ayush-56**, and anti-malarial herbal drugs.
15. **Water Chemistry — Hardness of Water**:
    - **Hard Water**: Water that does not produce lather easily with soap, but forms a sticky, insoluble white precipitate called **Scum** (calcium and magnesium stearates).
    - Caused by the presence of dissolved polyvalent metallic cations, predominantly **Calcium (Ca^{2+})** and **Magnesium (Mg^{2+})**.
16. **Temporary Hardness (Carbonate Hardness)**:
    - Caused by dissolved **Bicarbonates (Hydrogen carbonates) of Calcium and Magnesium**: Ca(HCO_3)_2 and Mg(HCO_3)_2.
    - Can be removed by simple physical or mild chemical methods: 1. **Boiling**: Converts soluble bicarbonates into insoluble carbonates/hydroxides that precipitate out: Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3\downarrow + H_2O + CO_2 \uparrow Mg(HCO_3)_2 \xrightarrow{\Delta} Mg(OH)_2\downarrow + 2CO_2 \uparrow 2. **Clark's Method**: Adding a calculated quantity of **Slaked Lime (Ca(OH)_2)**: Ca(HCO_3)_2 + Ca(OH)_2 \to 2CaCO_3\downarrow + 2H_2O Mg(HCO_3)_2 + 2Ca(OH)_2 \to 2CaCO_3\downarrow + Mg(OH)_2\downarrow + 2H_2O
17. **Permanent Hardness (Non-Carbonate Hardness)**:
    - Caused by dissolved **Chlorides and Sulphates of Calcium and Magnesium**: CaCl_2, MgCl_2, CaSO_4, and MgSO_4.
    - Cannot be removed by boiling. Requires chemical or ion-exchange treatment: 1. **Washing Soda Method**: Adding calculated sodium carbonate (Na_2CO_3): CaCl_2 + Na_2CO_3 \to CaCO_3\downarrow + 2NaCl MgSO_4 + Na_2CO_3 \to MgCO_3\downarrow + Na_2SO_4 2. **Calgon's Method**: Adding **Sodium Hexametaphosphate (Na_6P_6O_{18} or (NaPO_3)_6, Calgon = Calcium Gone)**: 2Ca^{2+} + Na_2[Na_4(PO_3)_6] \to Na_2[Ca_2(PO_3)_6]\text{ (Soluble complex)} + 4Na^+ The Ca^{2+} and Mg^{2+} ions are sequestered inside a soluble complex anion and cannot precipitate with soap. 3. **Permutit / Zeolite Process**: Passing hard water through a bed of hydrated sodium aluminium silicate (**Zeolite / Permutit**, Na_2O \cdot Al_2O_3 \cdot xSiO_2 \cdot yH_2O): 2Na\text{-Zeolite} + Ca^{2+} \to Ca\text{-(Zeolite)}_2 + 2Na^+ The exhausted zeolite bed is easily regenerated by backwashing with a **10\% aqueous common salt (NaCl) brine solution**. 4. **Synthetic Ion-Exchange Resin Method**: Uses cation-exchange resins (R-SO_3H) to exchange Ca^{2+}, Mg^{2+} for H^+, followed by anion-exchange resins (R-NH_3OH) to exchange Cl^-, SO_4^{2-} for OH^-. Yields completely demineralized, deionized pure water.
18. **Heavy Water (D_2O, Deuterium Oxide)**:
    - Discovered in 1932 by Harold C. Urey; first isolated in pure form by Gilbert N. Lewis (1933) by prolonged electrolysis of alkaline water.
    - Molecular mass = **20.03\text{ g/mol}** (compared to 18.015\text{ g/mol} for normal light water H_2O).
    - Physical properties: Denser than water (1.105\text{ g/cm}^3 at 20^\circ\text{C}); higher freezing point (**3.82^\circ\text{C}**); higher boiling point (**101.42^\circ\text{C}**); higher viscosity.

19. **Heavy Water in Nuclear Power Plants**:
    - Used as a **Neutron Moderator** and primary **Coolant** in **Pressurized Heavy Water Reactors (PHWRs)**, which form the bedrock of India's commercial nuclear power fleet (e.g., Rawatbhata, Kakrapar, Narora, Kaiga).
    - *Why Heavy Water is an Ideal Moderator*: Deuterium has a very low neutron capture cross-section (0.0005\text{ barns}) compared to normal hydrogen (0.33\text{ barns}), allowing PHWRs to operate efficiently using **natural, unenriched uranium (0.7\% {}^{235}U)** as fuel.
20. **Heavy Water Production in India**: Managed by the **Heavy Water Board (HWB)** under the Department of Atomic Energy (DAE). Major production facilities operate at **Manuguru (Telangana)** and **Thal (Maharashtra)** utilizing the Girdler-Sulphide (H_2S-H_2O) and ammonia-hydrogen exchange processes.
21. **Biological Toxicity of Heavy Water**: Ingesting high concentrations of D_2O is toxic to animals and plants. Deuterium forms stronger hydrogen bonds than protium, altering the 3D conformation of enzymes and significantly slowing metabolic biochemical reaction rates.
22. **Superheavy Water (T_2O, Tritium Oxide)**: Formed with radioactive Tritium (^3H); molecular mass \approx 22\text{ g/mol}. Tritium is a beta emitter with a half-life of 12.32\text{ years}.
23. **Hydrogen Peroxide (H_2O_2)**:
    - Discovered in 1818 by French chemist Louis Jacques Thénard.
    - Non-planar open-book structure with a dihedral angle of 94.8^\circ in gas phase.
    - Dilute 3\% solution (10\text{ volume } H_2O_2) is sold under the trade name **Perhydrol** as a mild antiseptic for cleansing septic cuts and bleaching delicate hair and ivory.
    - Concentrated (>90\%) H_2O_2 serves as a monopropellant and oxidizer in aerospace rocket engines.
24. **Lunar Caustic**: Chemically **Silver Nitrate (AgNO_3)**.
    - Discovered by medieval alchemists; named "lunar" because silver was historically associated with the Moon.
    - When applied to human skin in aqueous solution, it oxidizes epidermal organic matter and reacts with trace light to deposit a dark, indelible stain of **finely divided metallic Silver (Ag)**.
    - Essential chemical ingredient in **Indelible Electoral Voter Marking Ink** used by the Election Commission of India.
25. **Red Lead (Sindoor / Minium)**: Chemically **Trilead Tetroxide (Pb_3O_4)**.
    - A bright scarlet-red powder; mixed-valence oxide containing 2PbO \cdot PbO_2.
    - Used traditionally as vermilion/sindoor in Hindu rituals and industrially as an anti-corrosion rust-inhibiting primer paint for structural iron and steel bridges.
26. **Litharge**: Chemically **Lead Monoxide (PbO)**, yellowish-red powder used in lead glass and battery plates.
27. **Quicksilver**: Common historic name for elemental **Mercury (Hg)**, reflecting its fluid liquid mobility and bright silvery metallic luster.
28. **Vermilion (Cinnabar)**: Chemically **Mercuric Sulphide (HgS)**; deep crimson-red mineral pigment.
29. **Calomel**: Chemically **Mercurous Chloride (Hg_2Cl_2)**; white, sparingly soluble powder used historically as a medicinal purgative laxative and modernly as the standard **Calomel Reference Electrode** in pH meters and electrochemistry.
30. **Corrosive Sublimate**: Chemically **Mercuric Chloride (HgCl_2)**; an extremely toxic white crystalline poison used in taxidermy and wood preservation.
31. **Eosin**: A bright red water-soluble fluorescent dye (C_{20}H_6Br_4Na_2O_5, tetrabromofluorescein); standard coloring pigment in commercial **Red Ink** and histological biological tissue stains.
32. **Pearl Ash**: Commercial trade name for **Potassium Carbonate (K_2CO_3)**, historically extracted by leaching wood ashes in large iron pots (giving origin to the element name "potassium").
33. **Saltpetre (Nitre)**: Chemically **Potassium Nitrate (KNO_3)**; key oxidizer in gunpowder.
34. **Chile Saltpetre**: Chemically **Sodium Nitrate (NaNO_3)**; mined extensively in the arid Atacama Desert of Chile; used in nitrogen fertilizers and nitric acid manufacture.

35. **Sal Ammoniac**: Chemically **Ammonium Chloride (NH_4Cl)**; white subliming salt used as an electrolyte in Leclanché dry cells and as a soldering flux to clean metal surfaces.
36. **Laughing Gas**: **Nitrous Oxide (N_2O)**; discovered by Joseph Priestley (1772); induces euphoric mild hysteria and analgesia; used in dental minor anesthesia.
37. **Dry Ice**: **Solid Carbon Dioxide (CO_2)**; sublimes directly into gaseous CO_2 at **-78.5^\circ\text{C} (-109.3^\circ\text{F})** at 1\text{ atm} without leaving liquid residue; widely used for commercial cold-chain transport of frozen foods and vaccines.
38. **Baryta Water**: An aqueous solution of **Barium Hydroxide (Ba(OH)_2)**; used as an analytical reagent to test for and absorb carbon dioxide, forming insoluble white barium carbonate (BaCO_3\downarrow).
39. **Barium Meal (Barytes)**: Chemically **Barium Sulphate (BaSO_4)**. Completely insoluble in water and digestive acids; opaque to X-rays (high atomic number of Barium, Z=56). Administered orally to patients as a radiopaque contrast suspension for diagnostic **X-ray and CT imaging of the stomach and intestinal tract** without heavy metal toxicity.
40. **Blue Vitriol**: **Copper Sulphate Pentahydrate (CuSO_4 \cdot 5H_2O)**; agricultural fungicide (Bordeaux mixture) and copper electroplating bath.
41. **Green Vitriol**: **Ferrous Sulphate Heptahydrate (FeSO_4 \cdot 7H_2O)**; iron dietary supplement and black ink dye.
42. **White Vitriol**: **Zinc Sulphate Heptahydrate (ZnSO_4 \cdot 7H_2O)**; lithopone white pigment and ophthalmic astringent.
43. **Epsom Salt**: **Magnesium Sulphate Heptahydrate (MgSO_4 \cdot 7H_2O)**; saline osmotic laxative and soothing bath salts.
44. **Potash Alum (Fitkari)**: **K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O**; water-clarifying coagulant and hemostatic styptic.
45. **Glauber's Salt**: Chemically **Sodium Sulphate Decahydrate (Na_2SO_4 \cdot 10H_2O)**; named after German-Dutch alchemist Johann Rudolf Glauber (1625); medicinal purgative.
46. **Philosopher's Wool**: Chemically **Zinc Oxide (ZnO)**; fluffy white residue formed when metallic zinc burns vigorously in air; used in calamine ointments, sunscreen lotions (UV blocker), and white paints.
47. **Quicklime vs Slaked Lime vs Milk of Lime**:
    - Quicklime: Anhydrous Calcium Oxide (**CaO**).
    - Slaked Lime: Calcium Hydroxide (**Ca(OH)_2**).
    - Milk of Lime: A milky white suspension of slaked lime in water.
    - Lime Water: The clear, filtered aqueous supernatant solution of Ca(OH)_2 (turns milky with CO_2).
48. **Invert Sugar**: An equimolar mixture of **D-glucose and D-fructose** obtained by hydrolyzing cane sugar (sucrose); optical rotation inverts from +66.5^\circ (dextro) to -19.8^\circ (levo); present naturally in bee honey.
49. **Nobel Prize in Chemistry 2021**: Awarded to **Benjamin List** and **David MacMillan** for the development of **Asymmetric Organocatalysis** (using small, non-toxic organic molecules like proline instead of expensive heavy metal catalysts to construct chiral medicinal drugs).
50. **Nobel Prize in Chemistry 2022**: Awarded to **Carolyn R. Bertozzi**, **Morten Meldal**, and **K. Barry Sharpless** for the development of **Click Chemistry and Bioorthogonal Chemistry** (snapping molecular building blocks together cleanly inside living cells without disrupting cellular biology). K. Barry Sharpless joined the elite rank of two-time Nobel laureates (first in 2001).

One-Liner Revision – Chemistry Discoveries and Miscellaneous

#	Fast Fact
1	"Father of Modern Chemistry" = **Antoine Lavoisier** (France); named oxygen, hydrogen
2	Law of Conservation of Mass = **Lavoisier**; mass is neither created nor destroyed

#	Fast Fact
3	Dalton's Atomic Theory (1803) = atoms indivisible; each element has unique identical atoms
4	Avogadro's Number = **6.022 x 10^23** entities per mole
5	Haber Process: N2 + 3H2 = 2NH3; 400-500 degC; 200 atm; Fe catalyst; makes fertiliser
6	Contact Process: SO2 + O2 = SO3 (V2O5 catalyst); SO3 + H2SO4 = oleum -> H2SO4
7	Solvay Process = produces Na2CO3 (washing soda) and NaHCO3 (baking soda) industrially
8	Ostwald Process = 4NH3 + 5O2 = 4NO + 6H2O (Pt-Rh catalyst, 850 degC); makes HNO3
9	Oxygen discovered = **Carl Scheele (1772)** and Joseph Priestley (1774); named by Lavoisier
10	Hydrogen discovered = **Henry Cavendish (1766)**; named "inflammable air"
11	Nitrogen discovered = **Daniel Rutherford (1772)**; Lavoisier named it
12	Chlorine discovered = **Carl Wilhelm Scheele (1774)**; named by Humphry Davy
13	Dry ice = solid CO2 (-78.5 degC); sublimes (solid to gas directly); used in fog machines
14	Heavy water (D2O) = deuterium oxide; moderator in CANDU and Indian PHWR reactors
15	Laughing gas = N2O (nitrous oxide); anaesthetic; also greenhouse gas
16	Helium = lightest noble gas; non-flammable; used in balloons, MRI cooling
17	Noble gases discovered: Argon = Rayleigh + Ramsay (1894); Helium = Ramsay (1895)
18	Neon lights = neon gas in tube glows red-orange; other colours use different gases
19	Ozone (O3) = allotrope of oxygen; formed by UV + O2; both protective (stratosphere) and pollutant (troposphere)
20	Le Chatelier's Principle: equilibrium shifts against applied stress (T, P, concentration)
21	Law of mass action: rate = k[A]^m[B]^n (reactant concentrations)
22	pH of blood = 7.4; gastric acid = 1.5-3.5; milk = 6.5-6.7; lemon juice = 2-3
23	Galvanic cell = chemical energy -> electrical energy; Daniel cell = Zn-Cu
24	Electrolytic cell = electrical energy -> chemical energy; electroplating
25	Faraday's Laws of Electrolysis: 1st = mass proportional to charge; 2nd = mass proportional to equivalent weight
26	Electroplating = depositing thin layer of metal (Ag, Au, Cr, Ni) on another object
27	Anodising = electrochemical oxidation of aluminium surface; corrosion protection
28	Catalytic converter in cars = Pt-Rh-Pd catalyst; converts CO, NOx, VOCs to CO2, N2, H2O
29	Green Chemistry 12 principles = prevent waste; atom economy; safer chemicals; etc. (Paul Anastas)
30	India's 1st Chemistry Nobel (so far) = none; C.V. Raman (1930) was in Physics